

Diophantine Laws of Probabilistic, Quantum and Continuous Computation

Normalization budgets and unique quartic certificates,
four-dimensional quantum mortality, a continuous three-outcome trichotomy,
exact probabilities, exact gates and contracting dynamics

Research report merged from eleven manuscripts prepared for Vladimir Reshetnikov
(batch 60, manuscripts 07, 08 and 09; batch 62, manuscripts 06–09 and 11–14; all dated 30
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AI-assisted mathematical development and reproducible exact-arithmetic implementations; unrefereed

Abstract

[write] Which quantitative assertions about computation that is *not* a plain discrete execution log admit finite natural-number witnesses of integer polynomials? Part I (manuscript 08, the base text) represents the output law of every effective probabilistic or quantum program as the dyadic mass of a prefix-free projection of one fixed quartic zero set; determines the exact mass budget of a prescribed history of normalized output laws, the product of the successive largest coordinate ratios; compiles its feasibility into an explicit quartic with $5Nm + m + 4N + 3$ natural witnesses and exactly one witness per valid feasible input; and proves that normalized two-outcome probabilities with any success floor below one are exactly the weakly computable reals, whereas strict conditional comparisons are Σ_2^0 -complete. Part II adds exact Clifford+ T arithmetic and an operator-valued budget (08), and (manuscript 07) a rational Gram-packing construction under which deciding whether a seven-outcome rational quantum instrument in dimension four has an impossible outcome word is undecidable, given Neary’s published matrix-mortality theorem, with bijective quartics of $71n + 32$ witnesses for impossible words of length n . Part III (manuscript 09) computes the critical per-step noise radius of a continuous rational piecewise-affine map to within $(L + 1)/(2N)$ independently of time, gives complete quartic safety certificates for positive radius, and builds one map of $[0, 1]^3$ that realizes every disjoint pair of computably enumerable sets as exact acceptance and robust rejection, the rest being noise-fragile divergence. Parts V–VII (batch 62) turn fair-coin halting probabilities into root counts of uniformly well-conditioned quartics and exact stationary masses of rapidly mixing reversible chains into carriers of halting and of a rational/Liouville dichotomy (V); certify whole coherent tensor contractions with unique natural witnesses, and compile finite group presentations into rational four-dimensional rotations whose exact synthesis or state transfer is c.e.-complete (VI); and build strict contractions of $\mathbb{R}^4$ and of $\mathbb{Z}_2$ whose exact transients are c.e.-complete, and exact rational neural networks whose precision is unbounded memory (VII).

Status. [write] All eleven manuscripts are AI-assisted and unrefereed. They give ordinary mathematical proofs, apart from the external inputs they name (MRDP; Neary’s six-generator matrix-mortality theorem; Meyer’s and Lagrange’s theorems on quadratic forms; the existence of a universal Turing machine; the max–min representation of piecewise-affine functions; in Parts V–VII also Higman’s embedding theorem, Mihailova’s construction, Tarski’s decision method, Minsky’s counter machines and the Adamczewski–Bugeaud theorem), and finite exact-arithmetic checks. No result of this report is formalized in Lean or Rocq; placement beside ProveIt’s Lean project on Hilbert’s tenth problem confers no formal status (Section 1.3). No historical priority, minimal dimension or optimal constant is claimed, and no general single-fold or finite-fold MRDP question is settled. The status statements of the three batch-60 sources are printed in full in Section 1.6, and those of the eight batch-62 sources at the head of Parts V–VII.

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1 Research target and relation to ProveIt

1.1 What is being represented?

A computational substrate may manipulate tape symbols, lambda terms, fractions, rewrite graphs, amplitudes, or physical-looking states. If its finite executions are effective, existential arithmetic can describe their occurrence. This observation alone does not preserve the quantitative semantics of probabilistic computation. A polynomial may have many witnesses for one execution, and infinitely many distinct executions may contribute to an output probability. Quantum computational paths can interfere, so summing their squared amplitudes individually is generally wrong.

This article separates four questions. First, can one recover the *whole discrete output law* from integer polynomial solvability? Second, which probability comparisons are themselves existential Diophantine? Third, what is the exact probability mass required to realize a changing conditional output distribution? Fourth, can this last optimization problem have a small, explicit polynomial certificate with uniquely determined witnesses? The answers form a coherent extension of the finite-trace viewpoint.

Our strongest concrete construction is not merely an application of MRDP. It is a direct quartic compiler for optimal normalization feasibility, including precise witness and equation counts, invalid-input rejection, canonical tie resolution, and a uniqueness proof. The infinite and quantum extensions explain why that finite compiler is useful.

1.2 Repository boundary

The inspected ProveIt snapshot is

f608f1cb3c5be8a736df1328c01b293aabfccf4e.

Its Hilbert-tenth-problem development exposes the theorems `mrdp`, `mrdp_iff`, and `mrdp_dioph_iff` in the `Diophantine` namespace. In particular, the source of `Diophantine/MRDP.lean` states the equivalence between recursively enumerable subsets of $\mathbb{N}$ and projections of zero sets of integer polynomials with a finite natural witness dimension, chosen before the input. The statement includes input zero [1, 2]. We use exactly that mathematical interface, not the correctness of every other research claim in the repository.

The repository also supplies pairing, primitive-recursive graph, and trace modules adjacent to this interface. Those are natural integration points for the proposed compiler. This manuscript does not report a local rebuild of ProveIt or an axiom audit of newly supplied declarations: it supplies no new kernel-checked declarations. Its mathematical proofs are independent of a choice of proof-assistant syntax.

[write] **Manuscript 07's reading of the repository.** The inspected repository snapshot is

f608f1cb3c5be8a736df1328c01b293aabfccf4e.

The relevant material is in `Computability/HilbertTenthProblem`. Its MRDP guide states an interface that takes a recursively enumerable subset of $\mathbb{N}$ to the projection of the natural zero set of a fixed integer polynomial with finitely many witnesses. The guide explicitly distinguishes this existence theorem from a practical polynomial compiler with numerical resource bounds [14]. Its

shared trace module also exposes Diophantine closure under bounded universal quantification, exact iteration, and finite reachability [15].

This manuscript supplies an application at two levels. At the explicit level, matrix multiplication histories become quartic equations with counted, canonical witnesses. At the unbounded level, an enumerator of impossible words fits the existing MRDP interface. The rational Gram argument is a separate mathematical component and is not claimed to have already been formalized in the repository. No repository file has been changed, and no repository build has been run for this manuscript.

[write] Manuscript 09’s reading of the repository. The inspected ProveIt modules provide both directions of MRDP over natural inputs and natural witnesses. The declaration `Diophantine.mrdp` supplies an integer polynomial with finitely many witness coordinates, uniformly before the input is chosen; `Diophantine.mrdp_iff` supplies the converse [1, 14]. Thus, for a computably enumerable set $S \subseteq \mathbb{N}$,

$$n \in S \iff \exists w \in \mathbb{N}^k P(n, w) = 0, \quad P \in \mathbb{Z}[X, Y_1, \dots, Y_k]. \quad (1.1)$$

Input zero and zero-valued witnesses are allowed. Polynomial evaluation is in $\mathbb{Z}$, so subtraction never means truncated natural subtraction.

The additional work here lies before that interface: an exact rational abstraction, a complete finite safety checker, and a universal realization with quantitatively controlled noise. We do not use the repository’s optimized universal-equation operation counts or infer witness bounds from them. The guide explicitly distinguishes its existential MRDP endpoint from a practical polynomial-extraction algorithm. We preserve that distinction.

The repository was inspected on 30 September 2026. The head observed during inspection was `f608f1cb3c5be8a736df1328c01b293aabfccf4e`. The source declarations were read, but the repository was not rebuilt in this session. Accordingly, repository build and axiom-audit claims remain repository provenance, not a fresh verification receipt for this manuscript.

1.3 [write] What ProveIt has formalized, and what this report uses

This report continues the Lean project `Computability/HilbertTenthProblem` only through MRDP. Its fixed-arity statements (Theorems 4.3, 22.1 and 26.3, Corollaries 4.5 and 29.2) use `Diophantine.e.mrdp`, `Diophantine.mrdp_iff` and `Diophantine.mrdp_dioph_iff` (`Computability/HilbertTenthProblem/Lean/Diophantine/MRDP.lean`) as a mathematical interface. Manuscript 07 also names the trace interfaces `Diophantine.boundedForall_dioph`, `Diophantine.exactIter_dioph` and `Diophantine.existsExactIter_dioph` (`.../Lean/Diophantine/Common/DiophantineTrace.lean`) as a possible route; distinct declarations with the same short name `boundedForall_dioph` exist elsewhere in the project, so the full names are the ones meant. These interfaces assert that representations *exist*; they say nothing about how many witnesses a representation has. **None of the report’s own theorems is formalized**, and the Python programs are finite checks, not proofs.

Part II’s undecidability in dimension four (Theorem 18.2) rests on Neary’s published theorem that mortality of six integer 3×3 matrices is undecidable (STACS 2015, Corollary 12; our Theorem 18.1) and, for the $d + 3$ row bound, on Meyer’s theorem. ProveIt keeps Neary’s paper (`Computability/HilbertTenthProblem/Lean/LIPIcs.STACS.2015.649.pdf`) and formalizes, for its own purposes, the *binary-tag layer* of his construction: a binary tag system $0 \rightarrow 0, 1 \rightarrow u$ simulating

a tag system with deletion d , after his Lemma 9 (`TagBinary91.lean`), the universality of the resulting 91-operation tag certificate (`Jones1980.tag91_re`, `TagUniversal91.lean`), and the undecidability of the encoded tag family (`Jones1980.tag91_undecidable`, `Computability/HilbertTenthProblem/Lean/Diophantine/Paper1980/TagDecide91.lean`); see the Neary rows of `Computability/HilbertTenthProblem/Lean/STATUS.md` (lines 416–418 at the placement commit). The Post-correspondence and matrix-mortality steps of Neary’s Corollary 12, Meyer’s theorem, the rational Gram-packing theorem and everything downstream of them are *not* formalized. Lagrange’s four-square theorem, the only number-theoretic input of the elementary $4d$ route, is available in Mathlib as `Nat.sum_four_squares`, which the project already uses.

[write] The eight batch-62 sources of Parts V–VII likewise use only `Diophantine.mrdp` and its hypothesis `REPred` (with `Diophantine.mrdp_iff` and `Diophantine.mrdp_dioph_iff` as named context). None ships Lean or Rocq, and none of their theorems is formalized. ProveIt has no formal development of quaternion rotations, Mihailova or Miller groups, 2-adic dynamics, tensor contractions, reversible Markov chains or neural networks; source 12’s “existing 2-adic digit API” is Mathlib’s `PadicInt`, not a ProveIt module.

All three manuscripts inspected ProveIt at `f608f1cb3c5be8a736df1328c01b293aabfccf4e`. Nothing under `Computability/` changed between that commit and the placement commit `7498484af`, so every repository statement quoted in this report is still accurate; the sentences of manuscripts 07 and 08 that describe the repository were checked against the current files. [write] The eight batch-62 manuscripts of Parts V–VII pin `ccfb084ad`, `b6bf6406a`, `b998f70c6`, `526c2557f` or `e18718e83` (Appendix F.1); `Computability/` is unchanged from each of these pins to the placement commit `adeddcecc`, and their repository statements were checked against the current files.

1.4 Established ingredients and contribution ledger

The MRDP equivalence is an established theorem [9]. Formalized reductions through FRACSTRAN already exist [3]; hence a bare assertion that FRACSTRAN halting is Diophantine would not be a new result. Weakly computable reals and their bounded-variation characterization also predate this work [4, 5]. Max-relative entropy and its operator-order formula are established [8]. Finally, the arithmetic hardness of probabilistic termination and expectation comparison has substantial prior literature [6].

Component	Status in this manuscript
MRDP; prefix allocation; weak computability; max-relative entropy	Established ingredients, stated with attribution and used with explicit proofs of the needed consequences.
Exact normalization budget	Derived here as a finite/infinite optimization theorem, with a canonical minimal realization.
Unique quartic compiler	Explicit construction and proof here; implemented and checked on finite instances. Historical priority is not asserted.
Sharp binary and operator bounds	Proved here; the underlying entropy notions are not claimed as new.
High-success arithmetic boundary	Explicit realization and hardness reductions here, situated relative to known computability classifications.
Universal output-law polynomial	Existence and degree reduction proved; its full coefficients and witness count are not generated.

The questions answered here are precisely formulated mathematical questions about the proposed

extension. They are not presented as previously published open problems whose priority history has been exhaustively established.

1.5 [write] The three sources and the plan of this report

This report merges three manuscripts of batch 60 of ProveIt’s incoming-report intake. They were written independently, from the same repository snapshot, and none cites another.

Source	Archive and title	Printed here
08 (base)	ProveIt_Diophantine_Quantum_No rmalization: <i>Diophantine Laws of Probabilistic Computation</i>	Sections 1, 3–9 (Part I), 11–12 (Part II), 33, and its parts of Sections 34–37 and the appendices
07	quantum_diophantine_research: <i>Rational Gram Packing, Four-Dimensional Quantum Mortality, and Diophantine Certificates</i>	Sections 13–22 (Part II), and its parts of Sections 34–37 and Appendix A.2
09	ProveIt_Diophantine_Barriers: <i>Diophantine Barriers for Continuous Computation</i>	Sections 23–32 (Part III), and its parts of Sections 34–37 and the appendices

The three share no theorem. They meet in three places, each discussed where it occurs: two different *quantum models* (08’s exact Clifford+*T* amplitudes and operator budgets, 07’s rational real Kraus instruments; Section 10); three different *fixed MRDP polynomials* (Remark 2.1); and the absence of computable *witness or margin bounds* (Corollaries 18.4 and 30.3, Theorem 30.1). Section 2 fixes conventions and a dictionary of letters used differently by the sources. Appendix F records the provenance of the merge and every choice it made.

[write] **Batch 62: Parts V–VII.** Eight manuscripts of batch 62 were added as three new Parts, appended after Part IV so that nothing earlier rennumbers. They are numbered here by their shipped file prefixes, sources 10–17 (batch-62 manuscripts 06–09 and 11–14; Appendix F).

Source	Title	Printed in
13	<i>Probabilities from Well-Conditioned Quartics</i>	Part V
17	<i>Arithmetic of Rapidly Mixing Reversible Computation</i>	Part V
10	<i>Diophantine Certificates for Coherent Computation</i>	Part VI
11	<i>Decidable Gates, Undecidable State Transfer</i>	Part VI, with 16’s shared core printed once (Section 84)
16	<i>Dense Rotations, Undecidable Exactness</i>	Part VI
14	<i>Strict Contractions with Diophantine-Complete Transients</i>	Part VII
12	<i>Exact Diophantine Computation in Contracting 2-adic Dynamics</i>	Part VII, compared with 14 in Section 134
15	<i>Precision Is Memory</i>	Part VII

They re-derive three results of Parts I–III, which are cited rather than reprinted: source 13 the interior threshold classification of Theorem 9.1, source 17 the almost-sure cut classification of

Proposition 9.4, and source 14 the no-gap Theorem 30.1. Source 10 supplies the witness counts that Proposition 11.1 leaves open. None contradicts Parts I–IV. Each new Part opens with its own conventions, since their letters collide with those of Parts I–III in new ways (Sections 38, 70 and 110).

Abstract of manuscript 08. The statement that the halting set of an effective computational substrate is Diophantine follows from MRDP. For probabilistic and quantum computation, the more discriminating questions concern output *laws*, conditional probabilities, and the cost of realizing prescribed normalized histories. We develop a substrate-independent representation of an effective discrete output law as the dyadic mass of a prefix-free projection of a fixed quartic integer zero set. This construction counts projected events, not polynomial witnesses, and therefore requires no finite-fold hypothesis.

The central constructive result determines the exact mass budget for a sequence of strictly positive probability vectors. Its cost is the product of the successive largest coordinate ratios. For a finite rational history with N transitions and m outcomes, we construct one quartic polynomial with $5Nm + m + 4N + 3$ natural witnesses. A valid feasible input has exactly one witness tuple. A canonical earliest-maximum rule removes ties without increasing the degree. We also prove a sharp binary variation bound and an operator version governed by max-relative entropy.

Normalization changes the arithmetic boundary. Conditional two-outcome probabilities with any fixed success floor below one realize exactly the weakly computable reals in $[0, 1]$, whereas almost-sure return realizes exactly the computable reals. Strict conditional comparisons remain Σ_2^0 -complete even when success is guaranteed arbitrarily close to one. Explicit integer arithmetic for Clifford+ T circuits supplies a quantum front end. The accompanying code exports a concrete quartic and performs exhaustive, symbolic, and exact-arithmetic tests. We distinguish these proved constructions from established ingredients, unexpanded MRDP polynomials, and formalization work that has not been performed.

Abstract of manuscript 07. We develop a quantitative route from integer matrix mortality to exactly normalized rational quantum measurements and then to polynomial equations over the natural numbers. The central construction packs the rows of a rational Gram factor across existing Kraus operators. If an input consists of k rational $d \times d$ matrices, the resulting instrument has $k + 1$ outcomes and dimension

$$D = d + \left\lceil \frac{d + 3}{k} \right\rceil.$$

Mortality is preserved in both directions, arbitrary insertions of an invertible idle outcome do not create false witnesses, and the shortest nonempty mortal-word length increases by at most one. Combining this construction with the published six-generator, 3×3 matrix-mortality theorem gives undecidability for rational quantum instruments in dimension four with seven outcomes. An elementary four-square variant gives dimension five with seven outcomes without using Meyer’s theorem.

For fixed history length n , we give a fully specified quartic polynomial whose natural solutions are in bijection with the impossible outcome words. In dimension four with seven outcomes the construction uses $71n + 32$ natural witnesses and $72n + 48$ quadratic residuals. A separate MRDP argument gives a fixed-arity Diophantine representation of the unbounded property, without claiming that this compression preserves unique witnesses. We prove an exact length-dependent probability gap, explain its limited robustness meaning, and identify concrete interfaces to ProveIt’s

Diophantine library. The accompanying standard-library Python implementation passes 32,418 finite exact-arithmetic checks. The mathematical proofs are not Lean-certified, and priority of the resource improvement is not asserted.

Abstract of manuscript 09. We study which infinite-time properties of an unconventional universal computer admit finite Diophantine witnesses. For a continuous rational piecewise-affine map of a compact cube, we define the critical per-step perturbation needed to reach a closed rational target. A finite weighted grid computes this real number with an error bounded by $(L + 1)/(2N)$, independently of trajectory length. A bottleneck/cut duality converts positive critical radius into a complete family of explicit quartic sum-of-squares certificates. Encoding the finite checker and applying ProveIt’s MRDP interface then gives one fixed finite-dimensional Diophantine representation of robust safety.

We construct a single continuous rational piecewise-affine map $F : [0, 1]^3 \rightarrow [0, 1]^3$ and a fixed rational target box that realize every disjoint pair (A, B) of computably enumerable sets: A is exact acceptance, B is positive-radius avoidance, and the complement is nonacceptance vulnerable to arbitrarily small perturbations. The construction combines a packed Cantor-tape simulation, separated terminal slabs, and an expanding emergency coordinate. Rejection after T steps satisfies explicit exponential upper and lower bounds on the critical radius. Consequently the radius is a computable real-valued function, while its positivity is Σ_1^0 -complete and fragile divergence is Π_1^0 -complete. The two enumerable outcomes are even recursively inseparable. No existential Diophantine formula represents the third outcome, and no computable positive gap, certificate-size bound, or robustness-resolution bound exists in general.

1.6 [write] Status statements of the sources

Each source’s own status statement is kept verbatim.

Manuscript 08. Status. This is a research manuscript with complete ordinary mathematical proofs and executable finite checks, not a claim of independently established historical priority or a Lean/Rocq-verified extension. The most concrete deliverable is the explicit single-fold finite-history compiler in Section 6. No general finite-fold MRDP conjecture is assumed or settled.

Manuscript 07. Scope. This is a research contribution with explicit dependencies, not a claim to have newly discovered quantum undecidability or to have solved general single-fold Diophantine representability. The main contribution offered for further review is the packed normalization theorem and its quantitative consequences.

Manuscript 09. Status and scope. The manuscript supplies complete mathematical proofs for the core results, apart from classical universality and MRDP. The neural corollary also uses the standard max–min representation theorem. The proposed contribution is the joint realization–radius–certificate package, not the already known facts that continuous dynamics can be universal or that suitable robustness implies decidability. Global priority and minimal dimension are not established. The companion exact-rational code passes 5,660 checks; neither these results nor the compiler have been Lean-verified. No expanded universal machine table or fixed MRDP polynomial is claimed.

2 [write] Conventions and notation of the report

2.1 Domain

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Every polynomial witness ranges over $\mathbb{N}$; every polynomial has integer coefficients, and subtraction in a displayed polynomial is subtraction in $\mathbb{Z}$, never truncated subtraction. A *residual system* is a finite list of integer polynomials required to vanish; its single equation is the sum of their squares, of degree at most four when every residual is at most quadratic. Probabilities are exact rationals, exact elements of $\mathbb{Q}(\sqrt{2}, i)$, or computable reals, as stated in each Part; no floating-point quantity enters a theorem (the binary-sharpness diagnostics of manuscript 08’s software are illustrative only, Section 34).

A representation is *single-fold* if each parameter tuple has at most one witness, *finite-fold* if it has finitely many, and *parsimonious* (or *bijective*) for a set of semantic objects if its witnesses are in a specified bijection with them. The explicit certificates of Theorems 6.1, 21.1 and 26.2 are single-fold or bijective *for each fixed size parameter*; the fixed-arity MRDP polynomials of this report carry no multiplicity claim.

2.2 Words used in several senses

- *Certificate*. Here, a natural witness tuple of an explicit polynomial, or (Part III) the finite graph cut it encodes. In ProveIt’s Hilbert’s-tenth-problem project a “straight-line certificate” is a program whose arithmetic operations are counted; no result of this report concerns operation counts.
- *Quartic*. Total degree at most four. Part I’s quartic $F_{N,m}$, Part II’s $\mathcal{Q}_{D,K,n}$ and Part III’s Q_N are three unrelated explicit families; the fixed-arity polynomials obtained from MRDP are not explicit and have no stated degree unless Lemma 3.3 is applied.
- *Universal*. Theorem 4.3 (a fixed quartic for all effective output laws), Theorem 22.1 (a fixed polynomial for $\text{QMort}_{4,7}$) and Theorem 28.1 (a fixed three-dimensional map realizing all disjoint computably enumerable pairs) use “universal” for three different objects; none is a universal pair (ν, δ) in the sense of Jones’s articles.
- *Mortal / impossible / zero-probability*. In Part II a word is *mortal* for a matrix list when its ordered product is zero; for a positive definite input state this is the same as the outcome word having probability zero (Lemma 14.2). In Part I “nonreturn” is missing probability mass, not a zero-probability word.
- *Robust*. Part III’s robust avoidance is positivity of the critical *per-step* noise radius (Section 24); Part II’s robustness discussion (Proposition 20.2) is a perturbation of Kraus operators at fixed word length. Neither is a probabilistic noise model.
- *Budget*. Part I’s normalization budget G_N is a product of coordinate ratios; Part II’s operator budget is its noncommutative analogue. Part III’s radius is a noise tolerance, not a mass.

2.3 Letters used differently by the sources

Each Part keeps its source’s letters, so that a symbol is always read in the Part where it occurs; the table records the letters whose meanings are most easily confused across the sources. Where two

meanings met inside one Part, one was renamed (Section 2.4).

Letter	Part I (08)	Part II (08 in Sections 11–12; 07 from Section 13)	Part III (09)
N	horizon (number of transitions)	08: horizon in Theorem 12.1; 07: common denominator $H_j = NA_j$	mesh $N = D2^n$
m	number of output labels	07: number of active letters of a word; witness dimension in Theorem 22.1	number of interpreter states
K	—	07: number of outcome matrices in $\mathcal{Q}_{D,K,n}$ (so $K = k + 1$ for a packed instrument)	the cube $[0, 1]^d$; K_0 a halting set
k	witness dimension	07: number of active outcomes	witness dimension
D	D_s products of denominators; $D_\infty, D_{\max}$ divergences	07: quantum dimension; 08: $D_{\max}$	denominator-clearing integer
H	—	08: the Hadamard gate; 07: integer numerator matrices H_j, H_w	target set
A, B	A_s, B_s selected ratio witnesses	08: integer norm coefficients in $A + B\sqrt{2}$; 07: Kraus operators A_j , Gram factor B , blocks B_i	the disjoint computably enumerable pair (A, B) ; Lipschitz bounds in Theorem 31.1
T	—	08: the T gate; 07: the tensor matrices T_i	time; T_N grid targets
S	—	08: the S gate	a cut; $S_T(L)$ a geometric sum
μ	output law μ_e	07: shortest mortal length $\mu(\mathcal{M})$	envelope $\mu(n)$
δ	initial mass	08: initial mass; 07: gap δ_n , Kronecker δ_{ab}	a margin
α, β	input fraction α/β	07: $\alpha = 4/(5c)$, $\beta = 3/5$	α_j state offsets, $\beta = 1/(3m)$
L	—	08: number of gates	Lipschitz bound
Q	matrix of a Σ_2^0 normal form	07: positive definite form	quartic Q_N
R, G	R_s largest ratio, G_s product	08: operator ratio R_s and product G_s (renamed); 07: a residual block R	G_N the grid; r_N a bottleneck
ρ	—	density operator / accumulated ensemble	critical radius

The tempting false readings are these. Part II’s K is a count, never the accumulation factor (which is $R(\sigma, \tau)$ here). The symbols H , T , S without subscript in Section 11 are gates, not matrices of manuscript 07. Part III’s K is a set, and its N is a mesh, not a horizon. The single-fold quartic of Part I has $W(N, m)$ witnesses; Part II’s bijective quartic has $W(D, K, n)$; the two functions W and E are different, and Part III’s E_N is an edge set. Other letters (C , M , V , h , p , r , s , t , u , v , w , x , y , z among them) are local to the section that defines them; for example C_s are Part I’s numerator products while $C = L + 1$ in Part III, and h_s is a heralded mass in Part I while $h_N = 1/(2N)$ in Part III.

2.4 Renamed symbols and macros

- In Section 12 (manuscript 08’s operator budget), the domination factor written $K(\sigma, \tau)$, K_s in the source is written $R(\sigma, \tau)$, R_s here, and the product written H_s is written G_s here. The

renaming removes clashes with 07's outcome count K , 07's numerator matrices H_j and 08's own Hadamard gate H , and shows that the operator quantities specialize to Part I's R_s and G_s for commuting (diagonal) states. No normalization changed. The same renaming is applied to research question 6.

- In Section 16, manuscript 07's local $S = \sum_i M_i^T M_i$ and $T = \text{tr}(S)$ are written $S_{\mathcal{M}}$ and $t_{\mathcal{M}}$, to keep them apart from the S and T gates.
- Manuscript 09 defines `\ce` as “computably enumerable” while 08 defines it as “c.e.”; 09's macro is `\cenum` here, and the printed words are unchanged. The three definitions of `\code` (08, 07, 09) are unified as a breakable typewriter path; 09's escaped underscores inside it were unescaped. The pinned commit, typeset by three differently named macros in the sources, is printed by one.
- File names are the shipped names of this report (`code/NN-...`, `data/NN-...`); the delivered names are listed in the README, which also explains how to rerun each suite.

2.5 How the merge is marked

Text added in writing this report is marked [write]. Everything else is the text of the source named at the head of its Part or section, with labels, file names and the symbols of Section 2.4 changed. Cross-references between the sources were added only in [write] remarks. The letter table above covers Parts I–III; Parts V–VII have their own tables and renaming lists, in their opening sections. In Parts V–VII, which mix sources, an italic *[source NN]* under a section heading, added in writing, names the source of that section, and appendix sections of a source are titled “Source NN appendix”.

Remark 2.1 ([write] Three fixed polynomials). Theorem 4.3 (08) represents the relation “interpreter e allocates word code c to label z ”; Theorem 22.1 (07) represents the set $\text{QMort}_{4,7}$ of mortal four-dimensional seven-outcome instruments; Theorem 26.3 (09) represents positive critical radius and exact acceptance. All three come from MRDP applied to a semidecision procedure; none is expanded, none has a stated witness count, and none is claimed single-fold or finite-fold. The explicit, counted polynomials of the three Parts are the size-indexed families $F_{N,m}$, $\mathcal{Q}_{D,K,n}$ and Q_N .

Part I

Probabilistic computation (manuscript 08)

3 Effective output laws and quartic arithmetic

3.1 Conventions

We take $\mathbb{N} = \{0, 1, 2, \dots\}$. A relation on natural tuples is *Diophantine* when it has the form

$$R(\mathbf{x}) \iff \exists \mathbf{w} \in \mathbb{N}^k P(\mathbf{x}, \mathbf{w}) = 0, \quad P \in \mathbb{Z}[\mathbf{X}, \mathbf{W}]. \quad (3.1)$$

All polynomial witnesses in this article are natural numbers. Negative coefficients are allowed. This is not a statement about rational witnesses, or about unrestricted integer witnesses unless an explicit translation is given.

A real a is *left-c.e.* if a computable nondecreasing rational sequence converges to a . A discrete *subprobability law* is a function $\mu : \mathbb{N} \rightarrow [0, 1]$ with $\sum_z \mu(z) \leq 1$. Missing mass is allowed to represent nonreturn. An effective family of output laws is uniformly left-c.e. in its program and output arguments.

Definition 3.1 (Effective first-return presentation). An effective first-return presentation consists of an effective enumeration of finite terminal histories, a computable output label for each history, and uniformly computable nonnegative real history weights. Histories represent pairwise disjoint terminal outcomes, and the sum of their weights is at most one. Each actual finite terminal history occurs once in the presentation, or is canonically deduplicated.

For a quantum machine, a history in this definition is a sequence of actual measurement outcomes together with the classical control information. Interfering basis paths between measurements must already have been combined when computing its weight.

Lemma 3.2 (Effective first returns give lower approximations). *Every effective first-return presentation has a uniformly left-c.e. output law.*

Proof. For each history j , compute rational lower bounds on its weight, replace negative bounds by zero, and take running maxima. At stage s , sum the bounds for the first s enumerated histories with the specified output label. These sums are computable, nondecreasing, and bounded by the true output mass. Every finite collection of history weights is approximated arbitrarily well, so the limit is their full sum. The construction is uniform. Rounding down on successively finer dyadic grids gives dyadic lower approximations. For infinitely many labels, expose only finitely many labels at each stage; all lower masses still have sum at most one. $\square$

The assumptions permit an infinite state space and an unbounded running time. They do not permit noncomputable real transition constants hidden in the source language. They also do not turn an infinite limiting configuration into a finite first return.

3.2 A degree-reduction interface

Lemma 3.3 (Quartic lowering). *Every Diophantine relation has a representation by a single integer polynomial of total degree at most four. A fixed input polynomial gives a fixed finite new witness dimension.*

Proof. Evaluate the input polynomial by a finite arithmetic circuit. Introduce variables for its intermediate values. Each addition has a linear residual, and each multiplication has a quadratic residual. Signed intermediate integers can be represented by differences of natural variables; a product of two such differences is still quadratic. Equate the output difference to zero and sum the squares of all residuals. An integer sum of squares is zero exactly when every residual is zero. Conversely, a genuine evaluation supplies intermediate values and hence natural positive/negative parts. The resulting sum has total degree at most four. No uniqueness of the introduced witnesses is required for this lemma. $\square$

The polynomial in Lemma 3.3 can be much larger than the original one. Degree reduction is not an efficiency theorem. The direct construction in Section 6, in contrast, controls its arithmetic circuit size and natural witness multiplicity.

4 Output laws as masses of Diophantine projections

4.1 Canonical allocation of fair-coin events

A finite binary word σ specifies the cylinder of infinite fair-coin streams beginning with σ . Its probability is $2^{-|\sigma|}$. A prefix-free set of words specifies disjoint cylinders.

Lemma 4.1 (Effective labeled allocation). *Suppose a computable enumeration requests labeled binary words of lengths $n_1, n_2, \dots$, and $\sum_j 2^{-n_j} \leq 1$. There is a deterministic effective allocation of prefix-free words with exactly these lengths and labels.*

Proof. Maintain a finite set of free words, initially containing the empty word. Maintain the invariant that there is at most one free word of each length. For a request of length n , choose the longest free word of length at most n . Remove it. Repeatedly split its cylinder into its left child, which is continued, and its right child, which is put in the free set, until length n is reached. Allocate the final left descendant to the request.

There was no other free word with length strictly between the chosen length and n , by maximality. Thus the one-free-word-per-length invariant is preserved. Free and allocated cylinders remain pairwise disjoint and together have total mass one. If there were no suitable free word, all free words would have distinct lengths greater than n . Since the free set is finite, its total mass would be strictly less than $\sum_{k>n} 2^{-k} = 2^{-n}$. The requested mass would then exceed the remaining mass, contrary to the hypothesis. Every allocation therefore succeeds. $\square$

Theorem 4.2 (Fair-coin realization). *A uniformly given discrete subprobability law is realizable as the terminal output law of a fair-coin program if and only if it is uniformly left-c.e.*

Proof. A fair-coin machine supplies an effective enumeration of disjoint finite halting cylinders, so the forward implication follows directly, or from Lemma 3.2.

For the reverse implication choose monotone dyadic lower approximations $d_s(z)$, with finite support at each stage and total mass at most one. Each increment $d_s(z) - d_{s-1}(z)$ is a nonnegative dyadic rational and can be decomposed into finitely many requests for masses 2^{-n} , all labeled by z . The total requested mass is $\sum_z \mu(z) \leq 1$. Lemma 4.1 allocates their cylinders. A program dovetails this allocation enumeration with reading its random stream and returns z when it discovers an allocated cylinder containing that stream. Its output events are exactly the allocated cylinders, and their mass for label z is $\lim_s d_s(z) = \mu(z)$. $\square$

The program may read additional random bits before an allocated cylinder is recognized. The allocated word is an event code, not necessarily its literal minimal runtime leaf. The theorem preserves the terminal output law, not runtime, space, or interactive behavior.

4.2 One quartic zero set for all effective laws

Encode a binary word by

$$c(\sigma) = 2^{|\sigma|} + \text{value}_2(\sigma), \quad \ell(c) = \lfloor \log_2 c \rfloor \quad (c \geq 1). \quad (4.1)$$

The empty word has code one. Fix a universal allocation interpreter. It accepts requests only while preserving the prefix-free allocation invariant; malformed or overbudget requests are discarded or

leave the source stalled. Thus arbitrary program codes still specify a valid subprobability law. Valid source compilers from Theorem 4.2 never trigger this protection.

Theorem 4.3 (Universal projected-root representation). *There are a fixed k and a fixed integer polynomial $U(E, Z, C, W_1, \dots, W_k)$ of total degree at most four such that every effective discrete output law has, effectively from its presentation, an index e with*

$$\mu_e(z) = \sum_{c \geq 1} 2^{-\ell(c)} \mathbf{1}_{\{\exists \mathbf{w} \in \mathbb{N}^k \ U(e, z, c, \mathbf{w})=0\}}. \quad (4.2)$$

For fixed e , the word codes appearing in this sum across all labels are prefix-free, and each code has at most one label.

Proof. The relation “interpreter e allocates word code c to label z ” is c.e.: run the interpreter and enumerate its allocations. Apply MRDP to this relation, using an effective coding of its input tuple, and then apply Lemma 3.3. The resulting polynomial and its witness dimension depend only on the fixed interpreter. Prefix-freeness and label uniqueness are properties of that interpreter. Equation (4.2) is the disjoint-cylinder probability formula. Every law from Theorem 4.2 has a source for the interpreter. $\square$

Remark 4.4 (Projection is essential). Equation (4.2) assigns one weight to a word code for which a witness exists. It does *not* sum over witness tuples. A single word may have infinitely many arithmetic witnesses without changing its probability. Consequently this theorem does not assume a finite-fold or single-fold MRDP representation. Conversely, any effectively specified polynomial relation whose output-labeled projection is prefix-free defines a left-c.e. subprobability law by this same formula.

Corollary 4.5 (Ordinary strict thresholds). *For any effective output-law interpreter, the relation $b\mu_e(z) > a$, where $a \in \mathbb{N}$ and $b \geq 1$, is Diophantine and has a fixed quartic representation.*

Proof. A monotone rational approximation exceeds a/b at a finite stage exactly when the limit does. This is a c.e. relation in (e, z, a, b) . Apply MRDP and quartic lowering, including $b \geq 1$ in the decidable input guard. $\square$

Neither Theorem 4.3 nor Corollary 4.5 supplies the full expanded universal polynomial in the accompanying software. They give an existence construction through a fixed c.e. interpreter. The finite polynomial exported by the software is the substantially more explicit one constructed next.

5 The exact normalization budget

5.1 A mass-allocation problem

Let $p_0, p_1, \dots$ be strictly positive probability vectors on $\{1, \dots, m\}$, with $m \geq 2$. Suppose that by checkpoint s we have accumulated masses $a_{s,i}$ of disjoint returned outcomes. Those masses must be coordinatewise nondecreasing. Write

$$h_s = \sum_i a_{s,i}, \quad a_{s,i} = h_s p_{s,i}. \quad (5.1)$$

Here h_s is the probability already assigned to a successful return, also called the *heralded mass*. Fix an initial mass $h_0 = \delta > 0$ and a unit total budget $h_s \leq 1$.

The checkpoints are points in an effective enumeration of accumulated probability. They need not coincide with successive machine instructions or successive coin flips. This distinction is necessary: exact rational mass increments need not be realizable in a uniformly bounded number of fair coin flips.

Definition 5.1 (Normalization cost). For $s \geq 1$, define

$$R_s = \max_{1 \leq i \leq m} \frac{p_{s-1,i}}{p_{s,i}}, \quad G_0 = 1, \quad G_s = \prod_{j=1}^s R_j. \quad (5.2)$$

The path's total multiplicative cost is G_N for a finite horizon, and $G_\infty = \sup_s G_s$ for an infinite history.

Since both vectors sum to one, $R_s \geq 1$. Otherwise every old coordinate would be strictly smaller than the new coordinate, contradicting equal sums.

Theorem 5.2 (Optimal normalization budget). *A finite or infinite positive probability history admits masses satisfying (5.1), $h_0 = \delta$, $h_s \leq 1$, and $a_{s,i} \geq a_{s-1,i}$ if and only if*

$$\delta G_s \leq 1 \quad \text{for every checkpoint } s. \quad (5.3)$$

Its unique pointwise minimal feasible history is

$$h_s^* = \delta G_s, \quad a_{s,i}^* = \delta G_s p_{s,i}. \quad (5.4)$$

In particular, the largest feasible initial mass for a finite path is G_N^{-1} .

Proof. For a single transition, coordinatewise monotonicity is equivalent to

$$h_s p_{s,i} \geq h_{s-1} p_{s-1,i} \quad \text{for every } i,$$

and, because all quantities being divided by are positive, to $h_s \geq h_{s-1} R_s$. Iterating gives $h_s \geq \delta G_s$. This proves necessity and pointwise minimality of any possible candidate.

Conversely, the choice in (5.4) satisfies the single-step inequality with equality in the maximizing ratio and inequality in all other coordinates. Its coordinates are nondecreasing, its initial mass is δ , and the assumed budget condition gives total mass at most one. It is therefore feasible. Each of its total masses is forced for any pointwise minimal solution, and the prescribed vectors then force every coordinate. The argument applies to every finite prefix of an infinite history without any compactness or choice step. $\square$

Corollary 5.3 (Effective realization of a feasible history). *If δ and the $p_{s,i}$ are computable rationals, and the budget condition holds, the greedy masses define a computable nondecreasing rational approximation to an output subprobability law. A fair-coin program realizes that limiting law.*

Proof. Finite maxima and products of positive rationals are computable exactly. Theorem 5.2 supplies monotonicity and boundedness. Each coordinate therefore converges and is left-c.e.; the limiting sum is at most one. Apply Theorem 4.2. When necessary, round rational cumulative masses down to finer dyadic lower approximations before allocation. $\square$

Example 5.4 (An excursion costs mass). Consider

$$p_0 = (1/2, 1/2), \quad p_1 = (2/3, 1/3), \quad p_2 = (1/2, 1/2).$$

Then $R_1 = 3/2$, $R_2 = 4/3$, and $G_2 = 2$. Thus $\delta \leq 1/2$ is necessary and sufficient, although the initial and final conditional laws agree. At the maximum initial mass, the greedy cumulative masses are

$$(1/4, 1/4), \quad (1/2, 1/4), \quad (1/2, 1/2).$$

The first transition spends additional mass on the first output; the return transition spends it on the second. Previously assigned mass cannot be withdrawn.

5.2 An information-theoretic expression

For positive classical probability vectors, write

$$D_\infty(p||q) = \log \max_i \frac{p_i}{q_i},$$

with the natural logarithm throughout this article. Then

$$\log G_N = \sum_{s=1}^N D_\infty(p_{s-1}||p_s). \quad (5.5)$$

The divergence itself is standard, and is the commuting specialization of max-relative entropy [8]. Its role here is an exact additive budget for irreversible accumulation of terminal mass. The direction of the arguments is significant: the old law is compared with the new law. Reversing them generally changes the answer.

6 A single-fold quartic certificate with explicit size

6.1 Parameters and statement

Fix an external horizon $N \geq 0$ and an external number of labels $m \geq 2$. The free natural parameters are

$$(u_{s,i})_{0 \leq s \leq N, 1 \leq i \leq m}, \quad (v_s)_{0 \leq s \leq N}, \quad \alpha, \beta.$$

Call an input *valid* when

$$u_{s,i} \geq 1, \quad v_s = \sum_i u_{s,i}, \quad 1 \leq \alpha \leq \beta. \quad (6.1)$$

It specifies $p_{s,i} = u_{s,i}/v_s$ and $\delta = \alpha/\beta$. Denominators need not be reduced, and different rows may use independently rescaled denominators.

Theorem 6.1 (Explicit unique-witness quartic). *For each fixed N, m there is an explicitly constructible integer polynomial $F_{N,m}$ of total degree at most four with*

$$W(N, m) = 5Nm + m + 4N + 3 \quad (6.2)$$

$$E(N, m) = 5Nm + m + 6N + 4 \quad (6.3)$$

natural witnesses and squared residuals, respectively, such that:

- (i) a valid input has a witness if and only if its prescribed history is feasible with initial mass α/β ;
- (ii) every valid feasible input has exactly one natural witness tuple;
- (iii) invalid inputs have no natural witness tuple.

The residual system has an arithmetic-circuit implementation of size $O(Nm)$. This is not a claim that its fully expanded polynomial has only $O(Nm)$ monomials.

6.2 Validity, ratios, and selectors

Introduce positivity witnesses $g_{s,i}$ and g_α, g_β , with residuals

$$\begin{aligned} u_{s,i} - 1 - g_{s,i} &= 0, & v_s - \sum_i u_{s,i} &= 0, \\ \alpha - 1 - g_\alpha &= 0, & \beta - \alpha - g_\beta &= 0. \end{aligned} \tag{6.4}$$

For every transition $s = 1, \dots, N$ introduce $X_{s,i}, Y_{s,i}$ and enforce

$$X_{s,i} - u_{s-1,i}v_s = 0, \quad Y_{s,i} - v_{s-1}u_{s,i} = 0. \tag{6.5}$$

Their ratios are the candidates for R_s : $X_{s,i}/Y_{s,i} = p_{s-1,i}/p_{s,i}$, and their denominators are positive.

Introduce selectors $e_{s,i}$ satisfying

$$e_{s,i}(e_{s,i} - 1) = 0, \quad \sum_i e_{s,i} - 1 = 0. \tag{6.6}$$

They select exactly one coordinate. Define selected numerator and denominator witnesses A_s, B_s by

$$A_s - \sum_i e_{s,i}X_{s,i} = 0, \quad B_s - \sum_i e_{s,i}Y_{s,i} = 0. \tag{6.7}$$

Every residual displayed so far has total degree at most two.

6.3 Canonical earliest-maximum rule

Ordinary maximum constraints allow several selectors when two ratios tie. To eliminate this multiplicity, introduce natural slacks $t_{s,i}$ and use

$$A_s Y_{s,i} - B_s X_{s,i} - t_{s,i} - \sum_{j>i} e_{s,j} = 0. \tag{6.8}$$

If coordinate k is selected, the last sum is one for $i < k$ and zero for $i \geq k$. Thus the selected ratio must be *strictly* larger than every earlier ratio and at least as large as every later one. Integer strictness is encoded by subtraction of one, not by a new quantifier or a high-degree product.

Lemma 6.2 (Unique pivot). *For any positive integer lists $(X_i), (Y_i)$, the selector and comparison constraints (6.6)–(6.8) have exactly one solution in the selector, selected-ratio, and slack variables. Its selected index is the least index maximizing X_i/Y_i .*

Proof. The Boolean and sum constraints select an index k . Nonnegative slacks give $X_k Y_i - Y_k X_i \geq 1$ for $i < k$ and ≥ 0 for $i \geq k$. Positivity of all denominators makes these exactly the asserted strict and weak ratio comparisons. Hence k must be the least maximizing index. Conversely, that index satisfies those comparisons: a strictly positive integer cross difference is at least one. It determines all selectors, $A = X_k$, $B = Y_k$, and all slacks uniquely. $\square$

6.4 Canonical products and the terminal budget

Set $C_0 = \alpha$, $D_0 = \beta$ as expressions, not additional witnesses. For $s = 1, \dots, N$ introduce C_s, D_s and impose

$$C_s - C_{s-1}A_s = 0, \quad D_s - D_{s-1}B_s = 0. \quad (6.9)$$

Finally introduce a natural slack z and impose

$$D_N - C_N - z = 0. \quad (6.10)$$

For $N = 0$, this uses $D_0 = \beta, C_0 = \alpha$ directly. Products deliberately remain unreduced. There are no arbitrary common factors and no gcd certificates to create extra witnesses.

Let $\mathcal{R}_{N,m}$ be the complete residual list in (6.4)–(6.10), including every indicated index, and put

$$F_{N,m} = \sum_{r \in \mathcal{R}_{N,m}} r^2. \quad (6.11)$$

This is the promised explicit polynomial definition. It is a finite sum-of-squares specification, not an appeal to MRDP.

6.5 Proof of the compiler theorem

Proof of Theorem 6.1. Soundness. A natural zero of $F_{N,m}$ makes each residual zero. The validity residuals force (6.1). Equations (6.5) then force positive ratio denominators. Lemma 6.2 forces the least maximizing ratio at each stage, so $A_s/B_s = R_s$. The product residuals force

$$\frac{C_N}{D_N} = \frac{\alpha}{\beta} \prod_{s=1}^N R_s = \delta G_N.$$

The denominator is positive. The final natural slack forces $C_N \leq D_N$, hence $\delta G_N \leq 1$. Since G_s is nondecreasing, the budget condition holds for every earlier stage. Apply Theorem 5.2.

Completeness. For a valid feasible input, assign the positivity and input slacks their forced values, compute the X, Y lists, choose each least maximizing coordinate, and assign its uniquely determined nonnegative comparison slacks. Compute the unreduced products C_s, D_s . Feasibility makes $D_N - C_N$ nonnegative; assign it to z . Every residual is zero.

Uniqueness. The input fixes the validity slacks and all X, Y values. Lemma 6.2 fixes the selectors, A, B , and comparison slacks. The recurrence fixes the unreduced products, and the final equation fixes z . There are no other witnesses. The existence of other, nonminimal feasible mass histories is irrelevant: the certificate describes the canonical optimal budget, not an arbitrary feasible history.

Size and degree. The positivity slacks contribute $(N + 1)m$ witnesses, the two input slacks contribute two, and each transition contributes $4m + 4$ witnesses: X, Y, e, t for every coordinate and A, B, C, D . The terminal slack contributes one. This gives (6.2). Validity contributes $(N + 1)(m + 1) + 2$ residuals, each transition contributes $4m + 5$, and the terminal equation contributes one, giving (6.3). Each residual is quadratic or linear, so their squares have degree at most four. Shared suffix sums of the selectors evaluate all comparison residuals with $O(m)$ arithmetic operations per transition. $\square$

Remark 6.3 (The exact single-fold scope). The horizon N and label count m index a family of finite polynomials. They are not unbounded runtime variables inside one fixed-arity single-fold

universal halting polynomial. This construction therefore does not resolve a general finite-fold MRDP question. Its uniqueness is over natural witnesses. Replacing each natural variable by an arbitrary four-square representation would in general destroy that uniqueness.

6.6 The exported $N = 2, m = 2$ polynomial

For Example 5.4, take rows $(u_{0,1}, u_{0,2}) = (1, 1)$, $(u_{1,1}, u_{1,2}) = (2, 1)$, $(u_{2,1}, u_{2,2}) = (1, 1)$ and $(\alpha, \beta) = (1, 2)$. The denominators are $(2, 3, 2)$. The selected coordinate is two at the first transition and one at the second. Thus

$$(A_1, B_1) = (3, 2), \quad (A_2, B_2) = (4, 3), \quad (C_1, D_1) = (3, 4), \quad (C_2, D_2) = (12, 12).$$

The terminal slack is zero. This instance has 11 free input coordinates, 33 witnesses, 38 residuals, total degree four, and 165 expanded monomials. The bundle supplies the residual list, variable ordering, full natural assignment, and expanded polynomial. These values are computed by the included script rather than inferred from a numerical optimizer.

7 Variation bounds and a sharp binary constant

7.1 A general bound

For probability vectors define $\text{TV}(p, q) = \frac{1}{2} \sum_i |p_i - q_i|$.

Proposition 7.1 (Variation consumes mass). *For the positive histories of Theorem 5.2,*

$$\sum_{s=1}^N \text{TV}(p_{s-1}, p_s) \leq \sum_{s=1}^N (1 - R_s^{-1}) \leq \log G_N. \quad (7.1)$$

Every history feasible from initial mass δ therefore has total variation at most $\log(1/\delta)$, including an infinite history.

Proof. If $R_s = 1$, then $p_s \geq p_{s-1}$ coordinatewise and equal total sums imply equality. If $R_s > 1$, coordinatewise domination gives a probability vector q_s such that

$$p_s = R_s^{-1} p_{s-1} + (1 - R_s^{-1}) q_s.$$

Hence $\text{TV}(p_{s-1}, p_s) \leq 1 - R_s^{-1}$. For $R \geq 1$, $1 - R^{-1} \leq \log R$, as follows by integrating $1/t$ on $[1, R]$ or differentiating their difference. Sum over s and apply the budget theorem. For an infinite path pass to the supremum of the nonnegative partial sums. $\square$

7.2 The binary path identity

The binary case contains a stronger symmetric bound. Put $p_s = (x_s, 1 - x_s)$ with $0 < x_s < 1$, and define

$$\Phi(x) = -\frac{1}{2} \log(x(1 - x)). \quad (7.2)$$

Theorem 7.2 (Sharp balanced binary bound). *Every binary path satisfies*

$$2 \sum_{s=1}^N |x_s - x_{s-1}| + \Phi(x_N) - \Phi(x_0) \leq \log G_N. \quad (7.3)$$

In particular, a feasible path starting at $x_0 = 1/2$ and initial mass δ satisfies

$$\sum_{s=1}^N |x_s - x_{s-1}| \leq \frac{1}{2} \log(1/\delta). \quad (7.4)$$

For every $0 < \delta < 1$, the constant $1/2$ in this statement is sharp as a supremum over finite balanced-start paths.

Proof. If a step increases x from x to y , its largest ratio is $(1-x)/(1-y)$, so its logarithmic cost is $\int_x^y (1-u)^{-1} du$. If the step decreases x to y , the cost is $\int_y^x u^{-1} du$. Since

$$\Phi'(u) = \frac{2u-1}{2u(1-u)},$$

both cases have the exact common expression

$$\log R_s = \int_{\min(x_{s-1}, x_s)}^{\max(x_{s-1}, x_s)} \frac{du}{2u(1-u)} + \Phi(x_s) - \Phi(x_{s-1}). \quad (7.5)$$

Now $u(1-u) \leq 1/4$, so the integral is at least $2|x_s - x_{s-1}|$. Sum and telescope the potential to obtain (7.3). The potential is minimized at $1/2$, and $\log G_N \leq \log(1/\delta)$ for a feasible path, proving (7.4).

For sharpness, repeat the cycle $1/2 \rightarrow 1/2 + \varepsilon \rightarrow 1/2$ exactly n times. Its one-cycle multiplicative cost is $(1+2\varepsilon)/(1-2\varepsilon)$ and its variation is 2ε . Choose

$$r_n = (1/\delta)^{1/n}, \quad \varepsilon_n = \frac{r_n - 1}{2(r_n + 1)}.$$

The full cost is exactly $1/\delta$, while the full variation is $n(r_n - 1)/(r_n + 1)$, which tends to $\frac{1}{2} \log(1/\delta)$. Thus no smaller constant works. If rational paths are required, approximate each ε_n from below by rationals; feasibility is preserved and the same supremum is obtained. $\square$

This bound is a quantitative restriction on successive conditional estimates. Conditioning can hide noncomputable limiting information, but it cannot make a fixed positive amount of accumulated mass support an arbitrarily oscillatory normalized path. Small excursions near balance are the asymptotically least expensive way to accumulate binary variation.

8 Which real probabilities can normalization produce?

8.1 Weak computability and finite variation

A real is *weakly computable*, also called a difference of left-c.e. reals, when $c = A - B$ for finite left-c.e. reals A, B . The equivalence with computable rational approximation of bounded total variation is established in the literature [4, 5]. The elementary argument needed here is short.

Lemma 8.1 (Jordan decomposition of an approximation). *A real c is weakly computable if and only if there is a computable rational sequence (x_s) tending to c for which $\sum_s |x_{s+1} - x_s| < \infty$.*

Proof. If $x_s = A_s - B_s$ with monotone convergent computable rational approximations, then $|x_{s+1} - x_s| \leq (A_{s+1} - A_s) + (B_{s+1} - B_s)$, and the sum is finite. Conversely, set

$$P_s = \sum_{j < s} \max(x_{j+1} - x_j, 0), \quad Q_s = \sum_{j < s} \max(x_j - x_{j+1}, 0).$$

Both are computable increasing rational sequences with finite limits, and $x_s = x_0 + P_s - Q_s$. Their limits express c as a difference of left-c.e. reals. $\square$

8.2 An exact success-floor classification

Two disjoint successful terminal outcomes, A and B , have unconditional masses a, b and heralding probability $h = a + b$. Their conditional probability of A , given eventual successful return, is

$$c = \frac{a}{a + b}, \quad h > 0. \quad (8.1)$$

The denominator is an eventual-event probability. It is not assumed known or computable.

Theorem 8.2 (High-success normalization realizes exactly weak computability). *For a real $c \in [0, 1]$, the following are equivalent:*

- (i) *c is a conditional terminal-output probability of an effective process with positive heralding probability;*
- (ii) *c is weakly computable;*
- (iii) *for every rational $0 < \delta < 1$, there is such a process with heralding probability at least δ and conditional probability exactly c .*

The realization in (iii) may be chosen to have a computable rational accumulation checkpoint of total mass exactly δ .

Proof. (i) implies (ii). The masses a, b are left-c.e. by Lemma 3.2. If one is zero, c is an endpoint and is computable. Otherwise choose monotone rational approximations a_s, b_s and discard a finite initial segment until both are positive. Set $h_s = a_s + b_s$ and $x_s = a_s/h_s$. These give an effective positive binary history with initial mass $h_0 > 0$ and total budget at most one. Proposition 7.1 bounds its total variation by $\log(1/h_0)$. It converges to $a/(a + b)$, so Lemma 8.1 proves weak computability.

(ii) implies (iii). The endpoint cases are realized by a constant output. Suppose $0 < c < 1$, and take a computable rational bounded-variation approximation to c . Choose rational $\eta > 0$ with $c \in (\eta, 1 - \eta)$. Clamp the approximation to $[\eta, 1 - \eta]$; clamping is a contraction, preserves the limit, and does not increase total variation. Discard a sufficiently long finite initial segment so that the remaining variation is at most $\eta \log(1/\delta)$.

For a binary step confined to that interval, the integral formulas in the proof of Theorem 7.2 give $\log R_s \leq |x_s - x_{s-1}|/\eta$. Hence the remaining path has $\sum_s \log R_s \leq \log(1/\delta)$. Its greedy realization from initial mass δ is therefore feasible by Theorem 5.2. Its monotone rational coordinate masses converge to a, b with $a + b \geq \delta$. Because their normalized ratios are the chosen x_s , continuity at the positive denominator gives $a/(a + b) = c$. Apply Corollary 5.3 to realize the limiting law.

(iii) implies (i) is immediate. $\square$

Remark 8.3 (The nonuniform choice must not be hidden). The proof of (ii) $\Rightarrow$ (iii) chooses an interior interval and a late small-variation tail. It is an existence theorem for each real and each success floor; it is not an algorithm for discovering those choices from an arbitrary bounded-variation approximation index. The rational greedy compiler is uniform once a suitable tail and a valid budget bound are supplied. The finite-horizon quartic compiler is fully explicit and has no such unbounded tail choice.

Weakly computable reals are not all limit-computable reals; the distinction is established by computability-theoretic results such as those discussed by Ng, Stephan, and Wu [5]. Thus the theorem does not say that conditioning can realize an arbitrary Δ_2^0 real. It identifies an exact intermediate class using a concrete mass constraint.

Theorem 8.4 (The full-success boundary). *The conditional probabilities of effective two-outcome processes that return one of the two outcomes with probability one are exactly the computable reals in $[0, 1]$.*

Proof. Here $a + b = 1$. Increasing rational lower approximations give intervals $[a_s, 1 - b_s]$ containing a , whose widths tend to zero. To compute a to any required precision, search for an interval of sufficiently small width. The search terminates under the probability-one promise. Conversely, for a computable c , both c and $1 - c$ are uniformly left-c.e. Their two-label law has total mass one and is realized by Theorem 4.2. $\square$

There is therefore a genuine endpoint distinction: every fixed permitted failure probability greater than zero allows all weakly computable conditional probabilities; requiring failure probability exactly zero restricts the class to computable probabilities. This is an expressivity statement, not a physical procedure for obtaining a noncomputable real to certified arbitrary precision from experimental samples.

9 The arithmetic hierarchy of probability questions

9.1 Indexing and promises

A Σ_1^0 relation has an existential quantifier over a computable predicate; Σ_2^0 has an existential block followed by a universal block. The dual classes are Π_1^0 and Π_2^0 . Completeness below means computable many-one completeness. Inputs are program indices for a fixed effective universal model, with rational threshold $q \in (0, 1)$ fixed in advance.

When a success floor is required, use an effective family of wrappers that reserves a positive rational mass for guaranteed return before handing the remaining mass to an arbitrary source program. Restrict the syntax to wrappers whose reserved mass is at least the specified δ . All indices in that family satisfy the floor; no undecidable semantic validation is included in the language being classified. Equivalently, one may state the upper bounds as promise-problem bounds and ensure that every hardness reduction lands inside the promise. We will do both explicitly rather than silently conjoin an arbitrary-code problem with an additional floor predicate.

These distinctions matter because almost-sure termination itself has high arithmetic complexity, as is known from prior work [6]. Our proofs below establish the particular threshold claims directly.

9.2 Unconditional probabilities

Theorem 9.1 (Unconditional comparison table). *For a fixed interior rational q and a universal effective output-law model, the following classifications hold:*

$p > q$	$p \leq q$	$p < q$	$p \geq q$	$p = q$	$p \neq q$
$\Sigma_1^0\text{-c.}$	$\Pi_1^0\text{-c.}$	$\Sigma_2^0\text{-c.}$	$\Pi_2^0\text{-c.}$	$\Pi_2^0\text{-c.}$	$\Sigma_2^0\text{-c.}$

Here “-c.” means complete for the displayed class.

Proof. Let $p_s \uparrow p$ be computable rational lower bounds. Then $p > q \iff \exists s (p_s > q)$ and $p \leq q$ is its complement. Moreover,

$$p \geq q \iff \forall k \exists s (p_s > q - 2^{-k}),$$

so $p \geq q$ is Π_2^0 and $p < q$ is Σ_2^0 . Equality is the conjunction of $p \geq q$ and $p \leq q$, hence is Π_2^0 ; inequality is its complement.

For the first hardness claim, use a rational baseline q and add a small rational mass $\varepsilon > 0$ precisely when a given ordinary computation halts. The resulting mass exceeds q exactly in the halting case. Taking complements proves the Π_1^0 claim.

For the remaining claims, let φ_e be the e th ordinary partial computable function. A fair-coin program chooses $n \geq 0$ with probability $2^{-(n+1)}$, simulates $\varphi_e(n)$, and returns precisely if that simulation halts. Its return probability is

$$t_e = \sum_{n \geq 0} 2^{-(n+1)} \mathbf{1}_{\{\varphi_e(n) \downarrow\}}. \quad (9.1)$$

Thus $t_e = 1$ exactly when e belongs to the Π_2^0 -complete totality set. Set $p_e = qt_e$. Then $p_e < q$ and $p_e \neq q$ exactly for nontotal indices, whereas $p_e \geq q$ and $p_e = q$ exactly for total indices. These are the required reductions. $\square$

For completeness, totality has the form $\forall n \exists t \text{ “}\varphi_e(n) \text{ halts by } t\text{”}$, and every Π_2^0 predicate $\forall n \exists t R(e, n, t)$ with computable R reduces to it by the program that, on n , searches for such a t . No oracle is used by the probabilistic program in (9.1). [write] Part V re-derives this table for fair-coin programs with fixed quartic normal forms, and adds the rows $p = 1$, $p = 0$ and finite expected runtime (Theorem 46.3); Part VI’s Theorems 79.1 and 79.2 give the endpoint cases for algebraic quantum controllers.

9.3 Conditioning defeats the existential threshold boundary

Theorem 9.2 (Conditional comparison at any subunit success floor). *Fix rationals $q \in (0, 1)$ and $\delta \in (0, 1)$. In a universal effective wrapper family guaranteeing $a + b \geq \delta$, set $c = a/(a + b)$. Then*

$c > q$	$c < q$	$c \neq q$	$c \geq q$	$c \leq q$	$c = q$
$\Sigma_2^0\text{-c.}$	$\Sigma_2^0\text{-c.}$	$\Sigma_2^0\text{-c.}$	$\Pi_2^0\text{-c.}$	$\Pi_2^0\text{-c.}$	$\Pi_2^0\text{-c.}$

In particular, the strict conditional comparisons are not existential Diophantine relations in such a universal family.

Proof. Choose computable rational lower approximations a_s, b_s . Their normalized ratios $c_s = a_s/(a_s + b_s)$ are computable once the denominator is positive; use any fixed rational value before that point. They converge to c . Consequently

$$c > q \iff \exists k \exists n \forall s \geq n \quad c_s \geq q + 2^{-k}. \quad (9.2)$$

The analogous expression handles $c < q$. Thus these strict cuts are Σ_2^0 , and their complements are Π_2^0 . Equality has the more useful form

$$c = q \iff \forall k \forall n \exists s \geq n \quad |c_s - q| < 2^{-k}. \quad (9.3)$$

Convergence makes repeated arbitrary closeness equivalent to equality, so this is a Π_2^0 expression. Its complement is Σ_2^0 .

For hardness choose a positive rational $\varepsilon < \min(1 - \delta, q, 1 - q)$ and use t_e from (9.1). The first wrapper has masses

$$a_e = q, \quad b_e = 1 - q - \varepsilon + \varepsilon t_e. \quad (9.4)$$

It reserves mass $1 - \varepsilon$ for guaranteed return and assigns the remaining mass to the totality sampler. Its heralding probability is $h_e = 1 - \varepsilon + \varepsilon t_e \geq \delta$, and $c_e = q/h_e \geq q$. It follows that $c_e > q$ and $c_e \neq q$ hold exactly when e is nontotal, while $c_e \leq q$ and $c_e = q$ hold exactly when e is total.

For the other direction use

$$a_e = q - \varepsilon + \varepsilon t_e, \quad b_e = 1 - q. \quad (9.5)$$

The same denominator gives $c_e - q = -(1 - q)\varepsilon(1 - t_e)/h_e$. Thus $c_e < q$ exactly for nontotal indices and $c_e \geq q$ exactly for total indices. All masses in both wrappers are nonnegative and their sums are at most one. These reductions are effective, preserve the syntactic floor guarantee, and prove every hardness assertion. MRDP and the strict arithmetic hierarchy then exclude ordinary existential Diophantine representations for the complete Σ_2^0 cuts. $\square$

For example, the theorem applies with a success guarantee of 0.999999. The difficulty is not confined to rare postselection events: the unknown remaining denominator mass can encode a universal nontermination question. A finite lower bound for a does not certify that future additions to b will not reverse a conditional threshold comparison.

Remark 9.3 (A quantifier trap). The naive formula saying that c_s is eventually within every tolerance of q has the pattern $\forall k \exists n \forall s$ and only gives a Π_3^0 upper bound. Formula (9.3) is sharper because convergence is already known. Omitting that argument would obscure the exact boundary.

9.4 The probability-one promise

Proposition 9.4 (Comparisons under almost-sure return). *Under the promise $a + b = 1$, strict comparisons $c > q, c < q$ and inequality $c \neq q$ are Σ_1^0 -complete promise problems. Weak comparisons and equality are Π_1^0 -complete promise problems. The reductions can be chosen from a uniformly total family of samplers.*

Proof. Under the promise $c = a$ and $1 - c = b$, so $c > q$ and $c < q$ are witnessed by lower approximations to a and b , respectively. Their disjunction is inequality; taking complements within the promise gives the upper bounds.

Let $x_e = 2^{-t}$ if a designated computation halts at time t , and let $x_e = 0$ if it never halts. This is a uniformly computable real: after s steps, a still-undetected later halting time can contribute at most 2^{-s} . For a sufficiently small positive rational ε , the probabilities $q + \varepsilon x_e$ and $q - \varepsilon x_e$ lie in $[0, 1]$ and are uniformly computable. Their complementary two-label laws have uniform total samplers by Theorem 4.2. Strict comparison in the appropriate direction, or inequality with q , detects halting; equality and the opposite weak comparison detect nonhalting. $\square$

This is a promise classification. Deciding which arbitrary probabilistic programs satisfy the promise has not become easy. It is incorrect to erase that promise and infer a Π_1^0 algorithm for arbitrary exact-output questions. [write] Part V's Theorems 56.1 and 60.6 use the same halting-time device for exact stationary masses of reversible, uniformly mixing chains.

9.5 Polynomial quantifier normal forms

The hierarchy can itself be expressed using quartic integer equations. For a computable relation $R(e, u, v)$, MRDP applies both to R and to its complement. Consequently the four normal forms are

Class	Quartic arithmetic form, with natural quantified tuples
Σ_1^0	$\exists \mathbf{w} P(e, \mathbf{w}) = 0$
Π_1^0	$\forall \mathbf{w} P(e, \mathbf{w}) \neq 0$
Π_2^0	$\forall u \exists v, \mathbf{w} P(e, u, v, \mathbf{w}) = 0$
Σ_2^0	$\exists u \forall v, \mathbf{w} Q(e, u, v, \mathbf{w}) \neq 0$

For the last line choose Q to represent $\neg R$, so that universal nonvanishing expresses $\forall v R(e, u, v)$. Apply Lemma 3.3 to each matrix. These are arithmetic formulas *over* Diophantine relations, not all existential Diophantine representations. The strict hierarchy is exactly why the distinction cannot generally be removed.

All tables assumed $0 < q < 1$. At endpoints some classifications collapse: $p > 0$ is Σ_1^0 , $p = 0$ is Π_1^0 , $p = 1$ can be Π_2^0 -complete, and $p > 1$ is impossible. Endpoint statements must not be read off the interior tables without checking them.

Part II

Quantum substrates (manuscripts 08 and 07)

10 [write] Two quantum models

Part II joins two independent treatments of quantum computation. Manuscript 08 (Sections 11–12) works with exact amplitudes of Clifford+ T circuits over $\mathbb{Z}[1/\sqrt{2}, i]$, with the terminal classical output laws of measured programs, and with an operator-valued analogue of Part I's normalization budget. Manuscript 07 (Sections 13–22) works with *rational real* instruments $\sum_j A_j^\top A_j = I$ with one Kraus operator per outcome, and with the exact zero-probability (mortality) problem for their outcome words. The models overlap only in their common semantics of unnormalized branch states ($A_w \rho A_w^\dagger$ in 07, the unnormalized measured state in 08); there is no shared theorem. 08's corollary on quantum output laws in Section 11 makes every effective quantum program's terminal output law

a fixed-quartic projected-root law, while 07's Theorem 18.2 shows that the *zero set* of the output probabilities of rational instruments is undecidable already in dimension four; the two statements are compatible, since a strict threshold $p > q$ is Σ_1^0 while $p = 0$ is Π_1^0 (Theorem 9.1).

11 An explicit quantum computational front end

11.1 Exact amplitudes rather than floating-point paths

The Clifford+ T gate family admits exact arithmetic in $\mathbb{Z}[1/\sqrt{2}, i]$; the exact synthesis literature gives a much stronger structural account of this ring and its realizable operators [7]. Here we only need a direct forward evaluator for the gates H, T, S, X , and CNOT.

After L elementary gates, represent each amplitude as

$$\psi_j = \frac{a_j + b_j\sqrt{2} + i(c_j + d_j\sqrt{2})}{2^L}, \quad a_j, b_j, c_j, d_j \in \mathbb{Z}. \quad (11.1)$$

The common denominator is intentionally unreduced: each gate increments L by one, even when it could be implemented without a new denominator. For $z = a + b\sqrt{2} + i(c + d\sqrt{2})$ the relevant integer-linear maps are

$$\begin{aligned} \sqrt{2}z &\longleftrightarrow (2b, a, 2d, c), \\ iz &\longleftrightarrow (-c, -d, a, b), \\ \sqrt{2}(1+i)z &\longleftrightarrow (2(b-d), a-c, 2(b+d), a+c). \end{aligned} \quad (11.2)$$

For an H gate on an amplitude pair (z_0, z_1) , the new numerators are $\sqrt{2}(z_0 + z_1)$ and $\sqrt{2}(z_0 - z_1)$. For T , multiply the bit-zero numerator by two and the bit-one numerator by $\sqrt{2}(1+i)$. For S , use two and $2i$ respectively. The X and CNOT gates permute amplitudes and multiply all numerators by two. Dividing by the new 2^{L+1} gives precisely the usual gate actions.

The squared norm of one numerator is

$$|z|^2 = A + B\sqrt{2}, \quad A = a^2 + 2b^2 + c^2 + 2d^2, \quad B = 2(ab + cd). \quad (11.3)$$

Thus for a selected set J of measurement outcomes,

$$\Pr(J) = \frac{\sum_{j \in J} A_j + (\sum_{j \in J} B_j)\sqrt{2}}{4^L}. \quad (11.4)$$

The symbolic coefficient B can be negative, although the final probability is nonnegative.

Proposition 11.1 (Decidable finite-circuit probabilities). *For a finite circuit over these gates with a computational-basis input, comparison of a terminal basis-event probability with a rational threshold is decidable by exact integer arithmetic. Its graph and its comparison relations are Diophantine, with quartic representations.*

Proof. The gate recurrences compute the four integer coordinates exactly. After clearing positive denominators, a rational comparison reduces to the sign of $A + B\sqrt{2}$ with $A, B \in \mathbb{Z}$. If either coefficient is zero or they have the same sign, the sign is immediate. If they have opposite signs, compare A^2 and $2B^2$, reversing the resulting sign when $A < 0$. Equality $A^2 = 2B^2$ with both coefficients nonzero is impossible: cancelling common factors would give a reduced rational square

root of two. This supplies a terminating integer algorithm and therefore a computable relation. Apply MRDP and Lemma 3.3.

For a direct bounded evaluator, use linear gate residuals and quadratic norm residuals. Signed intermediate coordinates can be encoded by their natural positive and negative parts; adding $z^+z^- = 0$ makes that split canonical. The finite comparison algorithm can be lowered to the same arithmetic-gate format. This last observation does not assert the particular witness count of Theorem 6.1 for the quantum compiler. [write] Part VI gives explicit counts, a unique witness and heights for this evaluator (theorem 75.1). $\square$

Example 11.2 (Cancellation and an irrational output probability). Starting in $|0\rangle$, the circuit H^2 returns outcome zero with probability one. If its intermediate basis alternatives were treated as independent classical alternatives without recombination, the cancellation in the outcome-one amplitude would be lost.

The circuit HTH , acting right to left on the input as usual, gives

$$\Pr(0) = \frac{2 + \sqrt{2}}{4}, \quad \Pr(1) = \frac{2 - \sqrt{2}}{4}.$$

The accompanying sequential evaluator applies H , then T , then H , which is the same palindromic circuit. Both probabilities are represented exactly, not by decimal approximations. A finite fair-coin decision tree with a uniform depth bound has only dyadic output probabilities, so no such tree can realize this law exactly. The unbounded sampler in Theorem 4.2 can realize it and, because the two masses sum to one, returns almost surely.

11.2 Unbounded measured programs

Consider an unbounded classical controller with finitely described quantum gates and projective measurements, allowing only finitely many qubits at each finite execution stage. A finite measurement history has a weight computed by evolving an *unnormalized* state through its gates and measurement projections. Its squared norm is the history probability. Keeping the state unnormalized avoids division by a possibly zero branch probability.

Distinct recorded measurement histories are mutually exclusive classical events. Summing the weights of first-return histories therefore gives a presentation in Definition 3.1. Lemma 3.2 and Theorem 4.3 apply. The same argument works for uniformly computable gate entries when valid evolution and an effective finite history presentation are part of the model assumptions; exact algebraic zero tests are unnecessary for obtaining lower bounds.

Corollary 11.3 (Quantum output-law Diophantine representation). *Every such effective quantum program's discrete classical terminal output law has the fixed-quartic projected-root representation (4.2). Its ordinary strict output-probability thresholds are existential Diophantine. Its conditional eventual-return thresholds are subject to the distinctions in Section 9.*

Proof. The measurement-history presentation gives a left-c.e. law; apply the corresponding representation and threshold theorems. Classical fair-coin control is a special case, so the general hardness obstructions cannot be removed by admitting quantum operations. $\square$

This is an exact statement about a terminal *classical probability law*. It is not an efficient classical simulation of quantum computation, not a replacement for quantum states in interactive protocols,

and not a claim that all quantum behavior is captured by classical random variables. The reference state-vector implementation uses 2^n amplitudes for n qubits and is deliberately a small-instance exact checker. It does not implement the unbounded universal MRDP compiler.

12 An operator-valued normalization budget

12.1 Accumulated ensembles

A quantum analogue of coordinatewise accumulated output mass is a sequence of positive semidefinite operators ρ_s , with $\rho_s - \rho_{s-1} \succeq 0$ and $\text{Tr } \rho_s \leq 1$. For example, these may represent sums of subnormalized density operators over disjoint returned histories. Write

$$\rho_s = h_s \sigma_s, \quad \text{Tr } \sigma_s = 1.$$

This is an accumulation process, not the state trajectory of a single system undergoing arbitrary unitary evolution.

For positive definite density matrices define

$$R(\sigma, \tau) = \inf\{r > 0 : \sigma \preceq r\tau\} = \lambda_{\max}(\tau^{-1/2}\sigma\tau^{-1/2}). \quad (12.1)$$

The logarithm of this quantity is the established max-relative entropy $D_{\max}(\sigma\|\tau)$, with logarithm-base conventions adjusted to our natural logarithm [8]. The equality in (12.1) follows by conjugating the operator inequality by $\tau^{-1/2}$.

Theorem 12.1 (Exact operator accumulation cost). *For a finite or infinite sequence of positive definite density matrices $\sigma_0, \sigma_1, \dots$, put*

$$R_s = R(\sigma_{s-1}, \sigma_s), \quad G_s = \prod_{j=1}^s R_j.$$

There are accumulated ensembles $\rho_s = h_s \sigma_s$ with $h_0 = \delta > 0$, $h_s \leq 1$, and $\rho_s - \rho_{s-1} \succeq 0$ if and only if $\delta G_s \leq 1$ for every s . Their unique pointwise minimal total masses are $h_s^ = \delta G_s$. Equivalently, the logarithmic cost is*

$$\log G_N = \sum_{s=1}^N D_{\max}(\sigma_{s-1}\|\sigma_s). \quad (12.2)$$

Proof. A single monotonicity condition is $h_s \sigma_s \succeq h_{s-1} \sigma_{s-1}$, equivalent to $h_s/h_{s-1} \geq R_s$. Taking traces shows $R_s \geq 1$. The proof of Theorem 5.2 now applies verbatim to the scalar factors: iteration gives necessity, and choosing equality at every step gives positive semidefinite increments and sufficiency. $\square$

For singular density matrices, the same result holds with $R_s = +\infty$ when $\text{supp}(\sigma_{s-1})$ is not contained in $\text{supp}(\sigma_s)$, and with the inverse taken on the latter support otherwise. To see the obstruction, a vector in the new kernel must also lie in the old kernel whenever the old operator is bounded above by a multiple of the new one. Under support inclusion, the finite-dimensional restriction gives the same formula.

Corollary 12.2 (Data processing and trace variation). *For a completely positive trace-preserving map Λ ,*

$$R(\Lambda\sigma, \Lambda\tau) \leq R(\sigma, \tau).$$

Thus applying the same channel at each checkpoint cannot increase the optimal accumulation cost. Moreover, any feasible operator path satisfies

$$\sum_{s=1}^N \frac{1}{2} \|\sigma_s - \sigma_{s-1}\|_1 \leq \log(1/\delta).$$

Proof. A positive map preserves $\sigma \preceq r\tau$, so every feasible r before the map is feasible afterward. For the variation claim, if $R_s > 1$ then

$$\sigma_s = R_s^{-1} \sigma_{s-1} + (1 - R_s^{-1}) \omega_s$$

for a positive trace-one operator ω_s . The triangle inequality and $\|\omega_s\|_1 = \|\sigma_{s-1}\|_1 = 1$ give trace distance at most $1 - R_s^{-1}$. Handle $R_s = 1$ by equality, sum, and use the scalar logarithmic inequality from Proposition 7.1. $\square$

The data-processing property of max-relative entropy is known [8]. Here its immediate consequence is monotonicity of an entire accumulation budget, rather than a new entropy monotonicity theorem.

12.2 Two exact matrix examples

Example 12.3 (A coherent excursion erased by dephasing). Let

$$\sigma_0 = \sigma_2 = I/2, \quad \sigma_1 = \begin{pmatrix} 1/2 & t \\ t & 1/2 \end{pmatrix}, \quad 0 < t < 1/2.$$

The eigenvalues of σ_1 are $1/2 \pm t$. Thus

$$R_1 = \frac{1}{1-2t}, \quad R_2 = 1+2t, \quad \delta_{\max} = \frac{1-2t}{1+2t}.$$

At $t = 1/6$ the cost is two and the maximum initial mass is $1/2$. Dephasing in the displayed basis sends every matrix to $I/2$ and reduces the cost to one. These matrices commute because the endpoints are scalar; the example demonstrates a coherence-sensitive cost, not noncommutation.

Example 12.4 (A genuinely noncommuting cycle). Set

$$\rho = \begin{pmatrix} 2/3 & 0 \\ 0 & 1/3 \end{pmatrix}, \quad \sigma = \begin{pmatrix} 1/2 & 1/6 \\ 1/6 & 1/2 \end{pmatrix}.$$

Their commutator is nonzero. Both have determinant $2/9$, and

$$\det(r\sigma - \rho) = \det(r\rho - \sigma) = \frac{4r^2 - 9r + 4}{18}.$$

The least admissible domination factor in either direction is $R = (9 + \sqrt{17})/8$. Indeed, the determinant vanishes there and the Hermitian difference has positive trace, hence is positive semidefinite; the smaller root is below one and cannot dominate a trace-one operator. For the cycle $\rho \rightarrow \sigma \rightarrow \rho$,

$$G = R^2 = \frac{49 + 9\sqrt{17}}{32}, \quad \delta_{\max} = \frac{49 - 9\sqrt{17}}{32}.$$

Dephasing turns the middle state into $I/2$ and gives the strictly smaller cycle cost $(4/3)(3/2) = 2$. The symbolic checker verifies both determinant identities and the reciprocal cost exactly.

The operator theorem is a finite-dimensional order optimization theorem. It does not by itself compile every prescribed computable matrix sequence into a circuit over the discrete gate set of Section 11. Exact state preparation and a small Diophantine certificate for generalized eigenvalue maxima are separate problems.

13 Research target, provenance, and the precise contribution

The suggested research direction is particularly promising when “computation” is not identified with a discrete execution log. A quantum outcome word carries an ordered product of matrices. Its possibility depends on exact cancellation among amplitudes. Encoding only the existence of combinatorial paths can therefore give the wrong answer. This manuscript studies that cancellation directly and connects it to the natural-number polynomial interfaces already present in ProveIt.

The quantum measurement occurrence problem is already known to be undecidable. Eisert, Müller, and Gogolin established this for nine rational Kraus operators in dimension fifteen [10]. The research question pursued here is more specific: can the normalization of an arbitrary matrix-mortality instance be made much smaller while retaining *all* outcome words and a transparent relation between witnesses? The answer proved below is affirmative. This is a quantitative refinement of a known undecidability mechanism, not a new discovery of that mechanism.

13.1 Dependency and originality ledger

The logical dependencies should be kept separate from the proposed contribution.

Component	Status in this manuscript
Matrix mortality hardness	Imported from Neary’s published six-generator, 3×3 result; its proof is not reproduced [11].
Rational quadratic forms	Meyer’s theorem and Lagrange’s four-square theorem are classical inputs; the consequences needed here are proved explicitly [12].
Packed Kraus normalization	Full proof given here, including the erasure and one-step witness bounds. This is the principal contribution proposed for review.
Fixed-length quartics	Explicit formulas, correctness proof, exact counts, executable compiler, and sample expanded polynomial. The underlying arithmetic-circuit technique is standard.
Unbounded Diophantine form	Consequence of MRDP, not a new MRDP theorem and not a single-fold result [14, 13].
Validation	Exact finite computations; no machine-checked proof of the general theorems and no independent priority certification.

The literature check used the published STACS 2015 version of Neary’s result. In particular, the numerical input used throughout is *six* 3×3 matrices, as in its Corollary 12 [11]; no stronger bound from a differently worded early preprint is needed. The comparison with dimension fifteen is with the explicit theorem in [10]. A targeted literature search is not sufficient to declare the bound below the best previously unknown bound in every formulation of quantum mortality.

14 Words, mortality, and exact quantum outcomes

Throughout, $\mathbb{N}$ includes zero. Matrices have finite dimension, and an input rational number is specified exactly by integers. All products are ordered chronologically: for a word $w = i_1 \cdots i_n$,

$$M_w = M_{i_n} \cdots M_{i_1}, \quad M_\varepsilon = I_d. \quad (14.1)$$

This convention will also be used in the code. It eliminates the common ambiguity about whether appending a letter multiplies on the left or on the right.

Definition 14.1. A finite list $\mathcal{M} = (M_1, \dots, M_k)$ of $d \times d$ matrices is *mortal* if there is a nonempty word w with $M_w = 0$. Let $\mu(\mathcal{M})$ be the smallest such length, and put $\mu(\mathcal{M}) = \infty$ when there is no such word.

Repeated generators are allowed: labels are outcomes, not necessarily distinct matrices. Repeating a generator does not change whether a list is mortal. The empty word is never a witness because $d \geq 1$.

A rational, real, single-Kraus-per-outcome instrument on $\mathbb{C}^D$ is a list

$$\mathcal{A} = (A_0, \dots, A_k), \quad A_j \in \mathbb{Q}^{D \times D}, \quad \sum_{j=0}^k A_j^\top A_j = I_D. \quad (14.2)$$

The use of real entries is a restriction within ordinary complex quantum mechanics, not a different theory. For a density matrix ρ , the unnormalized state after the outcome word w is $A_w \rho A_w^\dagger$. Thus its probability is

$$p_\rho(w) = \text{tr}(A_w \rho A_w^\dagger). \quad (14.3)$$

Using unnormalized states avoids division by a zero probability when describing an impossible word. These are the instrument semantics used in the known occurrence problem [10].

Lemma 14.2 (Full-rank zero test). *If ρ is positive definite, then $p_\rho(w) = 0$ if and only if $A_w = 0$. In particular, for $\rho = I_D/D$,*

$$p(w) = \frac{\|A_w\|_F^2}{D}. \quad (14.4)$$

Proof. The expression in (14.3) equals $\|A_w \rho^{1/2}\|_F^2$. A sum of squared absolute values vanishes exactly when every entry vanishes. Since $\rho^{1/2}$ is invertible, this is equivalent to $A_w = 0$. The stated formula follows by substituting $\rho = I_D/D$. $\square$

The full-rank condition matters. For example, a nonzero projection can annihilate a particular pure state. Without full rank, an impossible word need not represent a zero matrix product. All hardness statements below use a full-rank state, and the explicit probability calculations use the maximally mixed state.

15 Rational Gram factors and their row budget

Let Q be a symmetric positive definite rational $d \times d$ matrix. We need a rational rectangular factor

$$Q = B^\top B, \quad B \in \mathbb{Q}^{s \times d}. \quad (15.1)$$

Over the reals one can take $s = d$, but the entries of a square factor need not be rational. Replacing an irrational square root by a decimal approximation would destroy exact normalization. The number of rational rows is therefore an actual resource in the construction.

15.1 The elementary four-square construction

Proposition 15.1 (An explicit $4d$ -row factor). *Every positive definite rational $d \times d$ matrix has a rational Gram factor with $4d$ rows. Such a factor can be found using rational elimination and finite integer searches.*

Proof. Rational elimination gives

$$Q = L \operatorname{diag}(q_1, \dots, q_d) L^\top,$$

where L is unit lower triangular over $\mathbb{Q}$ and every q_j is positive rational. This is the usual $LDL^\top$ factorization; positivity ensures that every pivot is positive and nonzero. Write $q_j = a_j/b_j$ with positive integers a_j, b_j . Lagrange’s four-square theorem gives integers $t_{j1}, \dots, t_{j4}$ such that

$$a_j b_j = \sum_{\ell=1}^4 t_{j\ell}^2, \quad q_j = \sum_{\ell=1}^4 (t_{j\ell}/b_j)^2.$$

For each (j, ℓ) , put a row $(t_{j\ell}/b_j) e_j^\top L^\top$ into B . Summing its outer products gives (15.1). The four-square search is finite: it suffices to enumerate integer coordinates of absolute value at most $\sqrt{a_j b_j}$. No efficient bit-complexity bound is being asserted. $\square$

Zero rows can be omitted in a concrete example. The uniform $4d$ bound deliberately does not rely on such simplification. This construction is implemented directly in the package, using exact fractions rather than floating-point Cholesky factorization.

15.2 A classical improvement to $d + 3$ rows

The stronger row count follows from the following standard rational quadratic-form theorem [12].

Theorem 15.2 (Meyer; classical input). *A nondegenerate indefinite quadratic form over $\mathbb{Q}$ in at least five variables has a nonzero rational isotropic vector.*

For clarity, indefinite means that the form takes both positive and negative values over $\mathbb{R}$. Theorem 15.2 is an imported theorem, not proved from elementary number theory here. It is a consequence of the Hasse–Minkowski theorem and the local isotropy theorem for rational quadratic forms of rank at least five. The weaker bound in Proposition 15.1 does not need this input.

Lemma 15.3 (Representation by a positive four-dimensional space). *Let V be a positive definite rational quadratic space of dimension at least four. Every positive rational number a is represented by its quadratic form.*

Proof. On $V \oplus \mathbb{Q}$, consider the form $q(v) - at^2$. It is nondegenerate, indefinite, and has dimension at least five. Meyer gives a nonzero rational solution to $q(v) = at^2$. If $t = 0$, positive definiteness forces $v = 0$, a contradiction. Therefore $q(v/t) = a$. $\square$

Proposition 15.4 (A $(d + 3)$ -row factor). *Every positive definite rational $d \times d$ matrix admits a rational Gram factor with $d + 3$ rows. There is a total search algorithm for such a factor.*

Proof. As above, write $Q = L \operatorname{diag}(q_1, \dots, q_d) L^\top$. In the standard positive quadratic space $\mathbb{Q}^{d+3}$, choose successively rational vectors $v_1, \dots, v_d$ satisfying

$$v_i^\top v_j = 0 \ (i \neq j), \quad v_j^\top v_j = q_j.$$

Suppose $v_1, \dots, v_{j-1}$ have been chosen. They are linearly independent because their norms are positive and they are orthogonal. Their rational orthogonal complement has dimension

$$d + 3 - (j - 1) \geq 4.$$

The restriction of the standard form is positive definite. Lemma 15.3 therefore supplies v_j . If V is the matrix whose columns are these vectors, then $V^\top V = \operatorname{diag}(q_1, \dots, q_d)$, so $B = VL^\top$ works.

For effectivity one does not need an efficient implementation of the local theory. Enumerate rational $(d + 3) \times d$ matrices by increasing common denominator and numerator height, and test $B^\top B = Q$ exactly. Existence guarantees termination. This is an algorithmic existence statement, not a claim that the search is practical on large inputs. $\square$

15.3 The general rational row bound is sharp

The next fact is included to identify a genuine arithmetic limitation of the method. It is not offered as a new theorem about quadratic forms.

Proposition 15.5 (Sharpness for arbitrary positive rational forms). *For every $d \geq 1$, the matrix $Q_d = \operatorname{diag}(1, \dots, 1, 7)$ has no rational Gram factor with $d + 2$ rows. Consequently $d + 3$ is the smallest uniform row bound valid for all positive definite rational $d \times d$ matrices.*

Proof. First, 7 is not a sum of three rational squares. Otherwise clearing denominators gives

$$x^2 + y^2 + z^2 = 7t^2,$$

with integers $t > 0$ and with no common factor among x, y, z, t . If t is even, reduction modulo 4 forces x, y, z all even, contradicting primitivity. If t is odd, the right side is 7 modulo 8, whereas a sum of three residues from $\{0, 1, 4\}$ cannot be 7 modulo 8.

Now suppose $B^\top B = Q_d$ with $B \in \mathbb{Q}^{(d+2) \times d}$. The first $d - 1$ columns are rational orthonormal vectors. Any rational unit vector u can be carried to e_1 by a rational orthogonal matrix: if $u \neq e_1$, use the Householder reflection

$$H = I - \frac{2(u - e_1)(u - e_1)^\top}{(u - e_1)^\top (u - e_1)}.$$

It is rational and orthogonal and satisfies $Hu = e_1$. If $u = e_1$, use the identity. Applying this procedure successively in orthogonal complements sends the first $d - 1$ columns to the first $d - 1$ coordinate vectors. The last column is then supported on the remaining three coordinates and has squared norm 7, a contradiction. The case $d = 1$ is exactly the first argument. A factor with fewer rows could be padded to $d + 2$ rows, so it is also impossible. $\square$

This does *not* prove that our final quantum dimension is optimal. The residual forms used below are special forms $c^2 I - \sum_i M_i^\top M_i$, and c is under our control. A better choice of c , a different embedding, or a different reduction could outperform the general-purpose bound.

16 The packed normalization theorem

We now state the main algebraic result. The row packing, rather than the existence of some normalization, is the important operation.

Theorem 16.1 (Mortality-preserving rational Gram packing). *Let $\mathcal{M} = (M_1, \dots, M_k)$ be rational $d \times d$ matrices, with $d, k \geq 1$. Choose a positive rational c such that*

$$Q = c^2 I_d - \sum_{i=1}^k M_i^\top M_i \quad \text{is positive definite.} \quad (16.1)$$

Suppose $Q = B^\top B$ with s rational rows. Put

$$r = \lceil s/k \rceil, \quad D = d + r.$$

Pad B with zero rows to obtain kr rows, and split it into k consecutive $r \times d$ blocks $B_1, \dots, B_k$. Define

$$A_i = \alpha \begin{pmatrix} M_i & 0_{d \times r} \\ B_i & 0_{r \times r} \end{pmatrix} \quad (1 \leq i \leq k), \quad A_0 = \begin{pmatrix} \beta I_d & 0 \\ 0 & I_r \end{pmatrix}, \quad \alpha = \frac{4}{5c}, \quad \beta = \frac{3}{5}. \quad (16.2)$$

Then $\mathcal{A} = (A_0, \dots, A_k)$ is an exactly normalized rational instrument, and

$$\text{Mort}(\mathcal{A}) \iff \text{Mort}(\mathcal{M}). \quad (16.3)$$

If mortal, the shortest witness lengths satisfy

$$\mu(\mathcal{M}) \leq \mu(\mathcal{A}) \leq \mu(\mathcal{M}) + 1. \quad (16.4)$$

Proof. We separate the normalization, soundness, and completeness arguments.

Normalization. Since the right block columns of A_i vanish,

$$\sum_{i=1}^k A_i^\top A_i = \alpha^2 \begin{pmatrix} \sum_i M_i^\top M_i + \sum_i B_i^\top B_i & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} (16/25)I_d & 0 \\ 0 & 0 \end{pmatrix}.$$

On the other hand, $A_0^\top A_0 = \text{diag}((9/25)I_d, I_r)$. Their sum is I_D . All entries are rational. The constants $3/5$ and $4/5$ are simply a rational point on the unit circle with both coordinates nonzero.

No false mortal word. Every generator is block lower triangular. Taking its upper-left $d \times d$ block is multiplicative within this block-triangular algebra. If w has m active letters from $\{1, \dots, k\}$ and z idle letters 0, and u is obtained by deleting all zeros, then

$$(A_w)_{\text{top}} = \alpha^m \beta^z M_u. \quad (16.5)$$

If $m = 0$, then $A_w = A_0^z$ is invertible and cannot vanish. If $A_w = 0$ and $m > 0$, the nonzero scalar in (16.5) implies $M_u = 0$. Thus every target witness yields a nonempty source witness, of length no greater than $|w|$.

Every source witness has a target extension. If u is nonempty and $M_u = 0$, the product of the active matrices has the form

$$A_u = \begin{pmatrix} 0 & 0 \\ R & 0 \end{pmatrix}$$

for some $r \times d$ matrix R . Multiplication on the left by any active matrix kills this residual:

$$A_i A_u = \alpha \begin{pmatrix} M_i & 0 \\ B_i & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ R & 0 \end{pmatrix} = 0 \quad (1 \leq i \leq k). \quad (16.6)$$

Therefore ui is a target witness. Applying the preceding two directions to shortest words gives (16.4). $\square$

16.1 Effective choice of the scalar

Let $S_{\mathcal{M}} = \sum_i M_i^T M_i$ and $t_{\mathcal{M}} = \text{tr}(S_{\mathcal{M}}) \geq 0$. Since $S_{\mathcal{M}}$ is positive semidefinite, $\lambda_{\max}(S_{\mathcal{M}}) \leq t_{\mathcal{M}}$. Hence any integer $c > \sqrt{t_{\mathcal{M}}}$ satisfies (16.1). An exact choice is

$$c = \left\lfloor \sqrt{\lfloor t_{\mathcal{M}} \rfloor} \right\rfloor + 1.$$

This can be computed with integer square root and rational arithmetic. No numerical eigenvalue calculation is required. A smaller user-supplied rational c is also allowed, provided positive definiteness of Q is checked exactly, for example by LDL^T pivots.

16.2 Why packing saves dimension

The identity $B^T B = Q$ constrains the *sum* of the contributions of its rows, not which Kraus operator receives each row. There is no requirement that every generator receive an entire $d \times d$ correction matrix. The s correction rows can instead be distributed among the k pre-existing active outcomes. Each added state-space coordinate provides k such row slots. Consequently $\lceil s/k \rceil$ additional coordinates suffice.

This allocation does not introduce a separate potentially dangerous “correction generator.” Each correction block remains attached to its original M_i . Soundness is then protected by the upper-left projection, while the zero right columns make one additional active outcome sufficient to erase the lower-left residual.

Corollary 16.2 (General dimension bounds). *An input list of k rational $d \times d$ matrices admits a mortality-preserving rational instrument with $k + 1$ outcomes and dimension*

$$D_{\text{Meyer}} = d + \left\lceil \frac{d + 3}{k} \right\rceil.$$

Using only the elementary construction of Proposition 15.1, one obtains

$$D_{\text{elem}} = d + \left\lceil \frac{4d}{k} \right\rceil.$$

Both constructions are effective, and both satisfy the witness-length bound (16.4).

Proof. Apply Theorem 16.1 with the factors from Propositions 15.4 and 15.1, respectively. Zero-row padding leaves the Gram identity unchanged. $\square$

17 A stronger word-level invariant

The projection argument is enough for undecidability, but a more precise formula is useful both for arithmetic certificates and for debugging an implementation.

Proposition 17.1 (Idle erasure and the final active letter). *Suppose a target word w contains the active subsequence $i_1 \cdots i_m$, with $m \geq 1$. Let z be its total number of zeros and z_- the number of zeros before its final active letter. Put*

$$P = M_{i_{m-1}} \cdots M_{i_1},$$

where $P = I_d$ if $m = 1$. Then

$$A_w = \begin{pmatrix} \alpha^m \beta^z M_{i_m} P & 0 \\ \alpha^m \beta^{z_-} B_{i_m} P & 0 \end{pmatrix}. \quad (17.1)$$

In particular,

$$A_w = 0 \iff \begin{pmatrix} M_{i_m} \\ B_{i_m} \end{pmatrix} P = 0. \quad (17.2)$$

Arbitrary insertion or deletion of idle letters does not change whether a word with an active letter has zero product.

Proof. Before the final active letter, the upper-left block is $\alpha^{m-1} \beta^{z_-} P$. Left multiplication by A_{i_m} replaces the lower-left block by αB_{i_m} times this upper-left block: the old lower-left block contributes nothing because the right block column of A_{i_m} is zero. Each remaining idle letter multiplies the upper-left block by β and leaves the lower-left block unchanged. This proves (17.1). The two scalar factors are nonzero, so they do not affect vanishing. $\square$

The proposition should not be misread as asserting that idle insertion multiplies the *whole* product by one scalar. It generally scales the two block rows differently. Only the zero/nonzero predicate is invariant. Keeping this distinction explicit prevents an otherwise tempting false identity.

17.1 Sharpness of the one-step interval

Both endpoints of (16.4) occur. For $d = k = 1$, take $M_1 = 0$, $c = 1$, and $B_1 = (1)$. The active matrix is

$$A_1 = \frac{4}{5} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

It is nonzero but its square is zero. Hence the source minimum is 1 and the target minimum is 2.

For equality, take $d = 1$, $M_1 = 0$, $M_2 = 1$, and $c = 2$. Put $r = 3$, $B_1 = (0, 0, 0)^\top$, and $B_2 = (1, 1, 1)^\top$. Their combined Gram matrix is $3 = c^2 - (0^2 + 1^2)$, while $A_1 = 0$. Both minima are 1. These are sharpness examples for the interval, not hard mortality instances.

18 Four-dimensional, seven-outcome undecidability

We import the following published result [11, Corollary 12].

Theorem 18.1 (Six-generator mortality; external input). *There is no algorithm deciding mortality for arbitrary lists of six integer 3×3 matrices. The proof proceeds through an effective reduction from halting, via binary tag systems and Post correspondence.*

Theorem 18.2 (Dimension four and seven outcomes). *The following decision problem is undecidable: given seven rational real 4×4 matrices $A_0, \dots, A_6$ satisfying $\sum_{j=0}^6 A_j^T A_j = I_4$, decide whether there exists a nonempty outcome word with probability zero for the maximally mixed input state.*

Undecidability already holds on the subfamily in which

$$A_0 = \text{diag}(3/5, 3/5, 3/5, 1),$$

all other matrices have a zero fourth column, and their upper-left 3×3 blocks are a common nonzero rational scalar times six integer matrices.

Proof. Given the six integer matrices, choose c as above and a six-row rational Gram factor of Q . Proposition 15.4 supplies exactly the needed row budget because $d + 3 = 6$. Thus $r = \lceil 6/6 \rceil = 1$ and $D = 4$.

The construction is a total computable transformation: a rational Gram factor may be found by exhaustive search, whose termination is guaranteed. By Theorem 16.1, the source is mortal if and only if the target is mortal. By Lemma 14.2, target mortality is exactly the existence of a zero-probability word. A decision algorithm for the stated target family would therefore decide the source problem, contradicting Theorem 18.1. $\square$

Corollary 18.3 (A more elementary resource bound). *The same occurrence problem is undecidable for seven rational outcomes in dimension five, using Lagrange’s four-square theorem in place of Meyer.*

Proof. Use $s = 4d = 12$ correction rows. Then $r = \lceil 12/6 \rceil = 2$ and $D = 5$. $\square$

Construction	Source (d, k)	Quantum dimension	Outcomes
Eisert–Müller–Gogolin [10]	(3, 8)	15	9
Packed, elementary	(3, 8)	5	9
Packed, Meyer	(3, 8)	4	9
Packed, elementary with [11]	(3, 6)	5	7
Packed, Meyer with [11]	(3, 6)	4	7

The table compares proved constructions, not known optimal lower bounds. The first two packed rows do not require the improved six-generator source theorem. A separate tradeoff is available by duplicating the source generators: twelve active outcomes suffice to fit an elementary 12-row factor into one extra dimension, yielding dimension four with thirteen outcomes without Meyer.

18.1 Computational completeness and the lack of a stopping bound

Finite products can be evaluated exactly, so the positive instances of the target problem are recursively enumerable. The effective halting reductions underlying Theorem 18.1, followed by Theorem 16.1, make this set recursively-enumerable complete under computable many-one reductions. No polynomial-time bound for the Gram-search portion of this reduction is asserted.

Corollary 18.4 (No total computable witness-length bound). *There is no total computable function of a valid rational four-dimensional, seven-outcome instrument description that bounds the shortest mortal-word length whenever the instrument is mortal.*

Proof. If such a bound were available, calculate it and test every nonempty word up to that length. There are finitely many, and every product is exactly computable. The bounded search would decide the problem in Theorem 18.2. $\square$

In particular, no total computable bound depending only on the bit length of the instrument description can serve this purpose. This concerns a bound guaranteed for every positive instance. It does not exclude useful bounds for restricted matrix classes or for individual examples with separately established invariants.

19 A fully explicit interference example

An example makes the distinction between a source zero product and its one-step target extension visible. It also provides a small exact regression test in the same dimension and outcome count as Theorem 18.2.

Let

$$\begin{aligned} M_1 &= \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & M_2 &= \begin{pmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & M_3 &= \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ M_4 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & M_5 &= M_4, & M_6 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}. \end{aligned}$$

Direct calculation gives $\sum_i M_i^\top M_i = 4I_3$. Choose $c = 3$, so $Q = 5I_3$, and take the six correction rows

$$B_1 = (1, 0, 0), B_2 = (2, 0, 0), B_3 = (0, 1, 0), B_4 = (0, 2, 0), B_5 = (0, 0, 1), B_6 = (0, 0, 2). \quad (19.1)$$

Their outer products sum to $5I_3$. Equation (16.2) now gives seven explicit rational 4×4 operators, with $\alpha = 4/15$.

The chronological source word $(2, 1)$ gives

$$M_{(2,1)} = M_1 M_2 = 0.$$

Its upper-left entry vanishes by the cancellation $1 - 1 = 0$. Taking absolute values of entries destroys this cancellation: $|M_1||M_2| \neq 0$.

The corresponding target product does *not* vanish yet. Its only nonzero block is the bottom-left row

$$\alpha^2 B_1 M_2 = \alpha^2 (1, 0, 0).$$

Consequently

$$p(2, 1) = \frac{\alpha^4}{4} = \frac{64}{50625} > 0, \quad p(2, 1, 1) = 0. \quad (19.2)$$

The appended active outcome removes exactly the lower-left residual predicted by the proof. This example disproves the stronger, generally false assertion $M_w = 0 \Rightarrow A_w = 0$ for the *same* word.

All operators have a common denominator 15. Writing $H_j = 15A_j$, the exact normalization becomes

$$\sum_{j=0}^6 H_j^\top H_j = 225I_4, \quad H_0 = \text{diag}(9, 9, 9, 15), \quad H_i = 4 \begin{pmatrix} M_i & 0 \\ B_i & 0 \end{pmatrix}. \quad (19.3)$$

The artifact `data/07-quantum-mortality-four_dimensional_instrument.json` contains these integer matrices. The example is not presented as a universal instance; it is a transparent test of the general compiler.

20 Exact probability gaps and their limits

Let $\mathcal{A}$ be any rational real instrument, not necessarily one produced by the packing construction. Choose a positive common denominator N and put $H_j = NA_j \in \mathbb{Z}^{D \times D}$.

Proposition 20.1 (Arithmetic separation at a fixed length). *For a word of length n and the maximally mixed state,*

$$p(w) = \frac{\sum_{a,b} (H_w)_{ab}^2}{DN^{2n}}. \quad (20.1)$$

Hence

$$p(w) = 0 \quad \text{or} \quad p(w) \geq \delta_n := \frac{1}{DN^{2n}}. \quad (20.2)$$

Proof. The common denominator factors out of the ordered product, giving $A_w = N^{-n}H_w$. Substitute this identity into (14.4). A nonzero integer matrix has at least one nonzero integer entry, so its squared Frobenius norm is a positive integer, at least one. $\square$

The denominator argument is the same basic mechanism behind length-dependent probability separation in the earlier quantum mortality work [10]. Here the formula is stated for the maximally mixed state so that its normalization factor and lower bound are explicit. For a different positive definite ρ , one obtains the lower bound $\lambda_{\min}(\rho)N^{-2n}$ instead. No rationality of ρ is required for that inequality.

Proposition 20.2 (A finite-length perturbation estimate). *Suppose A_j and $\tilde{A}_j$ are corresponding Kraus operators of two valid instruments of the same dimension, and $\|A_j - \tilde{A}_j\|_{\text{op}} \leq \varepsilon$. For every length- n word,*

$$|p(w) - \tilde{p}(w)| \leq 2n\varepsilon$$

for maximally mixed inputs.

Proof. Normalization implies $A_j^\dagger A_j \leq I$ for each j , so every Kraus operator is a contraction. The same is true of the tilded operators. Telescoping the difference of two ordered products yields $\|A_w - \tilde{A}_w\|_{\text{op}} \leq n\varepsilon$, and hence $\|A_w - \tilde{A}_w\|_{\text{F}} \leq \sqrt{D}n\varepsilon$. Each product has Frobenius norm at most $\sqrt{D}$. Therefore

$$\frac{|\|A_w\|_{\text{F}}^2 - \|\tilde{A}_w\|_{\text{F}}^2|}{D} \leq \frac{(2\sqrt{D})(\sqrt{D}n\varepsilon)}{D} = 2n\varepsilon.$$

$\square$

For a fixed exact rational reference instrument, a threshold of $\delta_n/2$ correctly distinguishes a zero reference probability from a nonzero one when the perturbed probability is known exactly and $\varepsilon < \delta_n/(4n)$. Any additional numerical or statistical error must also fit inside the remaining margin.

This is not a uniform experimental solution to the unbounded problem. The required error tolerance can decay exponentially with n , and Corollary 18.4 precludes a total computable stopping length in general. Exact rational input is a finite mathematical description; it is not an assertion that arbitrary real-world calibration achieves exact zero amplitudes.

21 A canonical quartic compiler for bounded histories

We now give an explicit Diophantine representation, rather than merely appeal to MRDP. Fix $D, K, n \geq 1$. Let

$$H_0, \dots, H_{K-1} \in \mathbb{Z}^{D \times D}$$

be the integer numerator matrices of an instrument. The polynomial construction also works for arbitrary integer matrix lists; normalization is only needed for the quantum interpretation.

21.1 Natural coordinates for signed entries

An integer x is encoded canonically by natural numbers x^+, x^- with

$$x = x^+ - x^-, \quad x^+ x^- = 0. \quad (21.1)$$

The pair is unique: $(x^+, x^-) = (\max(x, 0), \max(-x, 0))$. The complementarity equation is essential. Without it, a single integer would have infinitely many representations $(x^+ + t, x^- + t)$, destroying the witness-counting statement below.

Introduce natural selectors e_{tj} for $0 \leq t < n$ and $0 \leq j < K$, subject to

$$e_{tj}(e_{tj} - 1) = 0, \quad \sum_{j=0}^{K-1} e_{tj} = 1. \quad (21.2)$$

Thus exactly one outcome is selected at every time.

For each time introduce signed matrices Y_t and X_t , represented by canonical natural pairs. The selected one-step matrix and its accumulated product are enforced by

$$Y_{t,ab} = \sum_{j=0}^{K-1} e_{tj} H_{j,ab}, \quad 0 \leq t < n, \quad (21.3)$$

$$X_{t+1,ab} = \sum_{c=1}^D Y_{t,ac} X_{t,cb}, \quad 0 \leq t < n, \quad (21.4)$$

$$X_{0,ab} = \delta_{ab}, \quad X_{n,ab} = 0. \quad (21.5)$$

Every occurrence of X , Y , or H in these formulas denotes a difference of its positive and negative coordinates. The signed pairs for H are free *input* coordinates. They may be fixed canonically before constructing the witness; the representation is also valid for any fixed pair encoding the same integer inputs.

The intermediate variables Y_t are not optional if the matrix entries are treated as polynomial input variables and degree four is to be retained by this simple construction. Direct substitution of (21.3) into (21.4) would produce terms $e_{tj}H_{j,ac}X_{t,cb}$ of degree three. The separate Y_t gates keep every residual quadratic.

21.2 The polynomial and its exact count

Let $\mathcal{R}_{D,K,n}$ consist of the following residuals:

- all equations in (21.2), moved to the left;
- the complementarity products X^+X^- and Y^+Y^- for every entry;
- all equations in (21.3)–(21.5), moved to the left.

Define the integer polynomial

$$\mathcal{Q}_{D,K,n}(H^+, H^-, \mathbf{w}) = \sum_{R \in \mathcal{R}_{D,K,n}} R^2. \quad (21.6)$$

This is an explicit finite formula, not an appeal to an unspecified encoding theorem.

Theorem 21.1 (Parsimonious bounded-history representation). *The polynomial (21.6) has total degree at most four, including its free matrix input coordinates. For each fixed integer matrix list H and fixed encoding of that input, its natural-number solution tuples are in bijection with the length- n words w satisfying $H_w = 0$.*

It has

$$W(D, K, n) = nK + (4n + 2)D^2 \quad (21.7)$$

natural witness coordinates and is the sum of squares of

$$E(D, K, n) = n(K + 1) + (4n + 3)D^2 \quad (21.8)$$

quadratic-or-lower residuals. The $2KD^2$ natural coordinates describing H are free inputs and are not included in W .

Proof. Every residual has degree at most two, so its square has degree at most four. A sum of integer squares is zero if and only if each summand is zero.

Suppose $\mathbf{w}$ is a natural solution. The selector equations yield a unique label j_t at each time. Equation (21.3) forces $Y_t = H_{j_t}$. Starting with $X_0 = I_D$, induction in (21.4) gives

$$X_t = H_{j_{t-1}} \cdots H_{j_0}.$$

The final equations give $H_w = 0$ for $w = j_0 \cdots j_{n-1}$.

Conversely, for such a word choose the corresponding one-hot selectors, set $Y_t = H_{j_t}$, and let X_t be the actual prefix products. Encode every signed entry by its unique canonical pair. All residuals vanish. These choices are forced by the word: selectors are fixed, signed Y_t and X_t are determined successively, and (21.1) fixes their natural pairs. Thus the correspondence is a bijection, not merely a surjection.

For the count, there are nK selectors, $2nD^2$ coordinates for $Y_0, \dots, Y_{n-1}$, and $2(n+1)D^2$ for $X_0, \dots, X_n$. This gives (21.7). The residual count is displayed explicitly below and sums to (21.8). $\square$

Residual family	Number
Selector Boolean equations	nK
One-hot sums	n
Complementarity for all X_t	$(n+1)D^2$
Complementarity for all Y_t	nD^2
Matrix-selection equations	nD^2
Matrix-multiplication equations	nD^2
Initial identity	D^2
Final zero	D^2
Total	$n(K+1) + (4n+3)D^2$

Corollary 21.2 (Four-dimensional specialization). *For seven outcomes in dimension four, the explicit length- n certificate uses*

$$W(4, 7, n) = 71n + 32, \quad E(4, 7, n) = 72n + 48.$$

For $n = 3$, these numbers are 245 natural witnesses and 264 residuals.

Because the solutions are in bijection with words, a fixed input has at most K^n solutions to this particular bounded-history polynomial. This finite-fold property belongs to the fixed-length family. It is not a statement about the fixed-arity unbounded MRDP representation in Section 22.

21.3 Physical validity can be certified separately

If one also wants the polynomial input to certify a rational instrument, include a positive denominator $N = u + 1$ and the residuals

$$\sum_{j=0}^{K-1} \sum_{c=1}^D H_{j,ca} H_{j,cb} - N^2 \delta_{ab} = 0 \quad (1 \leq a, b \leq D). \quad (21.9)$$

After replacing signed entries by differences, these are quadratic. Adding their squares keeps the total degree at most four. The denominator does not enter the zero-product equations: multiplying every generator by the same nonzero scalar does not change mortality. The counts in Theorem 21.1 exclude these optional validity residuals and the denominator coordinate.

21.4 Fixed words and the companion nonzero certificate

A prescribed word can simply be substituted into the selectors. Alternatively, retain the selectors and add natural prefix variables $z_0, \dots, z_n$ with

$$z_0 = 0, \quad z_{t+1} = Kz_t + \sum_{j=0}^{K-1} j e_{tj}, \quad z_n = h.$$

For a fixed length, these equations encode the word as its base- K integer h , including leading zero labels. They add $n + 1$ natural witnesses and $n + 2$ linear residuals, and the extra witnesses are uniquely determined.

Proposition 21.3 (A bounded nonzero companion). *Replace the D^2 final-zero residuals by one residual*

$$\sum_{a,b} X_{n,ab}^2 - (v+1) = 0, \quad v \in \mathbb{N}. \quad (21.10)$$

Then the resulting sum of squares is still quartic, and its natural solutions are in bijection with the length- n words having $H_w \neq 0$.

Proof. The trace equations still force $X_n = H_w$. Its squared Frobenius norm is a nonnegative integer. It is positive exactly when it is $v+1$ for a natural v , and that v is unique. The new residual is quadratic in the signed-pair coordinates, so its square is quartic. $\square$

For a fixed word length, both possibility and impossibility therefore have completely explicit arithmetic certificates. This is consistent with the undecidability theorem: the difficult property is the existential search over *all lengths*. The assertion that every word of every length is possible has an unbounded universal quantifier.

22 Unbounded Diophantine representation and the computational substrate

22.1 The semidecider

Choose an effective encoding of finite rational matrix descriptions by naturals. Invalid encodings can be rejected. Define $\mathbf{QMort}_{4,7}$ to be the set of codes of valid rational four-dimensional, seven-outcome instruments for which some nonempty word has zero product. A semidecider is straightforward: first check exact normalization; then enumerate words in increasing length and evaluate their integer numerator products until a zero product is found.

The validity test is decidable, and every individual word test is decidable. If the input is mortal, its shortest witness is eventually reached. If it is not mortal, the word search continues indefinitely. The hard part is not evaluating a product but deciding that this unbounded search will never succeed.

Theorem 22.1 (A fixed-arity unbounded polynomial). *There exist a fixed $m \in \mathbb{N}$ and a fixed polynomial $U \in \mathbb{Z}[e, y_1, \dots, y_m]$ such that*

$$e \in \mathbf{QMort}_{4,7} \iff \exists \mathbf{y} \in \mathbb{N}^m U(e, \mathbf{y}) = 0. \quad (22.1)$$

The represented set is recursively-enumerable complete. Its complement is not Diophantine over $\mathbb{N}$.

Proof. The semidecider establishes recursive enumerability. MRDP supplies (22.1); this is exactly the form of the natural-input finite-polynomial interface described by ProveIt [14]. Completeness follows by composing the effective halting reduction into six-generator matrix mortality with the total computable construction of Theorem 18.2.

If the complement were also Diophantine, it too would be recursively enumerable: enumerate all natural witness tuples and evaluate its polynomial. Running the two enumerators in parallel would decide membership in $\mathbf{QMort}_{4,7}$, contradicting Theorem 18.2. $\square$

This theorem does not print an expanded universal polynomial U , nor claim a numerical witness bound for it. The explicit polynomials supplied in this package are $\mathcal{Q}_{D,K,n}$, a length-indexed family. One cannot obtain a fixed-arity polynomial merely by writing $\exists n \mathcal{Q}_{D,K,n} = 0$, because the number of coordinates in that notation grows with n . MRDP, or a genuine coded-trace construction, is the additional step that removes this changing arity.

22.2 What “Turing-complete substrate” means here

The chain of effective reductions is

$$\begin{aligned} \text{Turing-machine halting} &\longrightarrow \text{binary tag-system halting} \longrightarrow \text{Post correspondence} \\ &\longrightarrow \text{six integer } 3 \times 3 \text{ matrices} \longrightarrow \text{seven rational } 4 \times 4 \text{ Kraus operators} \\ &\longrightarrow \text{a natural-number Diophantine equation.} \end{aligned} \tag{22.2}$$

The first matrix-hardness portion is imported from the published reduction [11]. The new quantitative normalization is Theorem 16.1; the final unbounded polynomial step is MRDP. A binary tag system is a finite rewriting program acting on an unbounded word, whereas the quantum description has fixed matrix size but permits arbitrarily long outcome histories. These are different locations for the unbounded resource.

In particular, Theorem 18.2 does not say that one fixed four-dimensional device with a fixed finite set of coefficients is a universal programmable quantum computer. Its input is the *description of the instrument*: the rational coefficients vary with the encoded computation. Nor does the theorem show how to decide halting by physically waiting for an impossible outcome. Such an outcome never occurs. The decision problem concerns the exact mathematical description of the entire family of histories.

The substantive role of interference is that signed matrix products retain cancellation. Replacing the signed substrate by entrywise supports changes its computational behavior, as both the example and the following finite-state argument show.

Remark 22.2 ([write] What ProveIt formalizes of this chain). Of the chain (22.2), ProveIt formalizes in Lean, for its own purposes and independently of this report, a different route from Turing machines to *binary tag systems*: three-counter programs, single-register machines, a 30-tag system and Neary’s track-and-shift binary layer, ending in the undecidability of an encoded binary tag family (Jones1980.tag91_undecidable; Section 1.3). It does not formalize the Post-correspondence or matrix-mortality steps of Neary’s Corollary 12, nor Theorem 16.1. The source’s sentence that the first matrix-hardness portion is imported from the published reduction remains accurate for the matrix steps.

22.3 A decidable boundary: nonnegative entries

Proposition 22.3 (Nonnegative mortality is decidable). *Mortality for any finite family of entrywise nonnegative rational $D \times D$ matrices is decidable. A breadth-first search through at most 2^{D^2} Boolean support matrices suffices.*

Proof. For a nonnegative matrix M , let $\text{supp}(M)$ have entry 1 where M is positive and 0 otherwise. Nonnegativity implies

$$\text{supp}(MN) = \text{supp}(M) \odot \text{supp}(N),$$

where $\odot$ is Boolean matrix multiplication: an entry is 1 exactly when some intermediate index makes both factors positive. There is no cancellation. Thus reachable supports form a finite semigroup with at most 2^{D^2} elements. Starting with the identity support, multiply by the generator supports and explore every newly reached support. Mortality holds exactly when the zero support is reached. $\square$

This decidable contrast is already part of the quantum occurrence literature [10]; the proof is included to make the boundary explicit. It applies equally to real rational Kraus operators whose entries happen all to be nonnegative in the chosen representation. It does not apply merely because probabilities themselves are nonnegative: their underlying amplitude matrices can have signs.

22.4 A scalar matrix-product observable

There is also a precise tensor version of the same arithmetic invariant. For $M \in \mathbb{Q}^{d \times d}$ and $\Omega = \sum_{a=1}^d e_a \otimes e_a$, one has

$$\Omega^\top (M \otimes M) \Omega = \sum_{a,b} M_{ab}^2. \quad (22.3)$$

Consequently, with local matrices $T_i = M_i \otimes M_i$,

$$\Omega^\top T_{i_n} \cdots T_{i_1} \Omega = 0 \iff M_w = 0.$$

The proof is direct expansion and the identity $(M \otimes M)(N \otimes N) = (MN) \otimes (MN)$. This gives a scalar observable of a matrix-product representation that vanishes exactly on mortal words. It is an algebraic representation; additional normalization or boundary-state choices are required before identifying it with a particular normalized physical tensor-network state. No new matrix-product-state undecidability theorem is claimed here.

Part III

Continuous computation (manuscript 09)

23 The question beyond ordinary MRDP

The bare assertion that a Turing-complete substrate has a Diophantine halting set is not a new result. Once finite successful executions can be enumerated, MRDP applies. A useful research contribution must identify a property whose finite-witness structure is not already the finite execution itself, expose that structure, and delimit where it fails.

This article does that for *infinite-time robust avoidance* in a compact continuous-state computer. The finite witness is a cut in a rational graph, not a completed execution. Soundness rules out *all* finite perturbed executions, of arbitrarily large length. Completeness applies exactly when a strictly positive noise margin exists.

Three questions organize the development. Can an infinite-time safety property of a continuous universal machine be compiled into an ordinary integer polynomial? Can one fixed small-dimensional machine realize arbitrary independent enumerable acceptance and robust-rejection behaviors? Can its exact noise threshold be a computable real even though the sign of that threshold is undecidable? We answer all three under explicit hypotheses. These are the questions posed in this manuscript; we do not assert that they were previously advertised as named open problems.

[write] Reading guide for Part III

For the main realization theorem, read Sections 27 and 28; the crucial mechanisms are the enlarged rejecting slab and the expanding emergency coordinate. For finite certificates and their arithmetic meaning, read Sections 25–26, followed by Section 29. The fixed MRDP polynomial must be distinguished from the directly generated finite-grid quartics. For executable material, start with Section 32 and the report README. Section 35.3 describes the proof-assistant work still needed, and Appendix E records dependencies and non-claims. (The source’s own reading guide, of which this is an adaptation, preceded its first section; its repository baseline, Section 1.1 there, is printed in Section 1.)

23.1 Relation to existing research

Moore’s generalized shifts already connect symbolic computation with continuous dynamical systems and establish undecidability phenomena [16]. Our two tape coordinates are a direct relative of this classical Cantor encoding. Packing the finite control into one of those coordinates is an elementary space-saving step, not a claim that continuous universality begins here.

Bournez, Graça, and Hainry study adversarial perturbations and show that appropriate infinitesimal robustness forces decidability. Their bounded-domain arguments use finite spatial abstractions [17]. We credit that qualitative framework. The present proofs isolate a numerical bottleneck radius, its two-sided horizon-free approximation, an explicit quartic cut checker, and a simultaneous three-outcome realizer with time–radius bounds. Whether this exact package, or a stronger equivalent formulation, already appears elsewhere requires a wider priority audit.

Coarse chain recurrence studies stepwise perturbations, filtrations, and finite-error dynamical information [18]; the critical radius is naturally related to that viewpoint. The distinction between approximate numerical information and exact truth also appears in δ -decision procedures [19]. We do not identify our unbounded-time graph algorithm with bounded first-order δ -decision.

A recent preprint by Platzer and Qian treats axiomatization of robust safety for polynomial differential equations, with a bounded-horizon topological robustness notion [20]. Our model instead has discrete time, a continuous compact state space, and a uniform bound on adversarial error at *every* step. Neither theorem is automatically a theorem about the other noise model.

23.2 Principal statements

The quantitative core is

$$\rho \leq r_N \leq \rho + \frac{L+1}{2N}, \quad (23.1)$$

where r_N is a finite rational graph bottleneck. A Boolean cut is checked by an explicit degree-four polynomial. The universal core is a single three-dimensional map realizing, for every disjoint computably enumerable pair (A, B) ,

$e \in A$	$e \in B$	$e \notin A \cup B$
exact acceptance	positive noise radius	fragile divergence.

The radius of a rejecting input, when rejection first occurs at time T , obeys

$$\frac{1}{12m S_T(9m+6)} \leq \rho \leq \min\{1, 2 \cdot 3^{-(T-1)}\}, \quad S_T(L) = \sum_{j=0}^{T-1} L^j, \quad (23.2)$$

where m is the fixed number of interpreter states. The following sections prove these statements and their arithmetic consequences.

24 Noise semantics and the three outcomes

Fix $K = [0, 1]^d$ with the sup norm. Unless otherwise stated, $f : K \rightarrow K$ is a continuous rational piecewise-affine map, specified by finitely many rational polyhedra and rational affine formulas agreeing on overlaps. A rational number $L \geq 0$ is supplied with a proof that

$$\|f(u) - f(v)\|_\infty \leq L \|u - v\|_\infty \quad (u, v \in K). \quad (24.1)$$

For example, the largest row-sum norm of the affine linear parts is a valid bound: subdivide the segment from u to v at the finitely many cell faces and add the local estimates. Continuity is essential in this argument.

Let $H \subseteq K$ be a nonempty finite union of closed rational axis-aligned boxes. Degenerate boxes are permitted. Initially take $x \in K \cap \mathbb{Q}^d$ with $x \notin H$.

Definition 24.1 (Critical radius). For a finite path $\gamma = (x_0, \dots, x_T)$, where $T \geq 1$, $x_0 = x$, all $x_t \in K$, and $x_T \in H$, set

$$c_f(\gamma) = \max_{0 \leq t < T} \|x_{t+1} - f(x_t)\|_\infty.$$

The *critical noise radius* is

$$\rho_f(x, H) = \inf_{\gamma} c_f(\gamma). \quad (24.2)$$

If $x \in H$, define $\rho_f(x, H) = 0$ to include a time-zero hit.

The infimum is over real paths, not rational paths or symbolic encodings. It is finite because a one-step jump from $f(x)$ to any point of H costs at most one. In particular $0 \leq \rho_f(x, H) \leq 1$. An ε -pseudo-orbit is a path whose cost is at most ε . Errors may change at every step and may be selected adversarially using the previous states. The path must stay inside K ; there is no implicit extension outside the cube.

Definition 24.2 (Outcomes). Write

$$\begin{aligned} \text{Hit}(x) &\iff \exists t \geq 0 \quad f^t(x) \in H, \\ \text{Safe}(x) &\iff \rho_f(x, H) > 0, \\ \text{Frag}(x) &\iff \neg \text{Hit}(x) \text{ and } \rho_f(x, H) = 0. \end{aligned}$$

We call these *exact acceptance*, *robust avoidance*, and *fragile avoidance*, respectively. In the universal simulation, fragile avoidance will coincide with divergence of the interpreter.

An exact accepting orbit gives a zero-cost path, so **Hit** and **Safe** are disjoint. Every initial state has exactly one of these three outcomes. Furthermore,

$$\text{Safe}(x) \iff \exists \varepsilon > 0 \text{ no finite } \varepsilon\text{-pseudo-orbit from } x \text{ hits } H. \quad (24.3)$$

For the forward implication choose $0 < \varepsilon < \rho$. Conversely, if $\rho = 0$, paths of cost below any positive ε exist by the definition of infimum.

Remark 24.3 (The threshold itself). Neither $\rho = 0 \Rightarrow \text{Hit}$ nor attainment of the infimum is valid in general. A lower bound $\rho \geq \delta > 0$ certifies safety for all errors *strictly below* δ ; without additional information it does not certify safety at the endpoint δ . All later certificate statements respect this distinction.

Example 24.4 (An infinite horizon changes the answer). For $f(t) = t$, $x = 0$, and $H = \{1\}$, the exact orbit never moves. Nevertheless, $0, 1/T, \dots, 1$ is a path of cost $1/T$, so $\rho = 0$. No zero-cost path hits H . Thus an exactly nonaccepting state can be arbitrarily noise-sensitive.

25 Computing the radius without bounding time

Choose a positive integer D clearing all denominators of all endpoints of the boxes making up H . For $n \geq 0$, let

$$N = D2^n, \quad G_N = \{0, 1/N, \dots, 1\}^d, \quad V_N = G_N \cup \{x\}, \quad T_N = G_N \cap H.$$

The union defining V_N removes duplicates. In particular, a nongrid initial point is kept *exactly*; it is not silently rounded. The target set T_N is nonempty because all box endpoints are grid points.

On the complete directed graph with vertex set V_N , assign edge weight

$$w_N(u, v) = \|v - f(u)\|_\infty \in \mathbb{Q}_{\geq 0}. \quad (25.1)$$

Define r_N to be the minimum, over graph paths from x to T_N , of their largest edge weight. A repeated vertex can be removed together with the intervening cycle without increasing the largest weight. Thus a minimizing path exists and may be chosen with at most $|V_N| - 1$ edges.

Theorem 25.1 (Horizon-free grid approximation). *For every aligned mesh N , put $h_N = 1/(2N)$ and $C = L + 1$. Then*

$$\rho_f(x, H) \leq r_N \leq \rho_f(x, H) + Ch_N. \quad (25.2)$$

The numbers r_{D2^n} form a nonincreasing rational sequence. Consequently

$$[\max\{0, r_N - Ch_N\}, r_N] \quad (25.3)$$

is a computable interval containing $\rho_f(x, H)$ and of length at most Ch_N . The estimate does not involve the length of an accepting pseudo-orbit.

Proof. Every graph path is also a path of real states in K with the same cost. Taking infima gives $\rho \leq r_N$.

Conversely, take any real path $x_0 = x, \dots, x_T \in H$ of cost c . Keep $y_0 = x$ exactly. Round each coordinate of x_t for $t \geq 1$ to a nearest grid coordinate, obtaining $y_t \in G_N$ with $\|x_t - y_t\|_\infty \leq h_N$. The final state can be rounded inside H : choose a box containing x_T ; since both endpoints of each coordinate interval are on the grid, nearest-grid rounding stays in that interval. Thus $y_T \in T_N$.

For $1 \leq t < T$, the triangle inequality and (24.1) give

$$\|y_{t+1} - f(y_t)\|_\infty \leq \|y_{t+1} - x_{t+1}\|_\infty + \|x_{t+1} - f(x_t)\|_\infty + \|f(x_t) - f(y_t)\|_\infty \leq c + h_N + Lh_N.$$

The first edge has the better bound $c + h_N$, because $y_0 = x_0$. Therefore $r_N \leq c + Ch_N$. Taking the infimum over all real paths proves the upper bound even when that infimum is not attained.

The grid at mesh N is contained in the grid at mesh $2N$; the source and old targets remain available, with the same weights. Hence $r_{2N} \leq r_N$. All graph data are rational and finite, so the interval is effectively computable. No error was accumulated over time: each edge was estimated against the corresponding edge of the original path independently. $\square$

25.1 Algorithm, complexity, and what remains undecidable

A bottleneck version of Dijkstra's algorithm starts with distance zero at the source and relaxes an edge by

$$d(v) \leftarrow \min\{d(v), \max(d(u), w_N(u, v))\}.$$

For $M = |V_N| \leq (N + 1)^d + 1$, a simple implementation takes $O(M^2)$ rational comparisons and stores $O(M^2)$ edge weights. The companion implementation uses precisely this finite algorithm. The arithmetic bit costs also depend on the rational input and map descriptions; no unit-cost claim for arbitrarily large integers is intended.

Given an additive precision 2^{-p} , choose $N = D2^n$ with $C/(2N) \leq 2^{-p}$. The resulting interval proves that the critical radius is a *computable real*, uniformly from the rational data and a valid L . In fixed dimension, the explicit graph size is exponential in the number of precision bits. A four-dimensional or three-dimensional grid can already be impractical at moderate precision.

Corollary 25.2 (A promised threshold comparison). *Given rational ε and $\delta > 0$, under the promise $|\rho - \varepsilon| \geq \delta$, one can decide whether $\rho < \varepsilon$ or $\rho > \varepsilon$ by using an interval of width less than δ . No time cutoff is needed.*

This does not decide whether $\rho = 0$. Every interval in (25.3) may continue to contain zero. Later we construct a fixed map for which positivity is as hard as the halting problem.

25.2 Stability of the value, not of its zero test

Proposition 25.3 (Lipschitz dependence of the critical value). *For maps $f, g : K \rightarrow K$ and the same source outside H ,*

$$|\rho_f(x, H) - \rho_g(x, H)| \leq \sup_{u \in K} \|f(u) - g(u)\|_\infty.$$

If f is L -Lipschitz and $x, y \notin H$, then

$$|\rho_f(x, H) - \rho_f(y, H)| \leq L \|x - y\|_\infty.$$

Proof. For any fixed path, changing the map changes every edge cost, and therefore the maximum cost, by at most the uniform map distance. Take infima in both directions. To change the source, keep every vertex of a path except its first vertex. Only the first edge cost changes, by at most $\|f(x) - f(y)\|_\infty \leq L \|x - y\|_\infty$. Again take infima in both directions. $\square$

The exclusion of sources in H in the second statement is deliberate: our time-zero convention overrides the positive-length variational definition. Even on a region where the radius is Lipschitz and computable, equality to zero can be undecidable. Numerical regularity is not a decision procedure for exact membership in a zero set.

25.3 An effective-continuity extension

Proposition 25.4 (Beyond rational affine formulas). *Suppose instead that f is computable and has a known effective uniform modulus ω , with $\omega(h) \rightarrow 0$ and $\|f(u) - f(v)\|_\infty \leq \omega(\|u - v\|_\infty)$. At*

each graph vertex, compute a rational approximation $\tilde{f}(u)$ within $\sigma > 0$, and let $\tilde{r}_N$ be the bottleneck formed from $\|v - \tilde{f}(u)\|_\infty$. Then

$$\max\{0, \tilde{r}_N - \sigma - h_N - \omega(h_N)\} \leq \rho_f(x, H) \leq \tilde{r}_N + \sigma. \quad (25.4)$$

Thus the critical radius remains a computable real in this setting.

Proof. The rounding proof replaces Lh_N by $\omega(h_N)$. At fixed vertices, each approximate edge weight differs from the true edge weight by at most σ . A maximum along a path and then a minimum over paths preserve that additive error bound. Combine the two estimates. $\square$

This extension does *not* preserve every exact-acceptance assertion below. For a simple uniform counterexample, let $a_e = 0$ when machine e never halts, and $a_e = 2^{-(T+1)}$ when it halts at time T . The real a_e is uniformly computable: simulate for enough steps to make every later possible value smaller than the requested precision. For the computable constant map $f_e(t) = a_e$, source 1, and target $\{0\}$, exact acceptance is precisely nonhalting. Rational piecewise-affine formulas avoid this closed-target equality obstruction because every finite iterate is exactly rational.

26 Finite cuts as infinite-time safety proofs

For a finite complete weighted directed graph with source s and nonempty target set T disjoint from s , an admissible cut is a set S with $s \in S$ and $S \cap T = \emptyset$. Its bottleneck is

$$\kappa(S) = \min_{u \in S, v \notin S} w(u, v).$$

The minimum exists because S and its complement are nonempty.

Theorem 26.1 (Bottleneck/cut duality). *If r is the minimum path bottleneck from s to T , then*

$$r = \max_{S \text{ admissible}} \kappa(S). \quad (26.1)$$

An optimal cut is the set of vertices reachable from s using only edges of weight strictly less than r .

Proof. Every source-to-target path crosses every admissible cut. Its largest edge weight is therefore at least $\kappa(S)$, giving $r \geq \max_S \kappa(S)$. Let S be the set reachable below r . It cannot meet T , since that would produce a path with bottleneck strictly below r . Every edge from S to its complement has weight at least r , or its endpoint would also be reachable. Consequently $\kappa(S) \geq r$, which proves equality. This reasoning also covers $r = 0$, when there are no negative-weight edges. $\square$

Applying [theorems 25.1](#) and [26.1](#) gives the certificate interpretation

$$\kappa(S) \geq \eta > Ch_N \implies \rho_f(x, H) \geq \eta - Ch_N > 0. \quad (26.2)$$

The cut excludes paths of *every* length. This is the step that replaces an infinite-time universal assertion by a finite object.

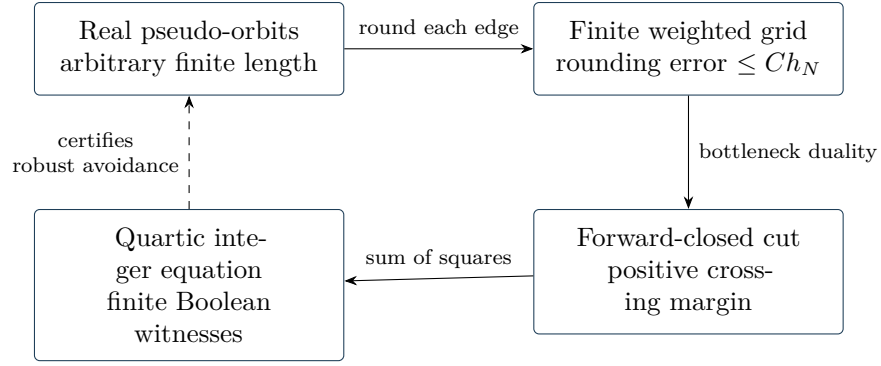


Figure 1: The certificate is a finite obstruction to all perturbed paths, not a single successful computation trace. The polynomial variables encode the cut; they are not the real state coordinates.

26.1 An explicit quartic polynomial

Set

$$\lambda_N = \frac{L+1}{N} = 2Ch_N, \quad E_N = \{(u, v) \in V_N^2 : w_N(u, v) < \lambda_N\}.$$

Introduce one variable b_v for each vertex. Define

$$\begin{aligned} Q_N(b) = & (b_x - 1)^2 + \sum_{t \in T_N} b_t^2 + \sum_{v \in V_N} (b_v(b_v - 1))^2 \\ & + \sum_{(u,v) \in E_N} (b_u(1 - b_v))^2. \end{aligned} \quad (26.3)$$

After the finite graph is computed, this is an ordinary integer polynomial of degree at most four. Its coefficients contain no real parameters. All rational comparisons have already been performed while constructing E_N .

Theorem 26.2 (Complete quartic safety certificates). *For each mesh N ,*

$$\exists b \in \mathbb{N}^{V_N} \ Q_N(b) = 0 \iff r_N \geq \lambda_N. \quad (26.4)$$

Every such zero certifies $\rho_f(x, H) \geq (L+1)/(2N) > 0$. Conversely,

$$\rho_f(x, H) > 0 \iff \exists n \geq 0 \ \exists b \in \mathbb{N}^{V_{D2^n}} \ Q_{D2^n}(b) = 0. \quad (26.5)$$

The same zero-set assertions hold if witnesses are allowed to range over $\mathbb{Z}$.

Proof. An integer sum of squares is zero exactly when every square is zero. The third group of terms forces $b_v \in \{0, 1\}$. The first two groups put the source in the set $S = \{v : b_v = 1\}$ and all targets outside it. The final group says that no edge of weight below λ_N leaves S . Thus S is an admissible cut with crossing weights at least λ_N , and [theorem 26.1](#) gives $r_N \geq \lambda_N$.

Conversely, if $r_N \geq \lambda_N$, let S consist of vertices reachable using edges below λ_N . No target is in S , and no such edge leaves S . Its indicator vector makes every term of Q_N vanish.

The lower bound follows from $\rho \geq r_N - Ch_N \geq \lambda_N - Ch_N = Ch_N$. If $\rho > 0$, choose n so large that $\lambda_{D2^n} \leq \rho$. Then $r_N \geq \rho \geq \lambda_N$, and a certificate exists. This proves completeness. $\square$

There are $M = |V_N|$ variables and at most $1 + |T_N| + M + M^2$ square terms. These are transparent size bounds, not optimized ones. A bit vector, together with the mesh, is a finite witness for an unbounded-time statement. The strict inequality defining E_N is part of the exact equivalence in (26.4); equality-weight edges must not be silently inserted.

26.2 From the explicit family to a fixed Diophantine equation

Let $e \mapsto x_e$ be a primitive-recursive rational input encoding for a fixed f, H, L, D . The number of vertices, the rational evaluations, the comparisons forming E_N , and the Boolean cut test are primitive recursive in their natural-number codes. For clarity, one can encode a cut by a bit mask less than 2^M . All graph loops are bounded by primitive-recursive integers, even though those integers grow very quickly.

Theorem 26.3 (Fixed-polynomial compilation). *The set $\{e : \rho_f(x_e, H) > 0\}$ is computably enumerable. Consequently there are fixed k and $P_{\text{Safe}} \in \mathbb{Z}[X, Y_1, \dots, Y_k]$ with*

$$\rho_f(x_e, H) > 0 \iff \exists w \in \mathbb{N}^k P_{\text{Safe}}(e, w) = 0. \quad (26.6)$$

For rational piecewise-affine f and rational boxes H , exact acceptance also has a fixed Diophantine representation P_{Hit} .

Proof. Enumerate meshes and check whether the finite below-threshold graph has a source-to-target path. Halt when it does not. By theorem 26.2, this semidecides exactly positive radius. Alternatively, enumerate mesh/bit-mask pairs and run the finite certificate checker. Apply (1.1) to this computably enumerable predicate. For exact acceptance, iterate the rational map exactly and test the rational target inequalities at every time; this is a semidecision procedure. Apply MRDP again. $\square$

There are two different levels of explicitness here. Formula (26.3) is a fully specified polynomial for each finite grid, with varying arity. Formula (26.6) is a fixed-arity existence theorem obtained from the finite-checker semidecider and MRDP. This article does not expand that second polynomial, establish its numerical degree or witness count, or claim single-fold or finite-fold representability. In particular, one may not transfer the word “quartic” and the count M directly from the first level to the second.

27 A three-dimensional continuous computer

We now construct the map used in the sharpness results. The construction is generic: a finite binary Turing-machine transition table determines a finite rational piecewise-affine map. Choosing a single universal interpreter table produces one fixed map. An expanded universal transition table is not needed for the existence proof, but producing and checking one is a separate implementation milestone.

27.1 The binary machine and its rational tape coordinates

Take a deterministic binary Turing machine with $m \geq 3$ states, numbered $0, \dots, m-1$. Its start state s , accepting state h , and rejecting state r are distinct. Every other state has an instruction

$$(j, a) \mapsto (j', b, \text{move}), \quad a, b \in \{0, 1\}, \quad \text{move} \in \{L, R, S\}.$$

The tape is two-sided, is indexed relative to the current head, and has blank symbol zero. Let a_i be the bit at relative location i . Define

$$u = \sum_{k \geq 1} \frac{2a_{-k}}{3^k}, \quad v = \sum_{k \geq 0} \frac{2a_k}{3^{k+1}}. \quad (27.1)$$

Both coordinates lie in the middle-third Cantor set. A finitely supported tape gives rational coordinates. On encoded tapes, the leading bit of a coordinate is identified by the disjoint closed intervals

$$J_0 = [0, 1/3], \quad J_1 = [2/3, 1].$$

There is no ambiguity between these two intervals, including at endpoints.

Let $c = a_{-1}$ be the bit immediately to the left of the head and $a = a_0$ the scanned bit. Writing b and moving gives

$$R: \quad u' = (2b + u)/3, \quad v' = 3v - 2a, \quad (27.2)$$

$$L: \quad u' = 3u - 2c, \quad v' = v/3 + 2c/3 + 2(b - a)/9, \quad (27.3)$$

$$S: \quad u' = u, \quad v' = v + 2(b - a)/3. \quad (27.4)$$

These are affine formulas on each digit rectangle $J_c \times J_a$.

Lemma 27.1 (Tape update on complete rectangles). *Equations (27.2)–(27.4) simulate the indicated tape operations on encoded tapes. In addition, each maps the entire closed rectangle $J_c \times J_a$ into $[0, 1]^2$, not merely its Cantor subset.*

Proof. Write $u = (2c + u_1)/3$ and $v = (2a + v_1)/3$, where $u_1, v_1 \in [0, 1]$. For a right move, the new left stack begins with b and then the old left stack, while the new right stack is the old right tail: this gives $u' = (2b + u)/3$ and $v' = v_1$.

For a left move, the left tail becomes u_1 . The new right stack begins with c , then b , then the old right tail. Its value is $2c/3 + 2b/9 + v_1/9$, which is precisely (27.3). The range is contained in $[0, 2/3 + 2/9 + 1/9] = [0, 1]$. For a stationary move, replacing the first right digit gives $v' = (2b + v_1)/3$. The remaining range assertions follow at once. $\square$

27.2 Packing the control state into the left coordinate

Set

$$\beta = \frac{1}{3m}, \quad \alpha_j = (3j + 1)\beta, \quad I_j = [\alpha_j, \alpha_j + \beta]. \quad (27.5)$$

Instead of retaining j and u as separate real coordinates, use

$$\xi = \alpha_j + \beta u. \quad (27.6)$$

The state intervals are separated: the gap between consecutive I_j is 2β . Consequently the encoded state can be read from ξ without confusing two different control states.

The three coordinates of the continuous machine are now $(\xi, v, z) \in [0, 1]^3$. The last coordinate z is an *emergency coordinate*; the exact simulation starts with $z = 0$ and keeps it zero until a terminal state is reached. Tiny positive values of z , however, can eventually trigger acceptance while the interpreter is still running.

Let

$$I_j^+ = [\alpha_j - \beta/2, \alpha_j + 3\beta/2], \quad p_j = (\alpha_j + \beta/2, 1/2, 1/2).$$

The enlarged terminal intervals I_h^+, I_r^+ remain inside $(0, 1)$ and are disjoint from each other and from all other state intervals. Define the accepting target box

$$H = I_h \times [0, 1] \times [0, 1]. \quad (27.7)$$

It is fixed, rational, closed, and full-dimensional. The accepting sink p_h lies in its interior relative to the cube.

27.3 Prescribed boxes and the emergency mechanism

Specify the map on the following finitely many closed rational boxes.

Terminal slabs. For $j = h, r$, set

$$F(\xi, v, z) = p_j \quad \text{whenever} \quad \xi \in I_j^+. \quad (27.8)$$

Here v, z range over the whole unit interval. The enlarged rejecting slab, not merely a single rejecting point, will provide robust rejection.

High emergency coordinate. For each active state $j \notin \{h, r\}$, set

$$F(\xi, v, z) = p_h \quad \text{on} \quad I_j \times [0, 1] \times [2/3, 1]. \quad (27.9)$$

Low emergency coordinate and valid leading digits. For each active j , each $c, a \in \{0, 1\}$, and the instruction $(j, a) \mapsto (j', b, \text{move})$, prescribe

$$F(\alpha_j + \beta u, v, z) = (\alpha_{j'} + \beta u', v', 3z) \quad \text{on} \quad u \in J_c, v \in J_a, z \in [0, 1/3], \quad (27.10)$$

where (u', v') is given by (27.2), (27.3), or (27.4). Since $u = (\xi - \alpha_j)/\beta$, this is affine in (ξ, v, z) on the prescribed box.

Every prescribed box maps into the cube by lemma 27.1. No two distinct prescriptions conflict. The terminal slabs are separated from active-state bands; low and high emergency intervals are separated by a gap; and the two digit intervals are separated by a gap. In particular, there is room to fill in all unprescribed regions continuously.

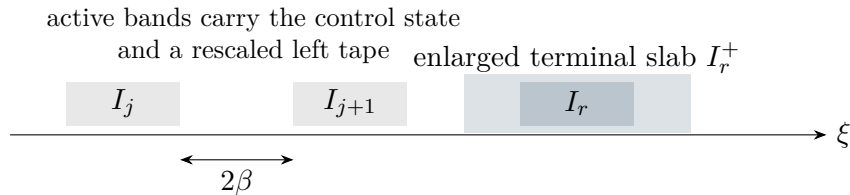


Figure 2: Schematic state packing. Each base interval has width β ; the terminal interval is enlarged by $\beta/2$ on either side. The drawing is schematic rather than a complete partition for a particular interpreter.

27.4 A constructive global extension

Lemma 27.2 (Continuous rational extension with an explicit bound). *The prescriptions (27.8)–(27.10) extend to a continuous rational piecewise-affine map $F : [0, 1]^3 \rightarrow [0, 1]^3$. One such extension has global sup-norm Lipschitz bound*

$$L = 9m + 6. \quad (27.11)$$

The extension can be computed from the finite transition table using rational arithmetic alone.

Proof. For the ξ axis, use all points $0, 1$ and $\alpha_j + k\beta/3$ for $j = 0, \dots, m-1$ and $k = 0, 1, 2, 3$, together with the endpoints of I_h^+ and I_r^+ . For each of the v, z axes use $0, 1/3, 2/3, 1$. Every prescribed box is a union of cells of this rectilinear grid.

At each grid vertex belonging to a prescribed box, assign the prescribed rational image. The disjointness just proved makes the assignment unambiguous. Assign p_r to every remaining vertex. Subdivide each rectangular cell by the same coordinate-ordered Kuhn triangulation: after normalizing its coordinates to a unit cube, use the six regions specified by an ordering of those coordinates. Interpolate the vertex values affinely on each simplex. Adjacent simplices and adjacent rectangular cells agree on their common faces, so the result is globally continuous. Convexity of the cube keeps every interpolated value in the cube. On any prescribed box, affine interpolation reproduces its prescribed affine formula.

Every interval in the ξ grid has length at least $\beta/3 = 1/(9m)$; the other two axis intervals have length $1/3$. Along the chain of vertices of a Kuhn simplex, successive vertices differ in exactly one coordinate. Hence each column of its affine linear part is an image difference divided by that coordinate's interval length. Each scalar image difference has absolute value at most one. Every row-sum norm is therefore at most $9m + 3 + 3 = 9m + 6$. The segment-subdivision argument following (24.1) makes this a global bound. $\square$

This extension is not claimed to be differentiable, reversible, injective, volume-preserving, or a polynomial map. Its continuity on the entire cube, including off-encoding states and guard boundaries, is part of the proved statement. Merely writing a discontinuous guarded update would not be enough for the robustness argument.

27.5 Exact simulation and finite input encoding

If the initial state is $(\alpha_s + \beta u, v, 0)$ with a finite encoded tape, then every active exact iterate satisfies (27.10), and its third coordinate remains zero. The first two coordinates perform exactly the binary machine computation. Entering state h places the image in H ; entering state r places it in the rejecting band, whose next image is p_r .

For a concrete primitive-recursive input coding, let $b_1 \dots b_k$ be the ordinary binary expansion of n (use $k = 1, b_1 = 0$ for zero), and take the self-delimiting word

$$\text{word}(n) = 1^k 0 b_1 \dots b_k 1. \quad (27.12)$$

These words are prefix-free and end in a one. Place the word starting under the head, put blanks elsewhere, and define

$$\iota(n) = \left(\alpha_s, \sum_{i=0}^{\ell(n)-1} \frac{2 \text{word}(n)_i}{3^{i+1}}, 0 \right). \quad (27.13)$$

The encoding is injective and primitive recursive as a rational map. Its values lie outside H . No uncomputable real parameter is present.

28 The universal trichotomy and its quantitative proof

Fix an effective enumeration $(W_a)_{a \in \mathbb{N}}$ of the computably enumerable subsets of $\mathbb{N}$. Fix also a primitive-recursive pairing bijection on natural numbers, with primitive-recursive decoding; the usual Cantor pairing suffices. We use $\langle a, b, e \rangle = \langle a, \langle b, e \rangle \rangle$.

Choose a *single* deterministic binary universal interpreter that, on input $\langle a, b, e \rangle$, dovetails the two semidecidings for $e \in W_a$ and $e \in W_b$. It accepts if the first succeeds first, rejects if the second succeeds first, and uses a fixed tie convention. When W_a and W_b are disjoint the tie convention is immaterial. A binary, one-tape, two-sided interpreter exists by the usual finite-alphabet and multitape simulations; those changes only replace one finite table by another. Its finite number m of states is fixed from now on.

Apply [section 27](#) to this table, and fix its map F , target H , input encoding ι , and bound $L = 9m + 6$ once and for all.

Theorem 28.1 (Universal three-dimensional trichotomy). *For every disjoint computably enumerable pair $(A, B) = (W_a, W_b)$ and every $e \in \mathbb{N}$, put $x_e = \iota(\langle a, b, e \rangle)$. Then*

$$e \in A \iff \text{Hit}(x_e), \quad (28.1)$$

$$e \in B \iff \rho_F(x_e, H) > 0, \quad (28.2)$$

$$e \notin A \cup B \iff \text{Frag}(x_e). \quad (28.3)$$

The map and target do not depend on a, b, e ; only the rational initial point changes. Moreover, on every $e \in B$, all sufficiently small perturbed trajectories enter the enlarged rejecting slab after finitely many steps and remain there.

We prove a quantitative rejection estimate and the emergency mechanism first.

Lemma 28.2 (Positive margin for a rejecting computation). *Suppose the exact interpreter first enters rejecting state r at time $T \geq 1$. Put $S_T(L) = \sum_{j=0}^{T-1} L^j$. No pseudo-orbit whose per-step error is at most*

$$\varepsilon_T = \frac{\beta}{4S_T(L)} \quad (28.4)$$

can hit H . Every such pseudo-orbit is in the enlarged rejecting slab at time T and stays there afterward. In particular, $\rho_F(x, H) \geq \varepsilon_T$.

Proof. Let $y_t = F^t(x)$ be the exact orbit and let x_t be a pseudo-orbit from the same source. Up to time T , global Lipschitz continuity gives

$$\|x_{t+1} - y_{t+1}\|_\infty \leq \varepsilon_T + L \|x_t - y_t\|_\infty.$$

Induction yields $\|x_t - y_t\|_\infty \leq \varepsilon_T S_t(L) \leq \beta/4$ for $1 \leq t \leq T$.

Before and at rejection, every exact packed coordinate is in a state interval other than I_h . Its distance from I_h is at least 2β . Thus no x_t for $t \leq T$ is in H . At time T , the exact packed coordinate belongs to I_r , so the perturbed one is in I_r^+ , because the enlargement has margin $\beta/2$ and the error is at most $\beta/4$.

On the entire rejecting slab, F is the constant p_r . Since $\varepsilon_T \leq \beta/4$, the next perturbed packed coordinate remains within $\beta/4$ of the center of I_r and hence within I_r^+ . This argument repeats forever and the slab is disjoint from H . It is important here that the terminal rule is constant for all values of the other two coordinates. $\square$

Lemma 28.3 (Arbitrarily small emergency acceptance). *If the exact interpreter never enters either terminal state, then $\rho_F(x, H) = 0$. If it first rejects at time $T \geq 2$, then*

$$\rho_F(x, H) \leq 2 \cdot 3^{-(T-1)}. \quad (28.5)$$

Proof. In an infinite active computation, take the first exact step, but change only the new emergency coordinate from zero to $2 \cdot 3^{-N}$, where $N \geq 1$. The cost of this single perturbation is $2 \cdot 3^{-N}$. Leave every subsequent step unperturbed. While $z \leq 1/3$, the first two coordinates continue the exact simulation and z is multiplied by three. At time N the emergency coordinate is $2/3$; the next step uses (27.9) and reaches p_h . Thus arbitrarily small costs suffice.

For a computation first rejecting at time $T \geq 2$, use the same construction with $N = T - 1$. At time $T - 1$ the interpreter is still active and $z = 2/3$. At time T , the emergency rule accepts instead of taking the rejecting instruction. The total path cost is the single initial perturbation $2 \cdot 3^{-(T-1)}$. $\square$

Proof of theorem 28.1. On exact encoded states the interpreter simulation is correct. If $e \in A$, the interpreter accepts, so $\text{Hit}(x_e)$ holds and the radius is zero. If $e \in B$, it rejects after a finite time, and lemma 28.2 gives positive radius and robust entry into the rejecting slab. If e belongs to neither set, the interpreter remains active forever. It never exactly accepts, while lemma 28.3 gives radius zero. These mutually exclusive cases exhaust all inputs and prove all three equivalences. $\square$

Corollary 28.4 (The time–radius sandwich). *For a first rejection at time $T \geq 1$,*

$$\frac{\beta}{4S_T(L)} \leq \rho_F(x, H) \leq \min\{1, 2 \cdot 3^{-(T-1)}\}, \quad \beta = \frac{1}{3m}, \quad L = 9m + 6. \quad (28.6)$$

Proof. The lower bound is lemma 28.2. For $T \geq 2$, combine lemma 28.3 with the universal one-step upper bound $\rho \leq 1$. For $T = 1$, that one-step bound gives the stated right side directly. $\square$

The constants are deliberately conservative. Their significance is the explicit dependence on the actual rejection time, not numerical optimality. A known lower bound $\rho \geq \delta > 0$ implies, for $T \geq 2$,

$$T \leq 1 + \log_3(2/\delta). \quad (28.7)$$

Thus a certified positive margin carries hidden information about how soon rejection must have occurred. It is not surprising that no computable inputwise positive margin can cover every rejecting input.

29 Arithmetic classification and limits of finite certificates

For the one fixed universal map, define sets of natural codes

$$\mathcal{H} = \{n : \text{Hit}(\iota(n))\}, \quad \mathcal{S} = \{n : \text{Safe}(\iota(n))\}, \quad \mathcal{F} = \{n : \text{Frag}(\iota(n))\}.$$

These definitions apply to all codes, even those decoding to overlapping pairs of enumerable sets. Their trichotomy follows from the dynamics, not from the disjoint-pair promise used for the universal realization theorem.

Theorem 29.1 (Exact computability classification). *The sets $\mathcal{H}$ and $\mathcal{S}$ are disjoint Σ_1^0 -complete sets. Their complement $\mathcal{F} = \mathbb{N} \setminus (\mathcal{H} \cup \mathcal{S})$ is Π_1^0 -complete. Completeness holds under primitive-recursive many-one reductions.*

Proof. Both positive sets are computably enumerable by [theorem 26.3](#). Let K_0 be a standard computably enumerable-complete halting set, with index k , and let o index the empty set. The map $e \mapsto \langle k, o, e \rangle$ reduces K_0 to $\mathcal{H}$, while $e \mapsto \langle o, k, e \rangle$ reduces K_0 to $\mathcal{S}$.

The complement of a union of two computably enumerable sets is co-computably enumerable. On the first slice, $e \notin K_0$ is equivalent to $\langle k, o, e \rangle \in \mathcal{F}$, giving Π_1^0 -hardness. All displayed coding maps are primitive recursive. $\square$

29.1 Two existential polynomials and one unavoidable negation

Corollary 29.2 (The Diophantine trichotomy). *There are fixed finite integer polynomials $P_{\mathcal{H}}, P_{\mathcal{S}}$ such that*

$$\begin{aligned} n \in \mathcal{H} &\iff \exists u \in \mathbb{N}^k P_{\mathcal{H}}(n, u) = 0, \\ n \in \mathcal{S} &\iff \exists v \in \mathbb{N}^\ell P_{\mathcal{S}}(n, v) = 0. \end{aligned}$$

There is no ordinary existential Diophantine representation of $\mathcal{F}$ over natural witnesses. It does, however, have the fixed universal nonvanishing representation

$$n \in \mathcal{F} \iff \forall u \in \mathbb{N}^k \forall v \in \mathbb{N}^\ell P_{\mathcal{H}}(n, u) P_{\mathcal{S}}(n, v) \neq 0. \quad (29.1)$$

Proof. Apply MRDP to the two computably enumerable sets. If $\mathcal{F}$ had an existential Diophantine representation, enumerating natural witness tuples would make it computably enumerable. Since it is already co-computably enumerable and is Π_1^0 -complete, this would decide the halting problem. The product in (29.1) is nonzero for every pair of tuples exactly when neither factor has a zero; the sets of tuples are nonempty, including when a witness dimension is zero. $\square$

The universal quantifier in (29.1) cannot be removed by the methods here. It expresses a genuine computability obstruction, not simply a missing clever encoding. This is a precise boundary for the proposed Diophantine treatment of unconventional computation: robust rejection has finite witnesses, but fragile divergence does not.

29.2 A stronger obstruction: no total separator

The simultaneous realization of arbitrary disjoint computably enumerable pairs gives more than separate completeness of the two positive outcomes.

Theorem 29.3 (Recursive inseparability). *There is no total computable Boolean classifier that outputs 1 on every $n \in \mathcal{H}$ and 0 on every $n \in \mathcal{S}$, even when either answer is allowed on $\mathcal{F}$.*

Proof. We first give the standard diagonal pair explicitly. Let (φ_e) be an effective enumeration of partial computable functions and set

$$A_* = \{e : \varphi_e(e) \downarrow = 0\}, \quad B_* = \{e : \varphi_e(e) \downarrow = 1\}.$$

These sets are disjoint and computably enumerable. Suppose a computable set C contained A_* and avoided B_* . Let d be an index for its total characteristic function χ_C . If $\chi_C(d) = 0$, then $d \in A_* \subseteq C$, a contradiction. If $\chi_C(d) = 1$, then $d \in B_*$ and hence $d \notin C$, again a contradiction. Thus no such C exists.

Use [theorem 28.1](#) for (A_*, B_*) . Composing the proposed total classifier with its primitive-recursive input coding would give exactly such a separator C . Therefore the classifier cannot exist. $\square$

An algorithm may still run the two semidecision procedures in parallel and halt on every input promised to be in $\mathcal{H} \cup \mathcal{S}$. The theorem says that this partial promise procedure has no total computable extension which remains correct on both positive outcomes.

29.3 Where decidability returns

Proposition 29.4 (A family without fragile cases). *For a computable input family of a fixed rational piecewise-affine map, suppose every input is either exactly accepting or robustly avoiding. Then exact acceptance on that family is decidable.*

Proof. Dovetail exact orbit simulation with enumeration of finite safety certificates. The two outcomes cannot both occur, and the assumption ensures that one semidecision halts. Its identity decides the answer. $\square$

This is a specialized form of the known robustness/decidability phenomenon [17]. The new universal construction shows how the missing third case can have full co-enumerable complexity, rather than being a negligible numerical inconvenience.

30 How the uncomputability hides in the noise scale

Restrict the universal map to the slice $(A, B) = (\emptyset, K_0)$, and write $\rho(e) = \rho_F(\iota(\langle o, k, e \rangle), H)$. Then $\rho(e)$ is a uniformly computable real in $[0, 1]$ and

$$\rho(e) > 0 \iff e \in K_0.$$

For $e \in K_0$, let $T(e)$ be the first rejection time of the fixed interpreter on that encoded input. These are interpreter times, not the step counts of the original machine whose halting is being tested.

Theorem 30.1 (No computable positive gap). *There is no total computable $g : \mathbb{N} \rightarrow \mathbb{N}$ such that*

$$e \in K_0 \implies \rho(e) \geq 2^{-g(e)}. \quad (30.1)$$

Indeed, for every total computable g , infinitely many $e \in K_0$ violate (30.1).

Proof. If such a bound existed, compute an interval for $\rho(e)$ of length less than $2^{-g(e)}/3$. Under the dichotomy $\rho(e) = 0$ or $\rho(e) \geq 2^{-g(e)}$, the interval distinguishes the two cases and decides K_0 . This contradicts undecidability.

If there were only finitely many violations, hardcode their membership answers and use the same decision procedure on all other inputs. A finite set is computable, irrespective of whether we have effectively found its list. This would again contradict undecidability. $\square$

[write] Part VII’s Corollary 116.4 proves the same absence of a computable positive gap for the peak value of a strict contraction.

30.1 A Busy Beaver-type envelope

For $n \in \mathbb{N}$, define two finite extrema with default values:

$$B(n) = \max(\{T(e) : e \leq n, e \in K_0\} \cup \{1\}), \quad (30.2)$$

$$\mu(n) = \min(\{\rho(e) : e \leq n, e \in K_0\} \cup \{1\}). \quad (30.3)$$

Thus $B(n)$ is finite and $\mu(n) > 0$ for every n , even when no relevant machine halts.

Proposition 30.2 (Quantitative finite-input envelope). *For every n ,*

$$\frac{\beta}{4S_{B(n)}(L)} \leq \mu(n) \leq \min\{1, 2 \cdot 3^{-(B(n)-1)}\}. \quad (30.4)$$

The function B has no total computable upper bound. The positive function μ has no total computable strictly positive rational lower bound.

Proof. The lower bound follows from corollary 28.4, the monotonicity of $S_T(L)$ in T , and the added default value one. If $B(n) > 1$, an input attaining that rejection time exists, and its radius gives the upper bound. If $B(n) = 1$, the upper bound is simply $\mu(n) \leq 1$.

A computable upper bound for $B(e)$ would let us simulate the interpreter long enough to decide whether input e is in K_0 . A positive computable rational lower bound for $\mu(e)$ would similarly separate $\rho(e) = 0$ from $\rho(e) > 0$ using its computable approximation. Both conclusions are impossible. $\square$

The phrase “Busy Beaver-type” refers to the absence of any computable upper bound on a finite maximum of halting times. We do not claim eventual domination of every computable function for an arbitrary numbering convention. The precise proved assertion is the one in proposition 30.2.

30.2 Consequences for certificate engineering

Corollary 30.3 (No uniform resolution or witness bound). *No total computable function of the input bounds the first mesh index at which a positive-radius input receives a certificate of theorem 26.2. More generally, no sound, complete decidable finite-certificate system for $\mathcal{S}$ has a total computable inputwise certificate-length bound. For any fixed Diophantine representation of $\mathcal{S}$, the height of its smallest natural witness tuple has no total computable inputwise bound on all positive instances.*

Proof. A computable mesh bound would permit checking all meshes up to that bound, thereby deciding $\mathcal{S}$. A computable certificate-length bound would permit checking all finitely many certificate strings up to that length. A computable natural-coordinate bound for a fixed polynomial would permit checking all witness tuples in a finite box. Each is a decision procedure for a Σ_1^0 -complete set, which is impossible. $\square$

This result does not say that certificates are usually large, or that useful subclasses cannot have excellent bounds. It says that a complete universal pipeline must expose the possibility of unbounded search rather than promise a computable worst-case stopping criterion.

31 Transferring the result to other substrates

A simulation of exact trajectories need not preserve robustness. Extra state coordinates can introduce shortcuts to acceptance which were absent in the original model. The following sufficient condition makes the noise issue explicit rather than hiding it behind the word “simulation.”

Theorem 31.1 (Noise-faithful retract transfer). *Let $f : K \rightarrow K$ and $g : M \rightarrow M$ be maps of metric state spaces. Suppose $E : K \rightarrow M$ and $\pi : M \rightarrow K$ are Lipschitz maps with positive bounds A, B and*

$$\pi E = \text{id}_K, \quad gE = Ef, \quad \pi g = f\pi. \quad (31.1)$$

For $H \subseteq K$, take the target $J = \pi^{-1}(H)$ in M . Then

$$\frac{1}{B} \rho_f(x, H) \leq \rho_g(E(x), J) \leq A \rho_f(x, H). \quad (31.2)$$

Exact acceptance, positive-radius avoidance, and fragile avoidance coincide on corresponding encoded inputs. Thus the three-way arithmetic classification transfers whenever the input encoding into M is effective.

Proof. Lift a path $x_0, \dots, x_T$ in K by E . Its final point belongs to J , and

$$d_M(E(x_{t+1}), g(E(x_t))) = d_M(E(x_{t+1}), E(f(x_t))) \leq A d_K(x_{t+1}, f(x_t)).$$

Taking infima gives the upper bound. Conversely, project any path $y_0 = E(x), \dots, y_T \in J$ by π . Its edge costs are bounded by

$$d_K(\pi(y_{t+1}), f(\pi(y_t))) = d_K(\pi(y_{t+1}), \pi(g(y_t))) \leq B d_M(y_{t+1}, g(y_t)).$$

Taking infima gives $\rho_f \leq B \rho_g$, proving the lower bound. The exact conjugacy on the embedded copy and the definition of J give exact acceptance equivalence. The two positive constants in (31.2) preserve the sign of the radius, and therefore the third outcome as well. $\square$

For example, adding auxiliary coordinates by $g(x, z) = (F(x), q(x, z))$, with $q(x, z_0) = z_0$, the product sup norm, and target $H \times Z$ preserves the radius *exactly*: use $E(x) = (x, z_0)$ and $\pi(x, z) = x$, both with bound one. Invertible affine changes of coordinates preserve the three outcomes and change the quantitative radius by at most the associated norm condition factors. These are concrete transfer results, not claims about arbitrary physical implementations.

31.1 A rational recurrent ReLU realization

There is also a direct neural-circuit interpretation. Let $\text{ReLU}(t) = \max\{0, t\}$. A standard max–min representation theorem states that a continuous piecewise-affine scalar function on a closed convex domain is a finite maximum of finite minima of its affine components [21]. Applying this theorem coordinatewise to F gives

$$F_i(x) = \max_{S \in \mathcal{A}_i} \min_{j \in S} \ell_{ij}(x), \quad i = 1, 2, 3,$$

where all ℓ_{ij} are rational affine functions. This cited representation theorem is an additional external dependency of the neural corollary, not of the main dynamical construction.

Corollary 31.2 (A fixed rational neural update block). *There exists a finite feedforward ReLU network with rational weights and biases and three outputs whose restriction to $[0, 1]^3$ is exactly the fixed map F . Repeatedly applying this one block to its three-dimensional state register realizes the trichotomy of [theorem 28.1](#), with exactly the same critical radius for adversarial perturbations of that state register after each complete network evaluation.*

Proof. Replace each maximum and minimum in the displayed representation by the identities

$$\max(a, b) = b + \text{ReLU}(a - b), \quad \min(a, b) = a - \text{ReLU}(a - b).$$

This produces a finite acyclic circuit of rational affine operations and ReLU gates. Signed intermediate wires can be carried through ordinary ReLU layers by the two nonnegative channels $\text{ReLU}(t)$ and $\text{ReLU}(-t)$, whose difference is t . Layering the finite circuit and carrying necessary wires forward therefore gives a standard finite feedforward network with rational parameters and an affine output layer. Its update function is F on the cube, so both the exact and the specified perturbed state trajectories are identical. $\square$

This is a three-dimensional *recurrent state*, not a claim of three neurons, three layers, or a small hidden network. The construction gives no optimized size bound and the network’s full weight list is not extracted in the companion code. Noise at every hidden neuron, fixed parameter errors, stochastic noise, finite floating-point hardware, and continuous-time neural circuits are different models. Their equivalence would require additional estimates or a transfer theorem such as [theorem 31.1](#). [write] Part VII’s Proposition 117.1 gives an explicit ReLU gate count, $9m + 2d_M - 3$, for a different map (a strict contraction of $\mathbb{R}^4$); the weight list of the network above is still not extracted.

32 A worked example: an unattained critical threshold

32.1 A contraction with an unattained critical threshold

Proposition 32.1 (An explicit radius). *For $f(t) = at$ on $[0, 1]$, $0 \leq a < 1$, source zero, and target $[b, 1]$ with $0 < b \leq 1$,*

$$\rho_f(0, [b, 1]) = (1 - a)b. \tag{32.1}$$

If $0 < a < 1$, no pseudo-orbit with error at most this exact critical value hits the target in finite time. If $a = 0$, the critical value is attained in one step.

Proof. Any error- ε path satisfies $x_{t+1} \leq ax_t + \varepsilon$, so

$$x_t \leq \varepsilon(1 + a + \cdots + a^{t-1}) = \frac{\varepsilon(1 - a^t)}{1 - a}.$$

When $0 < a < 1$ and $\varepsilon \leq (1 - a)b$, this is strictly less than b for finite t . Conversely, when $\varepsilon > (1 - a)b$, the positive-error recursion reaches or passes b after finitely many steps. One can stop at b on that step, or clip at one, without exceeding the allowed error. Thus the infimum is $(1 - a)b$ but is not attained. For $a = 0$, every step is within ε of zero, and the jump to b has exactly cost b . $\square$

For $a = 1/2$, $b = 3/4$, the exact value is $3/8$. The reference implementation gives the following rational enclosures using $L = 1/2$. Every entry was computed with integer fractions, without floating-point tolerances.

Mesh N	Vertices	r_N	Certified interval for ρ	Quartic zero found
4	5	$1/2$	$[5/16, 1/2]$	yes
8	9	$7/16$	$[11/32, 7/16]$	yes
16	17	$13/32$	$[23/64, 13/32]$	yes
32	33	$25/64$	$[47/128, 25/64]$	yes
64	65	$49/128$	$[95/256, 49/128]$	yes

Table 1: A computable critical radius need not be realized by a critical-cost finite path. All intervals contain $3/8$.

At mesh $N = 4$, the vertices are $0, 1/4, 1/2, 3/4, 1$, and $\lambda_N = 3/8$. The cut indicator

$$(b_0, b_1, b_2, b_3, b_4) = (1, 1, 0, 0, 0)$$

is a zero of (26.3). Its least outgoing edge has weight $3/8$, so the cut alone certifies $\rho \geq 3/16$. Optimizing the cut to obtain the actual bottleneck $r_4 = 1/2$ gives the stronger lower bound $5/16$ in the table. The full factored quartic and its witness are supplied as `data/09-continuous-barriers-small_quartic.txt` and `data/09-continuous-barriers-small_certificate.json`.

Part IV

Other substrates, implementation, formalization and questions

33 Other unconventional substrates

The decisive property for the output-law theorem is not a conventional instruction set. It is an effective, nonovercounting presentation of finite terminal histories with computable weights. Universality is needed for the completeness lower bounds, but not for the positive representation theorem. The following applications state their semantics explicitly.

Randomized combinator reduction. A lambda, SK, SKI, or Iota reducer with a specified effective reduction strategy and a fair-coin branching primitive has finite syntactic histories and dyadic history weights. Its first-return law therefore has (4.2). The random primitive and reduction strategy are part of the proposed extension; this is not a claim that the inspected repository already implements that probabilistic language. Adding probabilities does not excuse ambiguities between deterministic evaluation strategy and nondeterministic rewrite choice.

Stochastic fractional programs. One can give a FRACTRAN-style system an explicit rational random dispatcher between finite deterministic fractional programs, preserving each program's ordered divisibility semantics. A transition is then decidable arithmetic, with a rational branch probability. Finite histories fit our interface. This construction deliberately avoids saying “choose an applicable fraction randomly” without specifying how conflicts, priorities, and normalization are resolved. Deterministic FRACTRAN-to-Diophantine reductions are already present in the literature [3]; the additional object represented here is the probability law and its conditional boundary.

Random local rewriting and cellular systems. A finite description of each current active configuration, an effective finite set of next local choices, and rational transition weights are enough for an effective first-return presentation. Infinite configurations may also be admitted when a finite-history simulation and observable can be computed from their effective descriptions. Merely declaring an arbitrary infinite initial configuration does not meet that requirement.

Continuous-time jump systems. For a countable-state jump process with finitely many outgoing rational rates q_{ij} at each effectively represented state, the next-jump probability is $q_{ij} / \sum_k q_{ik}$ when the denominator is positive. A zero-rate state is treated as absorbing. Events that reach a designated terminal state after finitely many jumps therefore have an effective rational-weight history presentation. This application uses the embedded jump chain. It neither assigns a terminal value to an infinite explosion history nor claims a result about arbitrary deadline-constrained events. Those questions require additional semantics and analysis.

Robust observations of computable continuous trajectories. Suppose a trajectory is supplied with an algorithm for certified evaluation at rational times and is continuous. Reaching an effectively open target at some finite time is semidecidable: enumerate rational times and certified interior balls, and accept once an evaluation is certified inside one. Continuity ensures that an actual open-target visit contains an appropriate rational time, including a one-sided interval at time zero. This yields a Diophantine reachability predicate by MRDP. It says nothing about exact zero crossings with no robust margin, or about differential equations whose solutions are not uniformly computable from the supplied data.

The effective boundary is indispensable. An arbitrary real oracle parameter can encode a non-c.e. set, and an observation that asks for an exact infinite-time limit can move beyond existential arithmetic. Such a model is not covered merely because it is called Turing-complete. The theorem concerns effective computational substrates, not every imaginable extension that contains ordinary Turing machines as a submodel.

Remark 33.1 ([write] Relation to Part III). The paragraph on robust observations of computable continuous trajectories concerns reaching an effectively *open* target in continuous time, which is

semidecidable. Part III treats the complementary discrete-time question for *closed* rational targets under adversarial per-step noise: positive critical radius is semidecidable and has complete quartic certificates (Theorem 26.2), exact acceptance is semidecidable, and the remaining noise-fragile outcome is Π_1^0 -complete (Theorem 29.1). Theorem 31.1 states when such a trichotomy transfers to another substrate.

34 Implementation and validation

34.1 Manuscript 08: what the software actually verifies

The bundle contains an exact rational budget evaluator, a symbolic single-fold quartic compiler, a canonical prefix allocator, and an exact small quantum-circuit evaluator. The script `code/08-probabilistic-laws-verify.py` uses a fixed random seed, 20260930, for reproducible sampled tests. It also runs exhaustive tests that do not rely on randomness.

Check	Observed result
Exhaustive binary histories	780 histories, 3,120 initial-mass cases, and 22,220 forced-pivot candidates; exactly one valid pivot candidate precisely in each of the 1,330 feasible cases.
Multi-outcome and large-integer tests	192 sampled schedules, including 32 additional rescalings by 10^{100} ; all exact invariance and feasibility checks pass.
Symbolic quartic	$N = 2, m = 2$: 11 parameters, 33 witnesses, 38 residuals, degree four, 165 expanded monomials. All residuals vanish at the exported witness.
Mutation checks	Increasing each of the 33 witness coordinates separately by one is rejected. This is a finite robustness check, not a search of all wrong tuples.
Quadratic-field signs	25,921 small integer pairs cross-checked with high-precision decimal arithmetic; 200 near-cancelling Pell cases checked by exact identities.
Quantum circuits	100 sampled circuits of 30 gates; 3,000 exact norm checks pass, together with H^2 , HTH , and a Bell-state example.
Prefix allocation	12 dyadic approximation stages of the HTH output law; disjoint allocated mass is 4095/4096.
Operator budgets	Four exact coherent examples and one noncommuting cycle pass symbolic matrix checks.

The two allocated masses at the twelfth HTH stage are 437/512 and 599/4096. Their sum is 4095/4096, leaving exactly 1/4096 unallocated at that finite stage. No claim is made that twelve stages already represent the full irrational law. The script also includes floating-point diagnostics for the sharpness limit in Theorem 7.2; those diagnostics are explicitly labeled illustrative and do not support the exact proof.

The detailed results and software versions are in `data/08-probabilistic-laws-verification.json`. The full finite quartic is supplied both as a residual specification and as an expanded expression. The code uses Python integers and `Fraction` for exact rational arithmetic; SymPy is used for symbolic expansion and the matrix checks.

34.2 Manuscript 07: what the delivered programs do

The package uses only the Python standard library. Matrix entries are `fractions.Fraction`; all certificate computations are integer computations. The principal functions in `code/07-quantum-mortality-quantum_diophantine.py` are:

Function	Contract
<code>elementary_gram</code>	Rational LDL^T and four-square search; returns a $4d$ -row factor and checks its Gram identity.
<code>gram_search</code>	Height-ordered rational search, with an optional cap; succeeds on the included small examples with $d + 3$ rows. A cap failure is not an impossibility result.
<code>compile_instrument</code>	Checks the factor, packs its rows, builds all Kraus operators, and verifies normalization exactly.
<code>build_certificate</code>	Constructs the quadratic residuals with free signed-pair matrix inputs and asserts both symbolic counts.
<code>canonical_assignment</code>	Generates the uniquely determined natural assignment associated with a specified word.
<code>Certificate.expanded</code>	Expands the sum of residual squares into a sparse integer polynomial. Expansion is optional and can be large.

The test command, from the root of the delivered package layout (the README explains how to run it on a copy), is

```
python code/run_checks.py
```

It writes the example matrices, a canonical zero-word witness, the expanded small quartic, and a machine-readable result report.

34.3 Manuscript 07: observed test results

The delivered run passed 32,418 explicit checks. The total is a count of individual assertions, not a count of statistically independent experiments. It includes twelve seeded small random input lists and several deterministic examples. In particular, 3,460 outcome words were checked for the projection formula, exact integer probability formula, and arithmetic gap. The one-step flush implication was checked 12,449 times, using all active final letters when its premise held.

The suite verifies the four-square routine on every integer from 0 through 300, exercises the $(d + 3)$ -row rational search on three small positive forms, and tests the counterexample 7 to a three-square search within a specified finite height cap. The latter computational cap is only a regression test; the global impossibility is proved in Proposition 15.5.

For the uniform certificate with $(D, K, n) = (2, 2, 2)$, symbolic expansion yields 647 monomials and total degree four, with 44 witnesses and 50 residuals. For the dimension-four example and word $(2, 1, 1)$, the generated assignment satisfies all 264 residuals with 245 natural witnesses. Increasing any one of those witness coordinates by one makes the certificate nonzero. These mutation tests supplement, but do not replace, the inductive uniqueness proof.

The complete category counts are in `data/07-quantum-mortality-check_results.json`. A

local run took about five seconds; this is not a benchmark for the general rational Gram search. The reference implementation is intentionally transparent rather than optimized.

34.4 Manuscript 09: what was tested

The script `code/09-continuous-barriers-run_checks.py` reproduces 5,660 passing assertions. Numerical calculations use exact rational arithmetic through Python’s `fractions.Fraction`. The deterministic sampling seed is 20260930. The checks include the following groups.

Check group	Assertions
Finite-tape update formulas against independently encoded tape operations	768
Images of whole digit rectangles, including non-Cantor points	600
Bottleneck/cut duality and optimal strict-threshold cuts	240
All simplex Lipschitz bounds in the five-state sample compiler	1,350
Prescribed low boxes, emergency boxes, and terminal slabs	495
Quartic zero test on all assignments in $\{-1, 0, 1, 2\}^5$	1,024
Rejection-time upper schedules and lower-noise samples	656
Other enclosure, cube, emergency, and certificate checks	527
Total	5,660

Table 2: Implementation checks, not a formal verification of the theorems.

The sample machine for the exhaustive simplex check has five states and uses left, right, and stationary instructions. All 1,350 of its simplices satisfy the advertised rational row-sum bound. The tests also include rejecting countdowns of lengths one through nine and emergency perturbations with exponents one through fifteen.

The universal interpreter table itself is not supplied or tested, and finite tests cannot establish universality or an infinite-time theorem. Their role is narrower: detecting formula, indexing, boundary, and implementation errors in the constructive ingredients. The general results rest on the proofs in [sections 25 to 28](#).

Reproducing the artifact. [write] The source’s reproduction instructions (run `python code/run_checks.py results`, with Python 3.10 or newer and no third-party packages, which rewrites the JSON verification report, example certificate, factored quartic and contraction table in the directory named; then build the article twice with pdfLaTeX) apply to the delivered layout. The source uses standard T_EX packages, and no downloaded font files or third-party papers were included in its archive.

34.5 Manuscript 08: a trust-boundary summary

The ordinary proofs establish the general theorems. The finite tests provide evidence about the supplied implementations, including arithmetic signs, indexing, tie resolution, and degree accounting. They cannot replace an inductive proof of the compiler theorem. Conversely, the MRDP existence proof does not certify that the software has generated a universal polynomial: it has not. The delivered archive made this separation explicit in both its README and its machine-readable report; the report README repeats it.

34.6 [write] Rerunning the suites in this report

The programs are shipped byte-identical to the delivery but under prefixed names (`code/07-quantum-mortality-*.py`, `code/08-probabilistic-laws-*.py`, `code/09-continuous-barriers-*.py`), and they import one another by their delivered module names and write to delivered directories (`examples/` and `verification/` for 07, `artifacts/` for 08, the named directory or `results/` for 09). They must therefore be run on a copy with the delivered layout, never in place; the README gives the commands. Manuscripts 07 and 09 need only the Python standard library; 08 needs SymPy (`data/08-probabilistic-laws-requirements.txt` pins `sympy==1.14.0`). Rerun on copies on 30 September 2026 (07 and 09 with Python 3.14.4, 08 with Python 3.13.5 and SymPy 1.14.0), all three suites passed with the recorded counts, and every rewritten output equals the shipped file apart from line endings (the programs write CRLF on Windows) and 07's recorded runtime. 08's report records the interpreter version it runs under, so another interpreter changes that field too.

35 Formalization

[write] What ProveIt has formalized, and what it has not, is stated in Section 1.3. The three sources propose the following plans; none of them has been carried out.

35.1 Manuscript 08: proof obligations suitable for Lean or Rocq

A direct formalization should keep the finite arithmetic layer separate from the analytic probability layer. The following dependency chain avoids requiring a large quantum semantics development before obtaining a useful kernel-checked result.

Finite order and ratio layer. Formalize positive rational rows, least maximizing indices, and the single-step equivalence $h'q_i \geq hp_i \ (\forall i) \iff h' \geq h \max_i(p_i/q_i)$. Induct on the horizon to prove the optimal product theorem. This portion requires finite sums, ordered fields, and finite maxima, not MRDP.

Certificate layer. Define a finite index type for parameters and witnesses, an explicit residual family of multivariate integer polynomials, and its sum of squares. Prove that natural zeros force every residual to vanish. Prove the selector lemma, the product invariant, and witness uniqueness. Derive the cardinality formulas from the index constructors rather than from an external count. Prove a compiler correctness theorem and use the Python example only as an untrusted certificate generator.

Computability and MRDP layer. Encode the allocation interpreter and the threshold semidecision procedure on naturals. The inspected `Diophantine.mrdp` interface can then produce the finite polynomial existence theorem. A separate circuit lowering theorem establishes degree at most four. This layer should not claim witness uniqueness for a generic MRDP construction.

Measure and limit layer. Prove the prefix-cylinder allocation invariant and its measure identity; then identify limits of cumulative masses with output laws. Prove the variation bounds and the

weak-computability realization with its explicit nonuniform existential tail choice. Separate program extraction claims from existence in classical mathematics.

Quantum layer. First verify the integer-linear gate recurrences and the quadratic norm formula, with the state-vector dimension explicit. Then develop measured first-return semantics using unnormalized branch states. The operator-budget theorem can be formalized independently in a finite Hermitian matrix library.

This is a plan, not delivered Lean source or a report of successful Lean compilation. In particular, no guessed theorem names or placeholder `sorry` proofs are included as though they were checked results.

35.2 Manuscript 07: a realistic formalization decomposition

A useful implementation plan separates five proof obligations.

Matrix block algebra. Formalize the block matrices in (16.2), the sum-of-Grams identity, and the projection homomorphism. These are finite algebraic identities. The normalization and word-transfer theorem can first be proved under an explicit hypothesis $B^\top B = Q$; no number theory is needed at that stage.

Rational factor existence. Provide a constructive $4d$ factor from a rational $LDL^\top$ decomposition and four-square certificates. A separate theorem can improve this to $d+3$ rows when the appropriate rational quadratic-form theorem is available. This isolates the classical dependency rather than concealing it inside a tactic call.

Exact word evaluator. Encode an integer matrix by finitely many signed natural coordinates. Matrix multiplication and lookup in a finite generator list are computable operations. Their graphs can be handled directly by polynomial equations or through primitive-recursive graph interfaces. One should explicitly state the chronological multiplication convention in the evaluator theorem.

Quartic certificate correctness. Prove that the residual family is zero exactly when the selected matrices and products are correct. Then prove uniqueness of the natural sign split and count the coordinate families. A sum-of-squares theorem produces the single polynomial. This is a direct finite proof and need not invoke the entire MRDP construction.

Unbounded bridge. Prove that the exact enumerator semidecides $\text{QMort}_{4,7}$ and apply the public MRDP endpoint. An alternative route uses the existing `exactIter_dioph` and `existsExactIter_dioph` interfaces after representing the matrix-update relation [15]. These routes establish the same semantic property but need not preserve the same resource bounds or witness multiplicities.

No Lean source is supplied as “verified” in this package. A schematic statement with missing proofs would not add evidential value. The executable artifacts verify concrete instances and polynomial generation, while the general theorems are proved in the text.

35.3 Manuscript 09: a concrete formalization plan for ProveIt

The highest-value first formalization target is the finite checker and its soundness, not a large universal machine table. The mathematical architecture has clean trust boundaries.

35.3.1 Finite graph and polynomial layer

Represent a finite vertex type, a source, a finite target set, and rational weights. Prove bottleneck/cut duality with finite paths and cycle deletion. Then define (26.3) as an `MvPolynomial` over $\mathbb{Z}$. Its zero-set proof requires only finite sums, nonnegativity of squares, and the identity $b(b - 1) = 0 \Rightarrow b = 0 \vee b = 1$ over integers.

An external graph search may propose a bit mask. A proved finite checker verifies source membership, target exclusion, and every below-threshold edge. The proof assistant need not trust the search algorithm, its heuristics, or its report of an optimal bottleneck. When a claimed numerical lower radius is used, the checker must also verify the stated rational crossing margin.

35.3.2 Real rounding and geometric layer

Formalize nearest-grid rounding in the sup norm and its compatibility with closed rational boxes. Keep the nongrid source as a separate vertex. The critical-radius infimum requires explicit nonemptiness and bounded-below proofs; do not smuggle in the existence of an optimal real path. The rounded path argument then yields [theorem 25.1](#) independently of its length.

For the map compiler, certify a rational triangulation, agreeing affine restrictions on shared faces, cube-valued vertex data, and row-sum bounds. The critical global fact is continuity across faces. A source file with correct guarded formulas but an unspecified interpolation is not a proof of the required map properties.

35.3.3 Computability and MRDP layer

Choose one canonical coding for rationals, finite tuples, finite tables, and bit masks. Prove the finite graph/cut predicate primitive recursive in those codes. A semidecider then enumerates mesh/certificate pairs. At that point the existing theorem `Diophantine.mrdp` is the correct public interface. Its returned fixed polynomial certifies the encoded predicate, not a different informal notion of safety.

No modification to the MRDP core is mathematically necessary. Possible new modules in a `RobustReachability` namespace would be `Grid`, `CutCertificate`, and `Diophantine`. These names are proposals, not existing or compiled repository declarations.

35.3.4 Universal interpreter layer

The final layer needs a concrete universal binary transition table, a verified finite-tape semantics, and a proof that the chosen input coding matches the interpreter. One must separately establish exact simulation, terminal-slab robustness, emergency timing, and the global map's Lipschitz bound. The universal trichotomy then follows from ordinary dovetailing.

The present archive contained no Lean files, no claimed Lean build receipt, and no axiom audit for these new results. Its value for formalization is a small set of explicit intermediate statements

Layer	Artifact that can be generated	Fact that must be proved
Finite certificate	Vertex list, edge list, bit mask	The checked cut blocks all low-cost graph paths
Polynomial	Factored sum of squares	Its integer zeros are exactly admissible Boolean cuts
Geometry	Rational simplex table	Continuity, cube invariance, and a global error bound
Semidecider	Mesh/certificate enumeration	Acceptance iff the real critical radius is positive
Universal compiler	Interpreter table and its PWA image	Correct simulation and the three-outcome correspondence

Table 3: Generated data and proved semantics must remain separate.

rather than a large untested proof script.

36 Further research questions

The following questions are proposed research directions, not claims that their full priority or open-status history has been settled by the present literature search. They are ordered to put concrete extensions ahead of broader conjectural targets.

[write] The questions are printed by source: 08’s twelve in Section 36.1, 07’s ten in Section 36.2 (which the source posed as subsections and are numbered here), and 09’s ten, with its recommended order, in Section 36.3. How they overlap, and which one another source partly answers, is recorded in Section 36.4.

36.1 Manuscript 08: probabilistic laws and normalization

Research question 1 (Kernel-checked unique quartic compiler). Can Theorem 6.1, including invalid-input rejection and the exact count $5Nm + m + 4N + 3$, be formalized and connected to an executable polynomial exporter in ProveIt? A decisive deliverable would be a checked reflection theorem accepting the supplied $N = 2, m = 2$ residual certificate.

The finite positivity and tie-breaking lemmas make this the most sharply specified implementation target. It need not wait for quantum semantics or infinite probability theory.

Research question 2 (Smaller canonical arithmetic certificates). Can the X, Y cross-product witnesses or the selectors be eliminated while retaining degree at most four, natural-witness uniqueness, and a linear-size arithmetic circuit? What trade-off between witness count, degree, and coefficient size is optimal for this specific normalization predicate?

This is a finite compiler optimization problem. A mere MRDP existence argument cannot answer its size or uniqueness requirements.

Research question 3 (Sharp multi-outcome variation budgets). For a specified initial vector p_0 , determine the sharp supremum of $\sum_s \text{TV}(p_{s-1}, p_s)$ under $\sum_s D_\infty(p_{s-1} \| p_s) \leq B$. Does a

useful potential formula extend the balanced binary constant $1/2$ to $m \geq 3$, and what paths are asymptotically extremal?

Proposition 7.1 gives the universal upper bound B but Theorem 7.2 shows that it can be substantially nonsharp when the starting law is fixed.

Research question 4 (Uniform high-success realization). What effective additional data are necessary and sufficient to turn the existential high-success realization in Theorem 8.2 into a uniform compiler? Natural input formats include certified tail-variation bounds, an interior interval, or a computable bound on the remaining logarithmic cost.

The present theorem explicitly leaves tail selection nonuniform. A solution should specify its representation of real numbers and its promised inputs; otherwise an apparent uniformization may hide an oracle.

Research question 5 (Exact symbolic infinite histories). For which algorithmically described rational paths can G_∞ be computed, or bounded sharply enough to synthesize a desired success floor? Examples include periodic transformations, linear recurrences, and piecewise-rational paths with summable perturbations.

Finite prefix feasibility is decidable and uniquely certified here. Certifying every prefix with one finite object is an additional mathematical problem, not an automatic consequence of those certificates.

[write] *Batch-62 note (30 September 2026)*. Bears on it, not answered: source 17's Theorem 55.3 shows that a periodic binary environment gives a rational stationary root mass, computed exactly as the fixed point of an affine map. That is a different quantity from G_∞ .

Research question 6 (Operator certificates with controlled multiplicity). Can an explicit quartic natural-witness system certify the optimal factors R_s for algebraic density matrices, including support changes, without uncontrolled multiplicity from eigenvector choices? Can canonical principal minor or characteristic-polynomial data replace arbitrary spectral bases?

The scalar earliest-maximum rule has no immediate analogue for degenerate eigenspaces. Any proposed uniqueness theorem must address that issue rather than merely add spectral witnesses.

Research question 7 (Quantum front ends with structural compression). Which circuit classes admit polynomial-size exact Diophantine certificates that avoid the full 2^n state vector? Candidate interfaces include stabilizer structure, bounded tensor-network width, and a bounded number of non-Clifford gates, with all compression assumptions stated explicitly.

The current state-vector checker is exponential. A useful result would preserve exact zero tests and Born probabilities while making the source of any speedup mathematically visible.

[write] *Batch-62 note (30 September 2026)*. Partly answered by source 10 (Part VI) for bounded tensor-network width: for a supplied contraction tree of merge-union width w , Theorems 73.1 and 74.1 give unique-witness quartic certificates with at most $(72m - 64)D^w$ natural witnesses and polynomial bit length per coordinate, hence polynomial size when $w = O(\log m)$. Theorem 77.1 is the limit: a uniformly polynomial-size compiler for zero output probability would give $\text{NP} = \text{coNP}$. Stabilizer structure and few non-Clifford gates remain open (source 10's project 5).

Research question 8 (Interactive and compositional semantics). How much of the projected-root representation survives for interactive processes with adaptive inputs, correlated outputs, or sequential conditioning? Can a compositional representation preserve joint laws rather than only a terminal one-output marginal?

Independent samplers for marginals are not enough. A solution must preserve the common probability space or the appropriate quantum instrument.

Research question 9 (Direct universal substrate polynomials). Can one produce a fully expanded, audited quartic polynomial for a small measured quantum controller or a stochastic fractional language, avoiding a large generic interpreter? Which operation counts can be proved rather than estimated from informal circuit descriptions?

Theorem 4.3 gives a fixed polynomial in principle, not its coefficients or a competitive universal witness count.

Research question 10 (Counting executions without counting arithmetic witnesses). Can canonical finite source certificates be composed into useful resource-bounded counting reductions while preserving projected-event weights? Under what explicit restrictions can one obtain a genuinely parsimonious unbounded representation?

The finite single-fold construction here is not a general unbounded single-fold theorem. Any extension must confront that distinction directly.

[write] *Batch-62 note (30 September 2026)*. Partly answered by source 13 (Part V) for finite horizons: its compiler is parsimonious (natural zeros in bijection with accepted coin tapes, Theorem 41.2), and it proves a frontier: residual systems of interaction degree two are counted in polynomial time (Theorem 43.2), while degree three is already $\#P$ -complete (Corollary 43.3). The unbounded parsimonious question stays open.

Research question 11 (Continuous-time and robust physical semantics). Develop certified first-hit and deadline probabilities for effective jump systems and computable flows, with explicit treatments of explosion, Zeno behavior, exact equality observations, and robustness margins. Which observables remain c.e. threshold events, and which cross the normalization boundary proved here?

The finite-first-return interface identifies a tractable starting point; it does not prescribe a physically preferred semantics for limit events.

[write] *Batch-62 note (30 September 2026)*. Partly answered by source 17 (Part V) for exact stationary root masses of discrete reversible chains: strict rational cuts are c.e. and Diophantine, equality is co-c.e. and Π_1^0 -complete (Theorem 60.6), and arithmetic type is Σ_2^0 -complete (rational, algebraic) or Π_2^0 -complete (irrational, transcendental; Theorem 58.4). Continuous time, explosion and robustness margins remain open (source 17's Question 42).

Research question 12 (An algebra of probability specifications). Build a specification language that tracks whether a quantity is left-c.e., right-c.e., weakly computable, or computable, and automatically assigns a sound arithmetic quantifier class to each comparison. Can successful normalization-budget certificates improve the inferred class for useful restricted program families?

Such a system would prevent the common mistake of compiling every probabilistic specification to a single existential equation merely because its source language is Turing-complete.

[write] *Batch-62 note (30 September 2026)*. Partly answered by source 13 (Part V): quantifier normal forms with fixed quartics for probability thresholds and finite mean (Theorem 46.3, equation (47.3)), and a transfer theorem that separates preservation of probabilities from preservation of means (Theorem 48.2). Source 17 supplies a fully classified worked family, and source 10 (Theorems 78.1 and 79.2) the representation axis. No specification language is built.

36.2 Manuscript 07: quantum mortality and its certificates

The following are proposed investigations, not assertions that every formulation is a previously catalogued open problem. Establishing priority and the exact state of the literature is itself part of the first stage of each project.

Research question 13 (Smaller rational instruments). Can the same unrestricted zero-word decision problem be shown hard in dimension three, or with fewer than seven outcomes in dimension four? A successful construction must avoid spurious zero products from any added normalization operators. It is not enough to simulate a computation only on a regular language of “well-formed” histories while leaving other outcome words unchecked. The projection-and-idle structure here gives a useful baseline for such an audit.

Research question 14 (Residual-specific Gram compression). For $Q(c) = c^2I - \sum_i M_i^T M_i$, can one always choose rational c so that $Q(c)$ has fewer than $d + 3$ rational Gram rows? Proposition 15.5 does not answer this because it concerns arbitrary positive forms and does not respect the freedom to vary c . An arithmetic analysis of this one-parameter family could improve the quantum resource bound without changing the block argument.

Research question 15 (Height bounds and practical rational factors). The existence proof for a $d + 3$ factor is effective by search, but the search can be enormous. Develop height bounds in terms of the bit length of Q , and compare algorithms based on rational isotropy, lattice reduction, and specialized four-square manipulations. The relevant output complexity is not just row count: the numerator and denominator heights of the Kraus operators affect both file size and the gap in (20.2).

Research question 16 (Choosing the lower blocks to preserve a shortest word). Theorem 16.1 confines the shortest length to a two-point interval. For a known source witness ending in i , equation (17.2) says exactly when no extra letter is needed: B_i must annihilate the preceding product. Can one choose a Gram factor with prescribed annihilation subspaces while retaining the same row budget? For unknown shortest witnesses this interacts with an undecidable search; for specified witnesses it is a concrete constrained quadratic-form problem.

Research question 17 (Smaller explicit Diophantine certificates). The formulas $71n + 32$ and $72n + 48$ are transparent upper bounds, not claimed optima. Eliminating known-zero matrix coordinates, exploiting the block structure, and using the final-active-letter criterion could reduce constants substantially. Such reductions should preserve the distinction between free matrix inputs and existential witnesses and should state whether the word-to-witness bijection survives.

Research question 18 (A complete proof-producing compiler). Implement and certify the entire chain from a binary tag program and input word through integer matrix mortality, rational Gram packing, and polynomial generation. The first useful milestone is an independently checked rational Gram certificate plus a checked block-algebra theorem. A practical artifact should include exact source and target semantics, not merely display a polynomial with the desired informal interpretation.

Research question 19 (The fixed-arity uniqueness frontier). Can a useful restricted class of these instruments admit a fixed-arity Diophantine representation whose witnesses retain a controlled multiplicity? The bounded family already has unique arithmetic witnesses per outcome word. Unbounded trace compression can introduce arbitrarily many choices, and the general single-fold or finite-fold questions must not be claimed solved by the bounded construction. A sensible first target is a clearly specified instrument class with additional growth or normal-form constraints.

Research question 20 (Promise problems robust under perturbation). Study promises with a fixed separation from zero, bounded history length, or restricted coefficient heights. The estimate in Proposition 20.2 supplies an exact baseline for a finite-length promise, but not a length-independent robust undecidability result. A meaningful physical formulation must specify both the norm of the implementation error and what information about the exact reference model is available.

[write] *Batch-62 note (30 September 2026)*. Partly answered by source 16 (Part VI) in the exact-synthesis setting: positive-tolerance synthesis is computable with a computable length bound (Theorem 84.21), exact synthesis stays c.e.-complete inside every open ball (Theorem 92.3), and the length-dependent threshold $1/(qD^n)$ is exactly where approximation becomes exact synthesis (Theorem 93.1). Bearing on it: source 11 shows that positive-tolerance transfer is always possible while no positive margin separates its hard orbit (Section 90), and source 10 that exact zero is Π_1^0 -complete under Cauchy names (Theorem 78.1). A length-independent robust undecidability result for instruments is not given.

Research question 21 (Decidable signed subclasses). The nonnegative support argument gives one decidable boundary. Other restrictions worth investigating include simultaneous triangularization with arithmetic constraints, low-rank generated semigroups, or common invariant cones with an effectively checkable description. These proposals require care: positivity of operators in the Loewner order is not the same as entrywise nonnegativity, and either condition may fail to survive a change of basis.

[write] *Batch-62 note (30 September 2026)*. Bears on it: source 11 (Part VI) gives one signed (orthogonal) gate family in which rational-matrix membership and the orbit of e are polynomial-time while the orbit of ξ is c.e.-complete (Theorem 81.1); there the boundary is set by the source state, not by a matrix class.

Research question 22 (Other exact scalar fields and tensor observables). For a fixed algebraic number field with a chosen embedding into $\mathbb{C}$, exact field arithmetic can be represented by rational coordinate tensors. Determine how the Gram packing changes when conjugation and norms are expressed in those coordinates. The tensor identity (22.3) gives an elementary entry point for scalar matrix-product observables, but physically normalized tensor networks and general quantum channels with several Kraus operators per recorded outcome require separate semantic proofs.

[write] *Batch-62 note (30 September 2026)*. Partly answered by source 10 (Part VI): exact arithmetic in $\mathbb{Q}(\zeta_8)$ by four integer coordinates, with conjugation a signed coordinate permutation and exact norms (Section 74), certified tensor-network observables (Theorem 74.1), and a single-valued sign gadget for $p + q\sqrt{d}$ (Theorem 74.3 and the remark after it). How Gram packing changes is not addressed.

36.3 Manuscript 09: continuous computation

The following questions are concrete continuations of the results. Some are implementation or optimization problems; others may require new mathematical ideas. Their inclusion does not certify

that no answer exists in the wider literature. A focused prior-art audit should precede any priority claim.

Research question 23 (Extract a fixed safety polynomial). Can the finite cut checker be compiled through ProveIt’s existing machinery into a fully displayed polynomial with an audited degree, variable count, coefficient height, and witness-decoding procedure?

This is the most direct repository contribution. Start with a deliberately large but complete representation, then optimize. The milestone is an end-to-end proof that a zero decodes to a valid real safety statement. A small polynomial for just one grid does not meet the milestone. No single-fold or finite-fold conclusion should be assumed during optimization.

Research question 24 (Reduce the three-dimensional realizer). Can one fixed continuous rational piecewise-affine map of a square realize every disjoint computably enumerable pair with the same exact-acceptance/positive-radius trichotomy? What additional restrictions make such a realization impossible?

The current third coordinate provides an independently adjustable expanding emergency channel. Encoding that channel into the tape without allowing spurious robust rejections is the central challenge. Any lower-dimensional claim must retain a positive margin on every rejecting input and must specify a full continuous extension, not only a symbolic invariant subset.

[write] *Batch-62 note (30 September 2026)*. Posed again, for strict contractions in dimension four, by source 14’s Question 59 (dimension and observation minimality).

Research question 25 (Smooth and polynomial realizers). Which regularity classes support the same simultaneous realization theorem on a compact state space: C^∞ maps, analytic maps, or rational polynomial maps?

Smooth generalized-shift constructions suggest an accessible route for the first class. The polynomial and analytic cases cannot simply reuse the constant terminal slabs: a real analytic map constant on a nonempty open set of a connected domain is constant throughout. One would need contracting trapping regions instead of literal plateaus, with a new proof of the positive rejection margin and a computably recognizable accepting target.

[write] *Batch-62 note (30 September 2026)*. Partly answered by source 12 (Part VII) in the scalar 2-adic case: a scalar p -adic contraction analytic near its fixed point has a uniform finite extinction horizon (Theorem 129.1, with explicit bounds for rational polynomials in Corollary 129.2), so analyticity rules out universality there, while flatness to all orders does not (Proposition 128.2). Source 14’s Proposition 120.1 is the real homogeneous obstruction. The real C^∞ , analytic and polynomial cases of this question remain open.

Research question 26 (Extract and certify the rational neural circuit). Can the max–min representation in corollary 31.2 be generated as a concrete rational network, with a checked equivalence to the simplex table and useful size bounds?

The elementary max/min-to-ReLU identities supply a baseline. More important than reducing depth initially is keeping a machine-readable correspondence between the network, the PWA dynamics, and the certificate checker. Noise at hidden units should be treated as a separate theorem, not identified with noise at the recurrent state register.

[write] *Batch-62 note (30 September 2026)*. *Not answered*. Source 14 (Part VII) gives an explicit integer ReLU circuit, with $9m + 2d_M - 3$ gates, for its own contraction (Proposition 117.1), not for

the map F of Corollary 31.2. Source 15 gives a unique-witness trace quadratic that certifies the finite runs of any concrete rational clipped network (Theorem 137.1), and $\sigma(u) = \text{ReLU}(u) - \text{ReLU}(u - 1)$; that is the certification layer such an extraction would need, but neither source extracts this network.

Research question 27 (Quantitative transfers to genuinely different dynamics). Can discrete symbolic, chemical, mechanical, optical, or hybrid simulators admit effective noisy projection/lifting estimates analogous to (31.2)?

Exact Turing completeness alone is insufficient. A useful positive result would establish both directions of perturbation transfer, perhaps with a controlled time rescaling, and specify which target is observed. A useful negative result would identify a substrate for which an exact simulator necessarily destroys every positive rejection margin.

[write] *Batch-62 note (30 September 2026)*. Bears on it: in source 12 (Part VII) every finite quotient of the 2-adic contraction is nilpotent, so every positive-precision (clopen) observation is decidable while exact extinction is c.e.-complete (Proposition 127.1, Corollary 125.4): a data point for the negative direction, although source 12 does not formalize this Part’s noise radius on $\mathbb{Z}_2$.

Research question 28 (Replace dense grids by small inductive regions). Can positive-radius inputs be certified by adaptive polyhedral partitions, interval boxes, or algebraic invariants with substantially smaller witnesses than the complete grid?

The current grid proves completeness but is deliberately expensive. A natural certified search starts with a coarse cut, refines only ambiguous boundary cells, and carries a local Lipschitz bound for each relevant cell. The theorem to preserve is completeness for every positive margin, not merely success on benchmark examples. The no-bound theorem excludes a computable universal worst-case guarantee, but permits powerful subclass-specific bounds.

Research question 29 (Sharp radius–time tradeoffs). Can the lower and upper exponential estimates in (28.6) be made comparable up to fixed constants or a fixed exponent for a universal realizer?

The present lower bound uses a coarse global Lipschitz constant while the upper bound uses the emergency multiplier three. A construction whose simulation and emergency expansion scales are matched could make the smallest robustness radius a quantitatively sharper geometric encoding of halting-time complexity. Any such statement must specify the interpreter and input-length convention.

Research question 30 (Classify exact critical-level reachability). For which subclasses is reachability at error exactly ρ decidable or characterizable, assuming ρ is supplied exactly?

The contraction example shows that equality of the error budget to the critical value need not yield a finite accepting path. Compactness of the state space does not bound the time variable. Useful hypotheses might force a bounded witness length or turn the relevant limiting chain into a finite one. A classification here would complement, rather than repeat, the computability of the value itself.

[write] *Batch-62 note (30 September 2026)*. Bears on it: source 12’s Corollary 129.2 gives a decidable subclass, rational polynomial contractions of $\mathbb{Z}_p$, for exact reachability of the fixed point.

Research question 31 (Different noise semantics). What becomes of the Diophantine trichotomy for a single initial perturbation, a fixed parameter perturbation, stochastic errors, total-error budgets, or errors confined to specified coordinate directions?

Our fragile acceptance mechanism already uses only one perturbation in one coordinate, but the rejection theorem controls all coordinates at every step. This asymmetry provides a useful starting point. Probabilistic eventual acceptance, almost-sure avoidance, and existential noisy reachability must not be conflated; their quantifier structures can differ substantially.

Research question 32 (Geometry of the fragile boundary). What can be said about the topology, dimension, or effective descriptive complexity of the full set $\{x : \neg \text{Hit}(x), \rho_F(x, H) = 0\}$, beyond the Π_1^0 -complete classification on the encoded inputs?

On regions disjoint from H , the critical radius is Lipschitz by [proposition 25.3](#), so its positive set is open. The coded-input classification does not by itself determine the measure or fractal geometry of the full zero set. Studying that geometry could connect certificate complexity with dynamical notions of chain accessibility.

[write] *Batch-62 note (30 September 2026)*. Posed again, for different zero sets, by source 12’s Question [80](#) and source 14’s Question [65](#).

Recommended order. First formalize the finite graph and quartic checker, then the rounding soundness theorem, and only then extract a fixed MRDP polynomial. In parallel, make one concrete interpreter table and the neural representation explicit. Dimension reduction and polynomial realizations are better treated as separate mathematical projects after these certified baselines exist.

36.4 [write] How the questions overlap

- *Partly answered.* Question [11](#) (08) asks which observables of computable flows and jump systems remain c.e. threshold events once robustness margins are treated explicitly. For discrete-time rational piecewise-affine maps under adversarial per-step noise, Part III answers the robustness part: positive margin is Σ_1^0 -complete with complete quartic certificates, and noise-fragile nonacceptance is Π_1^0 -complete, hence not existential Diophantine (Theorems [26.2](#) and [29.1](#), Corollary [29.2](#)). The question stays open for continuous time, jump processes, deadlines, explosion and stochastic noise, which is also 09’s Question [31](#).
- *One frontier, two forms.* 08’s Question [10](#) and 07’s Question [19](#) both ask for a fixed-arity representation with controlled witness multiplicity; 09 (Section [26](#)) likewise declines to claim it. The sibling report `canonical-diophantine-certificates` of this category shows that a single-fold (finite-fold) universal halting polynomial exists exactly when every computably enumerable set has such a representation, an open problem.
- *Extraction of a fixed polynomial.* 08’s Question [9](#), 07’s Question [18](#) and 09’s Question [23](#) ask for explicit, audited forms of the three different fixed polynomials of Remark [2.1](#).
- *Smaller explicit certificates.* 08’s Question [2](#) and 07’s Question [17](#) ask the same kind of question for the two different explicit quartic families $F_{N,m}$ and $\mathcal{Q}_{D,K,n}$.
- *Formalization first.* 08’s Question [1](#), 07’s milestone in Question [18](#) and 09’s recommended order all put a kernel-checked finite certificate layer before any unbounded statement.
- [write] *Partly answered in Parts V–VII* (batch 62; the notes after each question give the details): Question [7](#) by source 10 (bounded tensor width) with the $\text{NP} = \text{coNP}$ limit; Question [22](#) by source 10 ($\mathbb{Q}(\zeta_8)$ coordinates and $\mathbb{Q}(\sqrt{d})$ signs); Question [20](#) by source 16 (exact synthesis

versus approximation); Question 10 and Question 12 by source 13; Question 11 by source 17 (stationary masses); Question 25 by source 12 (scalar 2-adic analytic contractions). Questions 5, 21, 24, 27, 30 and 32 are only borne on, and Question 26 is *not* answered: sources 14 and 15 supply a gate count for another map and a certification layer, not the network. Questions 2, 13, the variation-budget and high-success questions are not addressed.

37 Conclusions

37.1 Manuscript 08

The useful extension of Diophantine computation is not just another unconventional universal instruction set. It is a semantics-preserving account of which quantitative assertions admit finite arithmetic witnesses. Effective output laws admit a common projected-root representation, but normalization can raise a strict threshold from Σ_1^0 to Σ_2^0 , even with a success guarantee arbitrarily close to one.

The exact normalization budget supplies a constructive organizing theorem: accumulated mass can follow a prescribed positive conditional path exactly when its product of maximal coordinate ratios fits the available budget. That theorem yields a direct, unique-witness quartic compiler, a sharp binary oscillation bound, an exact weak-computability characterization, and a noncommutative operator-order extension. The explicit finite compiler and tests are ready to serve as a concrete formalization target; the broader universal and quantum results retain their stated existence and semantic boundaries.

37.2 Manuscript 07

The key quantitative step is to distribute rational Gram rows among the original active Kraus operators instead of allocating a large correction block to each one. Exact normalization then costs only $\lceil s/k \rceil$ additional state-space coordinates for an s -row factor. Classical rational quadratic-form theory supplies $s = d + 3$, while an elementary implementation supplies $s = 4d$.

This produces a proof of quantum zero-word undecidability in dimension four with seven rational outcomes, based on the published six-generator matrix theorem. Its witness transformation is unusually tight: erase idle outcomes in one direction, and append one active outcome in the other. The polynomial compiler subsequently records exact signed products, not merely legal paths, with quartic degree and explicit linear-in-length witness counts.

The work therefore extends the repository's Diophantine program toward an interference-based computational substrate without conflating exact mathematics, finite experimental precision, and formal proof status. The row-packing theorem is ready for independent mathematical review; the elementary branch and the bounded quartic compiler are ready for focused formalization.

37.3 Manuscript 09

The main contribution is not another bare invocation of MRDP on a halting set. It is a finite certificate architecture for a genuinely infinite-time property, together with a universal dynamical construction that proves the architecture's exact expressive boundary.

Positive critical radius is witnessed by a finite graph cut and hence by an explicit quartic equation

at some mesh. A fixed MRDP polynomial exists for the resulting enumerable predicate. One fixed three-dimensional continuous rational map realizes arbitrary disjoint enumerable acceptance and robust rejection sets, while the remaining inputs diverge without any positive noise margin. Its radius is numerically computable with an error bound that does not depend on trajectory length, yet deciding whether the value is positive is maximally hard among enumerable problems. An exact rational neural update block inherits the same statement under the specified output noise semantics.

The result is therefore both constructive and limiting: finite arithmetic witnesses can certify robust nonacceptance of a continuous universal computer, but they cannot provide a complete existential certification of its fragile divergence. The distinction survives even when the numerical value separating the outcomes is a computable Lipschitz function.

Part V

Exact probabilities: well-conditioned quartic landscapes and reversible equilibria (sources 13 and 17)

38 [write] Conventions of Part V

Part V prints two manuscripts of batch 62 in full: source 13, *Probabilities from Well-Conditioned Quartics* (batch-62 manuscript 09), and source 17, *Arithmetic of Rapidly Mixing Reversible Computation* (batch-62 manuscript 14). Both concern exact probabilities of effective stochastic systems, the subject of Part I. Source 13 turns the probability that a fair-coin computation halts by time t into 2^{-t} times the root count, component count or Euler characteristic of an explicit, uniformly well-conditioned quartic. Source 17 builds reversible birth–death chains with a sharp common spectral gap whose exact stationary root mass nevertheless carries halting, eventual periodicity and a rational/Liouville dichotomy. Source numbers are this report’s own: the shipped files of source NN carry the prefix NN-, and the batch-62 manuscript numbers are recorded in Appendix F. Source 13 was written from the pin `b998f70c6` and source 17 from `e18718e83`, the commit that wrote Parts I–IV; source 17 cites those Parts, source 13 does not.

38.1 What is printed once, and where Part I is re-derived

Neither source contradicts Part I. Where they re-derive it, Part I’s statement is cited and the source’s argument is kept as a second route:

- Source 13’s quantifier table re-derives Theorem 9.1 (the interior thresholds $0 < q < 1$), with the same totality sampler (9.1). That table is not reprinted; only its rows for $p = 1$, $p = 0$ and finite mean, which Part I does not print, are kept (Section 46). Its quartic normalization is Lemma 3.3, its fixed lower-cut quartic is Corollary 4.5 for this model, and the characterization half of its realization theorem is Theorem 4.2; the uniform geometric constants of that theorem are new.
- Source 17’s Theorems 56.1 and 60.6 (strict cuts Σ_1^0 -complete, equality Π_1^0 -complete) are

Proposition 9.4 transplanted to exact stationary masses, with the same halting-time device. The setting (reversibility, a sharp common spectral gap, rational-valued hard instances) is new.

Three finite-horizon probability representations. Three different ways of reading a finite-horizon probability off a polynomial occur in this category, and they must not be mixed. Part I writes an output law as the dyadic mass of the projection of *one fixed* quartic, $\mu_e(z) = \sum_c 2^{-\ell(c)} [\exists w U(e, z, c, w) = 0]$ (equation (4.2)); witnesses need not be unique and are never counted. Source 13 uses a horizon-dependent quartic $\mathcal{E}_{e,x,t}$ whose natural zeros are in bijection with the accepted t -bit coin tapes, so that $p_{e,x,t} = 2^{-t} \#\{\text{zeros}\}$ (Theorem 44.1); runs that halt early are padded with *random* bits. The sibling report **canonical-diophantine-certificates** (its Theorem with label `cdc:mem:thm:probability`) counts canonically stopped roots with weights $2^{T-\tau}$. The tempting false reading is to divide a *canonically padded* root count by 2^T : that report shows this is wrong, because a canonically stopped run has one root however early it halts. Source 13 pads with random bits, so every early-halting run contributes its 2^{t-h} extensions and its unweighted count divided by 2^t is right (Section 44).

38.2 Letters used differently

Each source keeps its letters inside its own sections. The table lists those that also occur in Part I with another meaning.

Letter	Part I	Source 13	Source 17
δ	initial mass; success floor	$\delta \leq \eta_*$ a sublevel parameter; $\eta_* = 1/4096$ (renamed, below)	the common spectral gap $(3 - 2\sqrt{2})/4$
a, b	return masses, $c = a/(a + b)$	$a_{e,x,t}$ accepted-tape counts; a threshold a/b	$b \in \{0, 1\}^{\mathbb{N}}$ the environment word, b_n its defect flags; thresholds a/v
D	D_s products of denominators	—	$D_n = \sum_{k < n} b_k$ the number of defects before n
α	input fraction α/β	a left-c.e. real; α_K	α_b the root mass $\pi_b(0)$
h	h_s heralded mass	halting tick of a run	$h(b)$, the binary word of $2 - Z_b$
K	K_0 (Part III)	a noncomputable c.e. set	—
N	horizon	—	prefix length
Z	—	$\mathcal{Z}$ zero sets	Z_b the normalizer
t, T	—	horizon t	time t ; halting time $T(e)$

The tempting false readings are these. Source 17's b is a binary *environment*, not Part I's rejection mass, and its α_b is a stationary mass, not an input fraction. Source 13's a/b is a rational threshold, not a ratio of return masses. Source 17's δ is a spectral gap, not a mass or a floor. Source 17's D_n counts defects; Part I's D_s is a product of denominators.

38.3 Renamed symbols and macros

- Source 13's low-energy level $\delta_* = 1/4096$ is written η_* , to keep it apart from Part I's δ and source 17's gap; the parameter $\delta \leq \eta_*$ of its sublevel sets keeps its letter. No normalization changed.

- Source 13 prints probabilities with a blackboard $\mathbb{P}$; this report uses Pr throughout, as Part I does. Its macros for $\Sigma_1^0, \dots, \Pi_2^0$ and for $\{0, 1\}$ are replaced by this report's, with identical output.
- Source 17 cites this report's README at its pin; those citations are internal cross-references here, and its bibliography entry for the README is omitted.
- File names printed inside the sources' text are their delivered names. Every shipped file carries the prefix `13-quartic-landscapes-` or `17-reversible-equilibria-` under `code/`, `data/` or the report root; the README maps each delivered name to its shipped name and gives rerun recipes that do not overwrite the shipped data.

38.4 Abstracts and status statements of the sources

Abstract of source 13. We construct a quantitative, witness-faithful translation from Boolean circuits to nonnegative integer quartics. A circuit with k input bits and s gates gives a coercive polynomial in at most $k + 5s + 5$ variables, with coefficient height at most 4 and variable-interaction degree at most 3. Its real zeros are Boolean and are in bijection with accepting inputs. Every zero is a nondegenerate minimum with Hessian spectrum in $[2, 20]$. Dimension-independent rounding and convexity estimates identify all sufficiently low sublevel components with those zeros; the components are contractible and remain count-faithful under specified smooth perturbations.

Applied to a fair-coin computation stopped at horizon t , the construction makes the probability of halting by t exactly 2^{-t} times a finite Diophantine root count, a low-energy component count, or an Euler characteristic. The limiting values are precisely the left-computably enumerable reals in $[0, 1]$. Thus noncomputable probabilities occur despite uniformly bounded coefficients, sparse interactions, separated minima, well-conditioned Hessians, and a fixed rounding gap. We prove a matching interaction-degree frontier: degree-two Boolean residual systems can be counted by path/cycle dynamic programming, whereas degree three already supports parsimoniously complete counting.

We also give fixed-arity quartic normal forms for strict probability lower bounds and their quantifier-optimal complements, explain the almost-sure and finite-mean termination boundaries, and identify exactly what transfers to coin-augmented unconventional computational substrates. The package contains a working compiler, a degree-two counter, explicit sparse polynomials, and 6,050 finite exact checks, supplemented by 60 numerical Hessian checks.

Status statement of source 13. Status. This is a self-contained research draft, not a Lean-checked or independently refereed paper. The simultaneous quantitative construction and its geometric consequences are developed here; the general MRDP and probabilistic-termination classifications are classical and are explicitly credited. Publication priority for the combined construction is not asserted. No finite-fold or single-fold MRDP conjecture is claimed solved.

Abstract of source 17. We construct a family of reversible birth–death chains on the nonnegative integers with transition probabilities in a fixed finite rational alphabet. Every chain has exponentially decaying stationary tails, a mean return time to the origin at most two, and the same explicit lower bound $(3 - 2\sqrt{2})/4$ on its spectral gap. From the origin, total-variation convergence has a uniform geometric rate. Nevertheless, exact stationary arithmetic retains the full difficulty of computation.

A single defect placed at a machine's first halting time makes its root mass a rational number whose equality to $1/2$ is equivalent to nonhalting. A finite guarded-affine history construction realizes this example

as a random walk along a reversible computational trajectory. More generally, the root mass is rational exactly when the binary environment is eventually periodic. Its full range is a Cantor set of dimension $\log_2((1 + \sqrt{5})/2)$. Sparse, factorially separated defects give a uniform rational-versus-Liouville dichotomy: rationality and algebraicity are Σ_2^0 -complete, while irrationality and transcendence are Π_2^0 -complete for primitive-recursive environments. A separate low-complexity criterion, using the theorem of Adamczewski and Bugeaud, gives an explicit transcendental Thue–Morse equilibrium.

Strict rational comparisons admit complete finite-prefix certificates. At horizon N , the arithmetic layer is a sum-of-squares polynomial of total degree at most four with $N + 1$ natural witnesses and $2N + 1$ residuals, with a unique witness tuple whenever the prefix certifies the comparison. A small-coefficient recurrence presentation is also given. ProveIt’s MRDP interface then supplies fixed-polynomial existence for the unbounded strict cuts, but not for the higher-level arithmetic predicates. The manuscript includes full proofs of the construction, reproducible exact-arithmetic checks, a formalization plan, and twelve further research questions.

Status statement of source 17. Status. These are proposed research contributions with ordinary mathematical proofs and finite computational checks. They have not been independently refereed or formalized in Lean or Rocq. Historical priority is not claimed. Classical inputs and exact boundaries of the results are identified below.

39 Question, contribution, and relation to ProveIt

[source 13]

39.1 Why a probability is not an accepting path

The usual Diophantine question asks whether an accepting computation exists. For a probabilistic machine, that is only a small part of the semantics. One must distinguish existence of a successful random tape, the measure of all successful tapes, termination with probability one, and finiteness of the expected running time. The first is existential. The others involve counting, limits, or additional quantifier alternation.

This distinction matters especially when a compiler introduces auxiliary witnesses. An equation can represent the right existential relation while having the wrong number of solutions. A single accepting tape might acquire many or infinitely many arithmetic witnesses. Dividing that witness count by 2^t would not recover a probability. The construction below avoids this problem at a fixed horizon: every input tape has exactly one assignment to all internal wires, including all copy and constant wires.

The stronger question addressed here is geometric. Can a computationally arbitrary probability be encoded by polynomial landscapes whose accepting minima are uniformly regular, rather than by a badly conditioned or singular numerical construction? The answer is yes, with explicit small constants. The unbounded complexity lives in the increasing family of finite computations, not in deteriorating local geometry.

39.2 The theorem package

The central result combines five properties that should not be conflated:

- (i) a parsimonious finite Boolean-to-quartic compiler;
- (ii) bounded coefficients and a subcubic interaction graph;
- (iii) uniform Hessian, rounding, and low-sublevel topology bounds;
- (iv) exact realization of stochastic halting probabilities as normalized component counts;
- (v) a proof of the logical boundary between existential Diophantine certificates and almost-sure or finite-mean claims.

Here “parsimonious” means that the relevant finite sets are in bijection, not merely equinumerous after an unexplained normalization.

39.3 Repository boundary and provenance

The inspected repository revision is

`b998f70c6886e6a00339a6f4a02ed8625324d357`.

The relevant foundation is `Computability/HilbertTenthProblem`, in particular its public MRDP interface and guide [46, 47]. The guide states the natural-number formulation, includes input zero, allows zero witnesses, and emphasizes that its public theorem is an existence result rather than a practical polynomial generator. The repository also documents corrected Jones papers and a universal degree-four construction. We do not re-audit those Lean builds here.

The finite compiler in this article does not use MRDP. It is a direct, executable construction. MRDP enters only when an unbounded semidecidable probability comparison is compressed into one fixed-arity Diophantine formula. This separation is important: the fixed-arity formulas do not inherit the finite compiler’s Boolean witnesses or its solution-count bijection.

Boolean arithmetization and parsimonious counting reductions are classical; Valiant’s work supplies the standard counting-complexity background [49]. The arithmetical classifications of probabilistic termination are due to earlier work, notably Kaminski and Katoen [6]. We give self-contained reductions suited to the present representation and do not present those classifications as newly discovered.

Quartic sum-of-squares reductions of satisfiability are also established; Duda gives an explicit such reformulation [50]. Ahmadi and Zhang study degree-sensitive complexity questions for critical points and local minima [51]. Our claim is not the first quartic encoding of SAT. Nor is our decision problem the existence of an arbitrary local minimum: every coercive polynomial here already has a global minimum. The problem is whether its *zero level* is attained, and the quantitative result combines counting-faithful witnesses, coefficient height four, subcubic interactions, and explicit uniform low-energy geometry. The cited literature provides context, not an exhaustive priority audit.

40 Conventions and probabilistic semantics

[source 13]

40.1 Numbers, circuits, and graph conventions

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. The coefficient height of an integer polynomial is the maximum absolute value of an expanded coefficient, including the constant coefficient. A circuit has k designated free Boolean input wires, s further gates, and one output wire. Gates belong to

AND, NOT, COPY, ZERO, ONE.

The gate graph is acyclic. Repeated operands are allowed initially; the compiler separates occurrences before producing polynomial residuals. Circuit parameters such as program text or an ordinary data input are fixed by the compiler, not counted as extra random input bits.

For a family of residuals r_j , its *interaction graph* has one vertex per variable and joins two distinct variables when they occur together in a residual. In our construction this is also the expanded polynomial’s monomial-interaction graph: no pair occurs in two residuals, so cancellation cannot remove the relevant edge. In the degree-two counting theorem, the residual presentation is part of the input.

40.2 One fair bit per clock tick

Fix an effective deterministic machine model with an additional instruction that branches on a fair bit. There is no scheduler or demonic nondeterminism. The accepting/halting state is absorbing. We may supply an independent fair bit at every clock tick, even if a deterministic instruction ignores that bit. This convention does not change the distribution of the bits that the program actually uses.

For a program e , ordinary input x , and infinite coin stream ω , let $\tau_{e,x}(\omega) \in \mathbb{N} \cup \{\infty\}$ be its first halting time. Put

$$p_{e,x,t} = \Pr(\tau_{e,x} \leq t), \quad p_{e,x} = \Pr(\tau_{e,x} < \infty). \quad (40.1)$$

Exactly t bits determine the event $\tau \leq t$. Thus, for a decidable Boolean predicate $A_{e,x,t}$ on $\{0,1\}^t$,

$$a_{e,x,t} = \#\{b \in \{0,1\}^t : A_{e,x,t}(b) = 1\}, \quad p_{e,x,t} = \frac{a_{e,x,t}}{2^t}. \quad (40.2)$$

A stream that has already halted stays successful under either next bit, so $a_{t+1} \geq 2a_t$ and $p_t \uparrow p$.

All finite-horizon predicates can be compiled to finite circuits. For a fixed local tape-machine model, an ordinary space-time simulation yields $O((t+1)(|x|+t+1))$ Boolean gates, with a model-dependent constant; the same statement holds for a fixed universal interpreter with its program text included in the input length. No assumption that an arbitrary mathematical function is a unit-cost machine instruction is made.

40.3 Three different representation levels

A finite circuit gives a finite polynomial whose dimension depends on the horizon. A probability is a limit of normalized finite counts. A fixed-arity Diophantine predicate is a single polynomial whose witness dimension is independent of the horizon and whose natural witnesses can be unbounded. The word “uniform” below always specifies which of these levels is meant.

41 A sparse, witness-faithful quartic compiler

[source 13]

41.1 Distributed constants and fresh occurrences

Introduce one new unit wire u_* , with residual $u_* - 1$. Every occurrence of a Boolean constant one is supplied by a copy of this unit wire. A NOT gate uses a local unit occurrence, rather than a new literal constant. The unrouted residual templates are

Operation	Residual	Defining interpretation
AND	$z - xy$	$z = xy$
NOT	$z + x - u$	$z = u - x$, with $u = 1$
COPY or ONE	$z - x$	$z = x$
ZERO	z	$z = 0$
Root unit	$u_* - 1$	$u_* = 1$
Final acceptance	$o - u$	require output one

The final acceptance comparison is not a definition. All other residuals define exactly one noninput wire.

We now route *every operand occurrence*, including the two operands of final acceptance and all unit occurrences. Suppose an original wire has $d > 0$ uses. Add one fresh copy wire as a trunk. If $d = 1$, the trunk itself is the occurrence wire. If $d \geq 2$, grow a binary copy tree below that trunk, with d fresh leaves, one per use. Every tree edge $v \rightarrow w$ has residual $w - v$. The root original wire therefore has only one outgoing copy edge, and the leaves used in a gate are all distinct.

The tree has $2d - 1$ new wires for every $d \geq 1$. This follows from the usual $2d - 2$ edges of a full binary tree with d leaves, plus the trunk edge; for $d = 1$ it gives the single trunk wire. Balanced splitting is convenient for the implementation but is not required for the proof.

Lemma 41.1 (Routing invariants). *The routed system has the following properties.*

- (a) *Every noninput variable has exactly one defining residual, and all definitions are acyclic.*
- (b) *Each non-Boolean residual involves at most three distinct variables.*
- (c) *Each variable occurs in at most three non-Boolean residuals.*
- (d) *Each pair of distinct variables occurs together in at most one non-Boolean residual.*
- (e) *The interaction graph has maximum degree at most three.*

Proof. Acyclicity follows by replacing each use edge of an acyclic circuit by a copy tree directed toward the use. The root unit has its own definition. A fresh occurrence leaf is consumed only at its assigned use; even two occurrences of the same original operand become different leaves. This proves the first two assertions.

An original gate output occurs in its defining residual and, if used, in one trunk-copy residual. A free input occurs only in its trunk-copy residual. An internal tree variable occurs in one incoming and at most two outgoing copy residuals. An occurrence leaf occurs in its incoming copy residual and one consuming residual. The root unit obeys the same count, with its pinning residual counted as its definition. Thus incidence is at most three.

A copy edge uses a unique child. An original gate's operands are private leaves, and its output is not one of those leaves. Two private leaves are never consumed together again. This proves pair uniqueness. For graph degree, an original gate output has at most two neighbors in its defining gate and one trunk neighbor; an internal copy vertex has at most three tree neighbors; a leaf has one parent and at most two consuming-gate neighbors. The remaining vertices have smaller degree. $\square$

41.2 The polynomial and exact size

Let $X_1, \dots, X_n$ be all routed variables, including the k original inputs. If $r_1, \dots, r_m$ are the non-Boolean residuals, define

$$\mathcal{E}_C(X) = \sum_{i=1}^n (X_i(X_i - 1))^2 + \sum_{j=1}^m r_j(X)^2. \quad (41.1)$$

Each summand has degree at most four, and the univariate Boolean penalties ensure that the degree is exactly four.

Theorem 41.2 (Quantitative compiler). *From a circuit C with k Boolean inputs and s gates one can effectively construct (41.1) with*

$$n \leq k + 5s + 5, \quad m = n - k + 1. \quad (41.2)$$

It has $2n - k + 1$ displayed squared residuals, coefficient height at most 4, constant coefficient exactly 1, and interaction degree at most 3. Projection to the k input coordinates is a bijection

$$\{X \in \mathbb{R}^n : \mathcal{E}_C(X) = 0\} \longrightarrow \{b \in \{0, 1\}^k : C(b) = 1\}. \quad (41.3)$$

The same zero set is obtained over $\mathbb{N}^n$ or $\mathbb{Z}^n$.

Proof. There are $k + s + 1$ original variables after the root unit is added. Let L be the number of unrouted operand occurrences. AND and NOT have two each; COPY and ONE have one; ZERO and the root unit have none; final acceptance has two. Therefore $L \leq 2s + 2$. Routing uses at most $2L$ new wires, giving $n \leq k + s + 1 + 4s + 4$. Every noninput variable has one definition, and final acceptance contributes one more residual, so $m = n - k + 1$.

Over $\mathbb{R}$, a sum of squares is zero exactly when every summand is zero. The Boolean residuals force $X \in \{0, 1\}^n$. The root pin and copy trees force every local unit to be one. Induction along the acyclic definitions then forces every gate and copy wire to be the unique circuit evaluation on the given input. The last residual vanishes exactly for output one. Conversely that evaluation makes every residual zero. This proves the bijection and the assertion about number domains.

It remains to check coefficients after expansion, rather than merely the small coefficients of the unexpanded residuals. The Boolean term is $X_i^4 - 2X_i^3 + X_i^2$. No non-Boolean square has a pure cubic or pure quartic term, since its distinct-variable templates are $z - xy$, $z + x - u$, $z - x$, z , and $u_* - 1$. Each contributes at most one to a pure quadratic coefficient, and at most three such residuals involve any variable. Consequently a pure quadratic coefficient is at most four. A mixed monomial can occur in at most one residual square by pair uniqueness, and its coefficient there has absolute value at most two. The only nonzero linear coefficient is $-2u_*$, and the only constant term is the one from $(u_* - 1)^2$. Thus the height is at most four. Graph degree is lemma 41.1(e). $\square$

Remark 41.3 (Why the trunk matters). Merely limiting fan-out to two gives a possible degree-four graph vertex: a gate output can have two gate neighbors and two outgoing copies. The single trunk edge reduces this to three without changing the conservative linear variable bound. Distributing the unit similarly prevents a high-degree global constant vertex and a growing constant coefficient.

42 Real geometry: rounding, conditioning, and topology

[source 13]

42.1 Coercivity and conditioning at zeros

Theorem 42.1 (Uniform minimum geometry). *The polynomial $\mathcal{E}_C$ is nonnegative and coercive on $\mathbb{R}^n$. At every real zero b , its Hessian satisfies*

$$2I \preceq \nabla^2 \mathcal{E}_C(b) \preceq 20I. \quad (42.1)$$

In particular each zero is an isolated nondegenerate minimum, with Hessian condition number at most ten. Distinct zeros have Euclidean distance at least one.

Proof. Nonnegativity is immediate. The lower bound $\mathcal{E}_C(X) \geq \sum_i X_i^2 (X_i - 1)^2$ tends to infinity when $\|X\|_\infty$ tends to infinity, proving coercivity in each finite dimension.

Let J be the Jacobian of the non-Boolean residuals at a Boolean zero. All residuals vanish there. The derivative of $X_i(X_i - 1)$ is $2b_i - 1 \in \{-1, 1\}$. Differentiating the sum of squares gives

$$\nabla^2 \mathcal{E}_C(b) = 2I + 2J^\top J.$$

Each row of J has at most three entries of absolute value at most one; each column has at most three such entries by incidence. Hence $\|J\|_\infty \leq 3$, $\|J\|_1 \leq 3$, and $\|J\|_2^2 \leq \|J\|_1 \|J\|_\infty \leq 9$. The claimed spectral interval follows. Distinct zeros are distinct Boolean vectors. $\square$

Remark 42.2 (Nondegenerate minimum versus smooth hypersurface). The nondegeneracy in theorem 42.1 is Morse nondegeneracy of the *energy minimum*. Necessarily $\nabla \mathcal{E}_C(b) = 0$. We do not claim that the hypersurface defined by the single equation $\mathcal{E}_C = 0$ is nonsingular at b in the algebraic-geometric gradient sense. Nor do we claim that its complex zero set is Boolean; sum-of-squares positivity was used over $\mathbb{R}$.

42.2 A dimension-independent rounding gap

Lemma 42.3 (Scalar Boolean rounding). *For every $x \in \mathbb{R}$, let $d(x) = \min(|x|, |x - 1|)$. Then*

$$d(x) \leq 2|x(x - 1)|.$$

Consequently, if b is obtained by nearest-Boolean rounding of X , then

$$\|X - b\|_2^2 \leq 4 \sum_i (X_i(X_i - 1))^2 \leq 4\mathcal{E}_C(X). \quad (42.2)$$

Proof. Inside $[0, 1]$, $|x(x - 1)| = d(x)(1 - d(x)) \geq d(x)/2$. Outside that interval it is $d(x)(1 + d(x)) \geq d(x)$. Squaring the scalar bound and summing proves (42.2). $\square$

Theorem 42.4 (Robust witness extraction). *If $\mathcal{E}_C(X) \leq 1/64$, nearest-Boolean rounding is unambiguous and yields a zero b of $\mathcal{E}_C$. In particular, if C has no accepting input, then*

$$\min_{X \in \mathbb{R}^n} \mathcal{E}_C(X) > \frac{1}{64}.$$

Proof. Write $\epsilon = \sqrt{\mathcal{E}_C(X)} \leq 1/8$. Each coordinate is within $\eta = 2\epsilon \leq 1/4$ of its rounded bit. Every non-Boolean residual has $|r_j(X)| \leq \epsilon$. A linear template changes by at most 3η under rounding. For an AND template, with $|x - b_x|, |y - b_y|, |z - b_z| \leq \eta$,

$$\begin{aligned} |(z - xy) - (b_z - b_x b_y)| &\leq |z - b_z| + |x| |y - b_y| + |b_y| |x - b_x| \\ &\leq (3 + \eta)\eta. \end{aligned}$$

The same bound dominates all templates. Therefore

$$|r_j(b)| \leq \epsilon + (3 + 2\epsilon)2\epsilon \leq \frac{1}{8} + \frac{13}{4} \frac{1}{4} = \frac{15}{16} < 1.$$

But $r_j(b)$ is an integer. Thus all those residuals vanish; the Boolean ones vanish by construction. Coercivity ensures that the global minimum is attained, and the contrapositive proves the strict gap when no zero exists. $\square$

The gap is an extraction theorem, not a polynomial-time optimization algorithm. Finding a point below the gap can still encode a hard search problem. The useful guarantee is that a sufficiently good approximate solution cannot be a numerically convincing false positive.

42.3 Uniform convex neighborhoods

For a Boolean zero b , set

$$Q_b = \{X : \|X - b\|_\infty \leq 1/32\}.$$

Lemma 42.5 (Convex well boxes). *Throughout Q_b ,*

$$I \preceq \nabla^2 \mathcal{E}_C(X) \preceq 24I. \quad (42.3)$$

Proof. More generally let $\|X - b\|_\infty \leq \rho \leq 1/32$. For the Boolean Jacobian contribution, $|2X_i - 1| \geq 1 - 2\rho$ and $|2X_i - 1| \leq 1 + 2\rho$. Its residual-Hessian correction has absolute diagonal entries at most $4\rho(1 + \rho)$.

For non-Boolean residuals, a Jacobian row has absolute sum at most $3 + 2\rho$, and a column has absolute sum at most $3(1 + \rho)$. Thus their positive semidefinite Jacobian contribution is bounded above by $6(1 + \rho)(3 + 2\rho)I$.

Only an AND residual has a nonzero second derivative: an off-diagonal -1 at its two operands. Since $r_j(b) = 0$, the preceding Lipschitz estimate gives $|r_j(X)| \leq (3 + \rho)\rho$. The absolute row sum of all non-Boolean residual-Hessian corrections is at most $6(3 + \rho)\rho$, using the incidence bound three. Therefore the full Hessian has lower spectral bound

$$2(1 - 2\rho)^2 - 4\rho(1 + \rho) - 6(3 + \rho)\rho = 2 - 30\rho - 2\rho^2 \geq \frac{543}{512} > 1.$$

Its upper spectral bound is at most

$$2(1 + 2\rho)^2 + 6(1 + \rho)(3 + 2\rho) + 4\rho(1 + \rho) + 6(3 + \rho)\rho \leq \frac{11215}{512} < 24.$$

These deliberately conservative estimates suffice uniformly in n . $\square$

42.4 Components and Euler characteristic

Put $\eta_* = 1/4096$, and for $0 < \delta \leq \eta_*$ define

$$\mathcal{K}_C(\delta) = \{X \in \mathbb{R}^n : \mathcal{E}_C(X) \leq \delta\}.$$

Theorem 42.6 (Low-energy topology). *Every connected component of $\mathcal{K}_C(\delta)$ is a compact convex set containing exactly one zero of $\mathcal{E}_C$. The components are in bijection with the accepting inputs of C . In particular,*

$$b_0(\mathcal{K}_C(\delta)) = \#\mathcal{Z}_C = \#\{b : C(b) = 1\}, \quad b_i(\mathcal{K}_C(\delta)) = 0 \quad (i > 0), \quad (42.4)$$

where homology may be taken with rational coefficients. The Euler characteristic is the same finite count. If the count is zero, the sublevel set is empty.

Proof. By theorem 42.4, a point in the sublevel set rounds to a zero b ; by (42.2), its sup-norm distance from b is at most $2\sqrt{\delta} \leq 1/32$. Thus the sublevel set is contained in the union of the pairwise disjoint boxes Q_b . Its intersection with each box is a convex sublevel set of a strongly convex function, by lemma 42.5, and contains b . It is compact by coercivity. Distinct boxes are a positive distance apart, so these nonempty intersections are exactly the connected components. They are contractible by straight-line contraction to b . The homology and Euler-characteristic assertions follow. $\square$

There are no additional positive-energy local minima inside this low sublevel set: within its box, strict convexity leaves b as the only critical minimum. No corresponding assertion is made about the landscape at larger energies.

Theorem 42.7 (A specified perturbation stability). *Let $h : \mathbb{R}^n \rightarrow \mathbb{R}$ be C^2 , with*

$$\sup_{\mathbb{R}^n} |h| \leq \eta_*/4, \quad \|\nabla^2 h(X)\|_2 \leq \frac{1}{2} \quad \text{on every zero box } Q_b.$$

Then $\{X : \mathcal{E}_C(X) + h(X) \leq \eta_/2\}$ has exactly one compact convex component for every zero of $\mathcal{E}_C$, and none else. Each component has a unique interior minimum.*

Proof. A point in this perturbed sublevel satisfies $\mathcal{E}_C(X) \leq 3\eta_*/4 < \eta_*$, and hence lies in a zero box. Each zero b itself belongs to the perturbed sublevel, since $h(b) \leq \eta_*/4$. On a box the perturbed Hessian is at least $I/2$, so each intersection is convex. The original energy on the boundary of a zero box is at least $\frac{1}{2}(1/32)^2 = 2\eta_*$, by integrating the Hessian lower bound from its zero. The perturbed boundary energy is therefore greater than the chosen level. Compactness and strict convexity now give the unique interior minimum. Disjointness of the boxes finishes the component argument. $\square$

This theorem concerns the stated bounded smooth perturbations. It does not assert stability under arbitrary changes of expanded polynomial coefficients, which need not stay uniformly small on $\mathbb{R}^n$.

43 Two sharp finite-computation frontiers

[source 13]

43.1 Degree four is necessary for a nonnegative multiwell model

Proposition 43.1. *A globally nonnegative real polynomial of degree at most three cannot have two isolated zeros. If it is also coercive, it has at most one zero. Consequently degree four is minimal for globally nonnegative coercive polynomial encodings capable of having two accepting minima.*

Proof. A nonzero odd-degree leading homogeneous part takes opposite signs on opposite rays, so a globally nonnegative polynomial of degree at most three actually has degree at most two. If such a quadratic vanishes at b , its gradient vanishes there, and it has the form $(X - b)^\top A(X - b)$ with A positive semidefinite. Its zero set is $b + \ker A$, an affine subspace. Coercivity makes A positive definite. $\square$

This is a statement about globally nonnegative real landscapes, not a claim that degree-three Diophantine existence has been classified.

43.2 Degree-two interactions can be counted exactly

Theorem 43.2 (Tractable interaction boundary). *Consider a presented system*

$$F(X) = \sum_{i=1}^n (X_i(X_i - 1))^2 + \sum_j r_j(X)^2,$$

with integer residuals, and suppose its residual-interaction graph has maximum degree at most two. The number of real, integer, or natural zeros can be computed in polynomial time from this presentation. After the local Boolean predicates have been tabulated, only linearly many arithmetic transfer operations are required.

Proof. Zeros are Boolean, so each residual defines a local allowed-assignment predicate. Its variable support forms a clique in the interaction graph. A degree-two graph has components that are isolated vertices, paths, cycles, or triangles. Any residual with three distinct variables fills a triangle, and that triangle is an entire connected component. No residual can have four distinct variables. Triangle components can therefore be enumerated on their eight assignments.

Every larger path or cycle has only unary and binary constraints. Combine all constraints on a vertex into $U_v(b) \in \{0, 1\}$, and all on an edge uv into $T_{uv}(a, b) \in \{0, 1\}$. Along a path $v_1, \dots, v_\ell$, let

$$D_1(b) = U_{v_1}(b), \quad D_{i+1}(b) = U_{v_{i+1}}(b) \sum_{a \in \{0,1\}} D_i(a) T_{v_i v_{i+1}}(a, b).$$

Its count is $D_\ell(0) + D_\ell(1)$. For a cycle, fix the first bit to zero and then to one, run the same recurrence, and impose the closing edge before summing. Multiply the component counts. Intermediate counts have at most $n + 1$ bits. Local truth tables require evaluation only at Boolean assignments on at most three variables, so their construction also has polynomial bit complexity. $\square$

Corollary 43.3 (Subcubic counting completeness). *Counting zeros, or the components in [theorem 42.6](#), of compiler-output quartics is $\#P$ -complete under parsimonious polynomial-time reductions. This remains true with coefficient height at most four, interaction degree at most three, and all the geometric guarantees above. Deciding whether a zero exists is NP-complete on this class.*

Proof. Any polynomial-time Boolean witness verifier can be unrolled into a polynomial-size Boolean circuit, with one input assignment per witness. The compiler preserves that count exactly. This gives the usual $\#P$ and NP hardness, in the counting framework of [49]. Membership holds because a Boolean zero is a polynomial-length certificate and can be checked by evaluating the presented residuals. Alternatively, on compiler outputs one may count only the k free input bits and reconstruct the unique remaining wires. $\square$

Together, [theorem 43.2](#) and [corollary 43.3](#) give a sharp interaction-degree frontier for these presented Boolean residual systems: degree two has an exact polynomial-time count; degree three already realizes all Boolean witness-counting problems. The statement is a reduction theorem, not an unconditional separation of FP from $\#P$.

44 Stochastic computation as a count of minima

[source 13]

Theorem 44.1 (Finite-horizon stochastic compiler). *For every fair-coin program e , ordinary input x , and horizon t , there is an effectively constructed polynomial $\mathcal{E}_{e,x,t}$ satisfying all properties of theorems 41.2, 42.1, 42.4 and 42.6, with $k = t$ free Boolean inputs, such that*

$$p_{e,x,t} = 2^{-t} \# \{X \in \mathbb{N}^{n_{e,x,t}} : \mathcal{E}_{e,x,t}(X) = 0\} \quad (44.1)$$

$$= 2^{-t} b_0(\{X \in \mathbb{R}^{n_{e,x,t}} : \mathcal{E}_{e,x,t}(X) \leq \eta_*\}) \quad (44.2)$$

$$= 2^{-t} \chi(\{X : \mathcal{E}_{e,x,t}(X) \leq \eta_*\}). \quad (44.3)$$

For a fixed local machine model, the dimension is polynomial in the horizon and the finite input length.

Proof. Construct the bounded-simulation circuit for $A_{e,x,t}$ and apply theorem 41.2. Its input bits are the t tick-indexed fair bits. The root bijection gives (44.1); the topology theorem gives the other equalities. Circuit size and (41.2) give the dimension bound. $\square$

44.1 Padding is necessary, not an arbitrary multiplicity

Suppose a run halts after $h < t$ ticks. It has 2^{t-h} extensions to a length- t coin tape. Those extensions are separate circuit inputs and must contribute separate roots. Their total normalized weight is $2^{t-h}/2^t = 2^{-h}$, the probability of the corresponding cylinder.

By contrast, duplicate auxiliary witnesses for the *same* tape would be an error. The copy trees and the common unit introduce no such duplicates. The distinction between padding multiplicity and arithmetic witness multiplicity is the reason that the exact bijection was proved before discussing probability.

44.2 Compatible injections across horizons

For an accepted t -bit tape b and a bit c , append c and reconstruct the unique root at horizon $t + 1$. This defines an effective injection

$$\mathcal{Z}_{e,x,t} \times \{0, 1\} \hookrightarrow \mathcal{Z}_{e,x,t+1}.$$

It is not a literal coordinate inclusion: the internal circuits and their copy-wire names can change with t . It is the canonical map induced by input-tape projection followed by unique extension. Hence the normalized root and component counts are monotone even though the ambient dimensions change.

44.3 What is being normalized

The denominator in (44.1) is 2^t , not $2^{n_{e,x,t}}$. Internal coordinates encode deterministic consequences and are not independently sampled. Nor is the formula a claim about the volume of attraction basins under gradient descent, or about a Gibbs distribution on $\mathbb{R}^n$. Equal counting weight on the free coin tapes is the precise probability model. Different well curvatures do not change that counting measure.

45 A geometric realization theorem for noncomputable reals

[source 13]

Definition 45.1. A real α is *left-computably enumerable*, or left-c.e., if it is the limit of a computable nondecreasing sequence of rational numbers. Equivalently, its strict rational lower cut is computably enumerable.

Theorem 45.2 (Uniformly regular quartic realization). *The limits in (44.1)–(44.3), as e, x vary over fair-coin computations, are exactly the left-c.e. reals in $[0, 1]$. In particular every such real is a limit of normalized component counts of coercive integer quartics having, simultaneously:*

- (i) *coefficient height at most 4 and interaction degree at most 3;*
- (ii) *only Boolean real zeros, separated by distance at least 1;*
- (iii) *Hessian spectrum in $[2, 20]$ at every zero;*
- (iv) *the fixed rounding threshold $1/64$ and low-energy level $1/4096$;*
- (v) *compact convex low-energy components with no higher homology.*

The construction is effective from an index for a computable increasing rational approximation to α .

Proof. For any program, (40.2) is a computable nondecreasing rational sequence converging to its halting probability. This gives one inclusion.

Conversely, let $a_s \uparrow \alpha \in [0, 1]$ be a computable rational approximation, clipped into $[0, 1]$ if necessary. A fair-coin program constructs an infinite uniform binary real $U \in [0, 1]$, one bit at a time. After s bits, write the current binary cylinder as

$$[u_s, u_s + 2^{-s}], \quad u_s = \frac{j_s}{2^s}.$$

The program computes a_s and halts when

$$u_s + 2^{-s} < a_s. \tag{45.1}$$

All comparisons are rational and all stage computations are finite. If it halts, then $U < \alpha$. If $U < \alpha$, choose a positive gap smaller than $\alpha - U$; eventually a_s is sufficiently close to α and 2^{-s} is sufficiently small that (45.1) holds. Thus the halting event agrees, up to the measure-zero endpoint convention, with $\{U < \alpha\}$ and has probability α . Unrolling this ordinary fair-coin program and applying theorem 44.1 gives every geometric property claimed. $\square$

Halting probabilities and left-c.e. reals have an extensive established theory; universal prefix-free halting probabilities provide the familiar algorithmically random examples [52]. The point here is not a new characterization of left-c.e. reals by computation. It is their simultaneous realization by the explicitly regular sparse quartics just constructed. [write] The characterization half of the theorem (the limits are exactly the left-c.e. reals in $[0, 1]$) is the one-output case of Theorem 4.2 and Lemma 3.2 of Part I; what is new here is the simultaneous realization with the uniform geometric constants (i)–(v).

45.1 An explicit noncomputable target without randomness theory

Let K be a fixed noncomputable c.e. halting set, and define

$$\alpha_K = \sum_{j \in K} \frac{2}{3^{j+1}}. \tag{45.2}$$

Finite stages of an enumeration give a computable increasing rational approximation, so theorem 45.2 applies.

To see directly that α_K is noncomputable, suppose the digits below j have already been recovered. Subtract their contribution. If $j \notin K$, the remaining tail is at most $3^{-(j+1)}$; if $j \in K$, it is at least $2 \cdot 3^{-(j+1)}$. An approximation with error smaller than one quarter of $3^{-(j+1)}$ distinguishes the cases. Induction would decide K from a computable real name for α_K , a contradiction.

Corollary 45.3 (Regular geometry does not give a convergence modulus). *There is a single computable family of the above quartics for which the normalized component counts have no computable modulus of convergence. Consequently no general modulus can be inferred solely from the displayed coefficient, interaction, gap, separation, or Hessian bounds.*

Proof. Choose the family realizing (45.2). A computable modulus, together with the computable finite counts, would compute its limit. $\square$

45.2 A useful obstruction to over-compression

Proposition 45.4 (Finite Boolean witnesses cannot be universally MRDP). *Suppose a fixed polynomial family $P(e, x, w)$ has a fixed number n of witness coordinates, and every natural witness satisfying $P = 0$ lies in $\{0, 1\}^n$. Then its existential input relation is decidable.*

Proof. Evaluate P on all 2^n Boolean witness tuples. More generally, any known computable finite witness bound gives a finite decision procedure. $\square$

Thus the increasing horizon or an unbounded witness domain is essential. Our regular Boolean quartics are not a counterexample to this elementary obstruction, and their dimensions are not fixed as $t \rightarrow \infty$.

46 Fixed-arity quartic probability predicates

[source 13]

46.1 A standard quartic reduction, stated precisely

We use MRDP in the form that every c.e. relation on a finite tuple of natural inputs is the existential projection of an integer polynomial zero set [48, 47]. The multi-input form follows from the one-input form by polynomial tuple coding. For example, $J(a, b) = (a + b)^2 + a$ is an injective natural-number pairing map: the intervals $[s^2, s^2 + s]$ for successive $s = a + b$ are disjoint. Iterating this pairing and substituting it into a representing polynomial suffices.

Lemma 46.1 (Quartic normalization). *Every c.e. relation has an existential natural-number representation by a single nonnegative integer polynomial of degree at most four, with a fixed finite number of witnesses.*

Proof. Start with an MRDP polynomial P . Write $P = P_+ - P_-$, where P_+ and P_- have nonnegative integer coefficients. Evaluate them by addition and multiplication circuits on natural inputs and witnesses. Introduce one natural wire value per arithmetic gate. Equations $z = x + y$ and $z = xy$ have degree at most two; equate the two final outputs. The sum of the squares of all residuals is a nonnegative integer polynomial of degree at most four. Its natural zeros are exactly the valid arithmetic evaluations of zeros of P . $\square$

This lemma preserves the multiplicity of the *pre-existing* MRDP witnesses, but does not make those witnesses unique. It also does not supply the small coefficients, Booleanity, coercivity, or sparse graph of theorem 41.2. [write] The lemma is Lemma 3.3 of Part I; the proof above is a second route to it.

46.2 Strict lower cuts are Diophantine

Write rational thresholds as $q = a/b$, with $a \in \mathbb{N}$ and $b \geq 1$. Denominator validity can be built into the c.e. relation; alternatively use $b = d + 1$ as a parameterization by all naturals.

Theorem 46.2 (Probability lower-cut polynomial). *There is one fixed nonnegative integer quartic D with a fixed witness tuple w such that, for every program, input, and valid rational threshold,*

$$p_{e,x} > a/b \iff \exists w \in \mathbb{N}^r D(e, x, a, b, w) = 0. \quad (46.1)$$

Proof. By monotone convergence,

$$p_{e,x} > a/b \iff \exists t \, b a_{e,x,t} > a 2^t.$$

The finite count $a_{e,x,t}$ is computable by enumeration of all t -bit tapes, regardless of efficiency. The displayed relation is c.e. Apply lemma 46.1 to the tuple (e, x, a, b) . $\square$

The program index is an input to a single universal c.e. predicate here; we are not silently choosing a different polynomial after learning a halting outcome. The present package outputs the finite-horizon quartics, not an expanded coefficient list for this much larger universal D . [write] For effective output laws this is Corollary 4.5 of Part I, proved by the same monotone-approximation argument.

46.3 One additional universal block is unavoidable

Apply the same construction to the c.e. relation

$$p_{e,x} > \frac{a}{b} - \frac{1}{k+1}.$$

It has a fixed quartic representation $Q(e, x, a, b, k, z) = 0$. Thresholds below zero cause no difficulty: the finite integer comparison remains computable and is automatically true when appropriate.

Theorem 46.3 (Exact quantifier normal forms). *For valid a, b the following equivalences hold:*

$$p_{e,x} > a/b \iff \exists w \, D(e, x, a, b, w) = 0, \quad (46.2)$$

$$p_{e,x} \leq a/b \iff \forall w \, D(e, x, a, b, w) \neq 0, \quad (46.3)$$

$$p_{e,x} \geq a/b \iff \forall k \exists z \, Q(e, x, a, b, k, z) = 0, \quad (46.4)$$

$$p_{e,x} < a/b \iff \exists k \forall z \, Q(e, x, a, b, k, z) \neq 0. \quad (46.5)$$

Equality is the conjunction of (46.3) and (46.4). Almost-sure termination is the special case

$$p_{e,x} = 1 \iff \forall k \exists z \, Q(e, x, 1, 1, k, z) = 0. \quad (46.6)$$

Proof. The first two statements are theorem 46.2 and its complement. For any real p , the assertion $p \geq q$ is equivalent to $p > q - 1/(k+1)$ for every k . Its complement says that for some k , $p \leq q - 1/(k+1)$. This gives the next two statements. Finally $p \leq 1$ is automatic for a probability. $\square$

46.4 Completeness and nonexistence of existential substitutes

The classical hierarchy results [6] can be recovered directly in this setting. We record the reductions because they determine which requested Diophantine representations are possible. [write] For a fixed interior threshold $0 < q < 1$ these are the classifications of Theorem 9.1 in Part I; the reductions below are second routes to it.

Sample $N \in \mathbb{N}$ with $\Pr(N = n) = 2^{-(n+1)}$, and then run an ordinary program M on N . Every finite n has positive probability, so this probabilistic program halts almost surely exactly when M halts on every input. The latter property is Π_2^0 -complete: any predicate $\forall n \exists t \, R(e, n, t)$ with decidable R is represented by the program that searches for a suitable t on input n . Thus AST is Π_2^0 -complete. [write] The sampler is the totality sampler (9.1) of Part I.

For any fixed rational $0 < q < 1$, a fair-coin program can enter that probabilistic computation with probability exactly q and otherwise loop forever. Its halting probability is qp . A rational gate can be implemented by finite-block rejection sampling, which terminates almost surely. Consequently $p \geq q$ and $p = q$ are Π_2^0 -hard, and $p < q$ is Σ_2^0 -hard. The normal forms give the matching upper bounds. To obtain Σ_1^0 -hardness of $p > q$, choose a dyadic $r > q$, enter a simulation of a fixed ordinary halting instance with probability r , and otherwise loop. The resulting probability is either zero or r . Complementation gives Π_1^0 -completeness of $p \leq q$.

[write] Source 13’s summary table also listed the five classifications for a fixed interior threshold $0 < q < 1$: $p > q$ is Σ_1^0 -complete, $p \leq q$ is Π_1^0 -complete, $p \geq q$ and $p = q$ are Π_2^0 -complete, and $p < q$ is Σ_2^0 -complete, with the normal forms (46.2)–(46.5). Those classes are exactly the ones of Theorem 9.1 in Part I, which is not reprinted. The rows that Part I does not print are kept:

Property	Exact class	Normal form
$p = 1$ (AST)	Π_2^0 -complete	(46.6)
$p = 0$	Π_1^0 -complete	(46.3) with $a/b = 0/1$
$\mathbb{E}\tau < \infty$ (scheduler-free)	Σ_2^0 -complete	(47.3); Theorem 47.1

Endpoint thresholds have exceptions: for example $p = 0$ is Π_1^0 -complete, not Π_2^0 -complete, and $p > 1$ is empty.

Since existential polynomial relations are c.e., hierarchy strictness rules out an exact existential Diophantine representation of AST on program indices. This is a proved obstruction, not a failure of a chosen coding technique. Bounded unique witnesses do not remove the extra universal quantifier.

47 Finite expected runtime and the danger of simulation slowdown

[source 13]

47.1 A root-count formula for the mean

For a nonnegative integer-valued stopping time, allowing infinity,

$$\mathbb{E}\tau = \sum_{t=0}^{\infty} \Pr(\tau > t) = \sum_{t=0}^{\infty} (1 - 2^{-t} \#Z_{e,x,t}). \quad (47.1)$$

Indeed $\tau(\omega) = \sum_{t \geq 0} \mathbf{1}_{\{\tau(\omega) > t\}}$ pointwise, and monotone convergence permits expectation to pass through the sum. Every finite partial sum is an exactly computable rational.

Finiteness is therefore

$$\mathbb{E}\tau < \infty \iff \exists M \in \mathbb{N} \forall N \in \mathbb{N} \sum_{t < N} (1 - p_t) \leq M. \quad (47.2)$$

The violation predicate “some partial sum exceeds M ” is c.e. By quartic normalization, there is a fixed quartic V such that

$$\mathbb{E}\tau < \infty \iff \exists M \forall w V(e, x, M, w) \neq 0. \quad (47.3)$$

This is the Σ_2^0 form of positive almost-sure termination in the scheduler-free model, consistent with [6].

47.2 Hardness even after a single random-sampling phase

The following proof makes the source of the second quantifier block particularly transparent.

Theorem 47.1 (Geometric-input finite-mean hardness). *There is an effective family of fair-coin programs with the form “sample one geometric natural number; then execute a total deterministic body” for which almost-sure termination always holds, but finite expected runtime is Σ_2^0 -complete.*

Proof. Use a standard effective enumeration W_e of c.e. sets, arranged to emit at most one previously unseen element at a stage. Let $I_e(s) \in \{0, 1\}$ indicate whether stage s emits a new element. Its computation is a bounded simulation with polynomial overhead in s for each fixed e .

Sample S with $\Pr(S = s) = 2^{-(s+1)}$. Compute $I_e(s)$. If it is one, perform a delay of $\Theta(2^s)$ elementary machine steps; then halt. Otherwise halt immediately. The body is total for every finite s . The sampling loop's infinite path has probability zero, so every program in the family is almost-surely terminating.

The expected bounded-simulation and sampling overheads are finite, since $\sum_s 2^{-s} s^d < \infty$ for every fixed d . The expected delay is finite exactly when

$$\sum_s 2^{-(s+1)} 2^s I_e(s) = \frac{1}{2} \sum_s I_e(s)$$

is finite. The constant-factor delay estimates suffice: each positive stage contributes an amount bounded above and below by positive constants. The last sum is finite exactly when W_e is finite.

For completeness, $\text{FIN} = \{e : W_e \text{ is finite}\}$ is Σ_2^0 -complete. Membership follows by asserting that after some stage no new element is emitted. For hardness, given $\exists n \forall t R(e, n, t)$ with decidable R , let n_s be the least $n \leq s$ satisfying every test $R(e, n, t)$ for $t \leq s$, or $s + 1$ if none exists. Emit the distinct values that occur in this sequence. If a least genuine witness n exists, every smaller candidate is eventually refuted and n_s eventually stabilizes at n . Only finitely many values are emitted. If none exists, every fixed candidate is eventually refuted, so $n_s \rightarrow \infty$ and infinitely many values are emitted. This is an effective reduction to FIN . Together with (47.2), it proves the claim. $\square$

A delay of $\Theta(2^s)$ can be implemented without a hidden huge-integer unit-cost instruction: traverse the full binary recursion tree of depth s , using a stack. Its number of nodes is $2^{s+1} - 1$, and a fixed local stack machine has constant work per node. Other standard models may add a polynomial factor; choosing a sampling law adapted to a documented time bound gives the same hardness, but the above explicit stack model avoids that detour.

47.3 Probability preservation is weaker than mean preservation

Proposition 47.2 (Polynomial slowdown can destroy finite mean). *A path-faithful simulation with polynomial time overhead need not preserve finite expected running time.*

Proof. Sample N by continuing with probability $1/4$ at each iteration. Then $\Pr(N = n) = 3/4^{n+1}$. A computation with delay $T = 2^N$ has

$$\mathbb{E}T = \frac{3}{4} \sum_{n \geq 0} 2^{-n} = \frac{3}{2} < \infty, \quad \mathbb{E}(T^2) = \frac{3}{4} \sum_{n \geq 0} 1 = \infty.$$

The sampling overhead also has finite expectation. A simulation that slows the delay quadratically has infinite mean, while preserving exactly which coin streams eventually halt. $\square$

An affine pathwise bound $\tau' \leq A\tau + B$ is sufficient to preserve finite mean in the forward direction. Two-sided suitable bounds are needed for equivalence. It would be incorrect to transfer the finite-mean classification to every Turing-complete substrate merely from an unspecified effective simulation.

48 Transfer to unconventional computational substrates

[source 13]

48.1 The precise transferable hypothesis

Definition 48.1. A coin-faithful simulation of a stochastic machine into a substrate is an effective translation with a deterministic evaluation strategy, finite simulation of each completed source step, and the following property: on the same stream of requested fair bits, the target eventually halts exactly when the source does. Extra target operations do not make independent probabilistic choices.

Theorem 48.2 (Stochastic substrate transfer). *A substrate admitting a coin-faithful simulation of fair-coin Turing computation inherits the full class of halting probabilities and the almost-sure termination hardness. If its bounded operational semantics is effective, it also has the finite-horizon quartic representations of theorem 44.1, using its own step convention. Finite-mean properties transfer only under additional time-overhead hypotheses.*

Proof. Coin faithfulness identifies the two halting events on the product probability space, so their measures agree. This also preserves the predicate that the measure is one. Effective bounded operational semantics gives a computable finite coin-tape predicate, which can be compiled into a Boolean circuit; the explicit circuit compiler then applies. The final qualification is the preceding slowdown proposition. $\square$

The bounded circuit may be much larger than a source-machine circuit. Effectiveness suffices for representability; a polynomial circuit-size bound requires a separate efficient-simulation argument.

48.2 A minimal randomized combinator example

Start with a deterministic evaluation strategy for a universal combinator calculus. Add a primitive $\text{coin}(A, B)$ that returns A or B with probability $1/2$, and insist that it reduces only when demanded by that strategy. An ordinary universal simulation can translate a source coin instruction to exactly one occurrence of this primitive and perform all other source instructions deterministically. That gives a coin-faithful simulation.

The demand restriction matters. Unrestricted reduction could duplicate a random expression before it is sampled, or sample it before duplication; those operations have different joint distributions. For example, copying one sampled Boolean value gives perfectly correlated copies, whereas reducing two independent coin expressions does not. The theorem requires operational semantics that fixes this distinction, not merely Turing completeness of the underlying rewrite rules.

48.3 Other legitimate targets and exclusions

The same interface can be attached to effective stack machines, pointer heap machines, finite-string rewriting evaluators, or cellular computational models with a specified random-bit input channel, provided a deterministic universal simulation and its coin discipline are supplied. These are applications of the transfer theorem, not claims that every chemical, physical, or biological system automatically has those properties.

In particular, a finite-state Markov chain has decidable rational linear algebra for its absorption probabilities; an unbounded effective state space is the source of the general computability phenomena here. State-dependent irrational transition laws need effective presentations and are outside the fair-coin assumptions as stated. Adding adversarial nondeterministic schedulers changes the problem substantially: work on positive almost-sure termination with such schedulers reaches Π_1^1 -completeness [53]. The scheduler-free Σ_2^0 classification must not be carried over to that different model.

49 Worked examples and exact artifacts

[source 13]

49.1 A four-variable one-bit polynomial

For the circuit $C(x) = x$, let u be the root unit, a the output occurrence, and b the unit occurrence. The compiler gives

$$F(x, u, a, b) = \sum_{v \in \{x, u, a, b\}} v^2(v-1)^2 + (u-1)^2 + (a-x)^2 + (b-u)^2 + (a-b)^2. \quad (49.1)$$

Its only real zero is $(1, 1, 1, 1)$. There is one accepted input out of two, so the corresponding probability is $1/2$.

The expanded polynomial is

$$F = x^4 + u^4 + a^4 + b^4 - 2(x^3 + u^3 + a^3 + b^3) + 2x^2 + 3u^2 + 3a^2 + 3b^2 - 2u - 2ax - 2bu - 2ab + 1.$$

Its coefficient height is three. At its zero, in coordinate order (x, u, a, b) , the Hessian is

$$\begin{pmatrix} 4 & 0 & -2 & 0 \\ 0 & 6 & 0 & -2 \\ -2 & 0 & 6 & -2 \\ 0 & -2 & -2 & 6 \end{pmatrix},$$

which is $2I + 2J^T J$ for the four displayed linear residuals.

49.2 Geometric waiting time

Consider a program that halts at the first tick with bit one. Its horizon- t predicate is the OR of its t input bits. Thus

$$a_t = 2^t - 1, \quad p_t = 1 - 2^{-t}, \quad p = 1.$$

For $t \geq 1$, the demonstration circuit uses $4(t-1)$ gates by iterating $x \vee y = \neg(\neg x \wedge \neg y)$. The implementation produces:

t	Gates	Variables	Non-Boolean residuals	Zeros	Probability
0	1	4	5	0	0
1	0	4	4	1	$1/2$
2	4	20	19	3	$3/4$
3	8	36	34	7	$7/8$
4	12	52	49	15	$15/16$
5	16	68	64	31	$31/32$
6	20	84	79	63	$63/64$
7	24	100	94	127	$127/128$
8	28	116	109	255	$255/256$

Here the $t = 0$ predicate is constant zero. Sparse expanded polynomials and residual lists for horizons $0, 1, 2, 3$ are included in the package.

49.3 A constant noncomputable limit with increasing finite objects

For the sampler of (45.2), every horizon still gives a finite exactly computable polynomial and a finite exactly computable count. What cannot be uniformly completed is the limiting probability. There is no contradiction between a finite exact artifact at each stage and a noncomputable limit. The present geometric theorem makes that separation explicit without appealing to fragile numerical encodings.

50 Executable verification and its limits

[source 13]

The package contains three Python modules. `quartic_compiler.py` implements circuits, occurrence routing, unique wire extension, exact sparse polynomial expansion, and JSON output. `degree_two.py` implements the transfer-matrix algorithm of [theorem 43.2](#). `verify.py` executes deterministic checks using seed 20260930. Core arithmetic uses Python integers and rational `Fraction` objects. NumPy is optional and is used only for separately labeled floating-point Hessian smoke tests.

The recorded run passed the following checks:

Exact finite check category	Count
Circuit structural audits	175
Input-tape unique extensions	1,262
Boolean-coordinate mutations	2,524
Exhaustive full Boolean assignments on small examples	864
Scalar rounding inequalities	321
Local gate rounding inequalities	468
Exact low-energy witness extraction	300
Exact analytic-constant checks	3
Geometric-waiting horizons	9
Degree-two dynamic-programming instances	122
Slowdown moment checks	2
Total exact finite checks	6,050

The structural audits check expanded degree, coefficient height, constant coefficient, residual counts, variable bound, incidence, pair uniqueness, and interaction degree. The observed maximum coefficient height was four and maximum interaction degree was three.

A further 60 floating-point Hessian tests examined 30 exact zeros and 30 points in their prescribed boxes. At the sampled zeros, the smallest computed eigenvalue was 2 up to floating-point error, and the largest was about 12.61545; the theorem’s bound is 20. These measurements are sanity checks, not a proof of the spectral inequalities. The exact rational bounds used in the proof were also checked separately: $543/512 > 1$ and $11215/512 < 24$.

The source does not implement a general Turing-machine-to-circuit frontend, the much larger fixed-arity MRDP polynomials D, Q, V , or a Lean proof of this article. Its executable boundary is the explicit Boolean circuit compiler and the degree-two counter. The machine simulation and fixed-arity representations are established by the mathematical arguments above, not by a claim that those absent frontends were run.

```
python code/verify.py
python code/quartic_compiler.py --or-inputs 3 --output landscape.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The verification report records exact counts, observed structural maxima, example sizes, the random seed, and the distinction between exact and numerical tests. Finite testing does not establish the unbounded theorems; its role is to exercise the implementation and expose concrete mistakes. [write] In this report the three modules are shipped as `code/13-quartic-landscapes-quartic_compiler.py`, `code/13-quartic-landscapes-degree_two.py` and `code/13-quartic-landscapes-verify.py`, and the recorded outputs of `results/` as `data/13-quartic-landscapes-*`. The commands below are the source’s, for its delivered layout; the README gives the rerun recipe on a copy with that layout.

51 A formalization plan for ProveIt

[source 13]

51.1 An appropriate module boundary

A natural new project is a stochastic compilation layer adjacent to the existing Hilbert-tenth development, rather than a modification of the MRDP core. Its first formal object should be an acyclic circuit with explicit input wires, gate definitions, and a distinguished output. A routed certificate should carry its defining equations and occurrence ownership, making incidence and graph bounds structural invariants.

The proof obligations separate into four layers. The finite algebraic layer proves unique evaluation, the sum-of-squares zero equivalence, and the expanded coefficient bound. The finite graph layer proves the trunk routing invariant and the degree-two count. The analytic layer proves rounding, Hessian bounds, and low-sublevel components. The measure layer relates cylinder probabilities to finite counts and then to the limit.

For the fixed-arity comparison predicates, the existing `Diophantine.mrdp` interface is the correct foundation [47]. One first proves that finite-horizon acceptance counts are computable, then that a strict threshold crossing is c.e., and only then invokes MRDP. This avoids attempting to formalize probability itself as an existential arithmetic witness.

51.2 Suggested first formal milestones

The shortest valuable milestone is the finite bijection, entirely over Boolean vectors and integer residuals. It already certifies every sparse JSON polynomial produced by a proved version of the compiler. Next comes the coefficient-height and subcubic graph theorem, followed by the real-valued rounding lemma. These are more local than the homology theorem and can be audited independently.

The low-energy topology proof can initially be stated without homology: a finite disjoint union of nonempty compact convex sets indexed by accepted inputs. Betti numbers and Euler characteristic are then derived consequences rather than prerequisites. Likewise, a first probability formalization can prove the rational finite-horizon identity before any infinite product-space construction is introduced.

A verified general-purpose frontend need not be the first deliverable. The elementary OR family, a bounded register-machine interpreter, or a fixed universal local machine would provide progressively stronger integration tests. The ordinary data input must remain distinguished from free random bits throughout.

51.3 Trust boundary

No new Lean file is presented as checked in this package. A downstream formalization should not import the mathematical theorems as axioms and then describe the compiler as verified. The repository's published MRDP interface, the present paper's informal proofs, executable finite tests, and any future proof-assistant certificates are four different evidence levels and should remain separately labeled.

52 The question: what does fast mixing fail to decide?

[source 17]

A quantitative convergence theorem answers an approximation question. It need not answer an exact equality question, and it gives even less reason to expect a decision procedure for rationality or algebraicity. This distinction becomes especially sharp when a system simultaneously has rational local rules, reversibility, strong drift, and a spectral gap bounded away from zero. The purpose of this article is to exhibit that

distinction in one elementary model, with explicit reductions and arithmetic certificates rather than a bare appeal to universality.

Our starting point is the following binary choice of birth probabilities:

$$P_b(n, n+1) = \frac{2-b_n}{8}, \quad P_b(n, n-1) = \frac{1}{2} \quad (n \geq 1), \quad b_n \in \{0, 1\}. \quad (52.1)$$

The remaining probability is assigned to staying put; there is no death move at zero. A flag $b_n = 1$ is called a *defect*. It reduces an already inward-biased birth probability. Defects can therefore record information without destroying confinement.

The results answer four concrete questions about this model. Can exact stationary equality still encode halting under a uniform mixing estimate? Yes, even when every stationary coordinate is rational. Can one characterize exactly which environments have rational root mass? Yes: precisely the eventually periodic ones. Can rationality or algebraicity be certified by one existential polynomial for all effectively specified environments? No: both properties already have second-level arithmetical complexity. Which equilibrium assertions do admit finite Diophantine certificates? Strict rational comparisons do, and their finite arithmetic layer has a small canonical quartic presentation.

52.1 Main results at a glance

The same stationary family underlies the following statements. The input model is part of each assertion; a statement about a known rational input is not a statement about a program producing an approximation to a real.

Property or task	Result proved here
Uniform analytic control	Reversibility, exponential tails, root return mean ≤ 2 , sharp common gap $\delta = (3 - 2\sqrt{2})/4$, root-start mixing $\leq \frac{1}{2}(1 - \delta)^t$.
Symbolic arithmetic	Rational root mass if and only if the environment is eventually periodic; the attainable-mass set has dimension $\log_2((1 + \sqrt{5})/2)$.
One-defect computation	All masses are rational, but exact equality to $1/2$ is Π_1^0 -complete. No computable description-length bound on reduced denominators exists.
Primitive-recursive environments	Rationality and algebraicity are Σ_2^0 -complete; irrationality and transcendence are Π_2^0 -complete.
Finite certificates	Sharp prefix bounds; a unique arithmetic witness with $N + 1$ coordinates and $2N + 1$ squared residuals in the compact presentation.

52.2 Relation to ProveIt and to earlier work

This manuscript is based on the repository snapshot `e18718e837d43e162252f9a314e8cb797fbd1a1f`. The relevant formal foundation is ProveIt’s natural-number MRDP interface, `Diophantine.mrdp`, together with its converse `Diophantine.mrdp_iff` [77]. It says that a computably enumerable subset of $\mathbb{N}$ is the projection of the natural zero set of one integer polynomial. It is an existence interface, not a stated practical compiler with numerical degree and witness bounds. We use it only with that scope.

The repository also contains a substantial informal report on probabilistic output laws, quantum mortality, and continuous-computation barriers ([write] Parts I–III of the present report; the source cites that report’s README at the pin). The present construction concerns stationary measures of countably infinite reversible chains, their arithmetic type, and a binary-environment coding. It does not reproduce that report’s normalization-budget or quantum-mortality constructions. This is a comparison with the inspected relevant material, not a claim that every file in the repository has been exhaustively audited.

Undecidability of exact stationary distributions is not new. Gamarnik proved it for constrained homogeneous random walks, even with an input linear Lyapunov function witnessing positive recurrence [78]. Our

contribution is an explicit combination of reversibility, a sharp common spectral gap, a uniform root-start mixing estimate, rational-valued hard instances, and a complete symbolic rationality criterion. A programmable one-dimensional environment and a constrained homogeneous random walk in a fixed-dimensional orthant are different input models. The result here is not asserted to strengthen all restrictions in Gamarnik's theorem.

History recording is a classical route to logically reversible computation [79]. We give the actual local history map needed here, rather than invoking reversibility as a black box. Counter-machine instruction sets must be specified carefully: reversibility can change halting decidability differently for different instruction sets [80, 81]. The lift below uses a history counter with multiplication by a fixed base, and does not claim a new theorem about a minimal reversible two-counter instruction set.

The self-similar-set dimension calculation is a specialization of classical separated-iterated-function-system theory [82]. The sparse-defect transcendence proof uses the elementary Liouville principle [83], which is proved in the needed form here. The low-complexity transcendence corollary uses the deeper theorem of Adamczewski and Bugeaud [84]; that theorem is explicitly imported, not reproved or presented as a new result.

52.3 Conventions and levels of effectivity

We write $\mathbb{N} = \{0, 1, 2, \dots\}$. An environment is an infinite binary word b . At first it need not be computable. A *primitive-recursive environment code* is a finite syntactic expression for a primitive-recursive function of one natural variable, followed by reduction modulo two. All such expressions can be enumerated and uniformly evaluated by a total computable interpreter; no universal primitive-recursive evaluator is assumed.

A real is *computable* when a program supplies rational intervals of arbitrarily small prescribed width containing it. This is not the same as producing its reduced numerator and denominator when it happens to be rational. A set is *Diophantine* here if it has a fixed existential integer-polynomial representation with natural witnesses. By MRDP this is equivalent to being computably enumerable [77]. Arithmetic hierarchy statements concern computable many-one reductions and the usual finite coding of tuples.

53 The chain and its exact stationary law

[source 17]

Fix $b \in \{0, 1\}^{\mathbb{N}}$, and define

$$r_n = \frac{2 - b_n}{4}, \quad D_n = \sum_{k < n} b_k, \quad w_n = \prod_{k < n} r_k = 2^{-n - D_n}, \quad Z_b = \sum_{n \geq 0} w_n. \quad (53.1)$$

Empty products are one. The kernel is given by equation (52.1) and its holding probability. Write $\alpha_b = \pi_b(0)$ for the stationary root mass.

Theorem 53.1 (Stationarity and uniform confinement). *For every environment b , the chain is irreducible, aperiodic and positive recurrent, with unique stationary distribution*

$$\pi_b(n) = \frac{w_n}{Z_b}, \quad \frac{4}{3} \leq Z_b \leq 2, \quad \frac{1}{2} \leq \alpha_b \leq \frac{3}{4}. \quad (53.2)$$

It is reversible. Moreover,

$$\pi_b(n) \leq \frac{3}{4} 2^{-n}, \quad \sum_{n \geq M} \pi_b(n) \leq \frac{3}{2} 2^{-M}, \quad \mathbb{E}_{\pi_b}[a^X] \leq \frac{3}{4(1 - a/2)} \quad (1 \leq a < 2). \quad (53.3)$$

The expected positive return time to zero is $Z_b \leq 2$.

Proof. Every birth and every available death move has positive probability; every state has a positive holding probability. Also $4^{-n} \leq w_n \leq 2^{-n}$, so summing the two geometric series gives $4/3 \leq Z_b \leq 2$. The identity

$$\pi_b(n)P_b(n, n+1) = \frac{w_n}{Z_b} \frac{2-b_n}{8} = \frac{w_{n+1}}{2Z_b} = \pi_b(n+1)P_b(n+1, n)$$

proves detailed balance and hence stationarity. An irreducible chain with a stationary probability distribution is positive recurrent and has that distribution uniquely. The displayed tail and moment bounds follow by geometric summation. The return identity $\mathbb{E}_0 \tau_0^+ = 1/\pi_b(0)$ is the usual cycle formula for a positive recurrent chain: successive visits to zero divide a path into identically distributed regenerative cycles, each with one initial visit to zero. Dividing the occupation count at zero by mean cycle length gives its stationary frequency. Thus the mean is Z_b . $\square$

Remark 53.2. There is no conflict between this mean return bound and a very late computational event. Most returns occur without visiting a remote defect. The stationary mass of a state carrying a long computation is exponentially small in its trajectory index.

The kernel is specified by a single bit at each state. All its nonzero entries belong to

$$\left\{ \frac{1}{8}, \frac{1}{4}, \frac{3}{8}, \frac{1}{2}, \frac{3}{4}, \frac{7}{8} \right\}.$$

No undecidable information has been put into an individual real-valued transition coefficient. When b is computable, every row is exactly computable with rational arithmetic.

54 A sharp common spectral gap and a uniform mixing estimate

[source 17]

Let $L^2(\pi_b)$ have norm $\|f\|_{2, \pi_b}$, and let

$$\mathcal{E}_b(f, f) = \langle f, (I - P_b)f \rangle_{\pi_b} = \frac{1}{2} \sum_{k \geq 0} \pi_b(k+1) (f(k+1) - f(k))^2.$$

The last expression counts each edge once after using detailed balance. For reversible chains, standard operator and Dirichlet-form background can be found in Aldous and Fill [85]; the estimate needed here has a short direct proof.

Lemma 54.1 (A weighted half-line Poincare estimate). *Suppose a birth-death chain has a strictly positive stationary distribution π , constant death probability $q > 0$, and ratios $\pi(n+1)/\pi(n) \leq r < 1$. Then*

$$\text{Var}_\pi(f) \leq \frac{1}{q(1-\sqrt{r})^2} \mathcal{E}(f, f) \quad (f \in L^2(\pi)). \quad (54.1)$$

Proof. Put $g_k = \sqrt{\pi(k+1)}(f(k+1) - f(k))$. Since the mean minimizes squared distance to constants,

$$\text{Var}_\pi(f) \leq \sum_{n \geq 1} \pi(n) (f(n) - f(0))^2.$$

The anchored differences obey

$$\begin{aligned} \sqrt{\pi(n)} |f(n) - f(0)| &\leq \sum_{k=0}^{n-1} \sqrt{\frac{\pi(n)}{\pi(k+1)}} |g_k| \\ &\leq \sum_{k=0}^{n-1} r^{(n-k-1)/2} |g_k|. \end{aligned}$$

The final expression is a convolution with the summable sequence $1, \sqrt{r}, r, \dots$. The triangle inequality for its shifted ℓ^2 vectors gives

$$\left(\sum_{n \geq 1} \pi(n)(f(n) - f(0))^2 \right)^{1/2} \leq \sum_{j \geq 0} r^{j/2} \|g\|_2 = \frac{\|g\|_2}{1 - \sqrt{r}}.$$

As $\mathcal{E}(f, f) = q\|g\|_2^2$, squaring proves the claim. The same argument applies to infinite sequences by convergence of the shift sum in ℓ^2 ; alternatively truncate the convolution and pass to the limit. The Dirichlet form is finite for $f \in L^2(\pi)$ since a stationary Markov operator is a contraction. $\square$

Theorem 54.2 (Uniform mixing, with the optimal common gap). *Let*

$$\delta = \frac{3 - 2\sqrt{2}}{4}, \quad \rho = 1 - \delta = \frac{1 + 2\sqrt{2}}{4}. \quad (54.2)$$

Every P_b has spectral gap at least δ , and this common lower bound cannot be increased. For every integer $t \geq 0$,

$$\|\delta_0 P_b^t - \pi_b\|_{\text{TV}} \leq \frac{1}{2} \sqrt{Z_b - 1} \rho^t \leq \frac{1}{2} \rho^t \leq \frac{1}{2} \left(\frac{23}{24} \right)^t. \quad (54.3)$$

The estimate is uniform in the entire environment.

Proof. Apply lemma 54.1 with $q = r = 1/2$ to obtain the gap. Each holding probability is at least $1/4$, so $P_b = \frac{1}{4}I + \frac{3}{4}K$ for a reversible stochastic kernel K . Consequently the spectrum of P_b is bounded below by $-1/2$. On the orthogonal complement of the constants it is bounded above by $1 - \delta$, and its absolute value is at most ρ .

The density of δ_0 with respect to π_b , minus one, has squared $L^2(\pi_b)$ norm $1/\pi_b(0) - 1 = Z_b - 1$. By reversibility, the density after t steps is obtained by applying P_b^t . Cauchy–Schwarz bounds half its L^1 distance by half its L^2 distance, proving the first two inequalities. Finally, $\delta > 1/24$: equivalently $17 > 12\sqrt{2}$, whose square is $289 > 288$. The optimality of δ is established in section 65 by test vectors for the defect-free chain. $\square$

The same constant appears in a drift computation. For $V(n) = (\sqrt{2})^n$ and $n \geq 1$,

$$\frac{P_b V(n)}{V(n)} = 1 + p_n(\sqrt{2} - 1) + \frac{1}{2}(1/\sqrt{2} - 1) \leq \rho, \quad p_n = \frac{2 - b_n}{8}.$$

Thus confinement is visible directly in the local rules as well as in the stationary formula. The root is a single exceptional small set.

Remark 54.3 (What uniformity means). The bound is uniform over environments when the initial state is zero. It is not uniform over all starting states. A walk begun at m cannot reach zero in fewer than m nearest-neighbor steps. In particular, no assertion of uniform ergodicity in the all-starting-states sense is being made. Also, a Markov step is not automatically a constant-time operation on the bit representation of a large computational state.

55 The equilibrium as a uniquely decodable binary object

[source 17]

The stationary formula has more structure than mere computability. It records the defect positions in a constrained binary expansion.

Theorem 55.1 (The defect expansion). *List the defect positions in increasing order as $d_1 < d_2 < \dots$, with a finite or empty list allowed. Then*

$$Z_b = 2 - \sum_j 2^{-(d_j+j)}. \quad (55.1)$$

The binary expansion of $L_b = 2 - Z_b$ is the infinite word

$$c = h(b), \quad h(0) = 0, \quad h(1) = 10. \quad (55.2)$$

In particular, this expansion contains no two consecutive ones.

Proof. Since $2w_{n+1} = (1 - b_n/2)w_n$,

$$w_n - 2w_{n+1} = \frac{1}{2}b_n w_n.$$

Summing over $n \geq 0$ gives $2 - Z_b = \frac{1}{2} \sum_n b_n w_n$. At the j th defect, $D_{d_j} = j - 1$, so its contribution is exactly 2^{-d_j-j} . The j th one in the morphic word appears at binary position $d_j + j$, where binary positions start at one. This proves both descriptions. Successive positions differ by at least two, so there is no ambiguity from a tail of all ones. In the finite-defect case the chosen expansion terminates in zeros. $\square$

55.1 The self-similar range

Let $\mathcal{K} = \{Z_b : b \in \{0, 1\}^{\mathbb{N}}\}$ and $\mathcal{C} = \{\alpha_b : b \in \{0, 1\}^{\mathbb{N}}\}$. Removing the first flag gives

$$Z_b = F_{b_0}(Z_{\sigma b}), \quad F_0(z) = 1 + \frac{z}{2}, \quad F_1(z) = 1 + \frac{z}{4}, \quad (55.3)$$

where σ is the left shift. The two first-level intervals are

$$F_1([4/3, 2]) = [4/3, 3/2], \quad F_0([4/3, 2]) = [5/3, 2]. \quad (55.4)$$

They are disjoint, separated by a gap of length $1/6$.

Theorem 55.2 (Geometry and faithful coding). *The map $b \mapsto Z_b$ is a homeomorphism from binary Cantor space onto $\mathcal{K}$. The latter is the compact invariant set $F_0(\mathcal{K}) \cup F_1(\mathcal{K})$, and*

$$\dim_H \mathcal{K} = \dim_H \mathcal{C} = \log_2 \left(\frac{1 + \sqrt{5}}{2} \right). \quad (55.5)$$

Proof. Agreement on N initial flags gives agreement on the first N maps in [equation \(55.3\)](#); their product contraction is at most 2^{-N} . This proves continuity. The disjoint first-level intervals, applied again after any common prefix, prove injectivity. Compactness of the domain and the Hausdorff property of $\mathbb{R}$ give the homeomorphism. The invariance equation follows by prepending either bit.

For completeness, the dimension argument can be given directly. Put $D = \text{diam}[4/3, 2] = 2/3$, and let $s > 0$ solve $2^{-s} + 4^{-s} = 1$. At depth k , the cylinder intervals have lengths DR_u , where R_u is the product of the contractions in the word u . Their s -powers sum to D^s . Their maximum length tends to zero, proving $\dim_H \mathcal{K} \leq s$.

Assign a cylinder the probability R_u^s . These assignments are consistent because $2^{-s} + 4^{-s} = 1$, and so define a probability measure on $\mathcal{K}$. For a small $\varepsilon > 0$, stop each branch when its cylinder interval first has length at most ε . Every stopped interval has length greater than $\varepsilon/4$. They are pairwise disjoint, and any interval of length 2ε can meet at most ten of them: except possibly the two end intervals, the others lie inside it and have disjoint interiors of length greater than $\varepsilon/4$. Each has mass at most $(\varepsilon/D)^s$. Hence every ball of radius ε has mass at most $10(\varepsilon/D)^s$. Covering the support and summing this inequality yields positive s -dimensional Hausdorff content up to a fixed normalization, and thus dimension at least s .

Writing $x = 2^{-s}$ gives $x + x^2 = 1$, which yields the displayed value. Finally $z \mapsto 1/z$ is bi-Lipschitz on $[4/3, 2]$:

$$\frac{1}{4}|z - z'| \leq |z^{-1} - z'^{-1}| \leq \frac{9}{16}|z - z'|.$$

It therefore preserves Hausdorff dimension. $\square$

This dimension calculation is an elementary instance of the separated self-similar-set theorem [82], not a new general fractal dimension theorem. Its significance here is that the same set carries a faithful arithmetic representation of all binary environments.

55.2 Rationality is exactly eventual periodicity

Theorem 55.3 (Exact rationality criterion). *For every binary environment b , the following are equivalent:*

$$\alpha_b \in \mathbb{Q} \iff Z_b \in \mathbb{Q} \iff b \text{ is eventually periodic.} \quad (55.6)$$

Proof. Reciprocation proves the first equivalence. If b consists of a finite prefix followed by repetitions of a finite nonempty word, the repeated word defines an affine map $z \mapsto A + Rz$ with rational A, R and $0 < R < 1$. Its fixed point is $A/(1 - R)$; applying the prefix map gives a rational Z_b .

Conversely suppose $Z_b = p/q$. Starting at $z_0 = Z_b$, repeatedly use the disjoint intervals in equation (55.4) to recover the bit and the next tail:

$$z_{n+1} = \begin{cases} 4(z_n - 1), & z_n \in [4/3, 3/2], \quad b_n = 1, \\ 2(z_n - 1), & z_n \in [5/3, 2], \quad b_n = 0. \end{cases} \quad (55.7)$$

Every $z_n = Z_{\sigma^n b}$ lies in $[4/3, 2]$ and in $q^{-1}\mathbb{Z}$. Only finitely many such rational values exist. The deterministic iteration must repeat a value, and from then on it repeats the same bits. Thus b is eventually periodic. $\square$

Corollary 55.4 (Decidable membership for a supplied rational). *Given an exact rational a , it is decidable whether $a \in \mathcal{C}$. When the answer is yes, an eventually periodic environment can be computed.*

Proof. Reject $a \notin [1/2, 3/4]$. Starting with $z = 1/a$, iterate equation (55.7). Reject on leaving the union of the two closed intervals, and accept on a repeated value. The fixed rational grid in the bounded interval makes one of these events inevitable. In the acceptance case the cycle gives a periodic tail; contraction proves that this itinerary has exactly the original value. $\square$

The number of visited rational grid points is at most $q + 1$ when q is the denominator of the initial z , so this is a numerical bound, not a claim of polynomial complexity in the bit length of q . For example, $a = 5/8$ is immediately rejected: $1/a = 8/5$ lies in the first gap. Later we show that determining whether the limit supplied by a program is rational is Σ_2^0 -complete. There is no contradiction: the exact rational is input in this corollary, rather than an unknown arithmetic property of a computable limit.

56 Halting hidden in a single rational defect

[source 17]

Fix a deterministic universal machine U with a chosen convention for its first halting time $T(e) \in \mathbb{N} \cup \{\infty\}$. The input e includes its program and data. Define

$$b_n^e = 1 \iff T(e) = n. \quad (56.1)$$

This is a uniformly primitive-recursive predicate: simulate only n steps and check whether the first halt is at that step. The rule does not ask an unbounded halting oracle whether to place a defect.

Theorem 56.1 (Hard exact arithmetic with rational equilibria). *For the family equation (56.1),*

$$\alpha_e = \begin{cases} \frac{1}{2}, & T(e) = \infty, \\ \frac{2^{T(e)+1}}{2^{T(e)+2} - 1}, & T(e) < \infty. \end{cases} \quad (56.2)$$

Every stationary coordinate is rational. Nevertheless, the set $\{e : \alpha_e > 1/2\}$ is Σ_1^0 -complete and Diophantine, whereas $\{e : \alpha_e = 1/2\}$ is Π_1^0 -complete and is not Diophantine. There is no uniform algorithm returning the exact reduced rational α_e for every input e .

Proof. There are either no defects or exactly one at T . By [theorem 55.1](#), $Z_e = 2$ or $Z_e = 2 - 2^{-T-1}$, respectively. Taking reciprocals gives [equation \(56.2\)](#). Every w_n is rational, so every $\pi_e(n) = w_n/Z_e$ is rational as well.

The strict inequality $\alpha_e > 1/2$ is equivalent to halting, and the equality is equivalent to nonhalting. The halting set is computably enumerable and complete at the first existential level; its complement is not computably enumerable. MRDP gives the claimed positive polynomial representation and rules out the negative one [\[77\]](#). An exact rational-output algorithm would decide equality to $1/2$ by ordinary integer arithmetic, deciding halting. $\square$

[write] Cf. [proposition 9.4](#) of Part I, which classifies strict comparisons and equality under the probability-one promise with the same halting-time device ($x_e = 2^{-t}$ at the halting time t); here the device places one defect in an exactly stationary, reversible and uniformly mixing chain.

For a halting computation the separation is explicitly

$$\alpha_e - \frac{1}{2} = \frac{1}{2(2^{T(e)+2} - 1)}. \quad (56.3)$$

A late halt leaves a correspondingly tiny equilibrium signature. This is not a slow-mixing phenomenon: the estimate [equation \(54.3\)](#) has no dependence on $T(e)$. It is an exact-symbolic phenomenon, because no prescribed finite precision distinguishes equality from every possible positive separation.

Corollary 56.2 (No computable symbolic-height bound). *For any ordinary finite description of machine inputs, there is no total computable function of description length that bounds the reduced denominator of α_e for all inputs of that length. There is also no such bound on the bit length of the denominator.*

Proof. For a halting input the numerator in [equation \(56.2\)](#) is a power of two and its denominator is odd, so the fraction is reduced. A computable denominator bound $B(\ell)$ would bound $2^{T(e)+2} - 1$ whenever a length- ℓ input halts. Computing an integer M with $2^{M+2} - 1 > B(\ell)$ and simulating for M steps would decide halting. A bit-length bound likewise bounds $T(e) + 2$. $\square$

This is a Busy-Beaver-type obstruction in symbolic output size, not in the mixing time. It also applies to the exact mean return time Z_e and, by restriction to the value $1/2$, shows that the graph of the exact rational-output function is not computably enumerable. The fraction $1/\alpha_e$ is easy to approximate but not uniformly easy to recognize as the correct finite fraction.

56.1 A transfer principle for other computational substrates

The machine model in this section is incidental. Suppose a substrate has finite encodings of initial states, deterministic effective steps, and a decidable halting flag. Suppose also that its halting problem is computably enumerable complete. Replacing U by its bounded simulation gives exactly the same reduction. When the bounded simulation is primitive recursive, the environment family above is jointly primitive recursive in its code and position. A more general computable bounded simulation still suffices for the equality undecidability and numerical approximation results.

Thus a proved simulation from lambda terms, register programs, tag systems, or another effective universal model is enough to transfer halting into [equation \(56.2\)](#). This statement is a general compiler principle, not a claim that each such front end was implemented in the accompanying code. The next section supplies a concrete local front end, so the interpretation is not limited to a row generator that reruns an entire computation from scratch.

57 A finite-rule reversible counter substrate

[source 17]

There are two different meanings of reversibility in this article. A *logically reversible* transition has at most one predecessor. A *reversible Markov chain* satisfies detailed balance with its stationary law. The following construction connects the two.

Consider a deterministic counter program with a finite control state q and counters $x = (x_1, \dots, x_m) \in \mathbb{N}^m$. Instructions are increments, decrement/zero-test branches with explicitly named successors, and halt. Such unrestricted finite-control counter programs are a universal model; care concerning more restrictive instruction sets is documented in [81].

Split each decrement instruction into two guarded branches. Assign a different positive tag j to every primitive branch, and choose a fixed base B larger than all tags. Add a natural history counter H . A branch with counter change v becomes

$$(q, x, H) \mapsto (q', x + v, BH + j). \quad (57.1)$$

Its guard is either always true, $x_i = 0$, or $x_i > 0$; in the last case a decrement is permitted. Add a fresh state **done**; a halt instruction takes one step into **done**, and **done** keeps appending its own tag forever. The original halting control states, not **done**, are the defect states.

Theorem 57.1 (Local reversible realization). *The finite guarded-affine rules [equation \(57.1\)](#) have a computable partial inverse and a unique forward successor from every well-formed state. From any initialized state with $H = 0$, the forward trajectory is an infinite ray with no repetitions. It contains one defect exactly when the original program halts.*

On this rooted trajectory, assign probability $(2 - b)/8$ to the forward successor, probability $1/2$ to the predecessor when present, and the remaining probability to holding. The resulting chain is isomorphic to the one-defect family of [theorem 56.1](#), with all the bounds of [theorems 53.1](#) and [54.2](#). A fixed universal counter program yields a fixed finite rule set with varying initial counter data.

Proof. For an alleged successor with history H' , compute its quotient and remainder on division by B . The remainder must be one of the positive branch tags. It identifies the branch uniquely. Subtract its counter change, verify the source guard and target control, and use the quotient as the preceding history. If any check fails there is no predecessor; if all pass there is exactly one. Equivalently, the inverse edge is the integer relation $H' = BH + j$ together with the corresponding finite guards. No unbounded search is required.

The original program selects exactly one enabled branch. The halt and **done** rules make the forward transition total. Since $BH + j > H$, the history increases strictly. An initialized $H = 0$ state has no predecessor, and its forward trajectory is therefore an infinite ray. Original execution before halting is preserved. On first reaching a halt control, the defect is present; the next step enters **done**, and no further defect is possible.

Number the successive states on the ray by $0, 1, 2, \dots$. The assigned probabilities then agree exactly with [equation \(52.1\)](#). The stationary law and the analytic estimates transfer through this bijection. Universality of a fixed ordinary counter interpreter, followed by this fixed history construction, gives the last assertion. $\square$

The probabilistic process need not advance monotonically through its computation; it repeatedly undoes and redoes history. Its equilibrium remembers whether a halting configuration exists arbitrarily far along the ray. Finite computation and random reversible motion have been separated without discarding either.

Remark 57.2 (Domain of the irreducibility claim). The full set of counter-history configurations contains many different components. Irreducibility is asserted on the component generated by the specified initialized state, not on their disjoint union. There is no hidden reachability test in the local inverse: tag, quotient, control and guard checks suffice. An arbitrary malformed-history state may simply have no predecessor. The globally defined local system is not claimed to be one irreducible chain.

Example 57.3 (The implemented countdown). Use control 0 to decrement counter x when positive and return to 0; when it is zero go to halt control 1. Let control 2 be **done**. With tags 1, 2, 3, 4 and base 5, an input $x = 3$ starts

$$(0, 3, 0) \rightarrow (0, 2, 1) \rightarrow (0, 1, 6) \rightarrow (0, 0, 31) \rightarrow (1, 0, 157) \rightarrow (2, 0, 788) \rightarrow (2, 0, 3944) \rightarrow \dots$$

There is one defect at trajectory index 4. Its root mass is $2^5/(2^6 - 1) = 32/63$. Each step is reversible using the last base-five tag. The code verifies this example and a nonhalting increment loop; it does not implement a full universal counter interpreter.

The history multiplication in [equation \(57.1\)](#) is a primitive operation of the stated guarded-affine substrate. Replacing it by unit-counter microsteps changes the trajectory clock and requires its own compiler analysis. Nothing here asserts universality of a homogeneous nearest-neighbor walk on $\mathbb{N}$, a finite-state Markov chain, or an ungarded one-species chemical reaction network.

58 Sparse defects and the rational–Liouville dichotomy

[source 17]

The one-defect example is enough for an equality obstruction. To make the arithmetic type itself encode a harder property, allow defects at widely separated computable positions.

Lemma 58.1 (Sparse defects force Liouville approximation). *Suppose infinitely many defects occur, and put $m_j = d_j + j$. If*

$$\frac{m_{j+1}}{m_j} \longrightarrow \infty, \quad (58.1)$$

then α_b is a Liouville number, in particular transcendental.

Proof. Let $L_j = \sum_{i \leq j} 2^{-m_i} = A_j/2^{m_j}$ and $L = 2 - Z_b$. The remaining terms satisfy

$$0 < L - L_j < 2^{1-m_{j+1}}. \quad (58.2)$$

The rational approximants

$$\alpha_j = \frac{1}{2 - L_j} = \frac{2^{m_j}}{2^{m_j+1} - A_j} \quad (58.3)$$

have denominators at most 2^{m_j+1} even after reduction. Also

$$0 < \alpha_b - \alpha_j = \frac{L - L_j}{(2 - L)(2 - L_j)} < 2^{1-m_{j+1}}.$$

For any fixed positive integer k , condition [equation \(58.1\)](#) eventually gives $m_{j+1} > k(m_j + 1) + 2$. Thus the error is smaller than the k th inverse power of the displayed denominator, and hence of the reduced denominator.

To check irrationality independently of these estimates, note that the defect gaps tend to infinity. An eventually periodic word with infinitely many ones has bounded gaps between ones. Hence b is not eventually periodic, and [theorem 55.3](#) makes α_b irrational. Its distinct rational approximants consequently have unbounded reduced denominators. The estimates just proved are the Liouville property. The elementary algebraic lower bound in [section 66](#) excludes algebraicity. $\square$

Fix a standard enumeration W_e of computably enumerable sets, with finite nested primitive-recursively decidable stages $W_{e,s}$ and $W_{e,-1} = \emptyset$. Let $E(e, s)$ hold when at least one new *distinct* element enters at stage s . Set

$$b_n^e = 1 \iff \exists s \leq n \text{ (} n = (s + 2)! \text{ and } E(e, s) \text{)}. \quad (58.4)$$

Both the factorial calculation and the finite-stage event test are bounded primitive-recursive computations. Consequently this is a jointly primitive-recursive family of environments.

Theorem 58.2 (The arithmetic-type dichotomy). *For the chains defined by equation (58.4),*

$$W_e \text{ finite} \iff \alpha_e \text{ rational} \iff \alpha_e \text{ algebraic}, \quad (58.5)$$

$$W_e \text{ infinite} \iff \alpha_e \text{ irrational} \iff \alpha_e \text{ transcendental}. \quad (58.6)$$

In the infinite case the mass is in fact Liouville. Every chain still satisfies the same confinement and mixing bounds.

Proof. A finite set has only finitely many stages at which a new distinct element appears. Hence there are finitely many defects, so theorem 55.1 gives a rational mass.

If W_e is infinite, list its event stages as $s_1 < s_2 < \dots$. Then $d_j = (s_j + 2)!$ and $m_j = (s_j + 2)! + j$. Since $j \leq s_j + 1$,

$$\frac{m_{j+1}}{m_j} \geq \frac{(s_j + 3)!}{(s_j + 2)! + s_j + 1} \rightarrow \infty.$$

Apply lemma 58.1. The analytic bounds hold for every binary environment, so they are unchanged. $\square$

58.1 The exact arithmetical-hierarchy classification

For clarity we include the elementary completeness fact used by the reduction. Write $\text{FIN} = \{e : W_e \text{ is finite}\}$ and $\text{INF} = \mathbb{N} \setminus \text{FIN}$.

Lemma 58.3. *The set FIN is Σ_2^0 -complete and INF is Π_2^0 -complete.*

Proof. Finiteness has the form $\exists M \forall s W_{e,s} \subseteq \{0, \dots, M\}$, which is Σ_2^0 . For hardness consider any predicate $A(e) \iff \exists u \forall v R(e, u, v)$ with decidable R . At stage s let a_s be the least $u \leq s$ satisfying all the finitely many tests $R(e, u, v)$ for $v \leq s$, and put $a_s = s + 1$ when none does. Enumerate the successive values a_s . The sequence is nondecreasing: a failed small candidate never becomes valid later. If there is a genuine witness u , the least genuine witness eventually becomes the permanently least candidate, and the enumerated range is finite. If there is none, every candidate is eventually rejected and the range is infinite. This gives a computable reduction to FIN; taking complements proves the other statement. $\square$

Theorem 58.4 (Complete classification for primitive-recursive codes). *Among all primitive-recursive binary-environment codes, rationality and algebraicity of α_b are Σ_2^0 -complete. Irrationality and transcendence are Π_2^0 -complete. Neither rationality, algebraicity, irrationality nor transcendence has a uniform existential Diophantine representation on these codes.*

Proof. By theorem 55.3, rationality can be written

$$\exists N \exists p \geq 1 \forall n \quad b(N + n) = b(N + n + p),$$

a Σ_2^0 formula under a total computable evaluation of primitive-recursive syntax.

For algebraicity, first note that α_b is uniformly computable from its code. For example, geometric tail bounds in equation (53.1) give arbitrarily narrow rational enclosures; section 60 will give sharp ones. For a nonzero $P \in \mathbb{Z}[X]$, evaluation of $P(\alpha_b)$ is therefore uniformly computable. Its equality to zero is Π_1^0 : compute an approximation r_k with error at most 2^{-k} and require $|r_k| \leq 2^{-k}$ for every k . Existence of a nonzero integer polynomial satisfying this condition is a Σ_2^0 definition of algebraicity.

The map $e \mapsto b^e$ in equation (58.4) is a computable map to primitive-recursive syntax. By theorem 58.2 and lemma 58.3, it reduces FIN to each of rationality and algebraicity, proving completeness. Complementation proves the two Π_2^0 classifications. Strictness of the arithmetical hierarchy implies that none of these complete sets is computably enumerable; MRDP therefore precludes the stated existential representations. $\square$

Only the standard hierarchy separation is used in the last sentence; no assertion that the four predicates are equivalent on arbitrary environments is intended. Their pairwise coincidence in equations (58.5) and (58.6) is a special feature of the sparse reduction family. In particular, this proof does not settle whether an algebraic irrational root mass occurs somewhere in the full Cantor set $\mathcal{C}$.

59 Low-complexity environments and an explicit transcendental mass

[source 17]

Sparse coding gives powerful uniform reductions, but one can also write a short, familiar formula for an individual environment whose root mass is transcendental. Let $p_b(n)$ denote the number of distinct length- n contiguous blocks occurring in b .

Lemma 59.1 (Complexity of the equilibrium digit code). *For $c = h(b)$ with $h(0) = 0$, $h(1) = 10$,*

$$p_c(n) \leq 2p_b(n+1) \quad (n \geq 1). \quad (59.1)$$

Proof. Any length- n block of c starts either at the beginning of an image word or at the second symbol of an image of 1. It is determined by that offset and the next $n+1$ input symbols: every image has length at least one, so this many symbols always suffice. There are at most two offsets per input block. $\square$

Theorem 59.2 (A low-complexity transcendence criterion). *Suppose b is not eventually periodic and*

$$\liminf_{n \rightarrow \infty} \frac{p_b(n)}{n} < \infty. \quad (59.2)$$

Then Z_b and α_b are transcendental.

Proof. Theorem 1 of Adamczewski and Bugeaud [84] says that the base- B digit complexity of every algebraic irrational satisfies $\liminf p(n)/n = \infty$, for each integer $B \geq 2$. By [theorem 55.3](#), $L_b = 2 - Z_b$ is irrational. Its binary expansion is $c = h(b)$ by [theorem 55.1](#), and [lemma 59.1](#) and [equation \(59.2\)](#) give finite liminf complexity ratio for that expansion. Thus L_b cannot be algebraic. Subtracting from two and taking a nonzero reciprocal preserve algebraicity in both directions, so Z_b and α_b are transcendental too. $\square$

This proof imports a deep digit-complexity theorem. The contribution here is the explicit equilibrium coding that makes it applicable; there is no new proof of the general transcendence theorem.

Corollary 59.3 (Thue–Morse equilibrium). *Let b_n be the parity of the number of ones in the binary expansion of n . Then*

$$\alpha_{\text{TM}} = \left(\sum_{n \geq 0} 2^{-n - \sum_{k < n} b_k} \right)^{-1} \quad (59.3)$$

is transcendental, and it is the root mass of a chain satisfying [theorem 54.2](#).

Proof. The Thue–Morse word satisfies the length-two substitution $0 \mapsto 01$, $1 \mapsto 10$. Its block complexity is $O(n)$ by a direct covering argument: choose k with $2^{k-1} < n \leq 2^k$. A length- n block lies in the images, under the k fold substitution, of at most two consecutive letters. It is determined by one of at most four letter pairs and an offset smaller than 2^k . Thus $p_b(n) \leq 4 \cdot 2^k < 8n$, with the small case adjusted trivially.

It is not eventually periodic. If a positive eventual period p existed, for all sufficiently large k one would have $b_{2^k-p} = b_{2^k} = 1$. But $2^k - p = (2^k - 1) - (p - 1)$ is the k -digit binary complement of $p - 1$, so

$$b_{2^k-p} \equiv k - s_2(p-1) \pmod{2},$$

which alternates as k increases. This is a contradiction. Apply [theorem 59.2](#) and then [theorem 54.2](#). $\square$

Unlike the sparse construction, this argument does not show that α_{TM} is Liouville. It establishes transcendence without claiming an irrationality exponent. The supplied code gives an exact rational enclosure from its first eighty environment bits.

60 Sharp finite-prefix bounds and canonical quartic certificates

[source 17]

The preceding obstructions concern exact arithmetic type. Strict rational inequalities have a different status: a finite prefix eventually certifies every true strict comparison. This section makes that statement completely explicit and separates its arithmetic layer from the computation of the prefix itself.

60.1 Exact enclosing intervals

For a given prefix $b_0, \dots, b_{N-1}$, introduce integer sequences

$$W_0 = S_0 = Q_0 = 1, \quad (60.1)$$

$$\begin{aligned} W_{i+1} &= (2 - b_i)W_i, \\ S_{i+1} &= 4S_i + W_{i+1}, \\ Q_{i+1} &= 4Q_i. \end{aligned} \quad (60.2)$$

By induction,

$$W_i = 4^i w_i, \quad S_i = 4^i \sum_{k=0}^i w_k, \quad Q_i = 4^i. \quad (60.3)$$

All quantities here are computed from only i flags.

Theorem 60.1 (Sharp prefix enclosure). *Every continuation of the prefix satisfies*

$$L_N := \frac{Q_N}{S_N + W_N} \leq \alpha_b \leq U_N := \frac{3Q_N}{3S_N + W_N}. \quad (60.4)$$

Both endpoints are attained by continuations: all zeros attain L_N , and all ones attain U_N . The enclosing intervals are nested as N increases, and

$$0 \leq U_N - L_N \leq \frac{2}{3} 2^{-N}. \quad (60.5)$$

They therefore converge to the single value α_b .

Proof. Let $s_N = \sum_{k=0}^N w_k$ and $w = w_N$. Ratios after the prefix lie between $1/4$ and $1/2$, so

$$\frac{w}{3} \leq \sum_{k>N} w_k \leq w.$$

The lower and upper bounds are attained by constant-one and constant-zero suffixes, respectively. Reciprocation reverses the inequalities, yielding [equation \(60.4\)](#). Appending a flag restricts the set of possible continuations, which proves nesting because the two extrema were attained. Finally,

$$U_N - L_N = \frac{(2/3)w}{(s_N + w/3)(s_N + w)} \leq \frac{2}{3}w \leq \frac{2}{3} 2^{-N}.$$

Here $s_N \geq 1$ and $w \leq 2^{-N}$. The bound proves convergence. $\square$

Remark 60.2. The interval is the smallest closed interval enclosing the attainable masses of the prefix cylinder. It is not asserted that every number between its endpoints is attainable: the actual set is a reciprocal affine image of the Cantor set from [section 55](#).

60.2 A small-coefficient recurrence presentation

Let $a \in \mathbb{N}$, $v \geq 1$ represent the rational threshold a/v . A lower certificate asserts that the whole prefix cylinder satisfies $\alpha_b > a/v$; an upper certificate asserts $\alpha_b < a/v$. The necessary and sufficient integer tests are

$$vQ_N - a(S_N + W_N) = 1 + \eta \quad (\text{lower}), \quad (60.6)$$

$$a(3S_N + W_N) - 3vQ_N = 1 + \eta \quad (\text{upper}), \quad (60.7)$$

with a natural slack η . These are strict inequalities because the left-hand sides are integers.

Theorem 60.3 (Canonical recurrence certificates). *Fix N . Take the residuals of equations (60.1) and (60.2), the N Boolean residuals $b_i(b_i - 1)$, and one threshold residual from equations (60.6) and (60.7). The sum of their squares is an integer polynomial of total degree at most four. Its free parameters are the flags and a, v , and its natural witnesses are*

$$(W_0, \dots, W_N, S_0, \dots, S_N, Q_0, \dots, Q_N, \eta).$$

There are $3N + 4$ witnesses and $4N + 4$ residuals. For valid binary flags and $v \geq 1$, it has exactly one natural witness tuple if the selected prefix bound strictly certifies the comparison, and none otherwise.

Proof. Each residual has degree at most two in all its variables, including parameters. Squaring and summing gives degree at most four. A sum of integer squares vanishes exactly when every residual vanishes. The three initial values and the deterministic recurrences force all W, S, Q coordinates uniquely. The final residual then forces η uniquely; it is natural exactly when the appropriate integer gap is positive. There are three initial residuals, four per flag, and one threshold residual. Counting coordinates gives $3(N + 1) + 1$. $\square$

For fixed numerical flags and threshold the degree in the auxiliary coordinates alone can be lower. The quartic claim refers to the polynomial with all input parameters left as variables. No optimality claim is made for this presentation.

60.3 Eliminating the redundant witnesses

When N is an external compiler parameter, the Q_i are constants and the S_i can be eliminated by a linear expression. Define

$$W_0 := 1, \quad A_N(W) = 4^N + \sum_{i=1}^N 4^{N-i} W_i. \quad (60.8)$$

This equals S_N whenever the weight recurrences hold.

Theorem 60.4 (Compact unique quartic certificates). *For every fixed horizon N , define*

$$\begin{aligned} C_i &= b_i(b_i - 1), \\ R_i &= W_{i+1} - (2 - b_i)W_i \quad (0 \leq i < N), \\ G_- &= v4^N - a(A_N(W) + W_N) - 1 - \eta, \\ G_+ &= a(3A_N(W) + W_N) - 3v4^N - 1 - \eta. \end{aligned} \quad (60.9)$$

Use G_- for a lower certificate and G_+ for an upper certificate. Then

$$\mathcal{P}_N^\pm = \sum_{i=0}^{N-1} (C_i^2 + R_i^2) + G_\pm^2 \quad (60.10)$$

is a polynomial of degree at most four with exactly $N + 1$ natural witnesses $W_1, \dots, W_N, \eta$ and $2N + 1$ squared residuals. For every valid prefix and threshold it has the same existence and unique-witness property as theorem 60.3.

Proof. The residuals are obtained by substituting $W_0 = 1$, $Q_N = 4^N$ and $S_N = A_N(W)$ into the full presentation. Each remains of degree at most two. The weight recurrence forces $W_1, \dots, W_N$ uniquely; the threshold forces η uniquely. Their forced values satisfy all residuals exactly when the corresponding strict prefix bound holds. The count is immediate, including $N = 0$, when the only witness is η and the only residual is the threshold residual. $\square$

This compression trades recurrence variables for coefficients as large as 4^N . Each coefficient has $O(N)$ bits. The unexpanded residual list has $O(N)$ terms and $O(N^2)$ coefficient bits with explicit binary storage; the geometric coefficient sequence can also be generated from N . The full presentation has uniformly small recurrence coefficients and is useful when composing with other arithmetic circuits. Neither presentation claims an optimal variable count.

Example 60.5 (An explicit four-witness quartic). For $N = 3$, a lower certificate uses witnesses W_1, W_2, W_3, η . The polynomial is

$$\begin{aligned} & \sum_{i=0}^2 b_i^2 (b_i - 1)^2 + (W_1 - 2 + b_0)^2 + (W_2 - (2 - b_1)W_1)^2 \\ & + (W_3 - (2 - b_2)W_2)^2 \\ & + (64v - a(64 + 16W_1 + 4W_2 + 2W_3) - 1 - \eta)^2. \end{aligned}$$

It has seven squared residuals and, when expanded with all parameters free, 51 monomials. For the prefix $(0, 1, 0)$ and threshold $1/2$, its unique natural witness is

$$(W_1, W_2, W_3, \eta) = (2, 2, 4, 15).$$

The sharp enclosure is $[4/7, 24/41]$, whose lower endpoint exceeds $1/2$. The archive includes the expanded polynomial and its residual specification, not merely a schematic existence assertion.

60.4 From bounded certificates to fixed-polynomial cuts

Theorem 60.6 (Diophantine strict cuts, and a boundary at equality). *For primitive-recursive environment codes e , the two relations*

$$\alpha_e > a/v \quad \text{and} \quad \alpha_e < a/v \quad (a \in \mathbb{N}, v \geq 1)$$

are computably enumerable and hence Diophantine after tuple coding. Both are Σ_1^0 -complete as relations with a rational threshold parameter. Exact equality to a rational threshold is Π_1^0 -complete and is not in general Diophantine.

Proof. Compute successive prefixes from the actual code e and their sharp bounds. If $\alpha_e > a/v$, eventually $L_N > a/v$, and if $\alpha_e < a/v$, eventually $U_N < a/v$. These tests never certify a false statement. They provide the two semidecision procedures and, by ProveIt's MRDP interface, fixed-polynomial representations.

The lower relation is complete by the one-defect family with threshold $1/2$. For the upper relation, set $b_n^e = 1$ exactly when the machine has not halted by time n . A nonhalting machine gives the all-one environment and mass $3/4$. A halting machine produces a zero suffix, strictly increases Z from $4/3$, and makes its mass strictly less than $3/4$. This is a reduction of halting to the upper relation.

For equality, the condition that a/v belongs to every computable interval $[L_N, U_N]$ is Π_1^0 and is equivalent to equality. The one-defect construction with threshold $1/2$ gives Π_1^0 -hardness and failure of computable enumerability. $\square$

[write] Cf. [proposition 9.4](#) of Part I (same halting-time device).

Two boundaries are essential. First, the flags in [theorem 60.4](#) are input parameters. They must be tied to the actual environment program by a verified prefix computation; allowing arbitrary flag witnesses would

certify statements about some different chain. A complete front end can use finite evaluator traces for the prefix predicate, followed by MRDP or a separate bounded trace compiler. The supplied implementation checks concrete supplied prefixes and does not implement this entire universal arithmetic front end.

Second, N is an external compiler parameter in [theorem 60.4](#). Passing from its horizon-indexed family to one fixed-arity polynomial is an MRDP step. The unique-witness assertion for a fixed prefix is not a single-fold representation of arbitrary unbounded computation. The existence theorem supplies no such multiplicity control, and nothing in this article settles the single-fold or finite-fold representability frontier discussed in ProveIt’s related reports ([write] Section 36.4 of this report, and the sibling report `canonical-d iophantine-certificates`).

61 Worked equilibria and reproducible verification

[source 17]

61.1 Exact examples

The defect-free environment has $Z = 2$ and $\alpha = 1/2$. The all-one environment has $Z = 4/3$ and $\alpha = 3/4$. For the periodic word $(01)^\omega$, its two-letter affine map is $z \mapsto 3/2 + z/8$, so $Z = 12/7$ and $\alpha = 7/12$. For a single defect the first few values are

Defect time T	0	1	2	5	10
Root mass	2/3	4/7	8/15	64/127	2048/4095

No finite truncation distinguishes the zero-defect case from a case whose one defect lies beyond the truncation, although their exact stationary masses differ.

For the Thue–Morse chain, eighty flags give the exact enclosure

$$\alpha_{\text{TM}} \geq \frac{664613997892457936451903530140172288}{1115932292267376191302758860221799791},$$

$$\alpha_{\text{TM}} \leq \frac{498460498419343452338927647605129216}{836949219200532143477069145166349843}.$$

The approximate display is 0.5955683893169543; the fractions, not the decimal display, are the certified numerical data. Their width is recorded exactly in `results/examples.json`.

61.2 What the programs do

The core module `code/equilibria.py` uses Python’s standard library and exact rational arithmetic. It implements transition rows, scaled prefix recurrences, sharp intervals, periodic-tail normalizers, the finite-defect formula, rational membership decoding, both certificate presentations, and the local counter-history lift. Invalid bit values, negative natural witnesses and nonpositive rational denominators are rejected explicitly. The rational decoder raises an exception on an imposed operational step limit rather than mistaking that limit for nonmembership; setting the limit to `None` runs the terminating mathematical procedure without a cap.

The verification driver `code/verify.py` uses SymPy only for symbolic expansion of the two horizon-three polynomials. Its random choices, limited to additional finite Poincare checks, use the fixed seed 20260930. The recorded run used Python 3.13.5 and SymPy 1.14.0 and passed **113,686 explicit checks**. Checks are performed by a custom assertion function, so Python’s optimization flag does not silently disable them.

The suite exhausts all 511 binary prefixes of lengths zero through eight for defect identities, endpoint bounds, interval nesting and several threshold tests. It also checks rational-decoder round trips for 930

supplied eventually periodic codes, halting formulas through $T = 100$, forward/backward counter histories, and coordinate mutations of valid certificates. There are 320 exact rational Poincare checks on finite reflecting chains. The full horizon-three polynomial has 63 monomials; the compact one has 51. All counts are emitted by the run rather than manually inferred from the output files.

These checks are finite regression tests, not proofs of the infinite-state spectral theorem, transcendence, hierarchy completeness, or universality. Those conclusions rest on the mathematical arguments and explicitly cited theorems. No Lean build was performed, no universal MRDP polynomial was expanded, and no proof-checker certification is claimed for the new results.

61.3 Reproduction

From the unpacked archive directory, run

```
python -m pip install -r requirements.txt
python code/verify.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The verification run rewrites only files in `results/` and its JSON report in `verification/`. The TeX source includes its bibliography and requires no external figures or data downloads. The README identifies the core-only and symbolic-check dependencies and the exact scope of every artifact. [write] In this report the programs are shipped as `code/17-reversible-equilibria-equilibria.py` and `code/17-reversible-equilibria-verify.py`, and the files of `results/` and `verification/` as `data/17-reversible-equilibria-*`; `requirements.txt` is `data/17-reversible-equilibria-requirements.txt`. The commands above are the source's, for its delivered layout; the README gives the rerun recipe (SymPy 1.14.0 is needed only by `verify.py`).

62 A formalization route for ProveIt

[source 17]

The natural integration point is the existing Hilbert-tenth/MRDP project, but the new analysis should be kept separate from the old formal theorem boundary. A useful dependency order is the following.

Finite arithmetic first. Define the W, S, Q recurrences over naturals and prove [equation \(60.3\)](#) by induction. Formalize polynomial evaluation for the residual lists and prove that a sum of integer squares vanishes if and only if every residual does. The canonical-witness and elimination proofs in [theorems 60.3](#) and [60.4](#) are finite algebra and induction, with no probability infrastructure. The exact horizon-three artifact can then be checked as an instance.

Infinite sums and interval semantics. Define the positive weights and prove the summability sandwich $4^{-n} \leq w_n \leq 2^{-n}$. Establish Z , its reciprocal, and [theorems 55.1](#) and [60.1](#). At this point a finite arithmetic certificate already has a theorem connecting it to the actual real root mass; that is a more important trust milestone than simply being able to evaluate a polynomial.

Computability and reductions. Specify a bounded machine evaluator and its first-halt flag. Prove the single-defect formula and the sparse-event compiler's correctness. Show that the strict-cut semideciders use the correct prefixes, then apply `Diophantine.mrdp`. Rationality's eventual periodicity characterization can be formalized without using transcendence. The sparse transcendence proof needs only a small algebraic-approximation lemma, whereas the low-complexity corollary requires a much larger external theorem.

Analysis as an independent layer. Construct the Markov kernel, verify detailed balance, and establish the weighted ℓ^2 convolution estimate. The elementary inequality in lemma 54.1 is the critical reusable lemma. The spectral and total-variation consequences can then be derived using the chosen operator and probability libraries. Their formalization should not be made an unnecessary dependency of the finite polynomial checker.

Local reversible semantics. For the counter lift, prove the quotient/remainder tag decoder is a left inverse of each forward rule, prove strict history growth, and identify the rooted orbit with $\mathbb{N}$. Only after those statements are proved should the chain on local configurations inherit the analytic results. This keeps logical reversibility, Markov reversibility and reachability-domain restrictions distinct.

All of these are proposed integration tasks, not existing source modules or completed Lean proofs. ProveIt’s own MRDP guide explicitly separates its existential polynomial theorem from numerical degree, witness and compiler guarantees [77]; the same separation is preserved here.

63 Source 13 appendix: a compact proof-dependency ledger

Result	Mathematical dependencies	Evidence in this package
Finite compiler	Acyclic evaluation; nonnegative sums of squares over reals	Full proof; executable compiler; exact sparse outputs
Height four and graph degree three	Trunk routing; occurrence ownership; pair uniqueness	Full proof; exact structural audits
Hessian interval [2, 20]	Jacobian identity; matrix norm inequality	Full proof; numerical smoke tests only as supplementation
Rounding gap 1/64	Scalar rounding; local Lipschitz bound; integrality	Full proof; rational finite tests
Convex low-energy components	Hessian estimate on boxes; disjoint Boolean neighborhoods	Full proof; no topology-checker implementation
Degree-two counter	Boolean truth tables; path/cycle transfer recurrence	Full proof; executable dynamic program; exhaustive comparisons
Probability realization	Finite tape counting; bounded circuit simulation; monotone limits	Full proof; OR-family examples; no universal frontend
Fixed-arity quartics D, Q, V	Classical MRDP; arithmetic quadratization	Existence/normal-form proofs; no expanded universal polynomial supplied
Hierarchy completeness	Universal halting; explicit sampling reductions	Self-contained arguments; classical status credited

64 Source 13 appendix: reproducibility and machine-readable data

The JSON polynomial format stores each monomial as a sorted list of variable indices. Repetition denotes exponentiation: the list [2,2,5] denotes $X_2^2 X_5$. Each entry records its integer coefficient. Residual lists, source circuit gates, and structural statistics accompany the expanded polynomial, allowing an independent reader to re-evaluate both representations.

The verification program reruns deterministically and rewrites `results/verification_report.json`. The optional NumPy check can produce insignificant last-bit differences across linear-algebra implementations; exact counts, exact polynomial coefficients, and rational bounds do not depend on NumPy. The supplied PDF was built from this L^AT_EX source. The archive excludes TeX auxiliary files and Python cache files.

65 Source 17 appendix: why the common spectral-gap bound is sharp

For the defect-free environment, $\pi(n) = 2^{-n-1}$. Under the unitary map $f \mapsto g$ with $g_n = \sqrt{\pi(n)}f(n)$ from $L^2(\pi)$ to $\ell^2(\mathbb{N})$, the Markov operator becomes a symmetric Jacobi operator A . Away from zero its diagonal entry is $1/4$ and its off-diagonal entries are

$$\sqrt{P(n, n+1)P(n+1, n)} = \frac{1}{2\sqrt{2}}.$$

At zero the diagonal is $3/4$. Its stationary unit vector is $\psi_n = \sqrt{\pi(n)}$, with $A\psi = \psi$.

Take a vector supported on L consecutive sites beginning at $M \geq 1$, with values proportional to $\sin(\pi j/(L+1))$, $j = 1, \dots, L$, and normalize it to unit ℓ^2 norm. The constant-coefficient tridiagonal calculation gives

$$\langle g, (I - A)g \rangle = \frac{3}{4} - \frac{1}{\sqrt{2}} \cos\left(\frac{\pi}{L+1}\right).$$

Subtract its projection onto ψ : $\tilde{g} = g - \langle g, \psi \rangle \psi$. The energy is unchanged because $(I - A)\psi = 0$, while

$$\|\tilde{g}\|_2^2 = 1 - |\langle g, \psi \rangle|^2 \rightarrow 1 \quad (M \rightarrow \infty),$$

since the ℓ^2 tail of ψ tends to zero. Thus the mean-zero Rayleigh quotient can be made arbitrarily close to the displayed value. Letting first M and then L tend to infinity yields

$$\text{gap}(P_0) \leq \frac{3}{4} - \frac{1}{\sqrt{2}} = \frac{3 - 2\sqrt{2}}{4}.$$

The reverse inequality is [theorem 54.2](#). Therefore the all-zero environment itself attains the asserted gap, and no larger common lower bound is possible. This argument does not require the spectral edge to be an eigenvalue.

66 Source 17 appendix: the elementary algebraic approximation bound

Suppose ξ is algebraic irrational of degree $d \geq 2$, and let $P \in \mathbb{Z}[X]$ be its irreducible polynomial. For a rational p/q in lowest terms, $P(p/q) \neq 0$ and $q^d P(p/q)$ is a nonzero integer. Hence $|P(p/q)| \geq q^{-d}$. If $|p/q - \xi| \leq 1$, the mean value estimate on $[\xi - 1, \xi + 1]$ gives

$$|P(p/q) - P(\xi)| \leq M|p/q - \xi|, \quad M = \max_{|x-\xi| \leq 1} |P'(x)| > 0.$$

Consequently

$$|p/q - \xi| \geq M^{-1}q^{-d}.$$

A Liouville number has rational approximations with arbitrarily large denominators and error smaller than their $(d+1)$ st inverse powers, contradicting this inequality for large denominators. This elementary argument is the only transcendence input needed for the factorially sparse family; the historical source is Liouville [83].

67 Source 17 appendix: source and claim ledger

The repository was inspected at the fixed commit `e18718e837d43e162252f9a314e8cb797fbd1a1f`, whose recorded commit timestamp is 30 September 2026, 19:50:53 UTC. The repository itself was not modified. The inspected MRDP guide, project README, related-report README and commit metadata are identified in

PROVENANCE.md. This pin controls claims about repository contents even if the default branch subsequently changes.

The following distinction should be retained in any merged report.

Layer	Status
ProveIt's MRDP interface	Existing repository theorem, used with its stated natural-witness existence scope; not rebuilt in this work.
Stationary coding, sharp gap, hierarchy reductions, finite certificates	Mathematical proofs in this manuscript; no independent proof-assistant verification.
Low-complexity transcendence	An application of Adamczewski–Bugeaud, with its imported theorem explicitly identified.
Python/SymPy artifacts	Executed finite exact-arithmetic and symbolic regression checks; not proofs of infinite or undecidable statements.
Local reversible counter front end	General semantic proof and executable small examples; no full universal interpreter supplied.
Historical novelty	The combination and explicit constructions are proposed contributions; no exhaustive priority certification.

68 Conclusions of Part V

68.1 Source 13

A finite probabilistic computation can be translated not just into an existential polynomial relation, but into a sparse quartic landscape with one explicitly controlled accepting minimum per successful coin tape. Fresh occurrence wires eliminate auxiliary multiplicity; a single trunk per used wire makes the interaction graph subcubic; one distributed unit keeps all expanded coefficients bounded. Boolean penalties give coercivity, separation, and a uniform Hessian contribution. These elementary pieces combine to yield fixed rounding, convexity, topology, and perturbation bounds.

The resulting normalized component counts realize every left-c.e. probability, including noncomputable ones. Their difficulty does not require poorly conditioned accepting minima. It arises from unbounded computation and from the distinction between finite evidence and a limiting measure. MRDP captures strict lower-cut evidence in fixed arity; almost-sure and finite-mean claims retain their required universal quantifiers. This is both a constructive extension of the repository's Diophantine program and a boundary that prevents misleading generalizations.

68.2 Source 17

Reversibility, a fixed rational transition alphabet, strong confinement and a uniform spectral gap do not remove computational complexity from exact equilibrium arithmetic. The mechanism is explicit: defects carry a binary message in the normalizing constant, while their inward bias preserves the analytic estimates. A first-halting defect produces only rational equilibria but destroys uniform symbolic extraction. Sparse defects raise arithmetic-type recognition to the second level of the arithmetical hierarchy, with a rational–Liouville dichotomy. Low symbolic complexity gives another route to individual transcendental examples.

At the same time, the model has an effective positive theory. Numerical enclosures are sharp, the root-mass range has a faithful self-similar coding, supplied rational membership is decidable, and strict inequalities have canonical finite quartic certificates. The useful Diophantine boundary is therefore not between conventional and unconventional computation. It is between properties witnessed by finite information and properties that quantify over an unbounded tail. ProveIt's MRDP interface is well suited to formalizing that boundary, provided the finite certificate, the environment semantics and the fixed-polynomial existence theorem remain explicitly distinguished.

69 Further research questions of Part V

[write] Source 13 posed ten numbered questions and source 17 twelve question environments; both sets are printed in full. A [write] note after a question names the question of Parts I–IV it duplicates or bears on. Which of Part I’s questions these sources partly answer is recorded after those questions and in Section 36.4.

69.1 Source 13: probabilities from well-conditioned quartics

1. Planar subcubic realization. Can the entire quantitative package be realized with a planar interaction graph, while preserving a unique extension for each input tape and linear or near-linear overhead? An ordinary Boolean crossover gadget is not enough: its witness multiplicity, unit distribution, and Hessian constants must all be audited. A positive answer would bring the construction closer to spatially local two-dimensional substrates.

2. Coefficient height three or two. The present height-four term occurs at a Boolean copy-tree vertex incident to three linear constraints. Can a different parsimonious fan-out gadget reduce the expanded height without losing subcubic interactions and uniform conditioning? Establish either a compiler with a better constant or a lower bound for a clearly specified local-gadget class.

3. Geometry-preserving composition. Our compiler rebuilds a circuit at each horizon. Can one construct a local extension rule for the energy itself, so that the injection $\mathcal{Z}_t \times \{0, 1\} \hookrightarrow \mathcal{Z}_{t+1}$ becomes an explicit embedding of well neighborhoods? This could give a more intrinsic topology of the increasing computation rather than merely a sequence of component counts.

4. Effective convergence from semantic restrictions. Which program classes yield computable upper as well as lower bounds for the normalized counts? Finite-state models, effective tail bounds, and suitable supermartingale certificates are plausible starting points. The target is an explicit modulus extracted from extra semantics, not from the universal geometric constants alone. Existing probabilistic termination proof systems provide relevant background [54]. [write] Compare Question 5.

5. Probability-preserving quantum analogues. For finite quantum computations, amplitudes of different paths interfere; one cannot replace amplitude addition by a count of successful classical tapes. An exact analogue should handle an explicitly presented algebraic gate field, interference, and Born probabilities without smuggling a counting oracle into an existential witness. The classical positivity argument identifies precisely which step needs replacement. [write] Partly answered for finite circuits by source 10 in Part VI: its cyclotomic contraction certificates (Theorems 74.1 and 75.1) keep interference and the Born rule, with one witness per correct output; unbounded quantum controllers are reached there only through MRDP (Theorem 79.1).

6. Resource-sensitive substrate transfers. For a chosen randomized combinator, tag, or heap machine, determine sharp pathwise simulation bounds. Probability and almost-sure termination need only coin faithfulness, but finite moments require quantitative overhead control. Establishing preservation of the r th moment under a specified simulation would be more informative than an undifferentiated claim of probabilistic universality. [write] Compare the question on cost-preserving compilation between substrates in the sibling report `canonical-diophantine-certificates`.

7. Fixed-arity versus finite counting interfaces. Can useful subclasses of threshold predicates admit explicit small fixed-arity quartics whose witnesses have a controlled, computable multiplicity? Such a result must state its subclass and must not infer a general finite-fold theorem from the bounded Boolean compiler. A good first target is a bounded-state process with an effective absorption-tail certificate. [write] Compare Questions 10 and 19.

8. Robustness under local coefficient perturbations. [Theorem 42.7](#) assumes a globally bounded smooth perturbation. Physical or numerical models often perturb local coefficients instead. Find size-independent hypotheses under which such local perturbations preserve the low-energy component count, possibly after shifting the reference energy separately in each well. Large ambient domains and accumulated constant shifts are real obstacles, not cosmetic technicalities.

9. Certified optimization-to-witness extraction. The rounding theorem can turn an approximate real point into an exact Boolean certificate. Develop a verified interval-arithmetic interface that checks the energy upper bound without trusting floating-point residual cancellation, then extracts and rechecks the discrete witness. This would exploit the proven gap without asserting that global optimization is easy.

10. Minimal structural obstructions at degree two. The degree-two counter works because residual scopes lie on paths, cycles, and isolated triangles. Study analogous exact frontiers for bounded treewidth, bounded feedback-vertex number, or constrained fan-out. The aim is a classification of which structural promises make well counting tractable while retaining the same local geometric bounds. [write] Compare the width-parameterized contraction certificates of source 10 ([Theorem 73.1](#)).

69.2 Source 17: rapidly mixing reversible computation

The following questions are proposed continuations of this construction. They are not all asserted to be previously published open problems. They specify what would constitute a further theorem or a useful implementation, rather than treating a finite experiment as a result.

Research question 33 (A small kernel-checked equilibrium certificate). Can the compact prefix checker, its unique-witness theorem and its real-interval semantics be formalized together as a short independent module? The first target is [theorem 60.4](#) linked to [theorem 60.1](#), not a full Markov-chain library. A useful deliverable would accept a concrete prefix certificate and return a theorem about the exact infinite equilibrium. [write] Compare Question 1.

Research question 34 (A reusable sharp half-line gap theorem). How generally can [lemma 54.1](#) be formalized and extended? For varying death probabilities bounded below by q , the same convolution estimate still suggests a direct lower bound. A sharper result could use nonuniform products of ratios rather than a single worst-case r , with mechanically checkable drift certificates.

Research question 35 (Compiling sparse arithmetic type into fixed local rules). The one-defect computation has an explicit finite guarded-affine realization. Can the rational–Liouville dichotomy be realized by one fixed universal local rule set, with a proved clock schedule preserving superlacunarity in the actual microstep index? Simply computing a factorial target in an auxiliary counter does not automatically give the required gap between defect times after compilation.

Research question 36 (Algebraic irrational points in the equilibrium Cantor set). Does $\mathcal{C}$ contain an algebraic irrational number? Equivalently, does $\mathcal{K}$ contain one, or can an algebraic irrational in $[0, 2/3]$ have a binary expansion containing no consecutive ones? The sparse and low-complexity results exclude large explicit classes but do not answer this full question. The relation to digit restrictions and the algebraic-number complexity theorem [\[84\]](#) is direct; no conclusion for arbitrary high-complexity expansions is claimed.

Research question 37 (Irrationality exponents of explicit equilibria). Determine the irrationality exponent of the Thue–Morse equilibrium in [equation \(59.3\)](#), or prove finite upper and lower bounds beyond irrationality. Transcendence from digit complexity does not itself supply the desired approximation exponent after the variable-length coding and reciprocal transformation.

Research question 38 (Decidable restricted environment languages). For specific finite descriptions, such as automatic sequences, substitutions or bounded-state transducers, determine the complexity of eventual periodicity and therefore of exact rationality of the root mass. The unrestricted Σ_2^0 result does not prevent strong algorithms on restricted descriptions. Any claimed algorithm must specify the environment language and its promise conditions.

Research question 39 (Coefficient size versus witness count). The full presentation uses $3N + 4$ witnesses and small recurrence coefficients; the compact presentation uses $N + 1$ witnesses and coefficients with $O(N)$ bits. Can one improve this tradeoff while retaining total degree four, a unique natural witness and a transparent prefix semantics? Lower bounds would require a clearly fixed encoding model, since substitutions and auxiliary coordinates can shift costs. [write] Compare Question 2.

Research question 40 (Beyond root mass). What are the exact arithmetic complexities of stationary entropy, resolvent entries, moments and asymptotic variance in the same family? Many such quantities have effective tail bounds, but their equality and arithmetic-type predicates need separate reductions. One should not infer their complexity merely from the root-mass construction.

Research question 41 (Sampling and rare defects). Quantify the sample or trajectory length needed to distinguish a zero-defect chain from a defect at time T with a prescribed error probability. The stationary root-mass separation is exactly [equation \(56.3\)](#); a path can additionally reveal a defect by visiting it. Comparing these two observation models could clarify which statistical costs coexist with the common mixing rate.

Research question 42 (Physical and continuous-time realizations). Which analogues persist for continuous-time birth–death chains with uniformly bounded rational rates, or for explicitly constrained reaction and queueing models? Uniformizing a specified continuous-time kernel is straightforward, but implementing the programmable environment using a physically restricted finite reaction language is a different problem. The cost of a zero test and of history arithmetic must be accounted for rather than concealed in a rate function. [write] Compare Question 11, which this source partly answers for discrete reversible chains.

Research question 43 (Fixed-arity multiplicity control). Can any nontrivial restricted class of these strict equilibrium cuts be compiled to fixed-arity polynomials with finite-fold or single-fold witness control? The fixed-horizon uniqueness result is insufficient: there can be arbitrarily many certifying horizons and evaluator encodings. A useful advance would either identify a restricted class with a complete construction or prove a precise obstruction. [write] Compare Question 10 and the fixed-arity compression question of the sibling report [canonical-diophantine-certificates](#).

Research question 44 (Robust arithmetic predicates). Which meaningful equilibrium properties admit decision procedures under an explicit separation promise? For example, a known positive gap from a rational threshold makes a strict comparison decidable by [theorem 60.1](#). Rationality and algebraicity are not open properties; formulating useful robust replacements, with certificate sizes and input representations specified, may be more fruitful than seeking an impossible exact decision procedure. [write] Compare Question 12.

Part VI

Exact gates: cyclotomic contraction certificates, exact synthesis and state transfer (sources 10, 11 and 16)

70 [write] Conventions of Part VI

Part VI prints three manuscripts of batch 62. Source 10, *Diophantine Certificates for Coherent Computation* (batch-62 manuscript 06), certifies whole coherent tensor contractions over $\mathbb{Z}[e^{i\pi/4}]$ with unique natural witnesses, and so supplies the witness counts that Part II’s quantum front end leaves open. Sources 11, *Decidable Gates, Undecidable State Transfer* (batch-62 manuscript 07), and 16, *Dense Rotations, Undecidable Exactness* (batch-62 manuscript 13), study exact computation with a fixed finite set of rational four-dimensional rotations. They were written independently, from the pins [b6bf6406a](#) and [e18718e83](#), and neither cites the other or Parts I–IV; source 10 was written from [ccfb084ad](#).

70.1 How sources 11 and 16 are merged

Sources 11 and 16 share one arithmetic core. Both evaluate the free group on two letters at the same quaternions $(3 + 4i)/5$ and $(3 + 4j)/5$, prove the same modulo-five denominator rigidity theorem, decode words by the same residue calculation, embed free groups of every finite rank by conjugates $a^j b a^{-j}$, and turn a finite group presentation into rotation gates by the same Mihailova fiber-product compiler. That core is printed once, in Section 84. Where the two sources state the same result, one statement carries both sources' labels, one proof is printed, and a genuinely different proof is kept as a marked second route. Where they differ in normalization (source 11 indexes generators from 0, source 16 from 1) or in the statement proved, both are printed.

The two sources then specialize the compiler in opposite directions, and both specializations are printed in full:

- Source 11 (Sections 86–90) compiles a Miller group, whose word problem is decidable in polynomial time and whose marked conjugacy is c.e.-complete: rational-matrix membership in the gate group is polynomial-time, while exact transfer from one fixed rational state is c.e.-complete, and the hard targets satisfy an order-five recurrence.
- Source 16 (Sections 91–95) compiles a group $G_0 * \langle t \rangle$ with c.e.-complete word problem: rational-matrix membership itself is c.e.-complete, even inside every open ball; the precision threshold $1/(qD^n)$ turns approximation into exact synthesis; the maximal synthesis depth has Turing degree $\mathbf{0}'$; and the commuting free-factor mechanism needs dimension four.

The two directions of the compiler are Theorem 84.16 (membership reduces to one word-problem query) and Theorem 84.12 (the word problem reduces to membership). Both certificate families are kept (Sections 96 and 98): source 11 certifies a *state* transfer with $(k + 5)T + 5$ integer variables, source 16 a *matrix* synthesis with $n(s + 17)$ natural witnesses and shifted entries in place of signed pairs.

Neither source gives an explicit gate count for its universal instance (both gate lists depend on an unprinted Higman presentation), so neither bears on the seven-outcome, dimension-four numbers of Part II's quantum mortality (Theorem 18.2). The predicates also differ: Part II asks for a word whose product is *zero*, which is impossible in an orthogonal group; here the question is membership of a target in the generated group, or reachability of a target state. Undecidability of rotation reachability on the three-sphere is not new (Bell and Potapov); both sources say so.

70.2 The exact–approximate boundary

Source 10's Theorem 78.1 and source 16's Theorems 84.21, 92.3 and 93.1 sit on opposite sides of one boundary. In source 10 a single rational rotation is supplied by fast Cauchy names, and exact zero probability is Π_1^0 -complete: the difficulty is in the *input representation*. In source 16 gates and targets are exact rationals; exact synthesis is Σ_1^0 -complete, positive-tolerance approximation is computable with a computable length bound, and the length-dependent threshold $1/(qD^n)$ is exactly where approximation becomes exact synthesis: the difficulty is in the *unbounded word search*. Source 11 adds that positive-tolerance transfer is always possible and that no positive geometric margin separates its hard orbit from nonreachability (Section 90).

70.3 Letters used differently

Letter	Part II	Source 10	Source 11	Source 16
L, T	L gate count; T the gate; T_i tensors	$L_{\text{leaf}}, T_{\text{mul}}$ profile counts (renamed); T the gate	T word length; L_k relator gates; $L(p)$ left multiplication	$L(p)$ left multiplication; T a numerator matrix
h, H	H Hadamard gate; H_j numerators	h number of Hadamards; H, H_R, H_* height bounds; $H_t(z)$ halting probability by step t	h_t scales; H an exponent	—
α, β	$\alpha = 4/(5c), \beta = 3/5$	sign Booleans in Theorem 74.3	α_τ Miller's action (the quaternions are a, b)	—
a, b	—	real coordinates of $U_B = a + b\sqrt{2}$	letters and quaternions $(3 \pm 4i)/5, (3 \pm 4j)/5$	the same
D	quantum dimension; D_{max}	edge dimension	common gate denominator	common gate denominator $D = 5^K$
K	outcome count	$K = v2^E + u2^h$	a c.e.-complete set	denominator exponent; $\mathcal{K}$ halting set
s	rows of a Gram factor	gate count in a network	number of relators	alphabet size $1 + 2(d + m)$
B	Gram factor	a Boolean output	$B(\varepsilon)$ approximation bound	$B(k)$ net bound; $B_j = DS_j$
E	equation count $E(D, K, n)$	edge count	forward shift	depth function $E(h)$
ρ, Φ, Ψ	ρ density operator	$R(a)$ Cayley rotation	Φ spin map (source: Ψ); η evaluation (source: ρ)	Φ spin map; ρ_d rank- d representation; Ψ_d product embedding

The tempting false readings are these. Source 10's T in $8L_{\text{leaf}} + 64T_{\text{mul}}$ is a count of contraction summands, not the T gate, which it also uses. The D of sources 11 and 16 is a common denominator, never Part II's quantum dimension, which is four for every gate of this Part. Source 16's ρ_d is a representation, not Part III's critical radius. The quaternions a, b are not Part II's scalars α, β , although both come from the 3–4–5 triple.

70.4 Renamed symbols and macros

- Source 10's contraction profile (L, T, O) is written $(L_{\text{leaf}}, T_{\text{mul}}, O)$, so that T stays the gate and L Part II's gate count. No normalization changed.
- Source 11's quaternions $\alpha = (3 + 4i)/5$ and $\beta = (3 + 4j)/5$ are written a, b , as in source 16 (source 11 already writes a, b for the free-group letters they evaluate); its spin map $\Psi(p, q)(v) = pv\bar{q}$ is written Φ , as in source 16, where $\Phi(p, q)(z) = pzq^{-1}$ is the same map on unit quaternions; its evaluation map ρ is written η , as in source 16. Its Euclidean norm macro is renamed internally (the printed $\|\cdot\|$ is unchanged), and Miller's action α_τ keeps its letter.
- Source 10's and source 16's probability and identity macros are unified with the report's; nothing printed changes except that Pr is upright.
- File names printed inside the sources' text are their delivered names; the shipped names carry the prefixes `10-coherent-circuits-`, `11-rational-rotations-` and `16-dense-rotations-`, mapped in the README.

70.5 Abstracts and status statements of the sources

Abstract of source 10. A certificate for one legal computational path does not, by itself, certify a quantum outcome: legal paths may cancel. We construct natural-number Diophantine certificates for the entire coherent contraction instead. A supplied binary contraction tree for an integer tensor network gives a quadratic system with a unique witness, and hence one sum-of-squares polynomial of degree at most four. An explicit lift to $\mathbb{Z}[e^{i\pi/4}]$ uses at most $8L_{\text{leaf}} + 64T_{\text{mul}}$ natural variables, where L_{leaf} counts leaf entries and T_{mul} counts contraction summands. Every coordinate has a proved height bound. A canonical quadratic comparison circuit for $\mathbb{Q}(\sqrt{2})$ adds at most 39 variables and 50 equations to certify a strict rational threshold for an exact quantum probability. We apply the construction to Clifford+T-type circuits and to adaptive equatorial measurements of graph states.

Two limitations make the representation meaningful. A uniformly polynomial-size, polynomial-bit witness compiler for zero output probability would imply $\text{NP} = \text{coNP}$. More fundamentally, with gate parameters supplied by fast Cauchy names, zero probability is not even computably enumerable on a uniformly valid family of single-qubit rational rotations. Finally, for unbounded algebraic quantum controllers, positive-probability halting is Diophantine by MRDP, whereas almost-sure halting is Π_2^0 -complete. The article gives full proofs, an executable exact-arithmetic prototype, reproducible tests, and a formalization and research agenda.

Status statement of source 10. Status. The contribution is an explicit compiler and a representation/resource analysis. Established path-sum, tensor-simulation, computable-analysis, and termination results are credited separately. Historical priority for the combined construction has not been established. No new Lean formalization or resolution of the general finite-fold or single-fold MRDP conjectures is claimed.

Abstract of source 11. We construct a fixed finite symmetric family of four-dimensional rotations with entries in $\mathbb{Z}[1/5]$ for which exact matrix membership is decidable in polynomial time, but exact transfer from one fixed rational unit vector is computably enumerable complete. For the same family, transfer from a second fixed rational vector is polynomial-time decidable. Both orbits are dense, the generated group is dense in $\text{SO}(4)$, and every positive-tolerance transfer is possible with a computable uniform bound on control length. The hard queries can be restricted to a rational target sequence whose four coordinates satisfy a common constant-coefficient recurrence of order at most five. In contrast to approximate control, neither the word length nor the matrix denominator needed for exact transfer admits a recursive bound in the target's bit length.

The proof gives a uniform compiler from finite group presentations to rational rotation gates. Its central identity identifies encoded state orbits with conjugacy classes in the presented group. A modulo-five quaternion argument supplies an exact decoder, and an explicit inverse spin calculation handles arbitrary rational matrix queries, including rotations without rational unit-quaternion lifts. A Miller free-by-free construction separates the word and conjugacy problems. Finally, every fixed-length transfer has an explicit quartic integer certificate with linear size in the length; MRDP gives a fixed unbounded Diophantine representation. We distinguish these statements from previously known rotation undecidability and from unresolved finite-fold questions. Exact Python implementations and finite verification results accompany the article; the new development has not been checked in Lean.

Status statement of source 11. Status. The contribution is a proved structural synthesis and explicit algorithmic refinement. Historical priority for the full combination of properties is not asserted. The fixed hard instance uses a classical finite-presentation existence theorem; its entire numerical gate list is not expanded in this package.

Abstract of source 16. We give a presentation-to-rotation compiler that turns the word problem of a finitely presented group into exact synthesis with finitely many rational four-dimensional rotations. All gate entries belong to $\mathbb{Z}[1/5]$, all gates are reversible, and the generated group is dense in $\text{SO}(4)$ whenever the presentation has a nontrivial relation. A universal specialization has a computably enumerable complete

exact-synthesis problem. After adjoining one free generator to the source presentation, completeness persists inside *every prescribed rational open ball* of rotations. In contrast, arbitrary positive-tolerance approximation is computable, even with a computable uniform length bound. A length-dependent rational separation threshold identifies precisely where approximation becomes exact synthesis.

The arithmetic core is an elementary modulo-5 proof that a freely reduced word of length n in $(3 + 4i)/5$ and $(3 + 4j)/5$ has least common denominator exactly 5^n . The same calculation decodes the word. For a fixed labeled alphabet of s gates and horizon n , we construct a quartic Diophantine polynomial with $n(s + 17)$ natural witnesses whose zeros are in bijection with the successful labeled gate words. This direct construction is distinct from the fixed-arity, unbounded representation supplied by MRDP.

The foundational inputs are classical: free-product normal forms, Higman’s embedding theorem, Mihailova’s fiber product, and MRDP. The contribution pursued here is their explicit arithmetic and local-topological combination, with full proofs of the stated refinements and executable finite checks. Historical priority for this package has not been established. The universal specialization is an existence theorem; a concrete small universal presentation, a fully expanded universal polynomial, and a Lean formalization are not supplied.

Repository baseline of source 16. **Repository baseline.** ProveIt, revision e18718e837d43e162252f9a314e8cb797fbd1a1f. The construction uses its MRDP interface as a proposed integration endpoint, not as evidence that the new group-theoretic arguments have been formalized.

71 The problem, the contribution, and the repository boundary

[source 10]

71.1 Why coherent computation is a different substrate

The question motivating this article is not merely whether quantum computation can be simulated by an ordinary effective machine. Once a finite gate alphabet and an effective observation semantics are fixed, such a simulation often gives a route to MRDP. The more informative question is whether arithmetic certificates can retain the actual algebraic semantics of coherence: addition of amplitudes, complex conjugation, destructive interference, and exact comparison of the resulting probabilities.

For a classical nondeterministic computation, exhibiting an accepting path is an adequate existential certificate of reachability. For a coherent computation, an accepting-looking path can have a nonzero amplitude while the total amplitude of its endpoint is zero. Conversely, a probability is a sum of squared *total* amplitudes, not a sum of the squared weights of its fine-grained paths. The order of summation and squaring is essential.

Dawson et al. already express amplitudes of Hadamard–Toffoli circuits through a difference of solution counts over $\mathbb{F}_2$ [24]. In the customary notation, an amplitude with h Hadamards has the form

$$2^{-h/2}(N_0 - N_1). \quad (71.1)$$

That connection is a starting point, not a novelty claim here. Our certificate records a complete arithmetic evaluation, including both signs, with exactly one natural witness for fixed data. It does not existentially choose one of the paths counted by N_0 or N_1 .

71.2 What was actually inspected in ProveIt

The repository was inspected at commit

ccfb084adaa2f32e8d2738a25f82a00377fb3a8c.

The relevant direct reads were the Hilbert-tenth-problem README and the Lean file `Diophantine/MRDP.lean` under `Computability/HilbertTenthProblem/Lean/` [22, 23]. The README describes corrected Jones articles, universal Diophantine systems, and a Lean development of both directions of MRDP. The inspected theorem file states a finite-polynomial equivalence, including input zero: every recursively enumerable subset of $\mathbb{N}$ is the projection of the natural zero set of one integer polynomial with finitely many witnesses, and conversely.

The repository’s theorem is used only in Section 79. The finite compilers in Sections 72–76 are proved directly and do not invoke MRDP. Repository assertions about axiom audits were not independently rechecked by a Lean build in this work. Code-search results included an older indexed revision; mathematical dependencies were taken from the pinned direct reads rather than silently equating that index with the inspected commit.

This is a selective extension of the repository, not an audit of its many other claims. In particular, no numerical universal-polynomial optimization from the repository is needed for any theorem below.

71.3 The main representation theorem in preview

Let a closed tensor network have m leaves, each edge taking D values. At a binary merge, let A, B be the sets of boundary edge labels of its children. Set

$$L_{\text{leaf}} = \sum_{\text{leaves } v} D^{\deg(v)}, \quad T_{\text{mul}} = \sum_{\text{merges}} D^{|A \cup B|}, \quad O = \sum_{\text{merges}} D^{|A \triangle B|}. \quad (71.2)$$

These quantities describe a *supplied* contraction tree. The merge-union width is $w = \max |A \cup B|$. It is a deliberately explicit resource measure; it is not being identified without proof with another author’s treewidth or contraction-complexity convention.

For integer entries, the compiler uses exactly $2(L_{\text{leaf}} + 2T_{\text{mul}} - O)$ natural variables before constant sharing. For entries in $\mathcal{R} = \mathbb{Z}[e^{i\pi/4}]$, it uses exactly $2(4L_{\text{leaf}} + 32T_{\text{mul}} - 4O)$ before constant sharing, and therefore at most $8L_{\text{leaf}} + 64T_{\text{mul}}$. All constraints are quadratic. Their sum of squares is a single quartic Diophantine equation. Fixed output assertions preserve the property that the witness set is either empty or a singleton.

For the doubled tensor network of an exact quantum circuit, a strict rational probability query requires at most 39 additional variables and 50 additional quadratic equations. The witness coordinates have polynomial bit length in the network and threshold description, even when their *number* is exponential in width. This last distinction is forced by the conditional obstruction in Section 77.

71.4 Novelty and scope

Tensor contraction as a simulation method, and its dependence on graph width, are established results; Markov and Shi are a central reference [25]. Clifford+T exact arithmetic and measurement-based universality are likewise established [7, 27]. The proposed contribution here is the assembled, fully specified *natural-witness* compiler: canonical signed coordinates, exact resource constants, a height proof that uses the contraction tree, a single-valued algebraic comparison gadget, and a sharp separation between exact finite descriptions and approximation-name inputs.

Theorems are proved below rather than asserted on the basis of tests. The implementation provides additional finite checks, not a formal proof of the compiler. The construction resolves the concrete representation problem posed here. It should not be advertised as a verified historically new solution to an established major conjecture.

72 Canonical arithmetic wires and a quartic normal form

[source 10]

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Integer coefficients are allowed in all polynomials, but existential variables range over $\mathbb{N}$. For a pair of natural variables z^+, z^- write $\Delta z = z^+ - z^-$. A *canonical signed wire* is such a pair together with the constraint

$$z^+ z^- = 0. \quad (72.1)$$

Lemma 72.1 (Unique signed representation). *For every $z \in \mathbb{Z}$, there is exactly one pair in $\mathbb{N}^2$ satisfying $\Delta z = z$ and (72.1). It is $(\max(z, 0), \max(-z, 0))$.*

Proof. The product condition says that at least one coordinate is zero. If the first coordinate is zero, the difference is nonpositive and determines the second; if the second is zero, the difference is nonnegative and determines the first. At zero both coordinates are zero. These alternatives also prove existence. $\square$

The product condition is indispensable for multiplicity control. The equation $x - y = 2$ alone has the infinitely many solutions $(2 + k, k)$, $k \in \mathbb{N}$. Introducing unrestricted positive and negative parts is therefore not a single-witness encoding.

Definition 72.2 (Arithmetic evaluation network). An arithmetic evaluation network is a finite directed acyclic graph with q source wires and s gates. Source values are fixed integers, or fixed integer parameters. Every gate is a binary addition, subtraction, or multiplication. Its output is a new wire; earlier wires may be reused. Let $N = q + s$.

For every wire impose (72.1). A source of value c has residual $\Delta z - c$. For a gate output z with inputs x, y , use respectively

$$\Delta z - \Delta x - \Delta y, \quad \Delta z - \Delta x + \Delta y, \quad \Delta z - (\Delta x)(\Delta y). \quad (72.2)$$

If natural input codes are required for signed parameters, write the latter as the difference of their supplied natural coordinates. The theorem is then understood separately for every fixed parameter tuple.

Theorem 72.3 (Canonical quartic compilation). *An arithmetic evaluation network with N wires has a system of $2N$ equations of degree at most two in $2N$ natural variables whose unique solution is its canonical evaluation. If the residuals are $F_1, \dots, F_{2N}$, then*

$$P = \sum_{j=1}^{2N} F_j^2 \quad (72.3)$$

has degree at most four and exactly the same natural solutions. Adding fixed linear output assertions gives either that unique solution or no solution. Before collection of monomials, a crude bound for the base polynomial is $37N$ monomial contributions.

Proof. Assign each wire the canonical pair of its actual integer value. All source, gate, and canonical residuals vanish, proving existence. Conversely, take any natural solution. The source equations and Lemma 72.1 uniquely fix every source. In a topological order, a gate equation uniquely fixes the integer difference of its output from already fixed input differences; the canonical equation fixes the two output coordinates. Induction proves uniqueness and correct evaluation at every wire.

Each residual is quadratic or linear. Over integers, a sum of squares is zero if and only if every summand is zero. This proves the quartic assertion and also shows that P is nonnegative on all real substitutions, although the uniqueness statement is specifically about natural substitutions. An output assertion only removes the unique solution when the asserted value is false.

A source or gate residual has at most six monomials before like terms are collected. Its square has at most 36 contributions. Each canonical residual has one monomial, and its square also has one. There are N residuals of each type, giving $37N$. This is a bound on a particular sparse expansion, not on the minimum number of terms in an equivalent equation. $\square$

The construction is syntactic: the residuals can be emitted without first knowing the final output. Evaluating the network supplies the witness but is not needed to justify the equations. Constants have their ordinary binary encoding; a large constant is not a unit-cost object in a bit-complexity claim. When desired, a constant of bit length b can instead be constructed from 0, 1 with $O(b)$ doubling and addition gates. The latter preserves quadratic constraints and canonicity. It does not establish a new optimum for the number of variables or coefficients.

Remark 72.4 (Why a general DAG has no small height bound). A chain $z_0 = 2$, $z_{i+1} = z_i^2$ has $z_i = 2^{2^i}$. The bit length is exponential in the number of gates. The polynomial height bound proved next is not a property of every arithmetic DAG: it comes from the disjoint-leaf structure of a tensor contraction tree.

73 Tensor contractions with exact size and height accounting

[source 10]

73.1 Networks and supplied contraction trees

A closed tensor network is a finite undirected multigraph without self-loops. Each edge has its own label and domain $[D] = \{0, \dots, D-1\}$. At vertex v , a tensor t_v assigns an integer to every assignment of values to its incident edge labels. Parallel edges are distinct labels. The network value is

$$Z_{\mathcal{T}} = \sum_{\sigma: E \rightarrow [D]} \prod_v t_v(\sigma|_{E(v)}). \quad (73.1)$$

Disconnected networks are allowed. Their values multiply.

For a subset S of vertices, let ∂S be the edge labels with exactly one endpoint in S , and let $e(S)$ count edges with both endpoints in S . The partial tensor on S sums over internal edges while keeping the values on ∂S fixed. A binary contraction tree has the vertices as its leaves. At a merge of disjoint subsets, put $A = \partial S_1$, $B = \partial S_2$. The shared labels $C = A \cap B$ are precisely the edges joining the two subsets. Their union has boundary $A \triangle B$, and

$$t_{S_1 \cup S_2}(z) = \sum_{c \in [D]^C} t_{S_1}(z|_{A \setminus B}, c) t_{S_2}(z|_{B \setminus A}, c). \quad (73.2)$$

All sums are finite; the formula follows directly by partitioning assignments in (73.1). When C is empty, the sum has one term, so no special empty-sum convention is needed.

Choose a fixed lexicographic order on assignments in every sum. Use one multiplication for each summand and a left-associated chain of additions to sum them. This convention makes the full witness, not only its output, well-defined relative to the tree.

Theorem 73.1 (Integer contraction compiler). *Let $\mathcal{T}$ have $m \geq 2$ tensors, E edges, edge dimension $D \geq 2$, and integer entries of absolute value at most $M \geq 1$. Fix a binary contraction tree and its profile $(L_{\text{leaf}}, T_{\text{mul}}, O)$ from (71.2). Then:*

- (i) *The prescribed evaluation has L_{leaf} source wires and $2T_{\text{mul}} - O$ arithmetic gates. Its canonical quadratic system has exactly $2(L_{\text{leaf}} + 2T_{\text{mul}} - O)$ natural variables and as many equations before sharing identical constants.*
- (ii) *Every fixed output-value query is represented by a single polynomial of degree at most four with at most one natural witness.*
- (iii) *Every coordinate of the base witness is at most $H = D^E M^m$. In particular, its bit length is at most $1 + \lceil E \log_2 D + m \log_2 M \rceil$.*
- (iv) *For merge-union width w , the number of base variables is at most $(6m - 4)D^w$.*

Proof. At a merge there are $D^{|A \triangle B|}$ output entries. Each has $D^{|A \cap B|}$ summands, so the total number of products is $D^{|A \triangle B| + |A \cap B|} = D^{|A \cup B|}$. The addition count is that number minus the number of output entries. Summing gives T_{mul} products and $T_{\text{mul}} - O$ additions. Theorem 72.3 proves (i) and (ii).

A partial tensor entry for S is a sum of $D^{e(S)}$ products of $|S|$ leaf entries. Consequently,

$$|t_S(z)| \leq D^{e(S)} M^{|S|}. \quad (73.3)$$

For two children, an individual product before summing shared labels has absolute value at most $D^{e(S_1) + e(S_2)} M^{|S_1| + |S_2|}$. A partial sum over any initial segment of shared assignments is bounded by the full sum of these absolute-value bounds, hence by (73.3) for the union. Thus every intermediate arithmetic wire, not only each completed tensor entry, has absolute value at most $D^E M^m$. Canonical coordinates have the same or smaller magnitude. This proves (iii).

At the first merge involving a given leaf, its entire incident edge set is contained in the merge union. Therefore every leaf degree is at most w . There are $m - 1$ merges, whence $L_{\text{leaf}} \leq mD^w$ and $T_{\text{mul}} \leq (m - 1)D^w$. Dropping the nonpositive term $-2O$ gives $2(L_{\text{leaf}} + 2T_{\text{mul}} - O) \leq 2mD^w + 4(m - 1)D^w$, proving (iv). $\square$

The case of a single scalar tensor is handled by a source wire without any merge. Non-scalar one-vertex closed networks would require self-loops, which were excluded; such loops may instead be contracted as an explicit preliminary finite sum.

73.2 What the height proof does and does not say

The important structural fact is that the two children of every merge depend on disjoint sets of leaf tensors. Their products expand into products in which each leaf is used exactly once. Reuse of a *table entry* in several output entries is harmless: it does not make a single output monomial reuse an entire subcomputation as a factor. This is why the repeated-squaring counterexample from Section 72 does not apply.

The theorem bounds the coordinates of the unique witness, not the computational effort needed to discover a favorable contraction tree. For a supplied tree, emitting the system takes time polynomial in its explicit arithmetic expansion and in the bit lengths of the leaf entries. In particular, for fixed D and $w = O(\log m)$ with polynomial-bit input entries, the certificate description, its witness, and its direct verification are polynomial in the original network size. This follows by the displayed counts and height bound; it is consistent with established low-width tensor simulation [25].

Proposition 73.2 (Independence of the chosen order). *Any two complete contraction trees for the same closed network produce the same output integer. The corresponding canonical witness sets for a fixed output query are either both empty or both singletons. A witness in one order can be replaced effectively by the canonical witness in the other order.*

Proof. Each order evaluates (73.1), by repeated use of (73.2). Theorem 73.1 gives the assertion about witness sets. Re-evaluating the other finite tree gives the stated effective replacement. There is no assertion that the replacement is local, or that the two witness spaces have the same dimension. $\square$

74 Cyclotomic arithmetic without noncanonical auxiliary choices

[source 10]

74.1 A four-coordinate integer algebra

Put $\omega = e^{i\pi/4}$ and $\mathcal{R} = \mathbb{Z}[\omega]$. Every element has a unique form

$$A = a_0 + a_1\omega + a_2\omega^2 + a_3\omega^3, \quad a_j \in \mathbb{Z}, \quad (74.1)$$

with $\omega^4 = -1$. For completeness, $X^4 + 1$ is irreducible over $\mathbb{Q}$: substituting $X + 1$ gives $X^4 + 4X^3 + 6X^2 + 4X + 2$, to which Eisenstein's criterion at 2 applies. This proves uniqueness of the four-coordinate form.

Addition uses four integer additions. Multiplication has coordinates

$$c_k = \sum_{\substack{0 \leq i, j < 4 \\ i+j=k}} a_i b_j - \sum_{\substack{0 \leq i, j < 4 \\ i+j=k+4}} a_i b_j, \quad 0 \leq k < 4. \quad (74.2)$$

Across the four output coordinates there are 16 products, four per coordinate, and 12 additions or subtractions. The first term for coordinate k may be chosen as the positive product $a_0 b_k$. Thus a direct scalar evaluation uses 28 arithmetic gates per ring multiplication, without a separate initial negation or accumulator.

For A as in (74.1), define $\|A\|_1 = \sum_j |a_j|$. Reduction modulo $X^4 + 1$ can only combine signed coefficients, so

$$\|AB\|_1 \leq \|A\|_1 \|B\|_1. \quad (74.3)$$

Theorem 74.1 (Cyclotomic contraction compiler). *For a closed $\mathcal{R}$ -valued tensor network and a supplied contraction tree, the base canonical quadratic system has, before constant sharing,*

$$V_R = 2(4L_{\text{leaf}} + 32T_{\text{mul}} - 4O) \quad (74.4)$$

natural variables and V_R equations. In particular,

$$V_R \leq 8L_{\text{leaf}} + 64T_{\text{mul}} \leq (72m - 64)D^w. \quad (74.5)$$

It has exactly one solution, encoding the entire contraction in the basis (74.1). Let

$$\lambda_v = \max(1, \max_{\text{entries } a \text{ of } t_v} \|a\|_1). \quad (74.6)$$

Every natural coordinate is bounded by

$$H_R = D^E \prod_v \lambda_v. \quad (74.7)$$

If every scalar coefficient in a leaf entry has absolute value at most $M \geq 1$, then also $H_R \leq D^E (4M)^m$.

Proof. There are $4L_{\text{leaf}}$ integer source wires. Each of the T_{mul} ring products uses 28 scalar gates; each of the $T_{\text{mul}} - O$ ring additions uses four. Thus the scalar gate count is $28T_{\text{mul}} + 4(T_{\text{mul}} - O) = 32T_{\text{mul}} - 4O$. Theorem 72.3 proves (74.4), uniqueness, and the quartic conversion. The width bound follows as in Theorem 73.1.

For a subset S , the ℓ^1 norm of a partial entry is at most $D^{e(S)} \prod_{v \in S} \lambda_v$, by (74.3). Each scalar product in (74.2) is bounded by the product of the input norms; the absolute value of any partial signed sum of its terms is bounded by the sum of the absolute values of all those products. This is at most that same product of norms. Subsequent partial sums over shared edge assignments are bounded exactly as in (73.3). Because every $\lambda_v \geq 1$, the bound for any subset or intermediate is at most (74.7). The lower bound 1 in the definition also avoids a zero leaf incorrectly forcing the height bound for unrelated intermediates to be zero. $\square$

74.2 Conjugation, real values, and exact norms

Conjugation is the signed coordinate permutation

$$\overline{(a_0, a_1, a_2, a_3)} = (a_0, -a_3, -a_2, -a_1). \quad (74.8)$$

An element is real precisely when $a_2 = 0$ and $a_3 = -a_1$. Since $\omega - \omega^3 = \sqrt{2}$, such an element is $a_0 + a_1\sqrt{2}$. These assertions follow by comparing coefficients in (74.8); the positive square root identity follows either from the chosen embedding or from squaring $\omega - \omega^3$ and checking its sign.

In particular, for an amplitude A ,

$$\begin{aligned} A\bar{A} &= a + b\sqrt{2}, \\ a &= a_0^2 + a_1^2 + a_2^2 + a_3^2, \\ b &= a_0a_1 - a_0a_3 + a_1a_2 + a_2a_3. \end{aligned} \tag{74.9}$$

This identity is obtained by multiplying (74.1) by (74.8) and reducing $\omega^4 = -1$. It also illustrates why ordinary integer signs of the two output coordinates do not determine the sign of the represented real number.

74.3 A unique quadratic nonnegativity gadget

An unrestricted difference of naturals is not canonical, and neither is an unrestricted four-square witness for nonnegativity. For this application we can avoid both sources of multiplicity.

Lemma 74.2 (Integer comparison with a unique witness). *Given a canonical signed wire z , introduce $b, s \in \mathbb{N}$ and impose*

$$b(b-1) = 0, \quad bz^- = 0, \quad z^- - (1-b)(s+1) = 0, \quad bs = 0. \tag{74.10}$$

There is exactly one solution in (b, s) . It has $b = 1$ if and only if $\Delta z \geq 0$; its slack is $s = \max(z^- - 1, 0)$.

Proof. The first equation forces $b \in \{0, 1\}$. If $b = 1$, the second forces $z^- = 0$ and the fourth forces $s = 0$, while the third is satisfied. If $b = 0$, the third forces $z^- = s + 1 > 0$, uniquely determining $s = z^- - 1$; the other equations hold. Canonicity of z makes $z^- = 0$ equivalent to $\Delta z \geq 0$. $\square$

Write $\text{NN}(z)$ for this uniquely computed Boolean, remembering that its two auxiliary variables and four equations are part of the system.

Theorem 74.3 (Single-valued comparison in $\mathbb{Q}(\sqrt{2})$). *Given canonical integer wires p, q , nonnegativity of $p + q\sqrt{2}$ has a quadratic Boolean-output circuit with a unique auxiliary assignment. With constant wires 0 and 2 already available, the circuit introduces 21 natural variables and 29 equations. A strict positivity assertion is obtained by applying it to $-p, -q$ and requiring its output to be zero.*

Proof. Compute $d = p^2 - 2q^2$ and $-d$. Let

$$\alpha = \text{NN}(p), \quad \beta = \text{NN}(q), \quad \gamma = \text{NN}(d), \quad \delta = \text{NN}(-d).$$

Introduce natural variables e, f, B with

$$e = \alpha(1 - \beta), \quad f = (1 - \alpha)\beta, \quad B = \alpha\beta + e\gamma + f\delta. \tag{74.11}$$

All three equations have degree at most two. The four sign cases prove the claimed interpretation. If $p, q \geq 0$, then $B = 1$. If $p, q < 0$, then $B = 0$. If $p \geq 0 > q$, the inequality is $p \geq |q|\sqrt{2}$, equivalent to $d \geq 0$; thus $B = \gamma$. If $p < 0 \leq q$, the inequality is $q\sqrt{2} \geq |p|$, equivalent to $d \leq 0$, and $B = \delta$. These arguments include zero coordinates and equality. The case $p = q = 0$ gives $B = 1$.

Every preceding scalar wire is unique by Theorem 72.3, the four comparators are unique by Lemma 74.2, and (74.11) then fixes e, f, B . The output is automatically Boolean, without another Boolean constraint.

The scalar operations $p^2, q^2, 2q^2, d, -d$ contribute ten natural variables and ten equations. Four instances of (74.10) contribute eight variables and sixteen equations. Finally e, f, B contribute three variables and three equations. This gives 21 and 29. For any real x , $x > 0$ is equivalent to the failure of $-x \geq 0$, proving the strict comparison claim. $\square$

This construction deliberately computes all integer sign tests *unconditionally*. Multiplying a quadratic residual by a Boolean branch guard would raise its degree to three; squaring that guarded residual would no longer give the promised quartic bound. Equations (74.10)–(74.11) avoid that mistake. The same proof works for $p + q\sqrt{d}$ with any fixed positive nonsquare integer d , replacing the constant 2 by d . No general number-field comparison compiler is asserted by this observation.

75 Exact quantum probabilities as single-witness Diophantine queries

[source 10]

75.1 The gate model and denominator discipline

Use a finite gate alphabet containing

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & \omega \end{pmatrix},$$

and the computational-basis permutations X , CNOT, and Toffoli. The gates $T^\dagger$, S , Z , and controlled- Z may also be included. These choices retain an exact cyclotomic arithmetic interface. The relationship between exact Clifford+T synthesis and $\mathbb{Z}[1/\sqrt{2}, i]$ is established in [7]; the compiler here only needs the displayed local matrices, not the converse synthesis theorem. Hadamard–Toffoli universality in the appropriate real-computation setting is discussed in [26].

Let C be a circuit with $n \geq 1$ qubits, g gates, and h Hadamards. Replace each H by its numerator matrix $H_0 = \sqrt{2}H$. Leave all other gates unchanged, obtaining a matrix $\tilde{U}_C$ over $\mathcal{R}$. Then

$$U_C = 2^{-h/2} \tilde{U}_C. \quad (75.1)$$

For a computational-basis input x , write $A_y = \langle y | \tilde{U}_C | x \rangle$. An output cylinder event B specifies fixed values on some subset of the output qubits and leaves the rest unobserved. Its probability is

$$p_C(B) = 2^{-h} U_B, \quad U_B = \sum_{y \in B} A_y \overline{A_y} \in \mathbb{Z}[\sqrt{2}]. \quad (75.2)$$

The physical normalization gives $0 \leq U_B \leq 2^h$ in the chosen real embedding. The symbol 2^h is a known integer determined by the finite circuit. It is not an exponentiation relation with an unconstrained existential exponent. Its binary description has $h + 1$ bits.

75.2 A doubled closed network

Construct one copy of the circuit numerator network and one entrywise-conjugate copy. Give their wire segments independent edge labels. Attach basis-input vectors on both copies. At each output qubit attach the two-index tensor

$$D_j(a, b) = \begin{cases} 1, & a = b \text{ and the equality agrees with the event, if specified,} \\ 0, & \text{otherwise.} \end{cases} \quad (75.3)$$

For an unspecified output qubit, this is just δ_{ab} . The resulting network is closed and has value U_B . Expanding all edge assignments first sums the ket amplitude and bra amplitude independently; the output tensors then identify their final basis labels and enforce the event.

If gate j has arity k_j , this particular dense-table construction has

$$m = 2g + 3n, \quad E = 2n + 2 \sum_{j=1}^g k_j, \quad L_{\text{leaf}} = 8n + 2 \sum_{j=1}^g 4^{k_j}. \quad (75.4)$$

There are $2n$ input vectors, $2g$ gate tensors, and n output tensors. Each side starts with n edge labels and creates k_j new labels at gate j , proving the edge count. The input and output tables together contribute $8n$ entries, while each of the two copies of an arity- k_j gate has 2^{2k_j} entries. This proves the last count. No sparsity optimization is hidden in it.

Every nonzero entry of these tensors is a signed power of ω with a single nonzero coefficient in the basis (74.1). Thus every λ_v in Theorem 74.1 equals 1, and every base witness coordinate is at most 2^E .

Theorem 75.1 (Canonical exact-probability certificates). *Fix a circuit C , basis input, cylinder event B , and contraction tree for the doubled network. Write its output in real coordinates as $U_B = a + b\sqrt{2}$. Then the following constructions are effective.*

- (i) *The graph of the exact output pair (a, b) has a natural-witness quartic representation with at most $8L_{\text{leaf}} + 64T_{\text{mul}}$ base witness variables. A fixed correct output pair has exactly one witness, and an incorrect pair has none.*
- (ii) *For fixed integers $u \geq 0, v > 0$, the assertion $p_C(B) > u/v$ has a quartic representation with at most $8L_{\text{leaf}} + 64T_{\text{mul}} + 39$ natural witness variables. The base system gains at most 50 quadratic equations. The witness set has cardinality one if the assertion is true, and zero otherwise.*
- (iii) *Let $H_* = 2^E$ and $K = vH_* + u2^h$. All witness coordinates, including the comparison auxiliaries in (ii), are at most $2K^2$.*

Proof. Theorem 74.1 gives a unique full contraction witness with output coordinates $(a, b, 0, -b)$. Appending linear assertions equating a, b to the supplied target differences proves (i). If targets are supplied by signed natural pairs, their interpreted values are those differences. The uniqueness is for each fixed target code; the output's own canonical pairs remain fixed.

For (ii), compute the signed integer wires

$$p = va - u2^h, \quad q = vb. \quad (75.5)$$

Because $v2^h > 0$,

$$p_C(B) > u/v \iff p + q\sqrt{2} > 0.$$

Compute the nonnegativity Boolean for $-p, -q$ from Theorem 74.3 and require it to be zero. Also impose the two linear reality checks on the full ring output, $c_2 = 0$ and $c_3 + c_1 = 0$. They are redundant for a correct doubled network but make the query interface explicit. All operations and sign choices are unique. Hence a true query has one witness and a false one has none. Squaring and summing all residuals gives the asserted polynomial of degree at most four.

Here is the full overhead audit. Constructing $p, q, -p, -q$ uses five scalar gates, hence ten variables and ten equations. The comparison circuit adds 21 variables and 29 equations. At most four new integer constants are needed: $0, 2, v, u2^h$, costing at most eight variables and eight equations. The two reality checks and the final Boolean assertion add three equations and no variables. The totals are 39 variables and 50 equations. Constant sharing can only lower these bounds.

For (iii), the base contraction coordinates are bounded by H_* , so $|p|, |q| \leq K$. During the comparison, the scalar squares and the twice-square are bounded by $2K^2$. Their difference satisfies $|p^2 - 2q^2| \leq 2K^2$, and the same bound holds after sign reversal. The integer comparison slacks are smaller than the corresponding negative parts. The remaining variables are Boolean or constants of size at most $\max(K, 2)$. As $K \geq 1$, all are bounded by $2K^2$. $\square$

[write] This is the explicit, counted form of the direct bounded evaluator sketched at the end of the proof of Proposition 11.1 in Part II, which states no witness count; it also supplies per-coordinate heights and a single-valued comparator in place of Part II's case analysis of A^2 against $2B^2$.

For fixed gate arity, the description lengths of C, u, v bound E, h and $\log_2 K$ polynomially. Therefore the height statement in (iii) is a *polynomial per-coordinate bit bound*. The number of coordinates still has its width-dependent factor. No general polynomial-time classical simulation of quantum computation follows.

Exact equality with u/v can be encoded more simply by $va - u2^h = 0$ and $vb = 0$. The equivalence uses the irrationality of $\sqrt{2}$. Non-strict inequalities use the nonnegativity output directly. For a general Boolean predicate of the output bits, its cost must be included: for example, compute its flag reversibly and then use a cylinder event on that flag. A large predicate table is not supplied for free by the cylinder theorem.

75.3 Two examples that expose the semantic issue

First take $C = HH$ and input $|0\rangle$. The unnormalized output amplitudes are $A_0 = 1+1 = 2$ and $A_1 = 1-1 = 0$, while $h = 2$. There are two nonzero fine-grained paths to output 1, but they cancel. Therefore

$$p_C(1) = 0, \quad p_C(0) = 1.$$

A Diophantine certificate for the existence of one path to 1 would answer the wrong question. The contraction certificate answers the right one, and an explicit eight-variable quartic for this calculation is given in Appendix 103.

Next take $C = HTH$, meaning that H , then T , then H is applied. Here

$$A_0 = 1 + \omega, \quad A_1 = 1 - \omega, \quad h = 2,$$

and (74.9) gives

$$p_C(0) = \frac{2 + \sqrt{2}}{4}, \quad p_C(1) = \frac{2 - \sqrt{2}}{4}. \quad (75.6)$$

Thus $p_C(0) > 1/2$ and $p_C(1) < 1/2$ are certified without a numerical approximation to $\sqrt{2}$. The example demonstrates both coherent addition and a genuinely nonrational probability. For the integer-entry Hadamard–Toffoli subcase, the numerator U_B is an ordinary integer and the quadratic-field comparator can be replaced by integer comparison. [write] Both examples are Part II’s Example 11.2, computed there in the coordinates $a + b\sqrt{2} + i(c + d\sqrt{2})$ over 2^L with L the number of gates; here they are computed in the four coordinates of $\mathbb{Z}[\omega]$ over $2^{h/2}$ with h the number of Hadamards. What source 10 adds is the explicit eight-variable quartic of Appendix 103 and the certificate sizes of Section 80.

76 An unconventional realization: adaptive graph-state measurements

[source 10]

76.1 Exact branch semantics

Measurement-based computation replaces a sequence of coherent gates on a moving register by measurements of an entangled resource, with later measurement choices depending on earlier outcomes. Its universality on suitable resource families is due to Raussendorf and Briegel [27]. We now provide a direct arithmetic interface for individual branches; no claim is made that every graph is a universal resource.

Let $G = (V, F)$ be a simple graph with n vertices and r edges. Its graph state is

$$|G\rangle = \prod_{\{i,j\} \in F} CZ_{ij} |+\rangle^{\otimes n} = 2^{-n/2} \sum_{x \in \{0,1\}^n} (-1)^{\sum_{\{i,j\} \in F} x_i x_j} |x\rangle. \quad (76.1)$$

Measure each vertex once, in the order $1, \dots, n$, in the equatorial basis whose outcome- s_i bra is

$$2^{-1/2} (|0\rangle + (-1)^{s_i} \omega^{-k_i} |1\rangle). \quad (76.2)$$

For a fixed transcript $s \in \{0,1\}^n$, the angle index is $k_i = k_i(s_1, \dots, s_{i-1}) \pmod{8}$. We assume these causal rules are explicitly given finite tables or Boolean circuits. This makes their evaluation effective with accounted-for finite input size, rather than hiding a possibly nonterminating classical subroutine in the word “adaptive.”

Theorem 76.1 (Canonical branch certificates). *For the complete measurement transcript s , define*

$$A_{G,s} = \sum_{x \in \{0,1\}^n} (-1)^{\sum_{\{i,j\} \in F} x_i x_j + \sum_i s_i x_i} \omega^{-\sum_i k_i(s_{<i}) x_i}. \quad (76.3)$$

Then its probability is

$$\Pr(s) = 4^{-n} A_{G,s} \overline{A_{G,s}}. \quad (76.4)$$

This numerator has a closed $\mathcal{R}$ -valued binary-edge network realization using only tensors of arity at most three, with at most $4n+6r$ tensors and $2n+8r$ edges. For every supplied contraction tree, it has the canonical quartic certificate of Theorem 74.1. Every base coordinate is at most 2^{2n+8r} . Exact rational threshold assertions use the comparator construction, with denominator 4^n .

Proof. Expand (76.1) and apply the product of the bras (76.2), evaluating the adaptive choices on the fixed transcript. The graph-state normalization contributes $2^{-n/2}$ and the n bras contribute another $2^{-n/2}$. Hence the branch amplitude is $2^{-n} A_{G,s}$, proving (76.4). This unnormalized branch calculation includes the product of all conditional measurement probabilities; one must not separately normalize intermediate branches and then forget the normalizing factors.

For the amplitude numerator, use one binary factor $(-1)^{x_i x_j}$ per graph edge and one unary factor $(-1)^{s_i x_i} \omega^{-k_i x_i}$ per vertex. A variable occurring in several factors is represented by a copy tensor: its value is 1 when all incident binary labels agree, and 0 otherwise. Summing the common value implements the sum over that variable. A rank- k copy tensor can be split into a tree of ternary copy tensors for $k \geq 3$, with rank-one and rank-two cases treated separately. Each satisfying external assignment extends uniquely to the internal labels, so this introduces neither a weight nor a multiplicity. These are equality constraints on tensor indices, not a physical operation that clones an arbitrary quantum state.

At vertex i the copy rank is $\deg(i)+1$. It needs at most $\max(1, \deg(i)-1)$ copy tensors and $\max(\deg(i)-2, 0)$ internal edges. Consequently the amplitude network has at most $2n+3r$ tensors: n unary factors, r edge factors, and at most $n+2r$ copy tensors. Its external factor-to-copy connections number $n+2r$, and its additional internal copy edges number at most $2r$. It therefore has at most $n+4r$ edges.

Take an independent conjugate copy and multiply the two scalar contractions. This doubles both bounds and gives the claimed numerator network, allowing disconnected components as in Section 73. Each nonzero tensor entry is a signed power of ω , so every $\lambda_v = 1$. Theorem 74.1 gives the height bound and canonical quartic representation. Replacing 2^h by 4^n in (75.5) proves the threshold statement. $\square$

76.2 Adaptivity, coarse events, and honest size accounting

For one fixed transcript, the angle rules can be evaluated first, so the coherent tensor construction has exactly the branch semantics above. A collection of accepted transcripts has probability equal to the sum of their individual probabilities. Explicitly enumerating those transcripts and adding the corresponding numerator networks is always valid, but can be exponential. One cannot claim a polynomial-size compilation of a coarse adaptive event merely because each local angle rule is small. A better construction must include the classical control factors and account for their induced width.

This gives a useful separation of responsibilities. The quantum part is certified by the ring-valued contraction. The classical causal rule is either evaluated on the fixed transcript or represented by its own deterministic arithmetic circuit. The final event predicate has its own counted cost. None of these components is replaced by an unexplained assertion that a chosen measurement branch is “legal.”

The graph-state result supplies a non-gate-level example of the same Diophantine interface. Together with an effective unbounded classical controller and an unbounded family of finite resources, such models fit the Turing-complete setting of Section 79. An individual finite graph, like an individual finite circuit, is only a bounded computation.

77 A certificate-size obstruction from Boolean unsatisfiability

[source 10]

The preceding construction has small witness *heights* but can have many witness *coordinates*. The distinction cannot simply be optimized away without a substantial complexity-theoretic consequence.

Theorem 77.1 (Zero-probability certificate obstruction). *Suppose a polynomial-time algorithm takes any finite circuit over $H, X, \text{CNOT}, \text{Toffoli}$ with a specified output event and produces a sparse integer polynomial P_C of some fixed bounded degree, such that*

$$p_C(B) = 0 \iff \exists z \in \mathbb{N}^{k_C} P_C(z) = 0.$$

Suppose further that the polynomial description has polynomial binary length, and that every true instance has a witness whose total binary length is polynomial in the circuit description. Then $\text{NP} = \text{coNP}$. Uniqueness of the witness is not required.

Proof. Let φ be a Boolean formula on n bits. Its Boolean evaluation has a reversible circuit of polynomial size over X, CNOT , and Toffoli, using clean ancillas. To see this directly, compute Boolean subexpressions into fresh work bits: Toffoli computes an AND into a zero target; CNOT copies a bit into a zero target; and NOT and a constant-one bit are supplied by X . OR can be expressed through AND and NOT. Copy the result into a flag and reverse the computation to clean the work bits. This implements

$$|x\rangle|0\rangle_{\text{work}}|0\rangle_{\text{flag}} \mapsto |x\rangle|0\rangle_{\text{work}}|\varphi(x)\rangle_{\text{flag}}.$$

Start the input register in $H^{\otimes n}|0^n\rangle$. Distinct inputs x remain orthogonal, so the flag-1 probability is

$$p_\varphi(1) = \frac{\#\{x : \varphi(x) = 1\}}{2^n}. \quad (77.1)$$

It is zero exactly when φ is unsatisfiable.

Apply the hypothesized compiler to this polynomial-size circuit. On an unsatisfiable formula, guess a witness of the promised polynomial binary length and check the resulting sparse polynomial equation. This verification is polynomial-time: the degree is fixed, the coefficients and variable values have polynomial bit length, and there are polynomially many terms. On a satisfiable formula no witness passes. Therefore UNSAT belongs to NP. Since UNSAT is coNP-complete under polynomial-time many-one reductions (the complement of the SAT completeness theorem [28]), $\text{coNP} \subseteq \text{NP}$, and complementation gives equality. $\square$

Corollary 77.2 (The obstruction also applies to strict thresholds). *A universal polynomial-size, polynomial-bit certificate compiler for the queries $p_C(B) > u/v$, with binary rational thresholds as part of the input and fixed bounded polynomial degree, would also imply $\text{NP} = \text{coNP}$.*

Proof. Use the complementary flag event in (77.1) and the threshold

$$r_n = 1 - 2^{-(n+1)}.$$

Its binary numerator and denominator have $O(n)$ bits. The probability $1 - \#\text{SAT}(\varphi)/2^n$ exceeds r_n exactly when the integer $\#\text{SAT}(\varphi)$ is zero. The same NP-verification argument applies. $\square$

The proof already uses nonnegative classical counting inside a quantum circuit. Thus interference is not the only reason that exact probability certification can be expensive. Coherent cancellation explains why a path witness is semantically inadequate; exact counting explains why a globally cheap replacement is implausible under standard complexity expectations.

These statements are conditional consequences, not proofs that $\text{NP} \neq \text{coNP}$. They do not exclude small certificates on restricted families, certificates with exponentially many variables, or fixed-arity MRDP representations whose least witnesses have no polynomial bit bound. They also require a genuinely polynomially checkable polynomial encoding; a succinct circuit that repeatedly squares gigantic integers is not a substitute for the sparse bounded-degree condition without another verification argument.

78 Exact descriptions versus approximation names

[source 10]

78.1 The representation of a parameter is part of the theorem

For an explicitly supplied rational number a/b , equality to zero is decided by its numerator. For an explicitly supplied element of $\mathcal{R}$, equality is decided by four integer coefficients. A *fast Cauchy name*, in contrast, is an algorithm producing a rational q_k with a guaranteed error $|q_k - x| \leq 2^{-k}$. Effective approximation does not by itself provide a test for exact equality. This is a basic distinction in computable analysis [29]; the following reduction shows exactly how it obstructs a Diophantine interpretation of a quantum observation.

Theorem 78.1 (A one-gate exact-zero obstruction). *There is a uniform computable family of fast Cauchy names for single-qubit unitary matrices R_e such that:*

- (i) *every actual matrix R_e has rational entries;*
- (ii) *the probability of observing 1 after applying R_e to $|0\rangle$ is zero if and only if Turing machine e never halts on blank input;*
- (iii) *this zero-probability index set is not computably enumerable, hence has no existential Diophantine representation over $\mathbb{N}$;*
- (iv) *there is no total computable procedure that, from e , produces an integer polynomial whose natural solvability is equivalent to this zero event.*

Proof. Number machine steps starting at 1 and define

$$a_e = \begin{cases} 2^{-t}, & \text{if machine } e \text{ first halts at step } t, \\ 0, & \text{if it never halts.} \end{cases} \quad (78.1)$$

To approximate a_e to error 2^{-k} , simulate e for k steps. If a halt is seen at step t , output 2^{-t} ; otherwise output zero. A later halt would give a value smaller than 2^{-k} , so this is a uniform total fast Cauchy-name algorithm. Every a_e is individually rational and lies in $[0, 1/2]$.

Now use the rational parametrization of a planar rotation,

$$R(a) = \frac{1}{1+a^2} \begin{pmatrix} 1-a^2 & -2a \\ 2a & 1-a^2 \end{pmatrix}, \quad R_e = R(a_e). \quad (78.2)$$

The identity $(1-a^2)^2 + (2a)^2 = (1+a^2)^2$ proves that $R(a)$ is orthogonal, hence unitary. Its entries are rational whenever a is rational. On the interval $[0, 1/2]$, the derivatives of both scalar entry functions are bounded in absolute value by 2. Thus approximating a_e at precision $k+3$ and substituting the rational approximation into (78.2) gives a rational approximation to each matrix entry with error less than 2^{-k} . This proves uniform validity of the gate names, not merely a promise on an unknown subset of names.

The required output probability is

$$p_e = \frac{4a_e^2}{(1+a_e^2)^2}. \quad (78.3)$$

Its denominator is positive, so $p_e = 0$ exactly when $a_e = 0$, which is exactly nonhalting. The nonhalting set is not computably enumerable: if it and the halting set both were, dovetailing their enumerations would decide halting, contradicting the diagonal halting argument. On the other hand, the projection of the natural zero set of an integer polynomial is computably enumerable, by exhaustive search of natural tuples. This proves (iii).

Finally, suppose the procedure in (iv) existed. On input e , compute its polynomial and enumerate all natural tuples of the returned finite dimension, evaluating the polynomial exactly. The search halts exactly on the zero-event instances, making the nonhalting set computably enumerable. This contradiction proves (iv). $\square$

The strength of the example is that *rational-valuedness alone does not repair the problem*. Each member has a finite exact rational description, but no uniform exact numerator/denominator extraction is available

from these approximation names. The obstruction lies in the effective input interface, not in the existence of transcendental numbers.

Nothing in the theorem assumes a physical device that can manufacture these rotations exactly from arbitrary machine indices. It is a theorem about mathematical specifications and exact predicates, not an experimentally available halting oracle. Finite noisy observations cannot certify the difference between probability zero and an arbitrarily small positive value with a uniform finite guarantee. [write] Source 16 in this Part sits on the other side of the same boundary; see Section 70.2.

78.2 What remains effectively observable

Proposition 78.2 (Strict inequalities survive approximation names). *For a uniformly computable family of finite circuits with computable entries, and a fixed finite measurement event whose probability has a computable error bound, $p > r$ is computably enumerable for rational r . Under these valid-name assumptions, equality $p = r$ is co-computably enumerable. Exact zero is Π_1^0 -complete already on the family in Theorem 78.1.*

Proof. Compute rational intervals $[l_k, u_k]$ containing p , of widths tending effectively to zero. For $p > r$, search for $l_k > r$. The search succeeds exactly for the strict inequality. The analogous search for $u_k < r$ decides strict inequality on the other side semidecidably. Dovetailing these searches semidecides $p \neq r$, proving the equality assertion. For finite circuits, finite additions and multiplications of certified entry approximations give such intervals; a coarse uniform entry bound is available from unitarity. The reduction in Theorem 78.1, and nonhalting's form $\forall t [e \text{ has not halted by } t]$, give completeness on that family. $\square$

MRDP may therefore supply an existential arithmetic representation of a strict observable for a uniformly valid approximation-name family. It need not supply one for exact equality. A proposed “universal substrate compiler” must state both its parameter representation and the logical form of the observation; Turing completeness by itself settles neither issue.

79 Unbounded algebraic controllers and the quantifier boundary

[source 10]

79.1 A genuinely unbounded computational substrate

Consider finite programs with an ordinary Turing-complete classical controller, a finite quantum register at each finite time, allocation of qubits in $|0\rangle$, gates from the exact alphabet of Section 75, and computational-basis measurements whose outcomes may affect the controller. There is an absorbing classical halting flag. One controller step performs a bounded number of these elementary actions. Arbitrarily long computations and arbitrarily large finite registers are permitted; an individual program still has a finite effective description.

This is Turing-complete even without using its coherent operations, because the classical controller can simulate an ordinary Turing machine. Its finite quantum prefixes also cover the coherent and measurement-based constructions above. No claim of super-Turing computational power is involved.

For a coded program/input pair z , let $H_t(z)$ be the probability that the halting flag has been set by step t . A finite prefix has finitely many measurement transcripts and a finite quantum register on each branch. Simulate unnormalized branch vectors instead of repeatedly normalizing conditional states. Their entries use exact cyclotomic arithmetic, and their squared norms have effectively known powers-of-two denominators. Summing halted branches therefore computes $H_t(z)$ exactly as an element of $\mathbb{Q}(\sqrt{2})$. In particular, rational comparisons with H_t are decidable. Also

$$0 \leq H_t(z) \leq H_{t+1}(z) \leq 1, \quad H_\infty(z) = \sup_t H_t(z). \quad (79.1)$$

The exact computation may be very expensive; only its effectiveness is used.

[write] The next two theorems are the quantum-controller cases of the endpoint thresholds $p > 0$ and $p = 1$ that Part I records after Theorem 9.1; the arguments are second routes, and the totality sampler (79.4) is (9.1).

Theorem 79.1 (Positive-probability halting is uniformly Diophantine). *The set of program/input codes z with $H_\infty(z) > 0$ is Σ_1^0 -complete. Consequently there are a fixed integer k and a fixed integer polynomial $P(z, w_1, \dots, w_k)$ such that*

$$H_\infty(z) > 0 \iff \exists w \in \mathbb{N}^k P(z, w) = 0. \quad (79.2)$$

The polynomial can also be replaced by a fixed degree-at-most-four polynomial with finitely many additional witnesses. No single-witness assertion follows for this uniform unbounded representation.

Proof. By monotonicity and the definition of supremum, $H_\infty(z) > 0$ is equivalent to $\exists t [H_t(z) > 0]$. The bracketed predicate is decidable by exact finite-prefix evaluation, proving membership in Σ_1^0 . Ordinary deterministic halting reduces to this problem by using only the classical controller: its eventual halting probability is 1 if it halts and 0 otherwise. Deterministic halting is Σ_1^0 -complete, as any predicate $\exists s R(z, s)$ with decidable R is implemented by searching for such an s .

Apply MRDP to this computably enumerable set of natural codes [48]. This is exactly the finite-polynomial interface stated in the inspected ProveIt theorem [23], and yields (79.2). To obtain degree at most four, express evaluation of the fixed polynomial P by a finite arithmetic DAG, introduce the signed-wire constraints of Theorem 72.3, require its output to be zero, and sum the squared residuals. The old witnesses w remain quantified, and the additional wire variables evaluate P canonically. For every fixed old tuple there is exactly one auxiliary evaluation, but MRDP may supply many different old witness tuples. Thus canonical evaluation does not turn a many-witness MRDP representation into a single-witness one. $\square$

The quantifier order in (79.2) matters: k and P are chosen before z , not separately after a finite running time is known. A family of finite trace polynomials with more variables for larger t is not, by itself, a fixed-arity Diophantine representation. MRDP supplies precisely that missing uniform compression; it does not supply the finite-circuit height or multiplicity bounds.

79.2 Almost-sure halting has a different logical form

Theorem 79.2 (Almost-sure halting). *For the algebraic quantum controllers above, the property $H_\infty(z) = 1$ is Π_2^0 -complete. It is not computably enumerable and thus cannot be the projection of the natural zero set of an integer polynomial.*

Proof. Using (79.1), almost-sure halting is equivalent to

$$\forall k \geq 1 \exists t \quad H_t(z) > 1 - 2^{-k}. \quad (79.3)$$

The finite-prefix comparison is decidable, giving the upper bound.

For hardness, take an index e for an ordinary partial computable function. Sample $n \in \mathbb{N}$ with probability $2^{-(n+1)}$, by tossing fair coins until the first zero and counting preceding ones. Fair coins are implemented by preparing $|0\rangle$, applying H , and measuring. The probability of sampling forever is zero. After choosing n , simulate machine e on input n and halt exactly when that simulation halts. The eventual halting probability is

$$\sum_{n \geq 0} 2^{-(n+1)} \mathbf{1}_{\{e(n) \text{ halts}\}}. \quad (79.4)$$

It equals 1 if and only if e is total: every omitted input has a strictly positive weight. Totality is Π_2^0 -complete. Indeed, for a predicate $\forall n \exists s R(x, n, s)$ with decidable R , the machine that on input n searches for an s satisfying $R(x, n, s)$ is total exactly for the desired x . This proves the lower bound.

For a direct non-enumerability argument, nonhalting reduces to totality. Given a blank-input machine e , construct a machine which, on input n , simulates e for n steps, halts if no halt was seen, and otherwise loops

forever. This machine is total exactly when e never halts. Combining this reduction with (79.4) shows that almost-sure halting cannot be computably enumerable. Polynomial solvability is computably enumerable, so no existential Diophantine representation of this index set is possible. $\square$

The classical probabilistic almost-sure-termination classification is already established by Kaminski and Katoen [6]. The theorem above gives its explicit embedding into the present algebraic quantum-controller model, together with an exact finite-prefix upper bound. It is not a claim to have newly discovered the classical Π_2^0 result.

79.3 Three interfaces that must not be conflated

The finite exact-circuit interface admits direct canonical quartic witnesses, with dimensions dependent on a contraction schedule. The unbounded positive-halting interface admits a fixed polynomial by MRDP, but loses the proved unique-witness and height guarantees. The almost-sure interface contains an essential universal quantifier and admits no such existential polynomial. The approximation-name exact-zero interface already fails enumerable observability with one gate. These are differences in predicates and effective representations, not contradictions between models of computation.

80 Implementation, exact experiments, and reproducibility

[source 10]

80.1 The delivered prototype

The package includes a standard-library-only Python implementation for Python 3.10 or later. The file `code/coherent.py` implements four-integer cyclotomic arithmetic, canonical signed wires, sparse quadratic residuals, exact quantum state evaluation, doubled probability networks, ternary-copy measurement networks, and the $\sqrt{2}$ comparison gadget. The tensor prototype uses binary edges ($D = 2$); the mathematical theorem permits general finite D .

Two deterministic contraction schedules are provided. The **greedy** policy chooses a pair minimizing the current merge-union size, breaking ties by output rank and current indices. The **queue** policy merges the first two current tensors and appends their result. Neither is an optimization certificate for the best contraction tree. In particular, “greedy” is a heuristic, not a theorem about the globally smallest width.

The prototype constructs witnesses while emitting their constraints, allowing an immediate exact evaluation check. Its scalar ring multiplication always uses the explicit 16-product, 12-addition/subtraction formula, even when many entries are zero. This makes it an auditable implementation of the proved count, not an optimized state-of-the-art simulator. Equal scalar constants are shared. If c distinct scalar constants occur in the leaf tables, the observed base count is

$$V_{\text{shared}} = 2c + 64T_{\text{mul}} - 8O, \quad (80.1)$$

which is bounded by (74.4) because $c \leq 4L_{\text{leaf}}$.

80.2 The finite verification suite

All arithmetic checks use integers or rational numbers; no floating-point tolerance is used to decide an identity or a sign. The run recorded in `data/verification.json` passed the following tests:

Check	Count
Exact circuit states, each with a normalization check	926
Doubled circuit-network values against direct state evaluation	3,966
Adaptive graph-state amplitudes against independent gate realization	296
Complete adaptive branch-distribution normalizations	44
Native graph-factor amplitude/probability network equalities	592
High-degree tests of ternary copy-tensor decomposition	2
Cyclotomic norm identities on coefficient tuples in $\{-2, \dots, 2\}^4$	625
Exact $\sqrt{2}$ sign-circuit checks on $[-24, 24]^2$	2,401
Integer-sign gadgets exhaustively checked in covering boxes	13
Quadratic-certificate instances, including false-query rejection	43
Single-coordinate witness mutations rejected	240
Contraction/comparison witness-height checks	42

The circuit states include every word of length at most four in the one-qubit alphabet H, T, X , for both basis inputs; every word of length at most three in a specified two-qubit alphabet, for all four basis inputs; and 60 seeded three-qubit circuits. The random seed is 20260930. The graph-state tests use all simple graphs on one through three vertices, all output transcripts, and four causal angle-rule families. Higher-degree copy decompositions are also exercised on a six-vertex star and a five-vertex complete graph.

For the sign tests, an independent rational enclosure of $\sqrt{2}$ is built by integer square root at denominator 10^{30} . This enclosure separates every nonzero tested expression, so the comparison test does not merely call the same sign routine twice. For norm checks, the integer formula (74.9) is compared to multiplication by the conjugate in the four-coordinate algebra.

A separate program, `code/check_certificate.py`, imports no generator code. It checks natural coordinates and the vanishing of all exported quadratic residuals. For the small expanded quartic, it also independently multiplies and collects the residual squares and verifies the polynomial identity, not just its value at the supplied point. This checker validates algebraic witnesses, while correspondence to the stated circuit remains the semantic compiler theorem and the cross-checks above.

80.3 Representative certificate sizes

The following are actual base counts from the delivered run. “Bits” is the largest coordinate bit length in that base witness. The two exported full circuit examples additionally include four linear output-value assertions; those four equations do not add variables and are not included in this table.

Circuit / observed event	Schedule	w	T_{mul}	Variables	Bits
HH , output 1	greedy	2	22	1,326	2
HH , output 1	queue	4	52	3,086	2
HTH , output 0	greedy	2	30	1,806	2
HTH , output 0	queue	4	64	3,806	2
HTH , output 1	greedy	2	30	1,806	2
HTH , output 1	queue	4	64	3,806	2
Bell preparation, qubit 0 equals 0	greedy	6	192	11,678	1
Bell preparation, qubit 0 equals 0	queue	5	140	8,414	1
Two Hadamards, Toffoli flag equals 1	greedy	8	860	53,342	1
Two Hadamards, Toffoli flag equals 1	queue	8	848	51,838	1

All examples start in the all-zero basis state. Bell preparation means H on qubit 0 followed by CNOT from 0 to 1. The Toffoli example applies H to each control, then Toffoli to a zero target; its flag-1 probability is $1/4$. The Bell event has probability $1/2$. The HTH values are (75.6). The queue schedule beating the

greedy heuristic on the Bell example is a useful warning against reading an optimization guarantee into local pair selection.

These counts are intentionally not advertised as competitive minimal Diophantine representations. Applying generic ring multiplication to every zero entry is wasteful. The value of the prototype is that the unoptimized construction, exact counts, output semantics, and height checks are all visible and reproducible. The much smaller hand-specialized quartic in Appendix 103 illustrates how large the optimization margin is.

80.4 Commands and trust boundary

From the package directory, run:

```
python code/verify.py
python code/check_certificate.py data/HH_expanded_quartic.json \
  data/HH_zero_certificate.json data/HTH_zero_certificate.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The first command recreates the JSON examples and test record. Its elapsed-time field is naturally machine-dependent. The tests use Python assertions and should not be run with the `-O` optimization flag. The PDF uses standard LaTeX packages and an embedded bibliography, so no external bibliography processor is needed.

No Lean files are supplied as supposedly completed proofs. A sensible formalization would separate canonical arithmetic evaluation, finite tensor semantics, the four-coordinate algebra, real comparison, and the operational controller model. The small independent checker is useful for auditing an individual witness, but it does not replace a proof that every emitted system has the intended meaning. General uniqueness follows from the topological induction, not from the 240 mutation tests.

[write] In this report the programs are shipped as `code/10-coherent-circuits-coherent.py`, `code/10-coherent-circuits-verify.py`, `code/10-coherent-circuits-check_certificate.py` and `code/10-coherent-circuits-build.py`, and the files of `data/` as `data/10-coherent-circuits-*`. The commands above are the source's, for its delivered layout; the README gives the rerun recipe on a copy.

81 The question and the main separation

[source 11]

A finite list of rational orthogonal matrices looks like an exceptionally benign computational substrate. Every individual operation is linear, invertible, norm-preserving, and exactly computable. Every finite execution can be checked using integer arithmetic. None of these properties, however, settles the existential question: does *some* finite execution move a specified state to a specified target?

The basic undecidability of rotation reachability on the three-sphere is not new: Bell and Potapov established it using quaternion encodings [35]. Our objective is a more discriminating question:

Can exact state transfer remain computationally universal when membership of a proposed *complete transformation* is efficiently decidable, all controls are reversible, all coefficients use a single denominator prime, and approximation is uniformly possible?

The answer proved here is yes. Moreover, one and the same control family has an easy orbit and a hard orbit, starting at two explicitly specified small rational vectors.

Write $S^3(\mathbb{Q}) = \{v \in \mathbb{Q}^4 : \|v\| = 1\}$. A rational tuple is supplied by binary numerators and positive binary denominators; equivalent reasonable canonical encodings differ only polynomially in length. For a finite symmetric set $\mathcal{S}$ of matrices, let $\Gamma = \langle \mathcal{S} \rangle$. The empty control word is allowed.

Theorem 81.1 (Main separation). *There exists a fixed finite symmetric list*

$$\mathcal{S} \subseteq \text{SO}(4, \mathbb{Z}[1/5])$$

with generated group Γ and fixed vectors

$$e = (1, 0, 0, 0), \quad \xi = \left(\frac{3}{5}, 0, -\frac{28}{125}, \frac{96}{125} \right) \quad (81.1)$$

such that the following hold.

- (i) *Given any rational 4×4 matrix A , deciding $A \in \Gamma$ takes polynomial time in its binary input length.*
- (ii) *Given $y \in S^3(\mathbb{Q})$, deciding $y \in \Gamma e$ takes polynomial time, whereas the set of rational targets $y \in \Gamma \xi$ is computably enumerable complete under computable many-one reductions.*
- (iii) $\bar{\Gamma} = \text{SO}(4)$. *The two displayed orbits are disjoint and dense in S^3 .*
- (iv) *There is a total computable function B on positive rational tolerances such that every $x, y \in S^3$ can be moved within ε using at most $B(\varepsilon)$ gates.*
- (v) *No total computable function of the target's bit length bounds either the shortest exact transfer word from ξ , or the least common denominator exponent of an exact transfer matrix in Γ .*

Here “complete” means that any computably enumerable set reduces to the reachability set. It does not mean that this is an autonomous differential equation simulating a Turing machine step for step. The computational semantics are *existential finite-word reachability*: the control word is the witness. The distinction matters physically, since exact equality and arbitrarily long control sequences are essential.

There is a second, unusually restricted form of the hard input family.

Theorem 81.2 (Fifth-order hard targets). *The family in Theorem 81.1 can be chosen with fixed rational unit quaternions R_0, S_0 such that, for a fixed computably enumerable complete set $K \subseteq \{1, 2, \dots\}$,*

$$y_n = \xi [S_0, R_0^n S_0 R_0^{-n}], \quad y_n \in \Gamma \xi \iff n \in K \quad (n \geq 1). \quad (81.2)$$

The commutator convention is $[u, v] = uvu^{-1}v^{-1}$. Every coordinate of (y_n) satisfies the same rational constant-coefficient recurrence of order at most five. The recurrence polynomial is given explicitly in Theorem 89.1.

The fixed quaternions R_0, S_0 describe the *input sequence*; their left multiplications are not added to the allowed gate list. Adding arbitrary extra gates can change the orbit problem and invalidate a hardness reduction.

81.1 Relationship to existing work and to ProveIt

Free rational rotation encodings go back to work such as Świerczkowski's, and appear explicitly in the quaternion and automata literature [34, 36]. The subgroup construction used below is Mihailova's fiber-product construction; modern treatments include Bogopolski and Ventura [72]. The connection between orbit problems and group conjugacy has an extensive literature [33]. Miller's free-by-free construction and its marked conjugacy property are classical [31]; Gillis provides a recent detailed analysis and quantitative refinements [32]. These components are credited rather than presented as newly discovered theorems.

The additional work here is the simultaneous arithmetic and algorithmic packaging: a full rational-matrix membership decoder, its polynomial-time bound, two contrasting dense orbits with fixed small starting points, a uniform approximation bound, a low-order target recurrence, and a sparse Diophantine certificate interface. The source search supporting this note was focused, not an exhaustive priority search.

ProveIt's Hilbert-tenth-problem development supplies a natural endpoint for the arithmetic part. At the inspected revision its public theorem `Diophantine.mrdp` states that every recursively enumerable subset of

$\mathbb{N}$ is the projection of the natural zero set of one integer polynomial, with the witness dimension fixed before the input [38]. ProveIt’s MRDP guide (`Computability/HilbertTenthProblem/Lean/MRDP.md`; [write] the source calls it “the companion guide”) explicitly distinguishes this existence theorem from an optimized polynomial generator or a numerical degree bound [39]. We use that distinction throughout: the bounded quartic below is explicit; the fixed unbounded polynomial is an MRDP consequence, not a newly printed universal polynomial.

82 Research question and precise scope

[source 16]

82.1 What is being represented?

A rational rotation is a matrix $M \in \mathrm{SO}(4) \cap \mathbb{Q}^{4 \times 4}$. Given a fixed finite alphabet $\mathcal{S} = (S_0, \dots, S_{s-1})$ of such matrices, its *exact synthesis problem* asks whether

$$M = S_{j_1} S_{j_2} \cdots S_{j_n} \quad \text{for some } n \geq 0. \quad (82.1)$$

The empty word has value I_4 . We include the identity as S_0 and include inverses, so the generated monoid is a group. Repeated or identity labels are retained as distinct symbols when counting witnesses.

This is a computational substrate in the following precise sense: a halting question can be transformed effectively into a target matrix, and the machine halts exactly when a finite gate sequence realizes that target. It is *not* an assertion that one fixed rotation, iterated without control, simulates a Turing machine. Nor does it assert that a physical experiment with finite precision decides halting. The obstruction is exact symbolic equality of rational transformations.

Three questions motivate the work. Can all state norms remain bounded while synthesis remains computationally universal? Can this happen even when every target is approximable? Can arithmetic certificates retain individual finite executions rather than hide them in an unspecified MRDP polynomial? The theorems below answer all three affirmatively, with explicit distinctions between finite horizons and unbounded search.

82.2 Relationship to existing research

Undecidability for quaternion and rotation semigroups is established prior work, not a new conclusion of this article. Bell and Potapov proved undecidability results for quaternion matrix membership and four-dimensional point rotation [35]. Their work also uses free rational quaternion encodings. The particular 3–4–5 pair used here appears again in Proposition 3 of the 2026 paper by Bell, Kenison, Niskanen, Potapov, and Semukhin [71]. That paper studies low-dimensional word embeddings; it is relevant to the representational boundary, not a certificate of novelty for the present results.

Mihailova’s construction converts equality in a finitely presented quotient into membership in a finitely generated subgroup of a direct product of free groups. We use the standard construction and reprove the required membership equivalence. Bogopolski and Ventura give a modern treatment and historical attribution [72]. Higman’s embedding theorem provides the finite-presentation input needed for a fixed complete example [30]. Products of free groups in Lie groups are also a developed subject [73]; our elementary dimension obstruction is specific to the mechanism used here.

The repository’s public MRDP guide states an equivalence between computable enumerability and existence of a finite-natural-witness integer polynomial representation, including input zero. It expressly does not claim a practical polynomial compiler or a numerical degree bound [76]. Our bounded construction therefore supplies information not contained in that existence interface: actual equations, a variable count, and an exact correspondence between words and witnesses.

82.3 Status of the claims

Claim	Status in this article
Mihailova, Higman, MRDP	Classical inputs, explicitly attributed.
Exact denominator and decoding	Proved by a complete finite-field calculation.
Dense rational compiler	Constructive for every supplied finite presentation.
Local completeness and precision boundary	Proved consequences, with all reductions specified.
Fixed universal gate set	Existence proved; no small universal presentation instantiated.
Bounded quartic certificates	Explicit polynomial scheme and witness bijection.
Unbounded quartic representation	Existence via MRDP and circuit reduction, not an expanded polynomial.
Machine verification	Exact finite Python tests only; no Lean build or formal proof.

The intended research output is a reusable theorem package, not a claim that a named longstanding conjecture has been settled. An independent literature review and proof review would be required before advertising historical novelty or a breakthrough.

83 Main theorem package

[source 16]

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. A language is *c.e.-complete* if it is computably enumerable and every computably enumerable language reduces to it by a total computable many-one reduction. Rational matrices have finite canonical encodings; any standard effective encoding gives the same computability conclusions.

Theorem 83.1 (Dense rational exact synthesis). *There exists a fixed finite symmetric labeled alphabet $\mathcal{S} \subseteq \text{SO}(4, \mathbb{Z}[1/5])$, including I_4 , with the following properties.*

- (i) *Its generated group Γ is dense in $\text{SO}(4)$, but membership in Γ for rational input matrices is c.e.-complete.*
- (ii) *For every rational $C \in \text{SO}(4)$ and rational $r > 0$, the language*

$$\{M \in \text{SO}(4, \mathbb{Q}) : \|M - C\|_F < r \text{ and } M \in \Gamma\}$$

is c.e.-complete. The hardness reductions can be obtained effectively from C, r after fixing the source group and its embedding constants.

- (iii) *A total algorithm produces an ε -approximation to every rational target rotation for every rational $\varepsilon > 0$. There is also a computable bound $B(k)$ such that words of length at most $B(k)$ form a 2^{-k} -net of all of $\text{SO}(4)$.*
- (iv) *Fix a common positive denominator D for the gates. If $M = T/q$ with $q \in \mathbb{Z}_{>0}$ and T integral, a word W of length n satisfies*

$$\|W - M\|_F < \frac{1}{qD^n} \iff W = M.$$

Thus searching for an approximation at this output-length-dependent precision is again c.e.-complete.

- (v) *For each horizon n , there is an explicitly specified polynomial P_n of total degree at most four, with $n(s + 17)$ natural witnesses, whose zeros correspond bijectively to successful labeled words of length n . It is a sum of $n(s + 18) + 16$ squares of quadratic or linear polynomials.*

The proof occupies Sections 84.2–98. Part (i) requires a group with complete word problem, while part (ii) uses its free product with one infinite cyclic group. Parts (iii)–(v) apply more generally to any suitable dense rational alphabet or, for (iv)–(v), to any finite rational rotation alphabet.

There are two useful supplementary conclusions. The smallest exact synthesis lengths have no computable bound in terms of rational height; a natural maximal-depth function has Turing degree $\mathbf{0}'$. And the faithful commuting-free-factor construction cannot be moved from real dimension four to dimension three. Neither statement should be confused with a classification of all lower-dimensional synthesis problems.

84 [write] The shared arithmetic core of sources 11 and 16

[write] This section prints once what sources 11 and 16 both prove. Each subsection names the source of each block. Source 11's section labels and source 16's are attached to the subsection that carries their content.

84.1 Quaternion conventions and the spin map

Source 11. Hamilton's algebra has basis $1, i, j, k$, with $i^2 = j^2 = k^2 = ijk = -1$. For $p = p_0 + p_1i + p_2j + p_3k$, put

$$\bar{p} = p_0 - p_1i - p_2j - p_3k, \quad N(p) = p\bar{p} = p_0^2 + p_1^2 + p_2^2 + p_3^2.$$

Quaternion conjugation reverses products and $N(pq) = N(p)N(q)$. Thus unit quaternions form the three-sphere as a group. We freely identify a quaternion with its four real coordinates.

For a rational quaternion, $\text{den}(p)$ denotes the least positive integer d for which $dp \in \mathbb{Z}^4$. This is the least common denominator, not the largest of four individual denominators.

Source 16 writes $|p|^2$ for $N(p)$; its unit quaternions form $\text{Sp}(1)$, isomorphic to $\text{SU}(2)$, and its den is the same least common denominator. Both sources fix

$$a = \frac{3+4i}{5}, \quad b = \frac{3+4j}{5}, \tag{84.1}$$

$$A = a^{-1} = \frac{3-4i}{5}, \quad B = b^{-1} = \frac{3-4j}{5}. \tag{84.2}$$

All four have norm one. Let $F(a, b)$ initially denote the *abstract* free group; evaluation in (84.2) will be denoted by η . For unit p, q both define the spin map

$$\Phi(p, q)(z) = pzq^{-1}, \quad z \in \mathbb{H} \simeq \mathbb{R}^4, \tag{84.3}$$

$$\Phi(p, q)(v) = pv\bar{q} \quad (\text{source 11's form; } \bar{q} = q^{-1} \text{ for unit } q). \tag{84.4}$$

This is a group homomorphism from $S^3 \times S^3$ to $\text{SO}(4)$: left and right multiplication commute, norm preservation gives orthogonality, and the determinant is 1 by continuity from the identity on the connected space $S^3 \times S^3$. This is real linear and norm preserving, and

$$\Phi(p, q)\Phi(p', q') = \Phi(pp', qq'). \tag{84.5}$$

The order qq' in the second coordinate is important: the inverse of qq' is $q'^{-1}q^{-1}$.

The matrix for left multiplication is

$$L(p) = \begin{pmatrix} p_0 & -p_1 & -p_2 & -p_3 \\ p_1 & p_0 & -p_3 & p_2 \\ p_2 & p_3 & p_0 & -p_1 \\ p_3 & -p_2 & p_1 & p_0 \end{pmatrix}. \tag{84.6}$$

The j th column of the full matrix $\Phi(p, q)$ is simply the coordinate vector of $pe_j\bar{q}$, with $(e_0, e_1, e_2, e_3) = (1, i, j, k)$. This column formula is also the implementation specification.

Lemma 84.1 (Kernel and surjectivity). *The kernel of Φ is $\{(1, 1), (-1, -1)\}$, and Φ is onto $\text{SO}(4)$.*

Proof. If $pv\bar{q} = v$ for all v , substitution of $v = 1$ gives $p = q$. Then p commutes with both i and j , so p is real and hence $p = \pm 1$. Conversely these two pairs are in the kernel.

For surjectivity let $A \in \text{SO}(4)$ and $a = A(1)$. Left multiplication by $\bar{a}$ followed by A fixes 1. It therefore restricts to an orientation-preserving rotation on the imaginary three-space. The axis-angle formula represents every such three-dimensional rotation as $v \mapsto qv\bar{q}$ for a real unit quaternion q . Consequently $A = \Phi(aq, q)$. $\square$

All coefficients of $\Phi(p, q)$ are bilinear integer expressions in the coordinates of p, q . In particular, rational unit quaternions give rational rotations. The converse at the level of rational points is false: a rational rotation can have only irrational unit-quaternion lifts. Section 84.7 handles this explicitly.

84.2 Denominator rigidity

Let $F(a, b)$ be the free group on two letters and evaluate a, b at the quaternions (84.1), writing η for the evaluation. The following elementary argument proves the required faithfulness and gives more than a nonvanishing test (source 11).

Theorem 84.2 (Denominator rigidity; sources 11 and 16). *If $w = s_1 \cdots s_n$ is a freely reduced word of length n in $a^{\pm 1}, b^{\pm 1}$ (source 11 writes ℓ for the length), then its evaluated quaternion satisfies*

$$\text{den}(\eta(w)) = 5^n. \quad (84.7)$$

Consequently η embeds $F(a, b)$ into $S^3(\mathbb{Z}[1/5])$, and its image contains no integral quaternion except 1; in particular -1 is not in it.

Proof (source 11). Reduce the integral quaternion algebra modulo 5 and use the algebra map

$$i \mapsto \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad j \mapsto \begin{pmatrix} 0 & 1 \\ 4 & 0 \end{pmatrix} \quad \text{over } \mathbb{F}_5.$$

The images square to $-I$, anticommute, and together with I and their product form a basis of $M_2(\mathbb{F}_5)$; hence this is an isomorphism. The four letter numerators have images

$$\begin{aligned} A_+ &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, & A_- &= \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \\ B_+ &= \begin{pmatrix} 3 & 4 \\ 1 & 3 \end{pmatrix}, & B_- &= \begin{pmatrix} 3 & 1 \\ 4 & 3 \end{pmatrix}. \end{aligned} \quad (84.8)$$

Each has rank one. Their image lines are distinct. The image of the inverse letter equals the kernel of the original letter. Inspection of these four lines therefore gives

$$M_x M_y = 0 \iff y = x^{-1}.$$

For rank-one matrices $M_t = u_t v_t^T$, the full product is

$$u_1 (v_1^T u_2) \cdots (v_{\ell-1}^T u_\ell) v_\ell^T.$$

It is nonzero when none of the adjacent scalar factors vanishes. A freely reduced word therefore has an integral numerator not divisible coordinatewise by 5. Its denominator before reduction is 5^ℓ , so this common denominator is already minimal. If $\ell > 0$ the result cannot be 1, or any integral quaternion. Applying this to the reduced word for uv^{-1} proves injectivity. $\square$

Second proof (source 16): the finite-field tables. The quaternion algebra over $\mathbb{F}_5$ has the following matrix realization:

$$i \mapsto \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad j \mapsto \begin{pmatrix} 0 & 1 \\ 4 & 0 \end{pmatrix}, \quad k \mapsto \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}. \quad (84.9)$$

These matrices satisfy the quaternion relations. The matrices for $1, i, j, k$ are linearly independent over $\mathbb{F}_5$, so the induced linear map $\sigma : \mathbb{H}(\mathbb{F}_5) \rightarrow M_2(\mathbb{F}_5)$ is an algebra isomorphism. Explicitly,

$$\sigma(p) = \begin{pmatrix} p_0 + 2p_1 & p_2 + 2p_3 \\ 4p_2 + 2p_3 & p_0 + 3p_1 \end{pmatrix} \pmod{5}. \quad (84.10)$$

For a letter $s \in \{a, A, b, B\}$, write $s = c_s/5$ with integral quaternion numerator c_s . Its residue matrix factors as $\sigma(c_s) = u_s v_s^\top$:

$$\begin{array}{c|c|c|c} s & \sigma(c_s) & u_s & v_s \\ \hline a & \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} & (1, 0)^\top & (1, 0)^\top \\ A & \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} & (0, 1)^\top & (0, 1)^\top \\ b & \begin{pmatrix} 3 & 4 \\ 1 & 3 \end{pmatrix} & (3, 1)^\top & (1, 3)^\top \\ B & \begin{pmatrix} 3 & 1 \\ 4 & 3 \end{pmatrix} & (3, 4)^\top & (1, 2)^\top \end{array} \quad (84.11)$$

The complete table of pairings, with rows and columns in the order a, A, b, B , is

$$(v_s^\top u_t)_{s,t} = \begin{pmatrix} 1 & 0 & 3 & 3 \\ 0 & 1 & 1 & 4 \\ 1 & 3 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix} \pmod{5}. \quad (84.12)$$

A pairing vanishes exactly when $t = s^{-1}$.

Second proof (source 16). For $n = 0$ the assertion is immediate. For $n > 0$ put $C = c_{s_1} \cdots c_{s_n} \in \mathbb{H}(\mathbb{Z})$. Multiplicativity of σ and (84.11) give

$$\sigma(C) = u_{s_1} \left(\prod_{j=1}^{n-1} v_{s_j}^\top u_{s_{j+1}} \right) v_{s_n}^\top.$$

Every scalar in the product is nonzero because the word is freely reduced. The exterior column and row are nonzero, so $\sigma(C) \neq 0$. Consequently at least one of the four integer coordinates of C is not divisible by 5. Since $\eta(w) = C/5^n$, its least common denominator is exactly 5^n .

Every nonidentity element of the abstract free group has a nonempty reduced representative, and hence its image cannot equal 1, whose denominator is one. The same argument excludes -1 for nonempty words; the empty word evaluates to 1. $\square$

Remark 84.3. The assertion concerns the length *after free reduction*. For example, aA evaluates to 1 and has denominator one, not 25. No lower bound is asserted in terms of the length of an unreduced spelling.

84.3 Decoding

Proposition 84.4 (Polynomial-time quaternion recognition; residue decoder; sources 11 and 16). *Membership of a rational quaternion in $\eta(F(a, b))$, together with recovery of the unique reduced word, is decidable in polynomial time in its bit length (source 11). For a member of least denominator 5^n , the reduced word is recovered by at most n residue-and-peeling steps (source 16).*

Proof (source 11). Check the norm and reject unless it is 1. Check that the least common denominator is a power 5^ℓ ; no general integer factorization is required. If $\ell = 0$, accept exactly the quaternion 1.

Otherwise the integral numerator has norm $5^{2\ell}$. Its image modulo 5 is a nonzero singular 2×2 matrix, and hence has rank one. Test the four right products $p\eta(x)^{-1}$. Keep a letter only when the least common denominator drops to $5^{\ell-1}$. There is at most one such letter: modulo 5, a drop requires the image line of its inverse numerator to equal the kernel line of the nonzero current numerator matrix. The four candidate image lines are distinct. For an actual encoded word its last letter does give the specified drop by Theorem 84.2.

Reject if no candidate works; otherwise record the letter and repeat. After ℓ steps accept exactly when the residual quaternion is 1. Soundness follows by multiplying back the recorded letters; completeness follows by stripping an actual reduced word. A denominator drop cannot be inferred merely from a zero product modulo 5: the algorithm tests the *exact* denominator.

For a unit quaternion with denominator 5^j , every numerator has absolute value at most 5^j . Thus all intermediate arithmetic has bit length bounded by a constant times the input length. The number of steps is $O(\ell)$ and ℓ is bounded by the input length. Exact rational arithmetic, greatest common divisors, and division by 5 are polynomial-time operations. $\square$

Second proof (source 16): peeling on the left. The four lines spanned by u_a, u_A, u_b, u_B are distinct in $\mathbf{P}^1(\mathbb{F}_5)$. For a nonempty reduced word, the proof of Theorem 84.2 shows that $\text{im } \sigma(C)$ is precisely the line of its first letter.

Here is an algorithm valid also on nonmembers. First check that the input has norm one. Compute its least common denominator; reject unless it is 5^n for some $n \geq 0$. Write the quaternion as $C/5^n$ with integer coordinate vector C . If $n = 0$, accept only $C = 1$. For $n > 0$, compute $\sigma(C)$. Reject unless its image is one of the four indicated one-dimensional lines; that line specifies a letter s . Compute $c_{s^{-1}}C$. Require coordinatewise divisibility by 25, replace C by $c_{s^{-1}}C/25$, and decrease n by one. Repeat and accept only when the final quaternion is 1.

For a true member, the identified letter is correct and stripping it leaves its reduced suffix. Its denominator decreases from 5^n to 5^{n-1} , so all tests pass. Conversely, every successful peeling step is the exact rational identity $q = sq'$. Acceptance at the final identity therefore reconstructs a word evaluating to the original input. This proves soundness independently of any promise that the input was a member. Termination follows from the decreasing nonnegative counter. $\square$

The denominator is thus an exact intrinsic length for this particular free-group coordinate system. The difficult membership problem constructed later does not come from an inability to read these ambient words. It comes from deciding which *pairs* of readable words agree in a finitely presented quotient.

84.4 Free groups of arbitrary finite rank

Source 11 indexes the generators from 0. For $r \geq 1$, let $F_r = F(x_0, \dots, x_{r-1})$ and define

$$\phi(x_j) = a^j b a^{-j}. \quad (84.13)$$

The conjugates $a^j b a^{-j}$ for $j \in \mathbb{Z}$ form a free basis for the kernel of the exponent-sum homomorphism $F(a, b) \rightarrow \mathbb{Z}$ that sends a to 1 and b to 0. For completeness, scan a binary word from left to right, recording the current a -height whenever a $b^{\pm 1}$ occurs. This writes the word as a product of indexed conjugates followed by its final power of a . A nonempty reduced product of indexed conjugates cannot expand to the empty binary word: after adjacent equal indices have been combined, the intervening nonzero powers of a separate nonzero powers of b . This proves the asserted free-basis property.

Corollary 84.5. *The map ϕ is an embedding. Membership in $\phi(F_r)$ and recovery of the reduced F_r word are polynomial-time decidable for fixed r . Furthermore $-1 \notin \phi(F_r)$.*

Proof. First use Proposition 84.4. The decoded binary word must have final a -height zero. Rewrite it in the indexed conjugate basis, freely reduce there, and require every remaining index to lie in $\{0, \dots, r-1\}$. These are exactly the words in the subgroup generated by the chosen finite part of that free basis. The number of recorded letters is no larger than the binary word length. The final assertion follows from denominator rigidity. $\square$

Source 16 indexes them from 1; the two embeddings are different maps with the same properties, and each is used in its source's instantiation and examples. The injectivity half of Lemma 84.6 is the free-basis

property just proved (source 11), and the block-expansion proof below is a second route to it; its density assertion is new relative to Corollary 84.5. For $d \geq 2$ let $F_d = F(x_1, \dots, x_d)$, and define

$$\iota_d(x_j) = a^j b a^{-j}, \quad \rho_d = \eta \circ \iota_d : F_d \longrightarrow \mathrm{Sp}(1). \quad (84.14)$$

For a word $w \in F_d$, let $\ell_d(w)$ be the length of the freely reduced a, b expansion $\iota_d(w)$.

Lemma 84.6. *The map ι_d is injective. Moreover,*

$$\mathrm{den}(\rho_d(w)) = 5^{\ell_d(w)}, \quad \ell_d(w) \leq (2d+1)|w|_{F_d}.$$

The group $\rho_d(F_d)$ is dense in $\mathrm{Sp}(1)$.

Proof. Combine consecutive powers of the same x_j in a reduced nonempty word. The resulting blocks have nonzero exponents and distinct adjacent indices. Their expansion has the form

$$a^{j_1} b^{e_1} a^{j_2-j_1} b^{e_2} \dots a^{j_r-j_{r-1}} b^{e_r} a^{-j_r},$$

where every e_i and every intervening $j_{i+1} - j_i$ is nonzero. No b block can disappear by free reduction, so the expansion is nonempty. This proves injectivity. The denominator assertion follows from Theorem 84.2; the length bound follows by expanding each letter before reducing.

The element b has infinite order by injectivity. Its powers lie in the circle $\{\cos u + j \sin u : u \in \mathbb{R}\}$ and are dense there: an infinite closed subgroup of the circle is the entire circle. The closure of $\langle \rho_d(x_1) \rangle$ is the conjugate circle with axis $a j a^{-1}$, and the closure for x_2 has axis $a^2 j a^{-2}$. These axes are not parallel. Indeed, writing $a = \cos \theta + i \sin \theta$ gives $\cos \theta = 3/5$, $\sin \theta = 4/5$, and $\sin(2\theta) = 24/25 \neq 0$.

The Lie algebra of any closed subgroup containing these two circles contains their two nonparallel imaginary-quaternion axes. It also contains their bracket, twice their cross product. These three vectors span the three-dimensional Lie algebra of $\mathrm{Sp}(1)$. Such a closed subgroup has full Lie algebra, hence is open; connectedness of $\mathrm{Sp}(1)$ forces it to be the entire group. This proves density. $\square$

For clarity, the only Lie-group fact used in that argument is that a closed subgroup has a Lie algebra closed under brackets, and a subgroup with the full Lie algebra contains a neighborhood of the identity. The same elementary consequence of the closed subgroup theorem will be used once more below.

There is also a simple membership test for $\iota_d(F_d)$ within $F(a, b)$. Read a reduced word while accumulating the exponent sum of a . Every encountered $b^{\pm 1}$ must occur at a height in $\{1, \dots, d\}$, and the final height must be zero. These conditions are necessary by (84.14) and sufficient by grouping the b letters at their successive heights. Combined with the residue decoder, the individual factors remain algorithmically transparent.

[write] The kernel and surjectivity assertions of the next lemma repeat Lemma 84.1; source 16's proof of them, by the differential of Φ , is a second route. Its injectivity assertion for Ψ_d is new.

Lemma 84.7 (Faithful product embedding). *The map*

$$\Psi_d : F_d \times F_d \longrightarrow \mathrm{SO}(4, \mathbb{Z}[1/5]), \quad (u, v) \longmapsto \Phi(\rho_d(u), \rho_d(v))$$

is injective. The map $\Phi : \mathrm{Sp}(1)^2 \rightarrow \mathrm{SO}(4)$ is surjective and has kernel $\{(1, 1), (-1, -1)\}$.

Proof. Norm preservation gives orthogonality. The determinant is 1: it is a continuous function into $\{\pm 1\}$ on the connected space $\mathrm{Sp}(1)^2$ and is 1 at $(1, 1)$. The entries lie in $\mathbb{Z}[1/5]$ when both quaternions do.

If $\Phi(p, q)$ is the identity, evaluation at $z = 1$ gives $p = q$. Then p commutes with every quaternion, so p is real and equal to 1 or -1 . Conversely both indicated pairs lie in the kernel. The image is all of $\mathrm{SO}(4)$: the differential sends imaginary quaternions (u, v) to the map $z \mapsto uz - zv$. Its kernel is zero, because $u = v$ would have to commute with all quaternions and be both real and imaginary. Domain and target Lie algebras both have dimension six. Thus the image contains a neighborhood of the identity, and it is all of the connected group $\mathrm{SO}(4)$.

Finally, neither free factor image contains -1 , by Theorem 84.2. The kernel on these images is therefore trivial, proving injectivity of Ψ_d . $\square$

Notice that for the special target $L(\rho_d(w))$ the matrix entries are signed copies of the quaternion coordinates. In particular,

$$\text{den}(L(\rho_d(w))) = 5^{\ell_d(w)}. \quad (84.15)$$

A reduced source word of length h produces a matrix whose standard binary encoding has length $O(h+1)$, with a constant depending on d . The unit-norm restriction bounds every numerator in absolute value by its common denominator.

84.5 The presentation-to-rotation compiler

Source 16 writes a finite presentation as

$$G = \langle x_1, \dots, x_d \mid r_1, \dots, r_m \rangle = F_d/N, \quad (84.16)$$

where N is the normal closure of the listed relators in the free group. A relator that is already freely trivial contributes nothing; the density statement below requires $N \neq \{1\}$. *Source 11* writes

$$G = \langle x_0, \dots, x_{r-1} \mid R_1, \dots, R_s \rangle, \quad \pi : F_r \longrightarrow G, \quad N = \ker \pi.$$

For the density results we pad the presentation by a dummy generator $x_0 = z$ with defining relator z and ensure $r \geq 2$. Such padding does not change the presented group. In the hard instance x_1 will be the Miller marker q .

Definition 84.8 (Mihailova fiber product; sources 11 and 16). The Mihailova fiber product associated with the presentation is

$$M_G = \{(u, v) \in F_d \times F_d : uN = vN\}. \quad (84.17)$$

Source 11 writes $\pi(u) = \pi(v)$ for $uN = vN$, and F_r for F_d .

Proposition 84.9 (Generators and membership). *The subgroup M_G is generated by*

$$(x_j, x_j) \quad (1 \leq j \leq d), \quad (r_l, 1) \quad (1 \leq l \leq m). \quad (84.18)$$

Furthermore, $(w, 1) \in M_G$ if and only if $w =_G 1$.

Proof. Every displayed generator belongs to M_G . Diagonal generators produce (u, u) for every $u \in F_d$. Conjugating $(r_l, 1)$ by (u, u) produces $(ur_l u^{-1}, 1)$; inverses and products therefore produce $N \times \{1\}$. For $(u, v) \in M_G$, the word uv^{-1} lies in N , and

$$(u, v) = (uv^{-1}, 1)(v, v).$$

Both factors have already been generated. The last equivalence is the definition of the quotient. $\square$

[write] *Source 11* states the generator half as its Lemma 84.10 (Mihailova generators), with the same proof; that lemma is printed next, since *source 11*'s Miller section applies it with the two factors exchanged.

Lemma 84.10 (Mihailova generators). *The group M_G is generated by*

$$(x_j, x_j) \quad (0 \leq j < r), \quad (R_k, 1) \quad (1 \leq k \leq s).$$

Proof. All displayed generators lie in M_G . They generate the diagonal subgroup. Conjugating $(R_k, 1)$ by (u, u) supplies every conjugate of every relator in the first factor, hence all of $N \times \{1\}$. For $(u, v) \in M_G$, write

$$(u, v) = (uv^{-1}, 1)(v, v).$$

The first factor belongs to $N \times \{1\}$ and the second is diagonal. $\square$

This is the classical part of the construction [72]. The rational compiler applies Ψ_d to the displayed finite list. Source 16 defines the positive gates as in the first line below; source 11 defines them, in its own indexing, as in the second line. Since $\Phi(p, 1) = L(p)$, they are the same gates up to that indexing.

$$A_j = \Phi(\rho_d(x_j), \rho_d(x_j)), \quad R_l = L(\rho_d(r_l)), \quad (84.19)$$

$$C_j = \Phi(\phi(x_j), \phi(x_j)), \quad L_k = \Phi(\phi(R_k), 1). \quad (84.20)$$

and the labeled alphabet

$$\mathcal{S}_G = (I_4, A_1, \dots, A_d, R_1, \dots, R_m, A_1^{-1}, \dots, A_d^{-1}, R_1^{-1}, \dots, R_m^{-1}). \quad (84.21)$$

It has $s = 1 + 2(d + m)$ labels. All entries are obtained by finite rational arithmetic from the relator words, and every inverse is a transpose. In source 11's indexing, let $\mathcal{S}_G$ be the symmetric list of the C_j , the L_k and their inverses, and $\Gamma_G = \langle \mathcal{S}_G \rangle$. Equations (84.13) and (84.20) constitute an explicit finite algorithm, implemented in the accompanying code.

Proposition 84.11 (Faithful image and arithmetic bounds). *The map*

$$(u, v) \longmapsto \Phi(\phi(u), \phi(v))$$

is injective on $F_r \times F_r$, and maps M_G onto Γ_G . The compiler uses at most $2(r + s)$ labelled gates, all in $\text{SO}(4, \mathbb{Z}[1/5])$. If the longest relator has length L , then

$$D = 5^H, \quad H = \max\{4r - 2, (2r - 1)L\} \quad (84.22)$$

clears every gate denominator.

Proof. The only possible extra kernel pair from Lemma 84.1 would be $(-1, -1)$, excluded by Corollary 84.5. Surjectivity onto the generated image is Lemma 84.10.

The binary expansion of x_j has length $2j + 1$. Each coefficient of C_j is bilinear in coordinates whose denominator divides 5^{2j+1} , so its denominator divides 5^{4j+2} . A relator expands to binary length at most $(2r - 1)L$, and left multiplication is linear in its quaternion coordinates. Transposition gives inverses without changing denominators. These observations prove the bound; cancellations often improve it. $\square$

Theorem 84.12 (Presentation-to-rotation compiler). *For the group $\Gamma_G = \langle \mathcal{S}_G \rangle$,*

$$w =_G 1 \iff L(\rho_d(w)) \in \Gamma_G. \quad (84.23)$$

If $N \neq \{1\}$, then Γ_G is dense in $\text{SO}(4)$.

Proof. Proposition 84.9 and the homomorphism property give $\Gamma_G = \Psi_d(M_G)$. The target $L(\rho_d(w))$ is $\Psi_d(w, 1)$. Injectivity of Ψ_d turns its membership in $\Psi_d(M_G)$ into $(w, 1) \in M_G$, proving (84.23).

For density, put $H = \text{cl}(\rho_d(N)) \leq \text{Sp}(1)$. This closed subgroup is normal in the dense subgroup $\rho_d(F_d)$. Continuity of conjugation then makes it normal in all of $\text{Sp}(1)$. A closed normal subgroup of $\text{Sp}(1)$ is either central or the entire group. Here is the needed argument. Its Lie algebra is an ideal in the imaginary quaternions with cross-product bracket; the only ideals are zero and the full space. In the full case it is the entire connected group. In the zero case the closed subgroup is discrete, hence finite by compactness. Conjugation by a connected group on that finite set is constant, so every element is central.

Choose $1 \neq n \in N$. The injective quaternion representation makes $\rho_d(n) \neq 1$, and its image cannot be -1 . Thus H is not central, so $H = \text{Sp}(1)$. The fiber product contains $N \times \{1\}$ and $\{1\} \times N$. Its image under $\rho_d \times \rho_d$ therefore has closure containing both full factors, and hence all of $\text{Sp}(1)^2$. Surjectivity of Φ gives density in $\text{SO}(4)$. $\square$

[write] The equivalence (84.23) is the converse comparison that source 11 records after Theorem 84.16; its density half holds for every nontrivial N and so generalizes Proposition 84.18, which uses the padding relator.

Evaluation is not synthesis. The group G itself is not being embedded into the rotation group. What is embedded is $M_G \leq F_d \times F_d$. In fact the word problem of Γ_G in its gate generators is decidable: multiply the given rational matrices and compare the result with I_4 . The hard problem is whether an arbitrary rational target has any gate-word representation. This distinction prevents a false inference that rational matrix multiplication somehow has an undecidable equality test.

84.6 Denominator and size bounds for the compiler

One can compute the least common denominator of the finite gate list exactly. A convenient upper bound avoids even this normalization. Set

$$K = \max \left(4d + 2, \max_{1 \leq l \leq m} \ell_d(r_l) \right), \quad D = 5^K, \quad (84.24)$$

where the maximum over an empty relator list can be omitted. Since $\rho_d(x_j)$ has denominator 5^{2j+1} , the entries of A_j have denominator dividing 5^{4j+2} . A relator gate has denominator $5^{\ell_d(r_l)}$. Therefore DS is integral for every labeled gate. The least such D may be smaller; either choice is valid in all later certificates and separation estimates.

Corollary 84.13 (Two dense classes). *Whenever $N \neq \{1\}$, both exactly synthesizable and nonsynthesizable rational rotations are dense in $\text{SO}(4)$.*

Proof. The first assertion is Theorem 84.12. The group $F_d \times F_d$ is torsion-free, and Ψ_d is injective. Thus $-I_4$, which has order two, does not belong to Γ_G . The coset $(-I_4)\Gamma_G$ is disjoint from Γ_G , consists of rational rotations, and is dense as a translate of a dense group. $\square$

Consequently no nonempty real neighborhood classifies all its rational points as yes-instances or all as no-instances. Topological density is not a substitute for algebraic equality.

84.7 Rational-matrix membership reduces to the word problem

Source 11. It is not sufficient to decide membership for matrices already promised to have the form $\Phi(\phi(u), \phi(v))$. The main theorem requires a decision procedure on *all* rational 4×4 matrices. This section supplies the missing arithmetic recognition step.

Let $e_0 = 1, e_1 = i, e_2 = j, e_3 = k$, and let

$$E_{ij}(v) = e_i v \bar{e}_j \quad (0 \leq i, j \leq 3).$$

Use the Frobenius inner product on real matrices.

Lemma 84.14 (Inverse spin tensor). *The sixteen matrices E_{ij} are an orthogonal basis of $M_4(\mathbb{R})$ with*

$$\langle E_{ij}, E_{kl} \rangle_F = 4\delta_{ik}\delta_{jl}.$$

For $A \in M_4(\mathbb{Q})$ define

$$T(A)_{ij} = \frac{1}{4} \langle E_{ij}, A \rangle_F. \quad (84.25)$$

If $A = \Phi(p, q)$ then $T(A) = pq^\top$ in coordinate notation.

Proof. The identity $\text{tr}(v \mapsto avb) = 4 \text{Re}(a) \text{Re}(b)$ follows by checking basis quaternions and extending bilinearly. Since the adjoint of left or right unit multiplication is multiplication by the inverse, this identity gives the displayed Gram matrix. In particular the sixteen matrices form a basis. Expanding p and q in the quaternion basis expresses $\Phi(p, q)$ as $\sum_{i,j} p_i q_j E_{ij}$; the Gram matrix recovers its coefficients exactly. $\square$

Proposition 84.15 (All rational unit lifts). *There is a polynomial-time algorithm that, given $A \in M_4(\mathbb{Q})$, returns the two pairs $(p, q), (-p, -q)$ of rational unit lifts when they exist, and otherwise rejects.*

Proof. Compute $T = T(A)$ and choose any nonzero column c ; reject if all columns are zero. Test whether $c^\top c$ has a positive rational square root t . In reduced form this amounts to testing its numerator and denominator for being integer squares. If not, reject. Put

$$p = c/t, \quad q = T^\top p. \quad (84.26)$$

Check $N(p) = N(q) = 1$ and $A = \Phi(p, q)$; reject if any check fails. Otherwise return the displayed pair and its simultaneous negative.

If A has rational unit lifts, Lemma 84.14 gives $c = q_j p$ with $q_j \neq 0$. Hence $t = |q_j| \in \mathbb{Q}$, and the procedure recovers precisely one of the two signs. Conversely the final reconstruction check proves soundness, even without assuming beforehand that A is orthogonal or that T has rank one. The kernel computation shows that there are no other lifts. All matrix dimensions are fixed and all integer arithmetic, including square-root testing by binary methods, has polynomial bit complexity. $\square$

For example the rational rotation

$$A_{90} = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

has no rational unit lifts: a nonzero tensor column has squared norm $1/2$. This is an implemented rejection test, not a promise excluded from the input domain.

Theorem 84.16 (Membership transfers the word problem). *For a fixed finite presentation of G , deciding rational-matrix membership in Γ_G reduces to polynomial-time preprocessing and at most one word-problem query in G , on a word of length $O(\text{bits}(A))$. The same is true of deciding $y \in \Gamma_G e$ for rational y . In particular, a polynomial-time word problem for G gives polynomial-time algorithms for both questions.*

Proof. Recover rational lifts using Proposition 84.15. For each sign pair, run the $\phi(F_r)$ decoder on both quaternions. At most one sign pair can pass both decoders, because $-1 \notin \phi(F_r)$. If neither passes, reject. Otherwise obtain $s, t \in F_r$ and ask whether $\pi(st^{-1}) = 1$. By Proposition 84.11 this is exactly matrix membership.

All recovered rational numbers have bit length $O(\text{bits}(A))$. Denominator rigidity bounds the decoded binary word lengths by that quantity, and the finite-rank decoder does not increase the number of letters. Thus the word-problem query has linear length in the matrix input size. For the orbit of e , decode y and test its resulting word in G , using (86.2). $\square$

The converse comparison is also useful: w represents 1 in G exactly when $\Phi(\phi(w), 1) \in \Gamma_G$. This is a polynomial-output-size reduction for a fixed presentation. Immediate rejection in the preceding algorithm is allowed, so no unjustified many-one reduction to the word problem of the trivial group is being asserted.

84.8 Density and approximation

Source 11. We first give a direct density argument so that density is not mistaken for a heuristic inference from numerical orbit plots.

Lemma 84.17 (Two distinct axes). *If two unit quaternions have irrational rotation parameters in their one-dimensional circles and have nonparallel imaginary axes, they generate a dense subgroup of S^3 .*

Proof. The closure of each cyclic subgroup is its entire circle. Write the corresponding imaginary unit axes as J, U . The closure therefore contains $\exp(tJ)$ and $\exp(tU)$ for every real t . It also contains the circle with axis $V = \exp(\delta J)U \exp(-\delta J)$. Choose a small nonzero δ so that J, U, V are linearly independent: conjugation rotates the component of U perpendicular to J out of the plane spanned by J, U . The derivative at zero of

$$(a, b, c) \mapsto \exp(aJ) \exp(bU) \exp(cV)$$

has full rank. The inverse function theorem gives a neighborhood of the identity in the closure. A subgroup containing such a neighborhood is open; a connected group has no proper open subgroup. The unit three-sphere is connected, proving the claim. $\square$

Set $c = aba^{-1} = \xi$. Direct quaternion multiplication gives

$$c = \frac{3}{5} - \frac{28}{125}j + \frac{96}{125}k, \quad \text{the } i\text{-coefficient of } cbc^{-1} = -\frac{2304}{3125} \neq 0. \quad (84.27)$$

Both b and any conjugate of it have real part $3/5$. Their circle parameter is an irrational multiple of π : otherwise $\exp(i\theta)$ would be a root of unity and $2 \cos \theta = 6/5$ would be a rational algebraic integer, hence an integer, which it is not.

Proposition 84.18 (Dense compiler image). *For the padded compiler with $z = x_0$ and $r \geq 2$, $\overline{\Gamma_G} = \text{SO}(4)$. The subgroup of left multiplications coming from $\phi(N)$ is already transitive up to arbitrary positive tolerance on S^3 .*

Proof. Because z is a relator, the gate $L_b : v \mapsto bv$ is present. The gate $C_c : v \mapsto cv\bar{c}$ is present because $c = \phi(x_1)$. Thus Γ_G contains the left multiplication

$$C_c L_b C_c^{-1} = L_{cbc^{-1}}.$$

The two axes are different by (84.27). Lemma 84.17 shows that these left multiplications are dense in all left unit multiplications. They belong to $\phi(N)$ because z and $x_1 z x_1^{-1}$ belong to N .

Similarly, $\phi(F_r)$ contains b and c , whose axes are nonparallel, so $\phi(F_r)$ is dense in S^3 . The closure of the fiber-product image contains both all left unit multiplications and all diagonal conjugations. Combining them gives all right unit multiplications as well. Lemma 84.1 then gives all of $\text{SO}(4)$. $\square$

Proposition 84.19 (A computable uniform approximation bound). *For the same fixed compiler there is a total computable bound $B(\varepsilon)$ on the number of gates required to move any $x \in S^3$ within ε of any $y \in S^3$, for every positive rational ε . For rational x, y , a successful word can be found effectively.*

Proof. Use only the two left multiplications in the preceding proof and their inverses. Their costs in the original gate list are at most one and three, respectively. Their unit quaternions generate a dense subgroup.

Here is an effective compactness argument with finite rational tests. Replace ε by $\min\{\varepsilon, 1\}$ and choose an integer $m > 8/\varepsilon$. On the rational grid of mesh $1/m$ in $[-1, 1]^3$, evaluate

$$\sigma(t) = \frac{(1 - \|t\|^2, 2t)}{1 + \|t\|^2},$$

and also take its negatives. This is a rational $\varepsilon/4$ -net of S^3 . Indeed, a point with nonnegative first coordinate has inverse stereographic coordinate t of norm at most one, and

$$\|\sigma(t) - \sigma(u)\| = \frac{2\|t - u\|}{\sqrt{(1 + \|t\|^2)(1 + \|u\|^2)}} \leq 2\|t - u\|.$$

Rounding to the grid gives error at most $\sqrt{3}/m < \varepsilon/4$.

For every point of this finite net, enumerate words in the two left multiplication generators until one is within $\varepsilon/4$. Each search terminates by density; squared-distance comparisons are rational and exact. The resulting finite list of unit quaternions is an $\varepsilon/2$ -net. The maximum original-gate length in the list is a computable integer $B(\varepsilon)$.

For any real $x, y \in S^3$, approximate $y\bar{x}$ by a member p of this list. Then $\|px - y\| = \|p - y\bar{x}\| < \varepsilon$. For rational input states, direct enumeration and rational distance comparison also give an effective successful word. $\square$

[write] Source 16's approximation statements follow. Its Proposition 84.20 is a weaker relative of Proposition 84.19 with a simpler proof; its Theorem 84.21 is a uniform bound for *matrices* (a net of all of $\text{SO}(4)$), where source 11's is a bound for *state transfer*, and it is proved by Tarski's decision method instead of an explicit stereographic net. The two are second routes to a computable uniform approximation bound.

Proposition 84.20. *For a fixed finite dense rational alphabet, rational target $M \in \text{SO}(4)$, and rational $\varepsilon > 0$, a gate word W with $\|W - M\|_F < \varepsilon$ can be found by a terminating algorithm.*

Proof. Enumerate words in increasing length and lexicographic order. Their matrix values and the squared Frobenius errors are rational, so the strict comparison with ε^2 is decidable. Density guarantees that some word passes. $\square$

This intentionally weak algorithm suffices for the computability contrast. It does not claim a useful running time. Efficient approximation for suitable universal gate systems belongs to the Solovay–Kitaev literature [74]; we do not need to transfer an efficiency bound or its implementation assumptions to the present gate set.

Theorem 84.21 (Computable uniform approximation bound). *For the fixed alphabet, there is a total computable function $B : \mathbb{N} \rightarrow \mathbb{N}$ such that every $M \in \text{SO}(4)$ lies within Frobenius distance 2^{-k} of a word of length at most $B(k)$.*

Proof. For each candidate length n , let $\mathcal{W}_n$ be the finite set of matrix values of words of length at most n . Test the first-order sentence over the real numbers

$$\forall X \left[X^T X = I_4 \wedge \det X = 1 \implies \bigvee_{W \in \mathcal{W}_n} \sum_{i,j=0}^3 (X_{ij} - W_{ij})^2 < 2^{-2k} \right]. \quad (84.28)$$

It has only rational coefficients and polynomial equalities and inequalities. Tarski's decision method for real algebra decides its truth [75]. Search successive n until the sentence is true.

The search terminates: the open 2^{-k} -balls around all gate words cover $\text{SO}(4)$ by density. Compactness gives a finite subcover. The maximum length of its finitely many center words is a successful n . Take the first successful n as $B(k)$. $\square$

Compactness alone would prove only existence of finite nets. The real algebra test is what makes the uniform bound computable. The argument does not claim that B grows slowly or that quantifier elimination is practical for these formulas.

85 [write] Source 11: completing the main separation

Proof of Theorem 81.1, parts (i)–(iv). Choose the fixed Q of Corollary 87.2, form its padded Miller group G , and apply the compiler. Theorem 84.16 and Lemma 88.1 prove polynomial-time matrix membership and polynomial-time membership of the orbit of e .

The marker $q = x_1$ is encoded by the fixed quaternion ξ in (81.1). For every word w on Y , Theorem 86.1 and Lemma 88.2 give

$$\phi(qw) \in \Gamma_G \xi \iff w = 1 \text{ in } Q. \quad (85.1)$$

This is a computable reduction from a complete word problem. The full rational target set is computably enumerable by enumerating all gate words and multiplying exactly, so it is complete.

The two orbits are disjoint because $q \neq 1$ in the free base subgroup of $M(Q)$ and hence cannot be conjugate to 1. Density and the uniform approximation bound are Propositions 84.18 and 84.19. Part (v) is proved in Section 90. $\square$

86 State orbits are conjugacy classes

[source 11]

Theorem 86.1 (Exact orbit–conjugacy identity). *For all $u, v \in F_r$,*

$$\phi(v) \in \Gamma_G \phi(u) \iff \pi(v) \text{ is conjugate to } \pi(u) \text{ in } G. \quad (86.1)$$

In particular,

$$\Gamma_G e = \phi(N). \quad (86.2)$$

Proof. A gate product has the form $\Phi(\phi(s), \phi(t))$ with $\pi(s) = \pi(t)$. It sends $\phi(u)$ to $\phi(sut^{-1})$. Injectivity of ϕ implies that reaching $\phi(v)$ means $v = sut^{-1}$ as a *free-group* equality. Passing to G gives

$$\pi(v) = \pi(t)\pi(u)\pi(t)^{-1}.$$

Conversely, if the last equation holds for a lift $t \in F_r$, set $s = vt u^{-1}$. Then $\pi(s) = \pi(t)$ and $sut^{-1} = v$ freely. The corresponding fiber-product element supplies the desired gate product. Taking $u = 1$ gives (86.2). $\square$

The difference between equality in F_r and equality in G is essential. The quaternion encoding is faithful to *syntax in the free group*; it does not impose the relators of G on the quaternions. Instead, those relators determine which *pairs* of left and right transformations are permitted. This is why all state coordinates remain explicitly rational while the orbit question can retain difficult group semantics.

For a fixed encoded source, every reachable target is still in $\phi(F_r)$. Thus a quaternion outside that decidable image can be rejected immediately. Within the image, orbit testing becomes a conjugacy question in the quotient. At e , it becomes only a word-problem question.

87 A finite presentation with complete word problem

[source 11]

We isolate the one deep group-theoretic existence theorem used below. Higman’s embedding theorem states that a finitely generated recursively presented group embeds in a finitely presented group [30]. Only this classical statement is needed; stronger embedding properties, such as those studied in recent refinements [37], play no role. The following construction makes the many-one completeness requirement explicit rather than silently replacing it by mere undecidability.

Fix a computably enumerable complete set $K \subseteq \mathbb{N}_{>0}$. Form

$$H_K = \langle d, s \mid [s, d^n s d^{-n}] = 1 \quad (n \in K) \rangle. \quad (87.1)$$

This is a two-generator recursively presented group.

Lemma 87.1. *For every positive integer n ,*

$$[s, d^n s d^{-n}] = 1 \text{ in } H_K \iff n \in K.$$

Proof. Let A_K have generators s_i , $i \in \mathbb{Z}$, and relations $[s_i, s_j] = 1$ whenever $|i - j| \in K$. Shifting every index by one is an automorphism. The semidirect product $A_K \rtimes \mathbb{Z}$ is generated by the shift d and s_0 , with $s_i = d^i s_0 d^{-i}$. It has exactly the presentation (87.1). More explicitly, the maps in both directions send s_i to $d^i s d^{-i}$ and preserve every relation, since each edge relation is a conjugate of a displayed relator or its symmetric version. Their composites fix the generators.

If $n \notin K$, map A_K to the free group $F(u, v)$ by sending $s_0 \mapsto u$, $s_n \mapsto v$, and every other s_i to 1. Every commutation relator survives: the only potentially obstructing edge would join 0 to n , and that edge is absent. The commutator $[s_0, s_n]$ maps to the nontrivial reduced word $[u, v]$. It is therefore nontrivial in A_K and in the semidirect product. For $n \in K$, the required equality is a defining relator. $\square$

Corollary 87.2. *There is a fixed finitely presented group $Q = \langle Y \mid \mathcal{R} \rangle$ with computably enumerable complete word problem. There are fixed words U, V on Y such that*

$$W_n = [V, U^n V U^{-n}] = 1 \text{ in } Q \iff n \in K \quad (n \geq 1). \quad (87.2)$$

Proof. Embed H_K in a finitely presented group Q by Higman's theorem and choose words U, V for the images of d, s . Injectivity and Lemma 87.1 give (87.2). The map $n \mapsto W_n$ is computable because U, V are fixed finite words. The word problem of a finitely presented group is computably enumerable by enumerating finite products of conjugates of relators. This supplies both halves of completeness. $\square$

The conclusion is an existence theorem about a *fixed* finite presentation, not a claim that its short numerical presentation has been printed here. The subsequent compiler is fully explicit once such a presentation and the two words are supplied. For the existence proof, no algorithm for discovering an unknown embedding is needed: its two fixed generator images already specify a computable reduction.

88 Miller's construction with an efficient word algorithm

[source 11]

Starting from any finite presentation $Q = \langle Y \mid \mathcal{R} \rangle$, introduce a new marker q and let $F = F(Y \cup \{q\})$. Introduce a free group on letters θ_y for $y \in Y$ and θ_R for $R \in \mathcal{R}$. They act on F by automorphisms fixing every Y -letter and satisfying

$$\alpha_{\theta_y}(q) = y^{-1} q y, \quad \alpha_{\theta_R}(q) = q R. \quad (88.1)$$

The inverse substitutions are $q \mapsto y q y^{-1}$ and $q \mapsto q R^{-1}$, respectively. Define the semidirect product

$$G = M(Q) = F \rtimes_{\alpha} F(\Theta).$$

A finite presentation consists of

$$\theta_y \theta^{-1} = y \quad (\theta \in \Theta, y \in Y), \quad (88.2)$$

$$\theta_y q \theta_y^{-1} = y^{-1} q y \quad (y \in Y), \quad (88.3)$$

$$\theta_R q \theta_R^{-1} = q R \quad (R \in \mathcal{R}). \quad (88.4)$$

We add the harmless generator z with relator z and order the presentation generators with $z = x_0$, $q = x_1$. This is the classical Miller construction [31, 32]; the proofs of the particular properties needed here are included.

Lemma 88.1 (Polynomial-time normal form). *For every fixed finite presentation of Q , the group $M(Q)$ has a polynomial-time word problem. It does not need an oracle for the word problem of Q .*

Proof. Every element has a unique normal form $f\tau$ with $f \in F$ and $\tau \in F(\Theta)$. The action of any τ has the form

$$\alpha_\tau(q) = u^{-1}qv, \quad u, v \in F(Y). \quad (88.5)$$

Under composition these pairs multiply componentwise. Indeed, applying the pair (s, t) outside the pair (u, v) gives $u^{-1}s^{-1}qtv = (su)^{-1}q(tv)$. The elementary pairs are (y, y) and $(1, R)$ and their inverses.

Scan an input word of length n . Keep the reduced Θ -word of the prefix, its action pair (u, v) , and its accumulated free part f . Appending a control letter updates the pair by one fixed pair and updates the reduced control word. Appending a Y -letter appends that letter to f . Appending q appends $u^{-1}qv$; appending q^{-1} appends $v^{-1}q^{-1}u$. Cancel freely using stacks. Ignore $z^{\pm 1}$.

The lengths of u, v are $O(n)$, with constants depending on the longest fixed relator. There are at most n marker letters, so at most $O(n^2)$ free letters are processed. Stack reduction gives a polynomial-time algorithm (quadratic in free-letter operations for this fixed presentation). The result is the identity exactly when both f and τ are empty. No relation of Q is used to reduce f : its Y -letters are genuinely free in F . $\square$

Lemma 88.2 (Marked conjugacy). *For every word w on Y ,*

$$q \text{ is conjugate to } qw \text{ in } M(Q) \iff w = 1 \text{ in } Q. \quad (88.6)$$

Proof. Let $\pi_Q : F(Y) \rightarrow Q$. The action pairs generated by (y, y) and $(1, R)$ are exactly

$$\{(u, v) : \pi_Q(u) = \pi_Q(v)\},$$

by the proof of Lemma 84.10, with the two factors exchanged. If $w \in \ker \pi_Q$, the pair $(1, w)$ occurs, and its control word conjugates q to qw .

Conversely, a general conjugator in $M(Q)$ has form $f\tau$. Conjugating q gives $f\alpha_\tau(q)f^{-1}$. With $\alpha_\tau(q) = u^{-1}qv$, where $\pi_Q(u) = \pi_Q(v)$, this says that $u^{-1}qv$ and qw are conjugate in the free group $F(Y \cup \{q\})$. Conjugating the first word by u reduces the question to $qv u^{-1}$ and qw .

After freely reducing their Y -tails, both words are cyclically reduced and contain exactly one occurrence of q , with positive sign. The cyclically reduced conjugacy criterion in a free group says that they must be cyclic permutations. The unique occurrence of q fixes the starting position, so $w = vu^{-1}$ freely. It follows that $\pi_Q(w) = 1$. $\square$

This elementary proof exhibits the exact information loss: the word algorithm can normalize a proposed conjugator efficiently, but the existential search for a conjugator encodes the word problem of Q . For the group in Corollary 87.2, the marked conjugacy class is computably enumerable complete even though the ambient word problem is polynomial-time decidable.

88.1 Compiler size for the Miller presentation

Suppose Q has m generators and s relators. The padded Miller presentation has

$$r = 2 + 2m + s, \quad s_G = 1 + (m + s)(m + 1) \quad (88.7)$$

generators and relators, respectively. Therefore the symmetric rotation list has at most

$$k = 2(r + s_G) = 2(m^2 + ms + 3m + 2s + 3) \quad (88.8)$$

labelled gates. If L_Q is the longest base relator, the new relator lengths are bounded by $\max\{6, L_Q + 4\}$, and Proposition 84.11 bounds the denominator exponent. These are uniform finite construction bounds, not estimates of the size of a smallest universal example.

89 An order-five recurrence carrying the hard inputs

[source 11]

Return to the fixed words U, V from Corollary 87.2. Regard them as words in the free Y -subgroup of the Miller group, lift them to the compiler alphabet, and define

$$R_0 = \phi(U), \quad S_0 = \phi(V), \quad y_n = \xi[S_0, R_0^n S_0 R_0^{-n}].$$

These are fixed rational quaternions and a computable rational target sequence. Theorem 86.1 and the marked-conjugacy lemma prove (81.2). At $n = 0$ the target is ξ and is reachable; only the positive indices encode K .

Theorem 89.1 (Explicit recurrence). *Let E denote the forward shift on sequences and put*

$$\tau = \operatorname{Re}(R_0), \quad A = 4\tau^2 - 2, \quad B = A^2 - 2.$$

Each of the four coordinates of y_n is annihilated by

$$P(E) = (E - 1)(E^2 - AE + 1)(E^2 - BE + 1). \quad (89.1)$$

Equivalently, with $S = A + B + 1$ and $T = AB + A + B + 2$,

$$y_{n+5} - Sy_{n+4} + Ty_{n+3} - Ty_{n+2} + Sy_{n+1} - y_n = 0. \quad (89.2)$$

All coefficients are rational; clearing denominators gives an integer constant-coefficient recurrence. The four coordinates remain rational and satisfy $\|y_n\| = 1$.

Proof. The assertions of rationality and norm follow directly from quaternion multiplication. If R_0 is real, the target is constant and the displayed polynomial still annihilates it. Otherwise write $R_0 = \cos \theta + J \sin \theta$ with $J^2 = -1$, and put $\zeta = e^{i\theta}$. Each coordinate of $R_0^{\pm n}$ is a linear combination of ζ^n and ζ^{-n} over a fixed extension field.

Expanding the commutator gives

$$y_n = \xi S_0 R_0^n S_0 R_0^{-n} S_0^{-1} R_0^n S_0^{-1} R_0^{-n}.$$

There are four power factors. Quaternion multiplication is bilinear in coordinates, so each resulting coordinate is a linear combination of ζ^{kn} with $k \in \{-4, -2, 0, 2, 4\}$. Thus it is annihilated by the product of the five corresponding shift factors. Since

$$\zeta^2 + \zeta^{-2} = 4\tau^2 - 2 = A, \quad \zeta^4 + \zeta^{-4} = A^2 - 2 = B,$$

that product is (89.1). Expanding it gives (89.2). □

This proves Theorem 81.2. It does *not* assert undecidability of a zero test for an order-five scalar recurrence. The undecidable predicate is membership of the vector y_n in a particular nonalgebraic orbit. The coordinates themselves are easy to generate. For fixed R_0, S_0 , their numerator and denominator bit lengths are $O(n)$; this does not mean polynomial output length in the binary length $\log n$ of the index.

As a numerical check independent of the hard group presentation, taking $R_0 = a$ gives

$$15625E^5 + 19475E^4 + 10906E^3 - 10906E^2 - 19475E - 15625.$$

The accompanying tests verify the recurrence for three different rational choices of R_0, S_0 , at twenty successive shifts in all four coordinates. Those finite examples test the algebraic identity; they are not claimed to instantiate the complete set K .

90 The exact-control precision barrier

[source 11]

Write $\ell_S(A)$ for the shortest gate-word length of $A \in \Gamma$. For a reachable rational target y , let

$$L_\xi(y) = \min\{\ell_S(A) : A\xi = y, A \in \Gamma\}.$$

All minima in this section are over nonempty sets.

Theorem 90.1 (No recursive exact-transfer budget). *For the group in Theorem 81.1, there is no total computable function f such that*

$$L_\xi(y) \leq f(\text{bits}(y))$$

for every reachable rational target y . There is also no total computable bound, in terms of $\text{bits}(y)$, on the smallest bit length of a matrix $A \in \Gamma$ satisfying $A\xi = y$.

Proof. A length bound would decide exact reachability: enumerate all gate words up to the proposed bound and compare their endpoints with y . This contradicts computable-enumerable completeness.

A matrix-bit bound would also decide reachability. Enumerate all rational 4×4 matrices of bounded encoding length, apply the total membership algorithm for Γ , and test $A\xi = y$ exactly for those that pass. Again this is a finite decision procedure, a contradiction. $\square$

Corollary 90.2 (Single-prime denominator version). *There is no total computable function bounding the least exponent h such that a transfer matrix $A \in \Gamma$ with $A\xi = y$ has common denominator 5^h , in terms of the target's bit length.*

Proof. Every group matrix has a power-of-five common denominator. Orthogonality bounds every entry in absolute value by one. A bound on h therefore bounds every numerator and hence the entire matrix's bit length, contrary to Theorem 90.1. $\square$

In contrast, there *is* a recursive bound on the shortest gate word for a *given member matrix* in terms of that matrix's bit length. For each n , enumerate all rational matrices with encoding length at most n , decide which belong to Γ , and enumerate gate words until a realization of every accepted matrix has been found. There are finitely many, and each search terminates. Take the maximum length. Thus the missing bound is genuinely about the projection from complete transformations to one state's endpoint, not merely an inability to recognize or enumerate the transformation group.

90.1 Consequences for invariants and nonreachability

Every continuous real-valued invariant of the action on S^3 is constant: it is constant on a dense orbit and therefore constant on its closure. Every nonempty closed set invariant under the gates is the whole sphere. In particular, polynomial conserved quantities restricted to the sphere cannot separate the easy and hard orbits.

The hard orbit and its rational complement are both dense. The first is dense by the theorem, while its complement contains the disjoint dense orbit of e . Consequently no positive geometric separation margin characterizes nonreachability from ξ .

Finally, nonreachability among valid rational targets is not computably enumerable. Otherwise simultaneous enumeration of reachability and nonreachability would decide the hard set. There can therefore be no complete system of finite, decidable checkable negative certificates for all such nonreachability statements. This is a computability obstruction, not a statement that individual negative instances lack proofs.

91 A fixed complete specialization

[source 16]

Undecidable word problem alone would not justify the stronger phrase “c.e.-complete.” We give a construction that obtains many-one completeness explicitly.

91.1 Coding the halting set into an amalgam

Let $\mathcal{K} \subseteq \mathbb{N}$ be a fixed c.e.-complete halting set. In two free groups $F(a, b)$ and $F(c, d)$ put

$$u_n = a^n b a^{-n}, \quad v_n = c^n d c^{-n}, \quad n \geq 0.$$

The u_n freely generate the subgroup they span, and the same holds for the v_n . The block-expansion proof of Lemma 84.6 works without change for every finite collection of distinct indices, including zero.

Let U be generated by the u_n with $n \in \mathcal{K}$, and V by the corresponding v_n . The basis correspondence $u_n \mapsto v_n$ defines an isomorphism $U \simeq V$. Form the amalgamated free product

$$Q_{\mathcal{K}} = F(a, b) *_{U=V} F(c, d) = \langle a, b, c, d \mid u_n = v_n \ (n \in \mathcal{K}) \rangle. \quad (91.1)$$

The presentation is recursively enumerable and has four generators.

Lemma 91.1. *For every $n \in \mathbb{N}$,*

$$n \in \mathcal{K} \iff u_n v_n^{-1} = 1 \text{ in } Q_{\mathcal{K}}.$$

Proof. For $n \in \mathcal{K}$ the equality is a defining relation. If $n \notin \mathcal{K}$, freeness of the indicated bases shows $u_n \notin U$ and $v_n \notin V$. The word $u_n v_n^{-1}$ is then a reduced alternating word with two syllables outside the amalgamated subgroup. It is nontrivial by the normal-form theorem for free products with amalgamation.

To specify the normal-form input: choose coset transversals for the common subgroup in each factor. Successively absorb subgroup elements and combine adjacent syllables from the same factor. The resulting alternating transversal form represents the identity only when no nontrivial transversal syllable remains. In particular, a two-factor word whose syllables are outside the common subgroup cannot be the identity. No algorithm for deciding membership in U is used in this existence argument. $\square$

91.2 Passing to a finite presentation

Higman’s embedding theorem states that every finitely generated recursively presented group embeds in a finitely presented group [30]. Apply it to $Q_{\mathcal{K}}$, obtaining a finite presentation of a group G_0 and an embedding $j : Q_{\mathcal{K}} \hookrightarrow G_0$. Fix once and for all finite words representing the images of a, b, c, d . Substitution in $u_n v_n^{-1}$ gives a total computable map $n \mapsto w_n$ such that

$$n \in \mathcal{K} \iff w_n =_{G_0} 1. \quad (91.2)$$

The word problem of a finitely presented group is c.e.: enumerate finite products of conjugates of defining relators and their inverses, freely reduce, and compare with the input word. Therefore $\text{WP}(G_0)$ is c.e.-complete under many-one reductions.

For the local theorem in the next section, fix instead

$$G = G_0 * \langle t \rangle. \quad (91.3)$$

It is finitely presented, its word problem is still c.e.-complete, and its presentation kernel is nontrivial. A free group has decidable word problem, so a presentation of a group with complete word problem cannot have trivial kernel. Take at least two generators, adding a redundant generator with a defining relation if necessary.

Corollary 91.2 (Fixed gates, varying target). *The finite alphabet $\mathcal{S}_G$ for (91.3) is fixed, dense, rational, and symmetric. Its rational-matrix membership problem is c.e.-complete.*

Proof. Theorem 84.12 gives the reduction from $\text{WP}(G)$ and density. Membership is c.e. by enumeration of all finite gate words with exact rational multiplication. Matrix validity and equality are decidable. $\square$

What is and is not explicit. The compiler (84.14), (84.19), (84.21) is an explicit algorithm on a supplied finite presentation. Its implementation is included. The preceding existence argument fixes a universal finite presentation and fixed embedding words, but this article does not print those potentially long constants. Thus it proves existence of a fixed universal alphabet without claiming a numerical universal gate count. The numerical example in Section 101 uses a different, decidable presentation and is labeled accordingly.

92 Completeness in every open neighborhood

[source 16]

The free generator t supplies a way to localize hard instances without changing their yes/no answer. Throughout this section $G = G_0 * \langle t \rangle$ and $\Gamma = \Gamma_G$ are fixed as above.

Lemma 92.1 (Commutator test). *For $w \in G_0$ and an integer $k \geq 1$,*

$$[w, t^k] = 1 \text{ in } G \iff w = 1 \text{ in } G_0,$$

where $[u, v] = uvu^{-1}v^{-1}$.

Proof. The forward implication fails for no nontrivial w : if $w \neq 1$, the four syllables w, t^k, w^{-1}, t^{-k} are all nontrivial and alternate between the two free factors. Their free-product normal form is nonempty. The reverse implication is immediate. $\square$

Lemma 92.2 (Uniform small commutators). *For unit quaternions p, q ,*

$$|pqp^{-1}q^{-1} - 1| \leq 2|q - 1|, \quad \|L(pqp^{-1}q^{-1}) - I_4\|_F \leq 4|q - 1|.$$

Proof. Right multiplication by q preserves the Euclidean quaternion norm, so

$$|pqp^{-1}q^{-1} - 1| = |pqp^{-1} - q| \leq |p(q - 1)p^{-1}| + |q - 1| = 2|q - 1|.$$

From (84.6), each coordinate of a quaternion appears four times up to sign, and hence $\|L(z)\|_F = 2|z|$. Apply this to $z = pqp^{-1}q^{-1} - 1$. $\square$

Theorem 92.3 (Local c.e.-completeness). *For every positive rational ε , there is a computable reduction from $\text{WP}(G_0)$ to exact membership in Γ all of whose output matrices satisfy $\|M - I_4\|_F < \varepsilon$. More generally, the same holds with any rational center $C \in \text{SO}(4)$ and any positive rational radius r .*

Proof. The quaternion $q_t = \rho_d(t)$ has infinite order. Its powers are dense in their circle, so there exists $k \geq 1$ with $|q_t^k - 1| < \varepsilon/4$. This k is computable by enumerating positive integers and comparing the exact rational squared norm with $(\varepsilon/4)^2$. For an input word w in the generators of G_0 , output

$$M_{w,\varepsilon} = L(\rho_d([w, t^k])). \tag{92.1}$$

Lemma 92.2 gives the required bound uniformly in w . Theorem 84.12 and Lemma 92.1 show

$$M_{w,\varepsilon} \in \Gamma \iff [w, t^k] =_G 1 \iff w =_{G_0} 1.$$

Thus hardness survives inside any ball about the identity.

For a rational center C and radius r , enumerate gate words until finding $g \in \Gamma$ with $\|g - C\|_F < r/2$; density guarantees termination. Use the preceding construction with $\varepsilon = r/2$ and output $gM_{w,r/2}$. Orthogonal invariance of the Frobenius norm gives

$$\|gM_{w,r/2} - C\|_F \leq \|M_{w,r/2} - I_4\|_F + \|g - C\|_F < r.$$

Since $g \in \Gamma$, multiplication by g preserves the membership answer. The restricted language is c.e., because rational ball membership is decidable and gate membership is c.e. Therefore it is c.e.-complete. $\square$

For each fixed ball, the power k and the translating word g are fixed constants independent of the hard input. The theorem asserts much more than that yes-instances and no-instances are both dense: a full complete family of instances can be placed inside the prescribed neighborhood. It still makes no claim that physically indistinguishable targets can be distinguished by finite-precision measurements.

93 Approximation, exact separation, and the precision boundary

[source 16]

[write] The first subsection of source 16's section, on ordinary approximation, is printed in the shared core (Section 84.8).

93.1 Exact rational separation

Theorem 93.1 (Length-height gap). *Let $S_0, \dots, S_{s-1} \in \mathbb{Q}^{4 \times 4}$ have a common positive integer denominator D , with $S_0 = I_4$. Let $M = T/q$, where $q > 0$ is an integer and T is an integer matrix. For every word value W of length at most n ,*

$$W \neq M \implies \max_{i,j} |W_{ij} - M_{ij}| \geq \frac{1}{qD^n}. \quad (93.1)$$

The same lower bound holds for its Frobenius distance.

Proof. Pad the word with identity gates to length n . Since each DS_j is integral, $W = A/D^n$ for an integer matrix A . If $W \neq M$, some entry of $qA - D^nT$ is a nonzero integer. That entry has absolute value at least one. Dividing by qD^n proves the maximum-entry bound, and the Frobenius norm is no smaller than that maximum. $\square$

In particular, for the *actual output length* $\text{len}(W)$,

$$\|W - M\|_F < (qD^{\text{len}(W)})^{-1} \iff W = M. \quad (93.2)$$

This gives an exact boundary between two different specifications. A requested tolerance ε fixed *before* searching is always attainable. A tolerance required to shrink with the length of the word *that is returned*, at the rate in (93.2), is attainable exactly for synthesizable targets. Density says nothing that removes this self-consistency requirement.

The strict inequality is essential to the statement as proved. The bound is a rational-lattice separation estimate; no optimality for a particular gate alphabet is asserted. Likewise the denominator q need not be minimal: a larger valid q yields a smaller, still valid threshold.

93.2 A simple exact length lower bound

For the gates with $D = 5^K$, the special target $L(\rho_d(w))$ has denominator $5^{\ell_d(w)}$. Any exact synthesis of length n therefore obeys

$$n \geq \left\lceil \frac{\ell_d(w)}{K} \right\rceil. \quad (93.3)$$

This linear lower bound is elementary. The next section shows why no computable upper bound in the same height parameter can work in the complete example.

94 A geometric synthesis-depth function of halting degree

[source 16]

For the fixed complete group Γ define $\lambda(M)$ to be the least length of a gate word equal to M , for $M \in \Gamma$. Set

$$E(h) = \max\{\lambda(M) : M \in \Gamma, 5^h M \in \mathbb{Z}^{4 \times 4}\}, \quad h \in \mathbb{N}. \quad (94.1)$$

The set is nonempty because it contains I_4 . It is finite because each entry of an orthogonal matrix has absolute value at most one; there are at most $(2 \cdot 5^h + 1)^{16}$ possible integer numerator arrays. Thus $E(h)$ is a well-defined total nondecreasing function.

Theorem 94.1 (Noncomputable exact depth). *The function E is lower semicomputable, has Turing degree $\mathbf{0}'$, and has no total computable majorant. For every total computable $f : \mathbb{N} \rightarrow \mathbb{N}$, the inequality $E(h) > f(h)$ holds for infinitely many h .*

Proof. Enumerate all labeled gate words in increasing length. For each new matrix of denominator dividing 5^h , its first occurrence gives its minimal length. The running maximum of these first-occurrence lengths increases to $E(h)$, since there are only finitely many eligible matrices. This proves lower semicomputability.

With a halting oracle, enumerate all integer numerator arrays bounded by 5^h , test orthogonality and determinant, and decide for each candidate whether it belongs to Γ . The latter is a c.e. question, hence decidable by that oracle. For the affirmative candidates, ordinary word enumeration computes their minimal lengths. Taking the finite maximum computes $E(h)$, so $E \leq_T \mathbf{0}'$.

Conversely, an oracle for E decides $\text{WP}(G)$. Given a source word w , compute its target $L(\rho_d(w))$ and $h = \ell_d(w)$. Search all gate words of length at most $E(h)$ for exact equality with that target. If the target belongs to Γ , its minimal length is at most $E(h)$; if it does not, no searched word equals it. By (84.23) this decides the complete word problem, so $\mathbf{0}' \leq_T E$.

The same decision argument works with any computable majorant in place of E , contradicting completeness. Finally, if some computable f exceeded or equaled E at all but finitely many arguments, those finitely many exceptions could be absorbed in one fixed integer constant, producing a total computable majorant. Therefore violations occur infinitely often. $\square$

This is a precise bounded-geometry analogue of a busy-beaver phenomenon. Every state lies on a fixed compact manifold, and every height-bounded family of rational targets is finite; nevertheless the maximal minimum exact description length cannot be bounded computably. The theorem does *not* prove that E eventually dominates every computable function. That stronger assertion is not needed and is not being claimed.

Together with Theorem 84.21, it separates two tasks on the same alphabet: a computable bound on the length required for any prescribed approximation scale, and no computable bound on the length required for exact synthesis at a prescribed rational height.

95 Why this commuting-factor construction needs dimension four

[source 16]

Theorem 95.1 (A mechanism-specific dimension obstruction). *There is no injective group homomorphism $F_2 \times F_2 \rightarrow \text{O}(3)$. Hence the faithful real orthogonal commuting-free-factor mechanism of this article has minimum dimension four.*

Proof. First suppose the target is $\text{SO}(3)$. Let a, b be free generators in the first factor. Their images have infinite order, since an injective map preserves the absence of nontrivial torsion. An infinite-order rotation in $\text{SO}(3)$ has an axis and an angle different from zero and π . Its centralizer consists exactly of the rotations around that same axis: a commuting orthogonal map must preserve its real fixed axis and act compatibly with its nonreal conjugate eigenvalues on the perpendicular plane.

The images of a, b do not commute, so their axes differ. The intersection of their two axial rotation groups is the identity. Every element of the second factor must commute with both, and hence has trivial image. This contradicts injectivity.

For an embedding into $\text{O}(3)$, take in each factor the subgroup generated by the squares of its two free generators. Each such subgroup is free of rank two, and the determinant of each of its image elements is positive. Restricting the proposed embedding would therefore give an embedding of $F_2 \times F_2$ into $\text{SO}(3)$, already excluded. Lower orthogonal dimensions embed into $\text{O}(3)$ by adjoining identity blocks. Lemma 84.7 supplies an embedding in dimension four. $\square$

This theorem does not decide arbitrary rational $\text{SO}(3)$ subgroup or semigroup membership. It also does not rule out an entirely different computational encoding in dimension three. The lower bound is for a faithful copy of two commuting nonabelian free factors. Broader embedding classifications, such as the Lie-group study in [73], should be consulted before treating it as a new general classification result.

96 Explicit quartic certificates for finite executions

[source 11]

We now turn the substrate into Diophantine equations without hiding the finite-execution construction inside MRDP. The construction applies to *any* fixed finite list of rational 4×4 gates, not just the hard family. Orthogonality is needed only for the subsequent height bound.

Let the gates be $G_1, \dots, G_k$, choose $D \geq 1$ with $A_j = DG_j \in M_4(\mathbb{Z})$, and write the source and target as

$$x = u/d, \quad y = v/e, \quad u, v \in \mathbb{Z}^4, \quad d, e \in \mathbb{Z}_{>0}.$$

Fix an external length $T \in \mathbb{N}$. Introduce integer variables

$$b_{tj} \ (0 \leq t < T, \ 1 \leq j \leq k), \quad z_t \in \mathbb{Z}^4, \ h_t \in \mathbb{Z} \ (0 \leq t \leq T).$$

The scale variables avoid inserting the exponentially large coefficient D^T directly into an endpoint equation.

Define the following residuals:

$$h_0 - d = 0, \quad z_{0,a} - u_a = 0 \quad (1 \leq a \leq 4), \quad (96.1)$$

$$b_{tj}(b_{tj} - 1) = 0, \quad \sum_{j=1}^k b_{tj} - 1 = 0, \quad (96.2)$$

$$h_{t+1} - Dh_t = 0, \quad (96.3)$$

$$z_{t+1,a} - \sum_{j=1}^k \sum_{c=1}^4 b_{tj}(A_j)_{ac} z_{t,c} = 0 \quad (1 \leq a \leq 4), \quad (96.4)$$

$$ez_{T,a} - h_T v_a = 0 \quad (1 \leq a \leq 4). \quad (96.5)$$

The step-indexed equations hold for $0 \leq t < T$; the Boolean equations hold for every selector in that range. Let $\mathcal{C}_T$ be the sum of the squares of all left-hand sides.

Theorem 96.1 (Exact finite-word quartic). *The polynomial $\mathcal{C}_T$ has integer coefficients and total degree at most four. Its integer zeros are in bijection with labelled gate words of length exactly T sending x to y . It uses*

$$(k + 5)T + 5 \text{ variables,} \quad (k + 6)T + 9 \text{ squared residuals.} \quad (96.6)$$

For a fixed gate list its sum-of-squares description and its expanded polynomial have size linear in T . With k variable, the unexpanded quadratic system has size $O(kT)$ and a direct expanded sum has $O(k^2T)$ terms before collecting cancellations.

Proof. All residuals have degree at most two. A sum of squares of integers is zero precisely when every residual is zero, so the degree and exact conjunction claims follow.

An integer root of $b(b - 1) = 0$ is 0 or 1. Together with the sum constraint, this selects exactly one gate at each time. The initial and scale equations uniquely force $h_t = dD^t$. The state equations uniquely force the integer recurrence $z_{t+1} = A_{j_t} z_t$ and $z_0 = u$. Therefore

$$\frac{z_{t+1}}{h_{t+1}} = G_{j_t} \frac{z_t}{h_t}.$$

The endpoint equations say exactly $z_T/h_T = v/e$. Conversely, any labelled gate word that makes the transfer supplies these uniquely specified selectors, scales and states. This proves the bijection, including the case $T = 0$.

There are kT selectors, $4(T + 1)$ state coordinates and $T + 1$ scales. The residual counts are five initially, $k + 6$ per step and four at the endpoint. Each quadratic step has $O(k)$ terms in the fixed dimension; squaring it gives $O(k^2)$ terms. The remaining claims follow. $\square$

The bijection is with *labelled words*, not with distinct matrices or distinct state trajectories. Different labels can represent identical gates, and different control words can have the same product. No canonicity of unbounded proofs is being inferred.

Corollary 96.2 (Natural witnesses and height). *The fixed-length certificate can be converted to natural witnesses without increasing its degree above four. For orthogonal gates and unit source, every genuine integer trace satisfies*

$$h_t = dD^t, \quad \sum_{a=1}^4 z_{t,a}^2 = h_t^2, \quad |z_{t,a}| \leq dD^t.$$

Consequently its witness coordinates have bit length $O(\log d + T \log D)$.

Proof. Replace each signed state coordinate z by $z^+ - z^-$ with $z^+, z^- \in \mathbb{N}$ and add the quadratic equation $z^+ z^- = 0$. This is the unique disjoint-support decomposition of an integer. The selectors and scales may themselves be taken natural: the equations already force selectors to be Boolean and h_t to be positive. Substitution preserves the degree-two residual bound and hence the quartic sum-of-squares bound. The displayed norm identity follows from orthogonality and z_t/h_t being the true unit state. $\square$

This finite-length height estimate does not conflict with the nonrecursive exact-transfer barrier: the missing quantity is a recursive bound on *which length T must be tried*.

97 The unbounded Diophantine representation

[source 11]

Let $\mathcal{O}_\xi \subseteq \mathbb{N}$ be the set of canonical codes of rational unit vectors in $\Gamma\xi$, rejecting invalid codes. Exact evaluation of a finite rational gate word is computable, and enumerating words semidecides this set. Thus $\mathcal{O}_\xi$ is computably enumerable.

Theorem 97.1 (Fixed polynomial, and fixed quartic after lifting). *There are a fixed integer m and a fixed polynomial $P \in \mathbb{Z}[n, w_1, \dots, w_m]$ such that*

$$n \in \mathcal{O}_\xi \iff \exists w \in \mathbb{N}^m P(n, w) = 0.$$

With more fixed auxiliary variables, P can be replaced by a polynomial of total degree at most four. The same statements hold for the positive indices in the hard recurrence family, with the convention at index zero specified in Section 89.

Proof. The first assertion is MRDP, applied to the described semidecider. It has exactly the finite-natural-witness shape of the public ProveIt endpoint [38].

For the quartic assertion, evaluate P by a fixed arithmetic circuit. Introduce an integer variable for each intermediate result. Each addition gate gives a linear residual and each multiplication gate a quadratic residual; require the output to be zero. Replace signed intermediate variables by disjoint-support differences of natural variables, as in Corollary 96.2, and square and sum every residual. The circuit is fixed independently of n , so the number of new witnesses is fixed, and the resulting degree is at most four. Finally, the recurrence-index language is computably enumerable by generating y_n exactly and using the same reachability semidecider. $\square$

97.1 What the two certificate levels do and do not say

The quartic $\mathcal{C}_T$ is an explicit sparse family whose number of variables grows with the externally fixed length T . The polynomial in Theorem 97.1 has a fixed number of variables and handles unbounded lengths, but its coefficients are not expanded here. The existence of universal quartic representations is classical; this argument does not improve the variable or operation counts of the Jones constructions documented in ProveIt [39].

The unique trace attached to a *chosen* bounded word does not imply single-fold or finite-fold representation of the unbounded orbit set. There can be many accepted words and arbitrarily many lengths. The MRDP witnesses themselves are not asserted to be unique or finite in number. Hence no finite-fold conjecture is settled by this construction.

Nonreachability from ξ cannot have a natural Diophantine representation, because such a representation would make it computably enumerable, contrary to Section 90. For the easy orbit of e , both membership and nonmembership are decidable, so both are Diophantine by MRDP after coding. This provides an arithmetic form of the two-orbit separation.

All the Diophantine statements here quantify over integers or naturals. Rational coordinates are supplied by finite integer encodings. Nothing in the construction assumes a solution to Hilbert's tenth problem over $\mathbb{Q}$ with unrestricted rational witnesses.

98 Explicit parsimonious quartic certificates

[source 16]

98.1 Natural variables without ambiguous signed encodings

Fix any finite rational rotation alphabet $\mathcal{S} = (S_0, \dots, S_{s-1})$, with a common positive integer denominator D , and let

$$B_j = DS_j \in \mathbb{Z}^{4 \times 4}.$$

Write the target matrix as

$$M = \frac{T}{q}, \quad q = Q + 1, \quad T = U - V, \tag{98.1}$$

where Q and all entries of U, V are natural-number *input parameters*. They need not be witnesses, and no uniqueness of the input encoding is assumed. A canonical external encoding may choose $U_{ik} = \max(T_{ik}, 0)$ and $V_{ik} = \max(-T_{ik}, 0)$.

For a horizon n , introduce the natural witnesses

$$e_{tj} \ (0 \leq t < n, \ 0 \leq j < s), \quad d_t \ (1 \leq t \leq n), \quad Y_{t,ik} \ (1 \leq t \leq n, \ 0 \leq i, k < 4).$$

The initial quantities are constants, not witnesses:

$$d_0 = 1, \quad Y_{0,ik} = \delta_{ik} + 1. \quad (98.2)$$

The intended meaning for a chosen word $j_0 \cdots j_{n-1}$ is

$$d_t = D^t, \quad W_t = S_{j_0} \cdots S_{j_{t-1}}, \quad A_t = d_t W_t, \quad Y_{t,ik} = A_{t,ik} + d_t. \quad (98.3)$$

Since W_t is orthogonal, $-d_t \leq A_{t,ik} \leq d_t$, so all $Y_{t,ik}$ are natural and at most $2d_t$. This shift avoids introducing two unconstrained natural numbers for every signed entry.

98.2 The full equations

The selector equations are

$$e_{tj}(e_{tj} - 1) = 0 \quad (0 \leq t < n, \ 0 \leq j < s), \quad (98.4)$$

$$\sum_{j=0}^{s-1} e_{tj} - 1 = 0 \quad (0 \leq t < n). \quad (98.5)$$

The scale and matrix recurrences are

$$d_{t+1} - Dd_t = 0, \quad (98.6)$$

$$F_{t,ik} := Y_{t+1,ik} - d_{t+1} - \sum_{j=0}^{s-1} e_{tj} \sum_{l=0}^3 (Y_{t,il} - d_t)(B_j)_{lk} = 0. \quad (98.7)$$

Finally impose

$$H_{ik} := (Q + 1)(Y_{n,ik} - d_n) - d_n(U_{ik} - V_{ik}) = 0. \quad (98.8)$$

The factor d_n multiplying the target numerator is indispensable: $Y_n - d_n \mathbf{1}$ represents $D^n W_n$, not W_n itself; here $\mathbf{1}$ is the all-ones matrix. Equation (98.8) is the cross-multiplied equality $qA_n = d_n T$.

Define one polynomial by summing the squares:

$$P_n(Q, U, V; e, d, Y) = \sum_{t=0}^{n-1} \left[\sum_{j=0}^{s-1} (e_{tj}(e_{tj} - 1))^2 + \left(\sum_{j=0}^{s-1} e_{tj} - 1 \right)^2 + (d_{t+1} - Dd_t)^2 + \sum_{i,k=0}^3 F_{t,ik}^2 \right] + \sum_{i,k=0}^3 H_{ik}^2. \quad (98.9)$$

All coefficients are integers. Every unsquared expression has total degree at most two, including the input parameters. Thus $\deg P_n \leq 4$.

Theorem 98.1 (Bijection with gate words). *For fixed input parameters Q, U, V , the natural zero set of P_n is in bijection with labeled words of length n whose value is the target matrix in (98.1). For every labeled word there is exactly one assignment to its selector, scale, and shifted-matrix variables that can be a zero.*

The number of witness variables is $n(s+17)$, and the number of squared equations in (98.9) is $n(s+18)+16$. In particular, each fixed input has at most s^n natural zero tuples.

Proof. Suppose first that a labeled word realizes M . Set its selectors to one at the chosen label and zero elsewhere, and use (98.3) for d, Y . These are natural witnesses by orthogonality. The one-hot and scale equations hold by construction. The identity

$$A_{t+1} = A_t B_{j_t}$$

gives (98.7). The endpoint $W_n = T/q$ gives $qA_n = d_n T$, proving (98.8). Hence every squared summand vanishes and $P_n = 0$.

Conversely, a sum of squares of integers is zero only when each square is zero. Equations (98.4)–(98.5) specify exactly one label j_t at each time. The scale recurrence forces $d_t = D^t$. Starting from (98.2), the matrix recurrence then uniquely forces

$$Y_{t,ik} - d_t = (D^t W_t)_{ik}$$

by induction on t . The terminal equations say $qD^n W_n = D^n T$. Since $q > 0$ and $D^n > 0$, this is exactly $W_n = T/q$. There are no unassigned slack variables, so the witnesses determined by a label sequence are unique.

There are ns selectors, n scales, and $16n$ shifted entries. At each of the n steps there are s Boolean equations, one one-hot equation, one scale equation, and 16 matrix equations. The 16 terminal equations complete the count. $\square$

The endpoint is not assumed in advance to be orthogonal. If the parameters encode a nonrotation, the soundness argument proves that no zero tuple exists. For $n = 0$, there are no witnesses and P_0 is the sum of the 16 squares of $(Q + 1)\delta_{ik} - (U_{ik} - V_{ik})$; it vanishes exactly at I_4 . Identity padding makes existence at horizon n equivalent to synthesis in at most n steps, but the bijection counts length- n labeled words, not equivalence classes under padding or cancellation.

98.3 Size, multiplicity, and computational use

At any zero, the maximum scale is D^n and the maximum shifted entry is at most $2D^n$. Thus the largest witness coordinate uses $O(1 + n \log D)$ bits. Summing over all times gives total witness bit length

$$O(ns + n^2(1 + \log D)).$$

The recurrence equations have $O(ns)$ terms before squaring, in fixed dimension. The displayed sum-of-squares circuit has linear size in n for fixed alphabet; fully expanding each square still has linear size in n with a constant depending on s . This does not imply that searching for a successful word is efficient.

The bijection is *parsimonious*: the number of natural zeros equals the number of successful labeled executions at the specified horizon. It is not a unique-witness theorem for the target. Different words can realize the same rotation, and repeated gate labels or cancellation loops can increase the count. This distinction matters in comparisons with single-fold and finite-fold Diophantine representations.

99 Unbounded Diophantine representation and its limits

[source 16]

99.1 The repository interface

Fix an effective natural-number code for rational 4×4 matrices, with malformed codes rejected. Let

$$\mathcal{E} = \{z \in \mathbb{N} : z \text{ codes a member of } \Gamma\}.$$

The set $\mathcal{E}$ is c.e. by exact word enumeration. ProveIt's public interface has the mathematical shape

$$S \text{ c.e.} \implies \exists k \in \mathbb{N} \exists P \in \mathbb{Z}[X, Z_1, \dots, Z_k] \forall x \in \mathbb{N}: x \in S \iff \exists z \in \mathbb{N}^k P(x, z) = 0. \quad (99.1)$$

The guide identifies `Diophantine.mrdp` as a public endpoint, together with `mrdp_iff` and `mrdp_dioph_iff` in the same namespace [76]. Applying the mathematical theorem to $\mathcal{E}$ gives one fixed polynomial with a fixed number of witnesses. This is an existence application, not an extraction of that polynomial from a Lean term in the present work.

99.2 A fixed quartic exists

Proposition 99.1 (Circuit degree reduction). *Every natural-witness integer-polynomial representation can be replaced by one of total degree at most four, with additional natural witnesses. For each original witness tuple, the extra circuit witnesses can be chosen uniquely.*

Proof. Evaluate the original polynomial by a fixed arithmetic circuit using integer constants, additions, subtractions, and multiplications. For every signed intermediate integer v , introduce two natural variables v^+, v^- with

$$v = v^+ - v^-, \quad v^+ v^- = 0.$$

The second equation makes this representation canonical: exactly one pair represents any specified integer. Express each circuit gate as an equation on these differences. Addition and subtraction give linear equations, and multiplication gives equations of degree at most two. Require the final difference to be zero and sum the squares of all gate, complementarity, and output equations.

The resulting polynomial has degree at most four. Circuit induction shows that every satisfying assignment computes the original polynomial, and that every original zero has exactly one extension by canonical signed intermediate values. Thus the projection and the claimed uniqueness of the extension are correct. $\square$

Consequently $\mathcal{E}$ has a fixed quartic Diophantine representation. No bound on its number of witnesses follows from this argument without first specifying the MRDP polynomial and the chosen circuit. The direct family P_n and the unbounded polynomial are different objects:

	Horizon-specific P_n	Unbounded MRDP polynomial
Arity	grows linearly with n	fixed
Equations	printed in full as a scheme	not expanded here
Witness meaning	individual gate word and its trace	MRDP-dependent
Fiber size	at most s^n	not controlled here

99.3 Why bounded canonicity does not settle finite-fold questions

The trace at a specified horizon is canonical once the labels are fixed. It does not follow that unbounded synthesis has one arithmetic witness, or finitely many arithmetic witnesses, per target. Identity padding and inverse pairs already produce infinitely many raw successful executions for every synthesizable target. Replacing arbitrary executions by a shortest execution still requires a separate arithmetic construction; an MRDP representation of that selection process need not have controlled auxiliary multiplicity.

There is a second logical pitfall. A computable many-one reduction between c.e. sets preserves ordinary Diophantine representability through MRDP. It does *not*, by itself, supply a finite-fold representation of the reduction's graph. Therefore c.e.-completeness of $\mathcal{E}$ should not be used here to claim equivalence with a general finite-fold conjecture. Any such transfer requires an additional finite-multiplicity argument. Proposition 99.1 preserves multiplicity *after* an original representation is given; it does not create finite-foldness that was absent or unproved before.

Finally, no fixed polynomial representing $\mathcal{E}$ can have a total computable bound $f(x)$ guaranteeing that every affirmative input x has some witness tuple with all coordinates at most $f(x)$. Such a bound would

allow finite exhaustive search and decide the complete set. This is a witness-size obstruction, not a denial of the existence of individual finite certificates.

100 Executable examples, checks, and trust boundaries

[source 11]

The package contains `code/rotations.py` and `code/verify.py`. They use only Python’s standard library, exact integers and `fractions.Fraction`. No floating-point approximation enters an acceptance decision. The implementation includes the binary and finite-rank quaternion decoders, the tensor lift reconstruction, the presentation compiler, Miller normal forms, full rational-matrix membership using that word algorithm, and finite quartic residual construction and evaluation.

100.1 A completely explicit small example

For a reproducible example rather than an unprinted universal presentation, take $Q = \langle x \mid x^2 \rangle$. The padded Miller presentation is

$$G = \langle z, q, x, t, u \mid z, txt^{-1}x^{-1}, uxu^{-1}x^{-1}, tqt^{-1}x^{-1}q^{-1}x, uqu^{-1}x^{-2}q^{-1} \rangle. \quad (100.1)$$

The compiler produces twenty labelled gates. Let $R = uqu^{-1}x^{-2}q^{-1}$ be the last relator. Starting from $\phi(q)$, apply C_u and then left multiplication by $\phi(R^{-1})$. The free word identity

$$R^{-1}(uqu^{-1}) = qx^2$$

shows that the endpoint is exactly $\phi(qx^2)$. The checked coordinates are

$$\begin{aligned} \phi(q) &= \left(\frac{3}{5}, 0, -\frac{28}{125}, \frac{96}{125} \right), \\ \phi(qx^2) &= \left(\frac{147}{3125}, \frac{2304}{3125}, -\frac{33044}{78125}, -\frac{40992}{78125} \right). \end{aligned}$$

The actual least common denominator of this twenty-gate list, as computed by the program, is $5^{20} = 95367431640625$. The two-step integer quartic uses 55 variables and 61 squared residuals. Its value on the constructed witness is exactly zero. This example illustrates the compiler and certificate; it is not itself the universal hard instance.

The following reproduces the exact checks from the package root:

```
python3 code/verify.py
```

The script writes `verification_results.json`. The checked counts in the distributed run are shown in Table 5.

100.2 Limits of the computational evidence

Finite enumeration cannot prove freeness at all word lengths, density, undecidability, or completeness. Those conclusions rest on the proofs above and the explicitly imported Higman and MRDP theorems. The supplied code does not implement Higman’s embedding construction or emit a fully expanded fixed universal Diophantine polynomial.

The inspected ProveIt revision is

b6bf6406a4c49017fedc6fba068b36c634987150.

Exact finite check	Cases
Modulo-five multiplication on quaternion basis pairs	16
Adjacent-letter zero-product criterion	16
All reduced binary words of length at most seven	4373
Finite-rank decoder round trips	200
Quaternion rejection examples	7
Entries of the inverse-spin tensor Gram matrix	256
Rational spin reconstruction and group-membership round trips	100
Matrix rejection examples, including a rational rotation without rational lifts	3
Miller defining relators across two examples	15
Miller inverse-word normal-form tests	60
Explicit density-subsystem identities	3
Exact trace certificates at lengths zero through eight	9
Corrupted trace or selector rejections	17
Compiled Miller gates accepted by the membership algorithm	20
Explicit two-gate marked-state transfer	1
Coordinate instances of the order-five recurrence	240

Table 5: Deterministic exact-arithmetic verification, random seed 20260930. Finite tests supplement, but do not replace, the mathematical proofs.

The MRDP source and guide were read, but this article’s quaternion, Miller, density and compiler proofs were not translated into Lean and no new Lean build or axiom audit was performed. Existing repository audits cannot be inherited merely by citing the repository. The distinction is between a mathematical dependency and an actually checked formal integration.

[write] In this report the programs are shipped as `code/11-rational-rotations-rotations.py`, `code/11-rational-rotations-verify.py` and `code/11-rational-rotations-build.sh`, and the results and source manifest as `data/11-rational-rotations-*`. The command above is the source’s, for its delivered layout; the README gives the rerun recipe on a copy.

101 Concrete implementation and finite verification

[source 16]

101.1 A fully instantiated, decidable example

The supplied program `verify.py` uses only Python’s standard library, integer arithmetic, and `fractions.Fraction`. Its finite presentation example is

$$G_{\text{ex}} = \langle x, y, t \mid xyx^{-1}y^{-1} \rangle \simeq \mathbb{Z}^2 * \mathbb{Z}. \quad (101.1)$$

This example has decidable word problem. It is *not* the universal specialization of Section 91. Its purpose is to test the same compiler and certificates on concrete matrices whose sizes remain manageable.

The rank-three embedding gives the nine labels

$$I_4, A_x, A_y, A_t, R, A_x^{-1}, A_y^{-1}, A_t^{-1}, R^{-1}, \quad R = L(\rho_3([x, y])).$$

The least common denominator of these matrices is

$$D = 5^{14} = 6\,103\,515\,625. \quad (101.2)$$

The three positive diagonal gates have denominators $5^6, 5^{10}, 5^{14}$, and the relator gate has denominator 5^{10} . Their transposes have identical denominators. All numerator matrices are printed in the machine-readable file `example_gates.json`.

For horizon two, the certificate has 52 natural witnesses and 70 squared equations. The word consisting of the relator gate followed by the identity is a direct successful two-step example for target R . The recurrence tests include every label sequence through horizon three, not only this particular sequence.

101.2 What was checked

The accompanying run record `verification_results.json` reports:

Exact finite check	Cases
All reduced a, A, b, B words through length eight, including empty	13 121
Norm, exact denominator, and full decoder on each such word	13 121
Explicit quaternion nonmembers rejected by the decoder	4
Seeded Spin(4) homomorphism, orthogonality, determinant tests	100
Valid example traces at horizons zero through three	820
Deliberately corrupted traces or targets rejected	3 277

The 820 successful trace checks equal $1 + 9 + 9^2 + 9^3$. For each nonempty trace, the tests perturb a shifted matrix entry, a scale, and a selected Boolean value separately; for every trace, they perturb a target numerator. The sum-of-squares residual must become positive. These tests catch, among other errors, a missing scale factor in the terminal equation.

A locality illustration chooses $\varepsilon = 1/5$. Enumerating powers of $\rho_3(t)$ finds $k = 61$ with $|\rho_3(t)^{61} - 1| < 1/20$. The script verifies using exact squared norms that the commutator target associated with x lies within Frobenius distance $1/5$ of the identity. This is a check of the quantitative inequality, not a numerical proof of undecidability.

101.3 Reproduction and limits

Run from the extracted archive directory:

```
python verify.py --output verification_results.json
pdflatex -interaction=nonstopmode -halt-on-error dense_rotations.tex
pdflatex -interaction=nonstopmode -halt-on-error dense_rotations.tex
```

The script uses a fixed pseudorandom seed for the 100 additional homomorphism tests. The exhaustive short-word checks do not depend on that seed. No floating-point comparison enters any assertion.

Finite tests cannot establish freeness for all lengths, density, Higman’s embedding theorem, or c.e.-completeness. Those claims rest on the mathematical proofs and their identified classical inputs. No Lean files are labeled as verified, and the ProveIt dependency graph has not been rebuilt for this article.

[write] In this report the program is shipped as `code/16-dense-rotations-verify.py` (with `code/16-dense-rotations-build.sh`), and its outputs as `data/16-dense-rotations-example_gates.json` and `data/16-dense-rotations-verification_results.json`. The commands above are the source’s, for its delivered layout; the README gives the rerun recipe on a copy.

102 A formalization route for ProveIt

[source 16]

The proposed integration should expose small independent interfaces rather than a single theorem whose computational and algebraic assumptions are hard to audit. The repository already supplies the final MRDP endpoint; the new work lies principally before that endpoint.

Layer 1: exact arithmetic. Formalize the quaternion multiplication table over $\mathbb{Z}$, $\mathbb{Q}$, and $\text{ZMod } 5$, the map (84.10), and the sixteen pairings in (84.12). A finite checked table plus an induction on reduced words proves that the numerator residue never vanishes. This is a compact starting target: it yields freeness, denominator growth, and decoder correctness without computability theory.

Layer 2: group compiler. Formalize ι_d , its reduced-block argument, the Φ action and its kernel, and the generator characterization of M_G . The main interface should state (84.23) for an arbitrary finite presentation. Density is logically independent of the membership reduction and should be in a separate module. That separation permits the arithmetic application to proceed even before the Lie-theoretic closure argument is available.

Layer 3: bounded polynomials. Define the one-hot selectors and the shifted integer recurrence. Prove Theorem 98.1 first as a system of equations, then as a single sum of squares. The exact endpoint must be $qA_n = d_n T$, not $qA_n = T$. Record the witness count as part of the interface, and use a datatype of labeled words so that duplicate gate values cannot silently invalidate the counting theorem.

Layer 4: unbounded semantics. Give rational-matrix coding and the exact finite-word evaluator an explicit computability proof. Its c.e. range may then be passed to `Diophantine.mrdp`. The repository's guide describes a one-natural-input interface, so multicoordinate matrices should be encoded before invoking it; simply writing many free matrix variables does not automatically match the existing theorem [76].

Layer 5: universality and local hardness. Treat the finitely presented complete group as an explicit imported mathematical dependency until a concrete presentation and reduction have been formalized. The local theorem needs only the free-product normal form and a computable small-power search on a unit quaternion. State that dependency plainly instead of inserting an untracked assumption that “the word problem is universal.”

These are proposed interfaces and proof tasks, not claims about code already present in the pinned repository. In particular, the source report of its MRDP audits should not be read as an audit of this article.

103 Source 10 appendix: a complete eight-variable quartic for destructive interference

This appendix gives a small explicit equation, not just a compiler existence statement. Let $u^+, u^-, v^+, v^-, w^+, w^-, z^+, z^- \in \mathbb{N}$ and write $\Delta u = u^+ - u^-$, with analogous notation for the other pairs. Define

$$\begin{aligned} P = & (u^+ u^-)^2 + (\Delta u - 1)^2 + (v^+ v^-)^2 + (\Delta v - 2\Delta u)^2 \\ & + (w^+ w^-)^2 + (\Delta w)^2 + (z^+ z^-)^2 + (\Delta z - (\Delta w)^2)^2 + (\Delta z)^2. \end{aligned} \quad (103.1)$$

This polynomial has integer coefficients and degree four. Its nine squared residuals describe

$$u = 1, \quad v = u + u = 2, \quad w = u - u = 0, \quad z = w^2 = 0,$$

together with the assertion $z = 0$. Here v and w are the two numerator amplitudes of $HH|0\rangle$; $z/4$ is the output-1 probability.

Because every summand is a square of an integer, $P = 0$ forces all nine residuals to vanish. Successively applying the canonical-pair lemma gives the unique solution

$$(u^+, u^-, v^+, v^-, w^+, w^-, z^+, z^-) = (1, 0, 2, 0, 0, 0, 0, 0). \quad (103.2)$$

This is a fully explicit single-witness Diophantine certificate of the zero probability, although it is intentionally specialized to a very small example. It is not a universal polynomial or a claim about minimal variable count.

The delivered file `data/HH_expanded_quartic.json` contains the complete 33-monomial sparse expansion, all nine quadratic residuals, variable names, and the witness. In that format a term is an integer coefficient and a list of variable indices; repeated indices represent powers. The empty list is a constant term. The independent checker verifies that the 33-term polynomial is identically the sum of the nine residual squares.

104 Source 10 appendix: proof-audit checklist and dependency ledger

The following checks delimit the claims proved in this manuscript.

Potential gap	Resolution or explicit limitation
Unrestricted signed pairs	The equation $z^+ z^- = 0$ is imposed on every integer wire; Lemma 72.1 proves uniqueness over $\mathbb{N}$.
Discarding interference	Both signed amplitude sums are retained inside a complete contraction before taking a squared norm.
Wrong scalar domain	All existential coordinates are natural numbers; negative integers occur only through differences and polynomial coefficients.
Hidden denominators	Hadamard factors are removed globally. The known integer 2^h , or 4^n for complete graph-state measurements, is retained explicitly and counted in binary length.
A cubic guarded residual	Sign tests are unconditional; the branch-combining equations themselves are quadratic. No Boolean is multiplied by a quadratic equality as an extra guard.
Arithmetic-DAG height	The small-height theorem uses disjoint leaf subsets. It is not asserted for unrestricted repeated squaring.
Tensor-width convention	The parameter is precisely $\max A \cup B $ of the supplied merge tree. No equality with a different treewidth convention is assumed.
A probability from one path	The HH example explicitly rejects that interpretation. The polynomial certifies the total numerator, including zero.
Uncounted adaptive logic	Branch rules are explicit finite descriptions. Aggregating many transcripts may cost exponentially many terms unless a separate causal compilation and width analysis is supplied.
Parameter representation	The one-gate obstruction uses Cauchy names, not explicit rational numerators and denominators. Every name in the reduction is uniformly valid.
Unbounded variable count	The direct finite compiler is not itself a fixed-arity unbounded polynomial. MRDP is invoked separately for enumerable positive-halting observations.
Single-fold MRDP claim	Canonical circuit evaluation supplies unique auxiliaries only after the original MRDP witnesses are fixed. It does not prove uniqueness of those witnesses.
Almost-sure versus possible	$\exists t [H_t > 0]$ and $\forall k \exists t [H_t > 1 - 2^{-k}]$ are distinguished and separately analyzed.
Finite tests as proof	Tests check examples and implementation invariants. General soundness, completeness, uniqueness, and height bounds are established by the proofs in the text. No Lean build was performed.

The only deep representation theorem used as a dependency is MRDP, and only in Theorem 79.1. The finite arithmetic compiler, tensor counts, cyclotomic comparison, graph-state factorization, and explicit

reductions are proved directly. Universality, path-count representations, low-width simulation, and the classical almost-sure-termination result are cited as established context, not claimed as discoveries of this work.

105 Source 11 appendix: a compact specification of the full compiler

Given a finite presentation $Q = \langle Y \mid \mathcal{R} \rangle$, the following specifies the complete construction without an informal choice of coordinates.

Order the padded Miller generators as

$$z, q, Y, \{\theta_y : y \in Y\}, \{\theta_R : R \in \mathcal{R}\}.$$

Number them $x_0, \dots, x_{r-1}$. Its relators are z together with the left side times the inverse of the right side of (88.2)–(88.4). Evaluate each x_j as $a^j b a^{-j}$, taking the quaternions a, b from (84.1). For every generator emit conjugation by its quaternion. For every relator emit left multiplication by its evaluated quaternion. Append the transposes of all emitted matrices. This order agrees with `MillerMachine.presentation()` and `Presentation.gates()` in the code, except that Python uses one-based signed letters for group words and zero-based gate labels.

For $p = (p_0, p_1, p_2, p_3)$, left multiplication is

$$L(p) = \begin{pmatrix} p_0 & -p_1 & -p_2 & -p_3 \\ p_1 & p_0 & -p_3 & p_2 \\ p_2 & p_3 & p_0 & -p_1 \\ p_3 & -p_2 & p_1 & p_0 \end{pmatrix}.$$

The matrix of $v \mapsto v\bar{q}$ is

$$R(\bar{q}) = \begin{pmatrix} q_0 & q_1 & q_2 & q_3 \\ -q_1 & q_0 & -q_3 & q_2 \\ -q_2 & q_3 & q_0 & -q_1 \\ -q_3 & -q_2 & q_1 & q_0 \end{pmatrix},$$

so $\Phi(p, q) = L(p)R(\bar{q})$. These formulas also make the integer coefficient and denominator claims directly checkable.

The membership routine first recovers rational unit lifts, then decodes both to free words, then applies the Miller word algorithm to their quotient. The orbit-of- e routine omits the lift step. A semidecider for the orbit of ξ simply enumerates chronological gate lists and tests the resulting endpoint exactly. The finite certificate evaluator uses the same chronological convention:

$$x_{t+1} = G_{j_t} x_t, \quad x_T = G_{j_{T-1}} \cdots G_{j_0} x_0.$$

Claim or artifact	Evidence and boundary
Free rational encoding	Full modulo-five proof in Section 84.2; the general technique has prior literature.
Full rational lift and membership decoder	Explicit algorithms and proofs in Section 84.7; exact finite tests supplied.
Fiber-product orbit identity	Full group-theoretic proof in Sections 84.5–86, using the classical Mihailova construction.
Finite group with complete word problem	Higman embedding theorem is imported; the complete recursively presented seed and reduction are proved in Section 87.
Polynomial-time word / hard conjugacy separation	Classical Miller construction, with explicit proofs and normal-form implementation in Section 88.
Simultaneous dense rational separation	Assembled and proved in Section 84.8; no priority certification for the combination.
Fifth-order hard target family	Reduction plus a direct harmonic calculation in Section 89; no scalar-recurrence undecidability claim.
Finite-length quartic	Explicit residuals, bijection proof and exact integer tests in Section 96.
Fixed unbounded polynomial	Imported MRDP plus routine circuit lifting; not expanded and not optimized here.
Lean verification	No new Lean proof or build included. Existing ProveIt MRDP source was inspected, not extended.

Table 7: The trusted mathematical dependencies and the implementation scope.

106 Source 11 appendix: dependency and claim ledger

107 Source 16 appendix: proof dependencies and reproducibility manifest

Result	Required mathematical input
Exact denominator and decoder	Quaternion multiplication; the displayed $\mathbb{F}_5$ tables.
Faithful $F_d \times F_d$ embedding	Reduced free words; quaternion left-right action.
Density	Closed subgroup theorem; elementary Lie algebra of unit quaternions.
Presentation compiler	Elementary normal closure and fiber-product identity.
Fixed completeness	Amalgam normal forms; Higman embedding; c.e. halting completeness.
Local completeness	Free-product normal forms; elementary commutator norm bound.
Uniform approximation bound	Density; compactness; real-algebra decision procedure.
Depth degree $\mathcal{O}'$	Completeness; finite enumeration of height-bounded rational matrices.
Bounded quartic and bijection	Integer arithmetic; one-hot selectors; orthogonality entry bounds.
Fixed unbounded quartic	MRDP; canonical signed arithmetic-circuit evaluation.

The archive contains:

dense_rotations.tex, dense_rotations.pdf

The source and compiled article.

verify.py

Executable exact finite verification and presentation-to-gate routines.

verification_results.json

The results from the supplied verification run.

example_gates.json

All nine rational matrices for the decidable example, including numerator arrays and denominators.

README.md, build.sh

Scope, reproduction instructions, and a minimal build command sequence.

108 Conclusions of Part VI

108.1 Source 10

A legal path is the wrong witness for coherent acceptance. A canonical arithmetic evaluation of the complete tensor contraction is the right one, and it can be expressed using natural variables and a single polynomial of degree at most four. The ring lift, exact comparison gadget, and resource counts make this statement constructive for gate circuits and adaptive graph-state branches.

The limits are equally informative. Width can force many coordinates even when every coordinate is short; approximation names can make exact zero non-enumerable even for individually rational gates; and almost-sure eventual behavior requires a different quantifier pattern from positive-probability halting. A Diophantine theory of unconventional computation should therefore organize itself around *effective observation semantics*, not merely around a list of Turing-complete substrates.

108.2 Source 11

The construction separates three notions that are easily conflated: recognizing an entire rational transformation, finding one transformation that satisfies a state constraint, and achieving a positive numerical tolerance. A fixed family of reversible rational isometries can make the first polynomial-time decidable, the second computably enumerable complete, and the third uniformly attainable. Two small rational starting states already distinguish an easy dense orbit from a hard dense orbit. The hard queries can lie in a fixed fifth-order rational recurrence family, and exact transfer requires no recursively bounded precision budget even though candidate matrices are easy to recognize.

For the Diophantine program in ProveIt, this gives a concrete substrate with an explicit finite-trace quartic compiler and a clearly delimited MRDP endpoint. Its value is the proved separation and the inspectable interfaces between group theory, rational dynamics, and arithmetic, not a claim to have rediscovered rotation undecidability or solved an unbounded witness-multiplicity conjecture.

108.3 Source 16

The finite-presentation compiler gives a precise sense in which a compact, norm-preserving, reversible system can carry full c.e. computational complexity. The geometry does not expand; exact rational descriptions and algebraic relations carry the complexity instead. The generated group can be dense, and the hard instances can be localized arbitrarily close to any rational rotation, while ordinary approximation remains a terminating computational task.

The most concrete arithmetic results are the exact denominator theorem, its residue decoder, and the explicit bounded quartic polynomial. They expose information that an undifferentiated appeal to MRDP would conceal: which word is represented, how its rational height grows, where precision forces equality, and how many natural witnesses encode a bounded execution. The classical foundations and the uninstantiated universal constants are kept separate from those explicit constructions.

This provides a research and formalization path for ProveIt with clear interfaces and testable claims. It also identifies the genuinely unfinished parts: an optimized numerical universal alphabet, a certified end-to-end implementation, historical novelty assessment, and stronger unbounded witness control.

109 Further research questions of Part VI

[write] Source 10 posed twelve numbered projects, source 11 seven and source 16 ten, both of the latter as subsections; all are printed. The two sets of subsections are numbered here as research questions, and three pairs that pose the same question are merged into one environment that prints each source's text. A [write] note names the question of Parts I–IV that a question duplicates or bears on.

109.1 Source 10: coherent computation

The following are proposed projects rather than additional claimed results. They are ordered roughly from immediately implementable proof infrastructure to more substantial representation questions.

1. **A verified canonical-wire backend.** Formalize Theorem 72.3 with an explicit evaluator, a polynomial compiler, and a theorem giving the unique extension of every fixed input. Prove the variable, degree, and sparse-size bounds as statements about the compiler output. This is a small foundation reusable outside quantum computation and does not depend on a formal theory of Hilbert spaces. [write] Compare Question 1.
2. **A formal finite-contraction interface.** Represent boundary labels, disjoint leaf subsets, and supplied binary trees so that (73.2) and the identity of the two full semantics are proved by finite-sum reindexing. The target is a machine-checked version of the height argument for *every scalar intermediate*, not only the final tensor. A separately verified tree checker should reject missing or repeated leaves.
3. **Reduce the ring arithmetic constants.** Search for lower-cost four-coordinate multiplication circuits, allowing a tradeoff between integer products and additions. An improvement must be measured after canonical wires are introduced; reducing multiplications alone need not reduce the total number of natural variables. Preserve an explicit bound on intermediate coefficient growth and prove every optimized circuit identity exactly.
4. **Certified zero and sparsity elimination.** The present implementation expends 28 scalar gates even on products involving zero entries. A compiler could propagate known zero patterns, permutation tensors, and shared phases before emitting residuals. Can one produce schedule-sensitive sparse profile bounds while keeping a deterministic, unique-witness evaluation trace? The optimization must be justified syntactically, rather than using an unverified approximate zero test.
5. **Structure other than contraction width.** Develop analogous certificate theorems for circuits decomposed into a small number of tractable components, or with a small number of non-Clifford gates. The objective is a transparent bound in the decomposition size, coefficient heights, and verification cost. Such a theorem should explain explicitly how it avoids the universal obstruction of Theorem 77.1. [write] Compare Question 7, which Theorems 73.1 and 77.1 partly answer.
6. **Coarse adaptive events without transcript enumeration.** Build one network for an entire adaptive success predicate, including Boolean angle circuits, outcome wires, and final acceptance logic. Determine the additional width forced by causal dependencies. This would strengthen Theorem 76.1, which certifies a fixed branch and deliberately does not hide the cost of aggregating exponentially many branches.
7. **Canonical comparisons in larger fixed number fields.** For a fixed ordered algebraic number field, can exact sign testing be compiled with comparably explicit quadratic degree and single-witness

bounds? A candidate uses a proved root-isolation algorithm with a canonical choice of interval and trace. Arbitrary existential isolating intervals would generally reintroduce multiple witnesses; this is the issue to solve, not just the existence of an order test. [write] Compare Questions 6 and 22.

8. **Exact channels and mixed-state programs.** Extend the backend to explicitly described Kraus operators or superoperators, with denominators cleared and conjugation handled algebraically. Separate the certificate for a probability from the certificate that a supplied operation is completely positive and trace preserving. Trusted finite gate alphabets avoid that second obligation; arbitrary user-specified channels do not. [write] Compare Question 22.
9. **Quantitative robustness under a supplied gap.** For promised instances satisfying $|p - r| \geq \varepsilon$ with an explicit rational $\varepsilon > 0$, compare exact cyclotomic certificates with certified interval evaluation. Determine which certificate dimensions and bit lengths can be bounded in terms of $\log(1/\varepsilon)$ and network structure. The promise must be supplied or separately certified; it cannot be inferred uniformly from an arbitrary Cauchy name. [write] Compare Question 20.
10. **Restricted unbounded canonical compression.** Identify useful controller classes for which a fixed-arity Diophantine representation retains a finite or unique witness property. Any proof must control the witnesses introduced when coding an unbounded computation and eliminating its time quantifier. Canonical finite traces alone are not such a proof. This project should be stated with restrictions rather than assuming the general finite-fold or single-fold MRDP conjectures. [write] Compare Questions 19 and 10.
11. **Representation-aware observational types.** A substrate interface could distinguish exact algebraic descriptions, computable approximation names, strict inequalities, exact equalities, and limit assertions. Prove a classification theorem for a chosen program language showing which interfaces yield enumerable observations and which require alternating arithmetic quantifiers. Theorems 78.1 and 79.2 supply mandatory counterexamples to an overbroad design. [write] Compare Question 12.
12. **Proof-carrying compilation into the repository.** Integrate the small backend with ProveIt's MRDP interface without confusing the direct finite theorem with the unbounded existential consequence. A useful deliverable would carry the source circuit, a checked contraction tree, polynomial residuals, and a theorem connecting their evaluation. Separate tests and axiom audits should record exactly which layers have been verified and which remain mathematical specifications. [write] Compare Question 9.

109.2 Sources 11 and 16: exact rational rotations

Source 11. The following are concrete next questions arising from the construction. They are not all asserted to be previously published open problems; in particular, dimensional minimality and optimization would require a broader literature audit before any priority claim. *Source 16.* The following are research directions arising from the construction. They are not all claims of independently established open problems in the literature. Where a status classification matters, it must be checked for the exact formulation before a publication makes an “open problem” assertion.

Research question 45 (Minimal dimension; lower dimensions). *Source 11* (“*Minimal dimension under the full separation requirements*”). Can a fixed symmetric family in $\text{SO}(3, \mathbb{Q})$ have polynomial-time rational-matrix membership but a computably enumerable complete rational state orbit? The four-dimensional proof uses independent left and right quaternion factors, so it does not descend automatically. The relevant question is not merely an unrestricted rotation-semigroup problem: the simultaneous membership, reversibility and fixed-source constraints are part of the target statement. The historical three-dimensional problem noted in the rotation literature should not be presumed to have remained unchanged through 2026 without a dedicated survey. *Source 16* (“*Explore lower dimensions without misusing the obstruction*”). Theorem 95.1 rules out the present product embedding in $\text{O}(3)$. It leaves other mechanisms untouched. A targeted study should separate fixed-subgroup membership, variable-generator semigroup membership, point reachability, and matrix reachability in rational three-dimensional rotations. These are different decision problems; an

answer for one cannot be transferred to another by terminology alone. [write] Compare Part II's Question 13 (smaller rational instruments), which concerns a different predicate.

Research question 46 (An expanded universal numerical example). Choose a concrete finite presentation with an explicit halting reduction, then run the Miller and rotation compilers to print every rational gate. The present theorem specifies a terminating finite compiler and proves existence of a suitable fixed input, but does not optimize or expand that input. A useful artifact would include the source machine model, exact presentation, reduction words, gate matrices, common denominator, and independent proofs that all links commute. Equation (88.8) provides a baseline against which smaller constructions can be measured. [write] Compare Questions 52 and 18.

Research question 47 (How low can the hard recurrence order be?). The commutator construction gives five frequencies $0, \pm 2\theta, \pm 4\theta$. Is there a fixed-source hard family generated by an order-three recurrence, or by one rational matrix power applied to a fixed vector, while retaining polynomial-time gate membership? The order-five bound is not claimed sharp. A promising route is to analyze whether a different recursively presented two-generator seed can encode the required membership test with fewer alternating power factors. Any proposed reduction must preserve both directions; an arbitrary simplification of the commutator may destroy independence of the defining relations.

Research question 48 (Quantitative exact control and group geometry; exact-versus-approximate comparisons). *Source 11* (“*Quantitative exact control and group geometry*”). The proof establishes nonrecursive worst-case precision, but does not identify a precise comparison between minimal transfer denominator, minimal conjugator length in $M(Q)$, and a geometric filling function for Q . Gillis’s quantitative study of Miller conjugators provides a useful starting point [32]. The arithmetic step is subtle: different long products can have small endpoint denominators even though full transfer matrices cannot be uniformly bounded. Sharp comparisons would clarify exactly which group-geometric complexity is visible in the rational control problem. *Source 16* (“*Sharpen exact-versus-approximate quantitative comparisons*”). The computable net bound $B(k)$ is deliberately inefficient. A next project could construct a practical approximation algorithm for the specific compiler output, while tracking the relators on which its constants depend. On the exact side, compare the depth function $E(h)$ with filling functions or subgroup distortion for the source presentation. The present proof establishes its halting degree and lack of computable majorants, but not an exact asymptotic correspondence with a chosen geometric group invariant.

Research question 49 (Effective approximation versus effective separation). The finite-net algorithm proves computability of $B(\varepsilon)$, not an attractive rate or practical synthesis method. One can study efficient approximation using the explicit dense two-generator left subsystem. A separate question is which additional promises make exact nonreachability decidable: a supplied separation margin, restrictions on the admissible target language, or bounded gate denominators could change the problem substantially. Noise models must be specified precisely; positive-tolerance reachability in the present system is always true, whereas repeated adversarial perturbations define a different semantics. [write] Compare Question 20, which source 16 partly answers.

Research question 50 (Other denominator primes and residue automata). *Source 11* (“*Other denominator primes and local decoders*”). The modulo-five proof turns a valuation into a syntactic length certificate. Which pairs of rational unit quaternions admit comparable single-prime decoders? One could seek a classification in terms of the image and kernel lines of numerator matrices over $\mathbb{F}_p$, together with an exact cancellation criterion over higher powers of p . A successful generalization would yield new arithmetic compiler families rather than merely another invocation of abstract free-subgroup existence. *Source 16* (“*Develop other residue automata*”). For which rational unit-quaternion generating sets does reduction modulo a prime give rank-one matrices whose zero pairings correspond exactly to inverse letters? Such a criterion could generate families of exact denominator encodings and permit optimization over primes and Pythagorean data. One should distinguish freeness, exact denominator growth, and deterministic first-letter recovery: none is automatically equivalent to the others for a different generator family.

Research question 51 (A modular Lean development). A manageable formalization can be divided into four dependency layers. First, formalize rational quaternion arithmetic, the modulo-five matrix isomorphism,

denominator rigidity, and soundness and completeness of the decoder. Second, formalize fiber-product generation and the orbit-conjugacy identity at the group level, keeping these proofs independent of analytic density. Third, verify the signed trace compiler and the quartic zero-set equivalence, including length zero, duplicate gate labels, common-denominator normalization, and natural signed witnesses. Fourth, connect the exact evaluator's semidecider to `Diophantine.mrdp`.

The existence of a finite presentation with complete word problem and the density argument can be tracked as separate dependencies until their own formalizations are available. This avoids contaminating the elementary arithmetic checker with unproved global claims. The most useful early milestone would be a kernel-checked theorem that every emitted finite certificate is equivalent to the actual rational execution relation. [write] Compare Question 53, Section 102 and Question 1.

Research question 52 (Instantiate a compact complete source presentation). The most valuable completion is a concrete, independently checked finite presentation G_0 , fixed words giving a many-one reduction from a chosen halting language, and the resulting numerical gate matrices. This would replace the existential specialization by a fully enumerated universal alphabet. The optimization objectives should be recorded separately: number of generators, number and total length of relators, expanded a, b lengths, denominator exponent K , and total bit size of the gates. Reducing one does not necessarily reduce the others.

Research question 53 (Verify the denominator decoder first). The modulo-5 proof is finite enough to be an attractive Lean entry point. A useful deliverable is not merely injectivity but a verified executable decoder with a bit-complexity analysis. The integer peeling step uses multiplication by a fixed numerator and division by 25; a careful implementation should avoid repeated general rational normalization. This would yield an efficient certified ambient word representation for later constructions.

Research question 54 (Classify length-dependent precision specifications). The threshold $(qD^n)^{-1}$ forces equality. What can be proved for larger thresholds $\tau(q, n)$? The right formulation must specify the norm, the actual output length, padding rules, and whether q is canonical. Since padding changes n without changing the matrix, such choices can alter the search problem. A classification of specifications between fixed tolerance and exact forcing would be more informative than a single undecidability example.

Research question 55 (Remove unrestricted inverse access). The compiler currently supplies every inverse as an available gate. For an application with only a one-sided semigroup, an encoding would need to simulate inverse use or modify the source computation. Compactness makes inverse approximation easy, but inverse approximation is not an exact inverse word. That distinction is precisely the kind of gap the current article exposes and should not be hidden in a semigroup variant.

Research question 56 (Build a fixed-arity arithmetic compiler with trace semantics). The bounded quartic family is transparent but has horizon-dependent arity. A direct fixed-arity compiler should retain a formally proved relationship between arithmetic witnesses and encoded gate traces, with explicit bounds on degree, variables, and coefficient complexity. The repository's bounded-quantifier and trace infrastructure may be a useful starting point, but importing MRDP as a black box does not supply these quantitative guarantees. [write] Compare Question 19.

Research question 57 (Investigate finite-foldness with the missing hypotheses visible). A controlled arithmetic representation of a canonical successful execution would be significant. The critical issue is auxiliary multiplicity, not merely uniqueness of the abstract execution code. A serious attempt should identify exactly which trace-coding, bounded-minimality, or computable-reduction graph needs a finite-fold representation. The parsimonious bounded theorem is a test case, not a solution to the general unbounded problem. [write] Compare Question 10 and the restricted-class question of the sibling report `canonical-diophantine-certificates`.

Research question 58 (Study robust operational questions rather than ideal equality). Because exact yes- and no-instances are both dense, a finite-precision experiment cannot decide arbitrary exact membership uniformly. More operational questions might promise a known separation margin, bound gate length, restrict denominators, or ask for a certified residual. The separation theorem immediately gives sound certificates under some such promises. A useful next step is to classify which promises restore decidability without making the model trivial. [write] Compare Question 20.

Part VII

Exact observation in continuous-state dynamics: strict contractions and neural precision (sources 14, 12 and 15)

110 [write] Conventions of Part VII

Part VII prints three manuscripts of batch 62 in full. Source 14, *Strict Contractions with Diophantine-Complete Transients* (batch-62 manuscript 11), builds a fixed rational continuous piecewise-linear $1/2$ -contraction of $\mathbb{R}^4$ whose exact half-space reachability is Σ_1^0 -complete while point-to-point reachability is decidable. Source 12, *Exact Diophantine Computation in Contracting 2-adic Dynamics* (batch-62 manuscript 08), builds a $1/2$ -contraction of $\mathbb{Z}_2$ whose exact extinction is Σ_1^0 -complete while every finite-quotient observation is decidable. Source 15, *Precision Is Memory* (batch-62 manuscript 12), gives unique-witness certificates for exact rational saturated-linear networks and shows that rational synthesis for a class of such networks is equivalent to Hilbert’s tenth problem over $\mathbb{Q}$. All three carry Part III’s contrast between exact and robust observation into regimes that Part III does not treat: global contractions, a non-Archimedean state space, and exact rational precision. Sources 14 and 15 were written from the pin 526c2557f, the parent of the commit that completed Parts I–IV, and source 12 from b6bf6406a; none cites Parts I–IV.

Sources 14 and 12 reach one phenomenon by independent routes, and both are printed in full; their comparison, term by term, is Section 134. Source 14’s Corollary 116.4 re-derives, for its own map, Part III’s Theorem 30.1; it is cited there. Source 14’s explicit ReLU gate count (Proposition 117.1) and source 15’s trace quadratic (Theorem 137.1) bear on Part III’s Question 26 without answering it: neither extracts the network of Corollary 31.2.

110.1 Letters used differently

One table covers the three sources and Part III, since sources 14 and 12 are compared symbol by symbol in Section 134.

Letter	Part III	Source 14	Source 12	Source 15
F	the fixed PWA map of $[0, 1]^3$	the contraction of $\mathbb{R}^4$	the radial map F_h of $\mathbb{Z}_2$	—
ρ	critical noise radius	(ReLU; renamed, below)	—	—
H	target set	the homogeneous map, $F = B^{-1}H \circ R$	halting time; the union of halting shells	height (reduced common denominator)
B	rejecting set; envelope $B(n)$	radial divisor, $B \geq 8$	—	common weight denominator
K	the cube $[0, 1]^d$; K_0	$K = 2n$; $\mathcal{K}$ a halting set	drift $K = r + 3$	—
D	denominator-clearing integer	$D = BK$	—	scales $D_t = D_0 B^t$
h	$h_N = 1/(2N)$	a candidate value function	the clock $h(t)$	—
E	edge set E_N	encoding E	encoding $E(q, L, R)$	—
G	grid G_N	ReLU gate count	the unit map G	—
L	Lipschitz bound	Lipschitz bound	left tape	lower excesses L_t
S, T	a cut; time	time T	input-bit bound S ; time T	horizon T
σ	—	—	—	the clip $\max(0, \min(1, u))$

The tempting false readings are these. Part III's **Hit** and **Safe** are *noisy* reachability and robust avoidance; source 14's half-space predicates are *exact* and are written **Hit**^{ex} and **Safe**^{ex} here. A zero of source 14's peak function $\mathcal{P}$ and a zero of Part III's radius ρ are different events: the first is exact non-arrival at an open half-space, the second is noise-fragility. Source 12's H is a time or a set of 2-adic integers, never a target box. Source 15's H is a height and its σ is the clip, not a ReLU.

110.2 Renamed symbols and macros

- Source 14 defines its ReLU macro as ρ and writes $\rho(a) = \max(a, 0)$. Since ρ is Part III's critical radius, the ReLU is printed $\text{ReLU}(a)$ throughout source 14. Its exact half-space predicates are printed **Hit**^{ex} and **Safe**^{ex} (source: **Hit**, **Safe**).
- Source 12's set $\{0, 1\}$ and indicator macros are replaced by this report's, with identical output. Its equation (131.4) is printed with R_{eq} : the source lost a backslash there and typeset R_{meq} (its later use has R_{eq}).
- Source 15's network macro $\mathcal{N}$ is renamed internally; its unused norm macro is dropped. Nothing printed changes.
- File names printed inside the sources' text are their delivered names; the shipped names carry the prefixes **14-strict-contractions-**, **12-twoadic-contraction-** and **15-neural-precision-**, mapped in the README.

110.3 Abstracts and status statements of the sources

Abstract of source 14. An effectively known attracting fixed point does not make exact finite-time observations decidable. We give a complete compiler from deterministic two-counter programs to rational, continuous, positively homogeneous piecewise-linear maps $F : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ satisfying $\|F(x) - F(y)\|_\infty \leq \frac{1}{2} \|x - y\|_\infty$. Every orbit converges to zero, yet for one fixed compiled map, reaching one fixed rational open half-space from a rational initial point is computably enumerable complete. In contrast, point-to-point reachability for every map produced by the compiler is decidable.

The construction yields a uniformly computable Lipschitz peak-value function that maps every rational point of a compact rational polytope to a rational number. Its zero test on rational points is nevertheless undecidable. Every strictly positive rational threshold is decidable, and the function is locally piecewise affine on its positive locus. The distinction is therefore not between computable and noncomputable real values, but between effective approximation and exact zero recognition.

We also compile the integer-scaled dynamics into a zero-bias integer ReLU circuit. A horizon- T first-hit trace has a unique natural-number certificate for one polynomial of degree at most four, with explicitly counted variables. For a three-state example, the supplied implementation exports an actual 2,105-term quartic with 400 witnesses. Unbounded reachability is connected to the ProveIt MRDP interface, with the fixed-horizon uniqueness claim kept separate from any unbounded finite-fold assertion. All new mathematical claims are proved in the text; exact-arithmetic tests and a reproducible compiler are included. Historical priority for the combined results is not asserted. Source 14’s status statement is its Section 111, subsection “Status and relationship to prior work”.

Abstract of source 12. We construct, from every deterministic binary Turing machine, a computable map $F: \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ that contracts all distances by at least a factor of $1/2$, has the unique fixed point 0, and simulates the machine through the odd part of each nonzero iterate. For a fixed universal machine, exact finite-time extinction on natural-number inputs is recursively enumerable complete, although every orbit enters the ball $2^k \mathbb{Z}_2$ within an explicit linear bound in k .

The construction is realized by a finite synchronous binary transducer, not merely an infinite-state Lipschitz machine. More generally, its radial clock may be replaced by any computable $h(t) \geq t + K$ without changing the extinction set or extinction times. Within this family, finite-state realizability is equivalent to eventual periodicity of $h(t) - t$. Arbitrarily fast prescribed convergence and infinite-order asymptotic flatness at zero are compatible with the same undecidable exact predicate.

A complementary rigidity theorem proves that a scalar p -adic contraction analytic near its fixed point has a uniform finite extinction horizon. For rational polynomials we give an explicit horizon and a finite bound on the extinction basin. Finally, we connect the construction to ProveIt’s MRDP interface and give an executable, clock-independent bounded-run compiler: its sum-of-squares quartic has a unique natural witness, with explicit linear counts of variables and equations. The accompanying implementation passes 121,822 deterministic finite checks. The unbounded representation is an MRDP consequence; the bounded unique-witness construction does not establish a fixed-arity single-fold MRDP theorem. The results below have mathematical proofs, but have not been Lean-verified or independently priority-verified.

Repository basis of source 12. **Repository basis.** ProveIt, inspected at revision b6bf6406a4c49017fedc6fba068b36c634987150. The repository’s MRDP theorem is inherited; the dynamical construction, clock classification, rigidity arguments, and bounded compiler are developed here. This is a research draft for mathematical review, not a claim of a settled literature-priority breakthrough.

Abstract of source 15. A bounded real state space can support universal computation when exact rational precision is an unbounded resource. We develop explicit Diophantine interfaces for this phenomenon in saturated-linear neural networks. A single natural-number quadratic equation represents a prescribed T -step run of an n -neuron rational network with exactly $4nT$ witnesses, all uniquely determined, including at the saturation thresholds. A projective variant represents bounded-time reachability from rational polyhedral input regions; real feasibility and rational feasibility coincide, with an explicit denominator bound.

The corresponding rational parameter-synthesis problem behaves differently. We prove that synthesis for a specified class of four-layer clipped networks, with bounded rational tied weights and one strictly positive output, is computably equivalent to Hilbert’s tenth problem over $\mathbb{Q}$. An explicit natural-number quartic certificate realizes that equivalence. In contrast, wholly strict output requirements give a decidable rational synthesis problem. Finally, we prove prime-support-sensitive bounds relating first acceptance time to denominator height, and show that a universal fixed rational network has no computable precision ceiling valid for all accepting inputs. Complete proofs, exact arithmetic implementations, examples, and a further

research agenda are included.

Status statement of source 15. **Status.** This is a research draft with proofs and executable checks, not a refereed or Lean-certified result. Priority for the specific refinements has not been established. Neural universality and complementarity representations are credited to earlier work. The synthesis equivalence does not decide $H10(\mathbb{Q})$.

111 The result and the question it answers

[source 14]

A common reason to expect a dynamical system to be tractable is that it is a strict contraction. There is a unique fixed point, convergence is effective, and initial errors do not amplify. These facts do indeed solve many approximation problems. They do not, by themselves, decide whether an orbit crosses a particular boundary at some finite time.

The distinction can be expressed as a precise question:

Does a rational, globally contractive, finite piecewise-linear description, together with an explicit exponential convergence rate, suffice to decide exact half-space reachability from rational initial data?

The answer is negative. Moreover, the same examples have a positive answer for exact point-to-point reachability. The observation being requested is as important as the rate of convergence.

Theorem 111.1 (Combined result). *There exist a fixed rational continuous piecewise-linear map $F : \mathbb{R}^4 \rightarrow \mathbb{R}^4$, an integer $B \geq 8$, a compact rational polytope*

$$\mathcal{C} = \{(z_0, z_1, z_2, r) : 0 \leq z_i \leq r \leq 1\},$$

and an integer linear functional ℓ with the following properties.

- (i) *F is positively homogeneous and globally $1/2$ -Lipschitz in the sup norm. It fixes zero and maps $\mathcal{C}$ into itself. On the unbounded cone $\mathcal{C}_\infty = \{0 \leq z_i \leq r\}$, the last coordinate is divided by B at every step.*
- (ii) *On rational initial points, the question $\exists t \in \mathbb{N} \ell(F^t(y)) > 0$ is Σ_1^0 -complete. The same statement holds for the closed half-space $\ell \geq 0$.*
- (iii) *Given arbitrary $y, z \in \mathbb{Q}^4$, whether $F^t(y) = z$ for some $t \in \mathbb{N}$ is decidable.*
- (iv) *The function*

$$\mathcal{P}(y) = \sup_{t \geq 0} \max\{0, \ell(F^t(y))\}, \quad y \in \mathcal{C},$$

is uniformly computable and Lipschitz, and $\mathcal{P}(\mathcal{C} \cap \mathbb{Q}^4) \subseteq \mathbb{Q}$. Its rational-input zero set is Π_1^0 -complete. Nevertheless, comparison with any strictly positive rational threshold, using either $>$ or $\geq$, is decidable on rational inputs.

- (v) *The restricted dynamical system $F|_{\mathcal{C}}$ has zero topological entropy and exactly one invariant Borel probability measure, namely the point mass at zero.*
- (vi) *Bounded first-hit trajectories have canonical natural-number quartic certificates. Unbounded half-space reachability has a fixed-arity Diophantine representation, and hence one of degree at most four after introducing arithmetic-circuit variables.*

The proofs are constructive at the level of a supplied counter program. The fixed universal instance uses the classical existence of a universal halting semidecider expressed as a two-counter program. We do not print a numerically optimized universal instruction table, nor the coefficients of the unbounded MRDP polynomial. We do print every formula needed to compile any instruction table, and supply an executable bounded polynomial compiler. These are different levels of explicitness.

111.1 What is substantive here

The useful strengthening is the conjunction of restrictions: four ambient coordinates; one globally continuous rational piecewise-linear map; a strict global Lipschitz bound; one attracting fixed point with an exact radial clock; a single linear observation; decidable exact point reachability; and Diophantine-complete region reachability. The peak-function theorem adds an exact-arithmetic consequence that is not merely the statement that halting can be translated into a polynomial.

The proof does not require chaotic dynamics, an undecidable limit, an uncomputable coefficient, an oracle, a discontinuous state update, or an infinite instruction table. It does require unbounded precision in exact observations and initial encodings. Computation survives in coordinate ratios while every absolute coordinate contracts.

111.2 Status and relationship to prior work

Moore established foundational examples of computation and undecidability in dynamical systems [44]. Koiran, Cosnard, and Garzon developed low-dimensional dynamical computation [56]. Those results make it inappropriate to present a generic piecewise-affine simulation of a counter machine as a new discovery. The present report instead supplies the contractive lift, the point/region separation, and the peak-value and certificate consequences as a single explicit development.

The distinction between point and region queries is also important when comparing with Varonka and Watanabe’s work on Bellman operators. Their reachability question asks whether an orbit reaches an exact rational target point; they give positive results, including the two-dimensional Bellman case [57]. Our map is not claimed to be a Bellman operator of a Markov decision process, and our undecidability result concerns a half-space, not a point. Thus these conclusions do not conflict.

Zero entropy alone is not a novelty claim either. Bruera, Cardona, Miranda, Peralta-Salas, and Salo explicitly construct universal Turing machines with zero topological entropy, while proving positive-entropy criteria under additional hypotheses [58]. Our zero-entropy proof is elementary because the ambient map is a contraction. Our decoding uses z/r and does not extend continuously to the cone apex; it is not a compact conjugacy to a full symbolic universal machine.

This report proves its construction without relying on informal research manuscripts in ProveIt. Its external mathematical inputs are classical counter-machine universality and MRDP. The repository connection is the actual finite-polynomial MRDP interface inspected at the pinned revision specified below. The new development has not been checked by Lean or Rocq. The finite Python checks are reproducibility evidence, not substitutes for the proofs or for the two external theorems.

112 Conventions and the repository boundary

[source 14]

All natural numbers include zero. Integer polynomials are evaluated over $\mathbb{Z}$, but their existential witnesses range over $\mathbb{N}$. A rational piecewise-affine map is described by finitely many rational polyhedral pieces and rational affine formulas agreeing on overlaps. The adjective *continuous* is essential. A piecewise-linear map in this report has linear, zero-constant formulas on conic pieces. Norms and Lipschitz constants are in the sup norm unless another norm is displayed.

A rational vector is supplied by finite integer numerator/positive-denominator codes. Such codes can be canonicalized effectively. Completeness means computable many-one completeness of the corresponding sets of codes. No polynomial-time reduction is claimed. Iterates include $F^0 = \text{id}$, so an initially accepting configuration has first-hit time zero.

112.1 The precise MRDP interface used

At repository revision

526c2557f2c6173054794f873a9624ebef616510,

`Diophantine.mrdp` states that for every computably enumerable $S \subseteq \mathbb{N}$, there are $k \in \mathbb{N}$ and $P \in \mathbb{Z}[X, W_0, \dots, W_{k-1}]$ such that

$$e \in S \iff \exists w \in \mathbb{N}^k P(e, w) = 0. \quad (112.1)$$

The source also exposes `Diophantine.mrdp_iff` and `Diophantine.mrdp_dioph_iff`. The guide explicitly distinguishes this existence theorem from a practical polynomial-generating algorithm or a numerical degree/witness bound [60, 61].

This is exactly the interface needed for the unbounded result. The fixed-horizon compiler below is independent of MRDP and actually exports its polynomial. Accordingly, a successful bounded certificate run must not be reported as a computation of the repository's unbounded universal polynomial. Likewise, the repository's documented formal status is not a claim that its entire development was rebuilt in the environment used for this report.

112.2 A source machine with exactly the needed instructions

A deterministic two-counter program has states $0, \dots, n$, with $n \geq 1$. State n is the unique halt state, made absorbing. Each other state has one of the following instructions:

$$\begin{aligned} \text{Inc}(i, k) : & \quad c_i \leftarrow c_i + 1, \quad j \leftarrow k; \\ \text{Dec}(i, k_0, k_+) : & \quad \begin{cases} j \leftarrow k_0, & c_i = 0, \\ c_i \leftarrow c_i - 1, \quad j \leftarrow k_+, & c_i > 0; \end{cases} \\ \text{Jump}(k) : & \quad j \leftarrow k. \end{aligned}$$

Here $i \in \{1, 2\}$. The optional jump instruction only simplifies example programs; universality does not depend on it.

Minsky's Theorem Ia uses exactly increment and conditional decrement instructions and represents a partial recursive function with initial counter values $(2^e, 0)$ [55, Theorem Ia, p. 449]. Apply it to the partial function that returns zero exactly on a fixed computably enumerable complete set $\mathcal{K}$. After relabelling states and making the halt state absorbing, we obtain one fixed program U with

$$e \in \mathcal{K} \iff U \text{ started at } (0, 2^e, 0) \text{ reaches its halt state.} \quad (112.2)$$

This precise input convention avoids assuming that an arbitrary two-counter program directly reads an ordinary binary Turing-machine tape.

113 A general contractive lift

[source 14]

We first isolate the geometric mechanism from counter-machine syntax. For $d \geq 1$, write

$$\mathcal{C}_\infty^{(d)} = \{(z, r) \in \mathbb{R}^d \times \mathbb{R} : r \geq 0, 0 \leq z_i \leq r\}, \quad \mathcal{C}^{(d)} = \mathcal{C}_\infty^{(d)} \cap \{r \leq 1\}.$$

Theorem 113.1 (Homogeneous contractive lift). *Let $f : [0, 1]^d \rightarrow [0, 1]^d$ be a rational continuous piecewise-affine map. One can effectively construct an integer $B \geq 2$ and a rational continuous piecewise-linear map $F : \mathbb{R}^{d+1} \rightarrow \mathbb{R}^{d+1}$ that is positively homogeneous, globally $1/2$ -Lipschitz, and satisfies*

$$F^t(rx, r) = rB^{-t}(f^t(x), 1) \quad (x \in [0, 1]^d, r \geq 0, t \in \mathbb{N}). \quad (113.1)$$

On rational inputs, point-to-point reachability for this F is decidable. Every rational polyhedral observation of $f^t(x)$ is transferred to a homogeneous rational polyhedral observation of $F^t(x, 1)$.

Proof. Define on the cone

$$H(z, r) = \begin{cases} (rf(z/r), r), & r > 0, \\ 0, & r = 0. \end{cases} \quad (113.2)$$

The value at $r = 0$ is forced, since the cone then contains only the origin. If $f(x) = A_a x + b_a$ on a rational polyhedron, its homogeneous lift has the linear formula $(A_a z + b_a r, r)$ on the corresponding rational polyhedral cone. These formulas agree on overlaps. Since $0 \leq rf(z/r)_i \leq r$, continuity at the apex follows too.

For each piece, form the $(d+1)$ -dimensional linear matrix of this formula. Let $M \geq 1$ be the largest absolute row sum among all these matrices. It is a computable rational bound. A line segment in the convex cone can be subdivided into finitely many segments on which a single linear formula applies. Summing the bounds on the subsegments gives $\|H(u) - H(v)\|_\infty \leq M \|u - v\|_\infty$. Agreement on boundaries handles segments that lie in more than one piece.

Put $\text{ReLU}(a) = \max(a, 0)$ and define the retraction

$$R(z, r) = (\min(\text{ReLU}(z_1), \text{ReLU}(r)), \dots, \min(\text{ReLU}(z_d), \text{ReLU}(r)), \text{ReLU}(r)). \quad (113.3)$$

Both minimum and maximum are nonexpansive in the sup norm on their input pairs. Thus R is 1-Lipschitz, positively homogeneous, and the identity on $\mathcal{C}_\infty^{(d)}$. Choose an integer $B \geq 2M$ and set

$$F = B^{-1}H \circ R. \quad (113.4)$$

All asserted algebraic and metric properties follow. On the cone the last coordinate becomes r/B , and positive homogeneity proves (113.1) by induction.

For a rational inequality $a \cdot x \triangleleft b$, where $\triangleleft$ is any of $<, \leq, =, \geq, >$, use $a \cdot z \triangleleft br$. For $r > 0$ this preserves truth, and finite Boolean combinations are preserved too. The point-reachability algorithm is proved in [theorem 115.2](#), using only the radial identity just established, so it applies to this general construction. $\square$

Remark 113.2 (What the lift preserves). The lift preserves one-step dynamics in projective coordinates, not absolute state coordinates. In particular, $x \mapsto (x, 1)$ is not a conjugacy onto an invariant slice: its image immediately moves to a smaller radial slice. The projection $(z, r) \mapsto z/r$ is defined only for $r > 0$ and is singular at the common limit. This is why the construction is compatible with strict contraction and with zero ambient topological entropy.

The theorem is useful beyond the counter encoding below. Any explicitly verified continuous piecewise-affine simulator on a cube can be used as its input. The extra radial coordinate supplies both a shrinking scale and a clock that makes exact target-point queries unusually simple.

114 An explicit four-coordinate counter compiler

[source 14]

Let M have $m = n + 1$ states. We now give all formulas directly, rather than requiring a polyhedral decomposition to be generated first.

114.1 Encoding, control hats, and local instruction maps

For a configuration $q = (j, c_1, c_2)$, set

$$E(q) = \left(\frac{j}{n}, 2^{-c_1}, 2^{-c_2} \right), \quad \mathbf{E}(q) = (E(q), 1). \quad (114.1)$$

A zero counter is represented by 1, while every positive counter is represented by a number at most $1/2$. There is therefore room to interpolate continuously between the zero and nonzero tests.

For $(s, x_1, x_2) \in [0, 1]^3$, define

$$g_j(s) = \text{ReLU}(1 - |ns - j|), \quad 0 \leq j \leq n.$$

At a represented control state $s = j/n$, exactly one hat has value 1 and every other hat has value 0.

To every instruction at state j , assign a local map $\phi_j : [0, 1]^3 \rightarrow [0, 1]^3$. For an increment of counter i with destination k , its control component is k/n , counter i becomes $x_i/2$, and the other counter is unchanged. For a jump to k , only the control component changes. At the halt state, put $\phi_n = (1, x_1, x_2)$.

For a zero/decrement instruction on counter i , put

$$b_i = \text{ReLU}(2x_i - 1), \quad (\phi_j)_0 = \frac{k_+ + (k_0 - k_+)b_i}{n}, \quad (\phi_j)_i = \min(2x_i, 1), \quad (114.2)$$

and leave the other counter coordinate unchanged. The control component is between k_+/n and k_0/n because $0 \leq b_i \leq 1$.

Define the global cube map componentwise by

$$f_a(s, x_1, x_2) = \max_{0 \leq j \leq n} \min(g_j(s), (\phi_j)_a(s, x_1, x_2)), \quad a = 0, 1, 2. \quad (114.3)$$

Using a maximum here, rather than a sum, keeps every component in $[0, 1]$ even between encoded control slices.

Lemma 114.1 (Exact cube simulation). *The map f is rational continuous piecewise affine, takes the cube to itself, and satisfies $f(E(q)) = E(Mq)$ for every configuration q .*

Proof. The formulas use only affine rational operations, absolute value, minimum, and maximum, so they have the asserted regularity. All arguments of the outer minima are nonnegative and at most one, proving cube invariance. At $s = j/n$, the j th term of (114.3) equals ϕ_j and all other terms vanish. The increment rule is $2^{-c_i}/2 = 2^{-(c_i+1)}$. If $c_i = 0$, then $x_i = 1$, $b_i = 1$, and the zero branch is taken while the counter remains 1. If $c_i > 0$, then $x_i \leq 1/2$, $b_i = 0$, and $2x_i = 2^{-(c_i-1)}$. The jump and halt cases are immediate. This checks every instruction and both decrement branches. $\square$

114.2 Division-free homogeneous formulas

Write $Y = (z_0, z_1, z_2, r) \in \mathcal{C}_\infty$. The homogeneous hats are

$$\theta_j(Y) = \text{ReLU}(r - |nz_0 - jr|). \quad (114.4)$$

Replace each cube local map by its homogeneous counterpart $V_j(Y)$. The increment rule has $V_{j,0} = kr/n$, $V_{j,i} = z_i/2$, and the other counter unchanged. A jump has control kr/n and unchanged counters. The halt rule is $V_n = (r, z_1, z_2)$.

For a decrement instruction, define

$$b_i(Y) = \text{ReLU}(2z_i - r), \quad V_{j,0} = \frac{k_+r + (k_0 - k_+)b_i(Y)}{n}, \quad V_{j,i} = \min(2z_i, r), \quad (114.5)$$

again leaving the other counter unchanged. Set

$$H_a(Y) = \max_j \min(\theta_j(Y), V_{j,a}(Y)) \quad (a = 0, 1, 2), \quad H_3(Y) = r. \quad (114.6)$$

These formulas contain no division by the radial variable. On $r > 0$ they are precisely the homogeneous lift of (114.3); at $r = 0$ they vanish. On the cone, every first component lies between zero and r .

Lemma 114.2 (An explicit global bound on the cone). *Let*

$$L = \max(2n + 1, 4), \quad B = 2L. \quad (114.7)$$

Then H is L -Lipschitz on $\mathcal{C}_\infty$.

Proof. For the scalar expression $nz_0 - jr$, the absolute coefficient sum is $n + j$. Absolute value and ReLU preserve a scalar Lipschitz bound. Adding the outside r in (114.4) gives $\text{Lip}_\infty(\theta_j) \leq n + j + 1 \leq 2n + 1$.

Increment, jump, unchanged-counter, and halt components have bounds at most one. For a decrement instruction, $\text{Lip}_\infty(b_i) \leq 3$ and

$$\text{Lip}_\infty(V_{j,0}) \leq \frac{k_+}{n} + 3 \frac{|k_0 - k_+|}{n} \leq 4.$$

The decremented counter has bound at most two, because $\min(2z_i, r)$ is the minimum of two scalar functions with bounds two and one. Finally, the minimum or maximum of finitely many scalar functions with a common Lipschitz constant has that same bound. Apply this first to the inner minima, then to the maxima in (114.6). The radial component has bound one. $\square$

Use the retraction (113.3) in dimension three and define

$$F_M(Y) = \frac{1}{B} H(R(Y)). \quad (114.8)$$

This is a finite rational continuous piecewise-linear formula on all of $\mathbb{R}^4$, not merely on a set of legal encodings.

Theorem 114.3 (Quantitative counter simulation). *The map F_M is globally $1/2$ -Lipschitz, positively homogeneous, and satisfies*

$$F_M^t(rE(q)) = rB^{-t}E(M^t q) \quad (r \geq 0, t \in \mathbb{N}). \quad (114.9)$$

It preserves $\mathcal{C}_\infty$ and $\mathcal{C}$. On $\mathcal{C}_\infty$, a point of radius r has norm r , and its image has radius and norm r/B .

Proof. The retraction is 1-Lipschitz and lemma 114.2 gives the required bound $L/B = 1/2$. Positive homogeneity follows from the zero-bias min/max/ReLU formulas. On a legal encoding the retraction does nothing, so lemma 114.1 proves the one-step identity. Induction and homogeneity prove (114.9). The inequalities $0 \leq H_a(Y) \leq r$ and $H_3(Y) = r$ give the invariance and radial claims. $\square$

115 Undecidable regions and decidable points

[source 14]

Define the fixed integer linear observation

$$\ell_M(z_0, z_1, z_2, r) = 2nz_0 - (2n - 1)r, \quad A_M = \{Y : \ell_M(Y) > 0\}. \quad (115.1)$$

On a legal configuration at positive radius,

$$\ell_M(rE(j, c_1, c_2)) = (2j - (2n - 1))r. \quad (115.2)$$

It equals r at the halt state and is at most $-r$ at every other state. There is no ambiguity on the boundary for legal finite-time configurations.

Theorem 115.1 (Half-space reachability is complete). *For the fixed program U in (112.2), the map $F = F_U$ and half-space $A = A_U$ have Σ_1^0 -complete rational-input reachability. A specific computable family of rational inputs witnessing hardness is*

$$Y_e = E(0, 2^e, 0) = (0, 2^{-2^e}, 1, 1). \quad (115.3)$$

The analogous finite-time reachability question for $\ell_U \geq 0$ is also Σ_1^0 -complete.

Proof. By theorem 114.3 and (115.2),

$$e \in \mathcal{K} \iff \exists t \, U^t(0, 2^e, 0) \text{ is halting} \iff \exists t \, \ell_U(F^t(Y_e)) > 0.$$

The rational tuple (115.3) is computable from e , which proves hardness. Rational evaluation of a rational CPWL expression is effective. Iterating and testing the strict rational inequality semidecides reaching A from any rational initial point. This gives the upper bound.

Along the hard input family, the radius is always positive at finite time and (115.2) never vanishes. Thus $\ell \geq 0$ has the same answers on this family. Its upper bound follows from exact rational non-strict comparison. The fact that the closed half-space contains the limiting fixed point does not mean it is reached in finite time. $\square$

The denominator in (115.3) has exponentially many bits in e itself. This is acceptable for computable many-one completeness but precludes reading the proof as an efficient binary-input simulation theorem. A different source encoding might improve this overhead; none is assumed.

Theorem 115.2 (Point-to-point reachability). *For every map obtained by theorem 113.1, and in particular for every F_M , the relation*

$$\{(y, z) \in \mathbb{Q}^{d+1} \times \mathbb{Q}^{d+1} : \exists t \in \mathbb{N} F^t(y) = z\}$$

is decidable. From a nonzero cone point, a specified nonzero target point can occur at most once.

Proof. Check $y = z$ first to handle time zero. Evaluate $y_1 = F(y)$. Every point from time one onward lies in the cone, so a target outside the cone can now be rejected. If $y_1 = 0$, the only subsequent point is zero.

Otherwise let $r > 0$ be the radial coordinate of y_1 . The radius after another t steps is rB^{-t} . A target of radius zero cannot be reached. For a target of radius $s > 0$, success requires $s = rB^{-t}$. If $s > r$ reject. Otherwise repeatedly divide the rational r by B until it is at most s . This loop terminates because $s > 0$. Strict inequality rules out success; equality determines a unique candidate time. Evaluate that many iterates exactly and compare the resulting rational vector with the target.

All exceptional cases were decidable, and all loops terminate. Strict radial decrease also proves the uniqueness assertion. $\square$

Proposition 115.3 (Targets separated from the apex). *Suppose D has decidable rational membership and a supplied rational $\delta > 0$ such that $\|z\|_\infty \geq \delta$ for every $z \in D$. Reachability of D from a rational point under a compiled map is decidable.*

Proof. The global contraction bound gives $\|F^t(y)\|_\infty \leq 2^{-t} \|y\|_\infty$. Choose T effectively so that the right side is less than δ for $t \geq T$, and test the finitely many earlier points. On the cone one may replace 2^{-t} by B^{-t} . $\square$

The undecidable half-space reaches all the way to the apex. There is no positive separation constant to feed into proposition 115.3.

115.1 Asymptotic dynamics really are trivial

Proposition 115.4. *For a compiled map, zero is the only fixed point and the only periodic point. On $\mathcal{C}$, topological entropy is zero and the only invariant Borel probability measure is δ_0 .*

Proof. Since $F(0) = 0$, contraction gives $\|F^t(y)\|_\infty \leq 2^{-t} \|y\|_\infty$. A fixed or periodic point must therefore be zero. For the entropy claim, let $d_T(x, y) = \max_{0 \leq t < T} \|F^t(x) - F^t(y)\|_\infty$ be the Bowen metric. Contraction and the presence of $t = 0$ give $d_T(x, y) = \|x - y\|_\infty$. Thus the maximum size of an ε -separated subset of the compact polytope is bounded independently of T , and its exponential growth rate is zero.

If μ is invariant and h is continuous on $\mathcal{C}$, invariance yields $\int h d\mu = \int h \circ F^t d\mu$. Uniform convergence of F^t to zero on $\mathcal{C}$ makes the right side tend to $h(0)$. Therefore integration against μ agrees with integration against δ_0 for every continuous function, which identifies the measures. $\square$

116 A rational-valued computable function with an undecidable zero test

[source 14]

Set $g(Y) = \text{ReLU}(\ell_M(Y))$ and, on $\mathcal{C}$, define

$$\mathcal{P}(Y) = \sup_{t \geq 0} g(F_M^t(Y)), \quad \mathcal{P}_T(Y) = \max_{0 \leq t < T} g(F_M^t(Y)) \quad (T \geq 1). \quad (116.1)$$

On the cone, $0 \leq g(Y) \leq r$. This upper bound uses $\ell_M(Y) \leq 2nr - (2n - 1)r = r$, not the larger generic Lipschitz bound.

Theorem 116.1 (Effective approximation and regularity). *For $Y \in \mathcal{C}$ of radius r and every $T \geq 1$,*

$$\mathcal{P}_T(Y) \leq \mathcal{P}(Y) \leq \max\{\mathcal{P}_T(Y), rB^{-T}\}. \quad (116.2)$$

In particular, $0 \leq \mathcal{P} - \mathcal{P}_T \leq B^{-T}$ uniformly on $\mathcal{C}$. The function $\mathcal{P}$ is uniformly computable, is $(4n - 1)$ -Lipschitz, and takes rational values at every rational point of $\mathcal{C}$. On its positive locus it is locally rational piecewise affine.

Proof. For $t \geq T$, radial decay bounds $g(F^t(Y))$ by $rB^{-t} \leq rB^{-T}$. Taking the supremum proves (116.2). Each $\mathcal{P}_T$ is a finite rational continuous piecewise-affine expression, so it is computable; the explicit uniform error bound supplies a computable approximation algorithm for $\mathcal{P}$ on computable real inputs.

The absolute coefficient sum of ℓ_M is $4n - 1$. Consequently $g \circ F^t$ has Lipschitz constant at most $(4n - 1)2^{-t}$, and all these functions have the common bound $4n - 1$. Taking a supremum preserves this bound: apply the individual inequality to every t and then interchange x and y .

Now let Y be rational. If the peak is zero, it is rational. If it is positive, some rational iterate has a rational value $a = g(F^s(Y)) > 0$. Choose $T > s$ such that $rB^{-T} < a$. Then the tail cannot exceed the finite prefix maximum, and $\mathcal{P}(Y) = \mathcal{P}_T(Y) \in \mathbb{Q}$.

Finally let $\mathcal{P}(Y) > 0$, without assuming Y rational. Choose T so that $B^{-T} < \mathcal{P}(Y)/3$. The error bound gives $\mathcal{P}_T(Y) > 2\mathcal{P}(Y)/3$. By continuity, on a relative neighborhood of Y in $\mathcal{C}$, $\mathcal{P}_T$ remains greater than B^{-T} . There the tail is smaller than the prefix and $\mathcal{P} = \mathcal{P}_T$. This proves local rational piecewise affinity on the positive locus. $\square$

The rationality proof is existential in its zero branch. It does not give a total algorithm that recognizes when the peak is zero. On positive rational inputs it does give a terminating exact algorithm: search for one positive iterate, use it to bound the remaining tail, and take a finite maximum.

Theorem 116.2 (A computable peak with undecidable exact recognition). *For F_U and the inputs Y_e in (115.3), let $\tau(e)$ be the first halting time when it exists. Then*

$$\mathcal{P}(Y_e) = \begin{cases} B^{-\tau(e)}, & e \in \mathcal{K}, \\ 0, & e \notin \mathcal{K}. \end{cases} \quad (116.3)$$

On rational points of $\mathcal{C}$, positivity of $\mathcal{P}$ is Σ_1^0 -complete and its zero test is Π_1^0 -complete. No total algorithm returns an exact rational numerator/denominator pair for $\mathcal{P}(Y)$ for every rational $Y \in \mathcal{C}$.

Proof. Before the first halt, (115.2) is negative, so $g = 0$. At the halt and at every later time it equals the radius B^{-t} , because the halt state is absorbing in the simulated configuration. The largest positive value is therefore $B^{-\tau(e)}$. If no halt occurs all values are zero. This proves (116.3) and hardness.

For a rational point, a positive iterate semidecides positivity. Its complement is exactly the zero set because $\mathcal{P} \geq 0$, proving the upper bounds as well. A hypothetical total exact-rational output algorithm could be run on Y_e and its numerator compared with zero, deciding $\mathcal{K}$, a contradiction. $\square$

Proposition 116.3 (Every positive rational threshold is decidable). *Given rational $Y \in \mathcal{C}$ and rational $\varepsilon > 0$, each of $\mathcal{P}(Y) > \varepsilon$ and $\mathcal{P}(Y) \geq \varepsilon$ is decidable.*

Proof. Choose $T \geq 1$ such that $rB^{-T} < \varepsilon$. Every tail value is then strictly below the threshold. Both questions reduce to the corresponding comparison with the rational number $\mathcal{P}_T(Y)$. Equality at the threshold is therefore handled exactly, not by an approximation convention. $\square$

The restriction to rational initial data in this proposition matters. Exact comparison of a finite piecewise-affine expression at an arbitrary computable real input can itself require a nontrivial zero test.

Corollary 116.4 (No computable positive separation bound). *There is no total computable function $d : \mathbb{N} \rightarrow \mathbb{Q}_{>0}$ such that $e \in \mathcal{K}$ implies $\mathcal{P}(Y_e) \geq d(e)$.*

Proof. Such a function, followed by [proposition 116.3](#) with $\varepsilon = d(e)$, would decide $\mathcal{K}$. $\square$

[write] Cf. Theorem 30.1 of Part III, the same absence of a computable positive gap for the critical noise radius of Part III's map, where infinitely many violations are also proved.

More precisely, there is no total computable function $h : \mathbb{N} \rightarrow \mathbb{N}$ such that every positive peak in the hard family is at least $2^{-h(\text{length}(\text{code}(Y_e)))}$. Otherwise that expression would give the function forbidden by [corollary 116.4](#). In the formula $B^{-\tau}$, halting time has become signal amplitude rather than distance from the fixed point.

116.1 A value equation and its boundary condition

Proposition 116.5. *The peak is the unique continuous function $h : \mathcal{C} \rightarrow \mathbb{R}$ with $h(0) = 0$ satisfying*

$$h(Y) = \max\{g(Y), h(F(Y))\}. \quad (116.4)$$

Proof. Splitting the supremum at time zero verifies the equation for $\mathcal{P}$. Conversely, iterating (116.4) gives

$$h(Y) = \max\{\mathcal{P}_T(Y), h(F^T(Y))\}.$$

Since $F^T(Y) \rightarrow 0$ and h is continuous with value zero there, the second term tends to zero. The first tends to $\mathcal{P}(Y) \geq 0$, so $h = \mathcal{P}$. $\square$

Thus even a uniquely specified continuous value function with an effective geometric approximation scheme can have an undecidable zero test on rational arguments. The state transition is contractive; the max operator in (116.4) is not being asserted to be a strict contraction in the supremum norm on functions.

117 An integer ReLU compiler for the shrinking dynamics

[source 14]

The real orbit contracts, but its integer certificate need not become smaller. We exploit a deterministic denominator convention to avoid introducing arbitrary representations of rational numbers at every step.

Put

$$K = 2n, \quad D = BK, \quad S(Y) = KH(Y) \quad (Y \in \mathcal{C}_\infty). \quad (117.1)$$

On the cone, $S = DF_M$. We will express S as a finite integer-coefficient zero-bias ReLU circuit. A circuit here permits integer linear expressions in the four inputs and earlier ReLU outputs, followed by ReLU gates; the final outputs are integer linear expressions. No sign restriction is imposed on weights, but ReLU outputs are nonnegative.

117.1 Explicit integer formulas

First compute θ_j from (114.4) using integer linear forms and ReLU. Its scaled version is $\widehat{\theta}_j = K\theta_j$. For an increment instruction,

$$\widehat{V}_{j,0} = 2kr, \quad \widehat{V}_{j,i} = nz_i, \quad \widehat{V}_{j,a} = Kz_a \quad (a \neq 0, i).$$

For a jump, use control $2kr$ and unchanged scaled counters. For the halt, use (Kr, Kz_1, Kz_2) . For a decrement instruction,

$$\widehat{V}_{j,0} = 2k_+r + 2(k_0 - k_+) \operatorname{ReLU}(2z_i - r), \quad \widehat{V}_{j,i} = K \min(2z_i, r), \quad (117.2)$$

with the other scaled counter unchanged. Every coefficient is an integer. Then

$$S_a(Y) = \max_j \min(\widehat{\theta}_j(Y), \widehat{V}_{j,a}(Y)), \quad S_3(Y) = Kr. \quad (117.3)$$

The equalities use positive homogeneity of minimum and maximum under multiplication by the positive integer K .

The elementary identities

$$|a| = \operatorname{ReLU}(a) + \operatorname{ReLU}(-a), \quad \min(a, b) = a - \operatorname{ReLU}(a - b), \quad \max(a, b) = b + \operatorname{ReLU}(a - b) \quad (117.4)$$

turn these formulas into the required integer ReLU circuit.

Proposition 117.1 (Gate count). *Let d_M be the number of decrement instructions. The direct compiler in `verify_contraction.py` uses exactly*

$$G = 9m + 2d_M - 3 \leq 11m - 5 \quad (117.5)$$

ReLU gates for S . Its linear forms have bounded support and coefficients with $O(\log m)$ bits.

Proof. For each of the m control states, an absolute value uses two ReLU gates and the outside positive part of the hat uses one, giving $3m$ gates. Clipping each of three local components by its scaled hat uses $3m$ more. Taking the componentwise maximum of the m clipped vectors uses $3(m-1)$ gates. A decrement instruction additionally uses one gate for its zero-test bump and one for $\min(2z_i, r)$. This totals $9m + 2d_M - 3$. There is one halt state, so $d_M \leq m - 1$.

The stated construction retains previous ReLU outputs as wires rather than substituting their formulas recursively. Each local expression has bounded support. In the running maximum, writing the new result as $b + \operatorname{ReLU}(a - b)$ keeps the linear output support bounded as well. Coefficients are bounded by a polynomial in m , which proves the bit bound. $\square$

This is a circuit-size statement, not a claim that a full polyhedral partition into all activation regions has only linearly many cells. The partition can be much larger; the compiler does not enumerate it.

[write] This is an explicit gate count for this source's contraction, not for Part III's map F : the extraction of the rational ReLU network of Corollary 31.2 (Question 26) is not answered by it.

Theorem 117.2 (Canonical integer scaling). *Let $Y_0 = W_0/d_0 \in \mathcal{C}_\infty \cap \mathbb{Q}^4$ with $W_0 \in \mathbb{N}^4$ and $d_0 > 0$. Define the integer sequence $W_{t+1} = S(W_t)$. Then*

$$F_M^t(Y_0) = \frac{W_t}{d_0 D^t}, \quad (W_t)_3 = K^t(W_0)_3. \quad (117.6)$$

Every W_t is a nonnegative integer cone point, and

$$\ell_M(F_M^t(Y_0)) > 0 \iff \ell_M(W_t) > 0. \quad (117.7)$$

Proof. All integer circuit operations preserve integrality. Since $S = KH$ and H preserves the cone, every output remains a nonnegative cone point. The radial formula follows from $S_3 = Kr$. If $F^t(Y_0) = W_t/(d_0 D^t)$, positive homogeneity gives

$$F^{t+1}(Y_0) = \frac{F(W_t)}{d_0 D^t} = \frac{S(W_t)}{d_0 D^{t+1}},$$

which proves the orbit identity by induction. Finally, ℓ_M is linear and the denominator is strictly positive, so it does not affect the sign. $\square$

The denominator convention is intentionally not reduction to lowest terms at every step. It is one fixed deterministic representation of the orbit. Its benefit is that the certificate only needs integer circuit transitions; no exponential Diophantine formula for D^t is needed at a fixed horizon.

118 Canonical quartic first-hit certificates

[source 14]

The central local device is complementarity with natural variables.

Lemma 118.1 (A unique ReLU witness). *For every $v \in \mathbb{Z}$, the equations*

$$y - z = v, \quad yz = 0, \quad y, z \in \mathbb{N} \quad (118.1)$$

have exactly one solution: $(y, z) = (\text{ReLU}(v), \text{ReLU}(-v))$.

Proof. If $v > 0$, the equation $y - z = v$ implies $y > 0$, so $yz = 0$ forces $z = 0$ and $y = v$. If $v < 0$, the symmetric argument gives $y = 0, z = -v$. If $v = 0$, then $y = z$ and $y^2 = 0$, so both are zero. Each displayed solution satisfies the equations. $\square$

The nonnegative domain is indispensable. Over unrestricted integers, $y - z = v$ and $yz = 0$ also permit the wrong-sign branch. A real-field certificate would require explicit nonnegativity inequalities. Here it is built into Diophantine witness quantification.

118.1 The complete polynomial schema

Number the ReLU gates of S in topological order. At gate a of step t , let $v_{t,a}$ be its integer linear preactivation in W_t and previous positive gate outputs of the same step. Introduce natural variables $y_{t,a}, z_{t,a}$ and impose

$$E_{t,a} = y_{t,a} - z_{t,a} - v_{t,a} = 0, \quad J_{t,a} = y_{t,a} z_{t,a} = 0.$$

For each of the four outputs, impose the linear equation equating $W_{t+1,i}$ to its circuit output expression; denote this residual by $O_{t,i}$. These equations include the exact radial recurrence.

For a prescribed horizon $T \geq 0$, add natural slacks $a, b_0, \dots, b_{T-1}$ with

$$\ell_M(W_T) - 1 - a = 0, \quad \ell_M(W_t) + b_t = 0 \quad (0 \leq t < T). \quad (118.2)$$

They assert acceptance at time T and nonacceptance at every earlier time. The strict inequality at the final time is an integer inequality, hence $\ell_M(W_T) > 0$ is equivalent to $\ell_M(W_T) \geq 1$.

Define the following one integer polynomial, presented as a finite sum of squares rather than as a large expanded expression:

$$\begin{aligned} Q_{M,T} = & \sum_{t < T} \sum_{a < G} (E_{t,a}^2 + J_{t,a}^2) + \sum_{t < T} \sum_{i=0}^3 O_{t,i}^2 \\ & + (\ell_M(W_T) - 1 - a)^2 + \sum_{t < T} (\ell_M(W_t) + b_t)^2. \end{aligned} \quad (118.3)$$

Here W_0 is the free input tuple. The later W_t , the gate pairs, and the slacks form the witness tuple. All linear preactivations remain linear in these variables: gate formulas are not recursively substituted into them.

Theorem 118.2 (Unique-witness bounded representation). *Fix M, T and an integer cone input W_0 . The polynomial $Q_{M,T}$ has degree at most four and has a natural witness exactly when the orbit of W_0 under S first satisfies $\ell_M > 0$ at time T . When a witness exists, it is unique. Its number of coordinates, and the number of equations squared in (118.3), are both*

$$N_T = (2G + 4)T + (T + 1) = (2G + 5)T + 1. \quad (118.4)$$

The sum-of-squares arithmetic description has $O(mT)$ size and $O(mT \log m)$ coefficient-description length.

Proof. Every $E_{t,a}$ and $O_{t,i}$ is linear and every $J_{t,a}$ is quadratic. The slack equations are linear. Their squared sum therefore has degree at most four. Since it is evaluated over integers, it vanishes if and only if each squared residual is zero.

Given W_t , apply lemma 118.1 in topological order to each gate. Every positive gate output and negative-part witness is forced uniquely. The output equations then force exactly $W_{t+1} = S(W_t)$. Induction on t forces the entire trace. The last equations force $a = \ell_M(W_T) - 1$ and $b_t = -\ell_M(W_t)$; these are natural exactly for the stated first-hit condition. Thus soundness and uniqueness hold. Conversely, an actual first-hit trace supplies all these forced natural values and makes every residual zero, proving completeness.

There are $4T$ later-state variables, $2GT$ gate variables, and $T + 1$ slacks. The equation count is the same: $4T$ output equations, $2GT$ gate equations, and $T + 1$ slack equations. The circuit bound in proposition 117.1, with bounded-support linear forms, gives the size bounds. When $T = 0$, the schema reduces to one accepting slack and one squared linear residual, so the edge case is included. $\square$

Remark 118.3 (Other bounded specifications). Omitting the earlier nonacceptance slacks represents acceptance at exactly time T , with a unique witness. Prescribing an output tuple instead of acceptance represents a bounded endpoint graph. For legal machine encodings, absorbing halting makes acceptance by time T equivalent to acceptance at time T . That equivalence is not asserted for arbitrary interpolated cone points.

118.2 Why this does not prove an unbounded single-fold theorem

Although the first-hit time is unique when it exists, the number of variables in (118.3) grows with T . One cannot replace the family by a single fixed polynomial merely by writing “ $\exists T$ ” in front of it: the displayed formula would then have variable arity. Encoding a variable-length trace and eliminating its computational predicates requires an additional representation theorem.

MRDP supplies that representation, but its auxiliary witnesses need not be unique even when the simulated computation is deterministic. Our conclusion is therefore a canonical finite-horizon quartic, together with a separate fixed-arity unbounded representation. It is not a resolution of the single-fold or finite-fold Diophantine representation problem.

119 The unbounded Diophantine consequences

[source 14]

Fix an effective natural-number coding of rational cone points. Invalid codes can be rejected decidably; this convention does not affect any hardness assertion. For a fixed compiled program M , define

$$\text{Hit}_M^{\text{ex}}(e) \iff e \text{ codes } Y \in \mathcal{C} \cap \mathbb{Q}^4 \text{ and } \exists t \ell_M(F_M^t(Y)) > 0.$$

Theorem 119.1 (The MRDP connection). *For each fixed M , there are k and an integer polynomial P_M such that*

$$\text{Hit}_M^{\text{ex}}(e) \iff \exists w \in \mathbb{N}^k P_M(e, w) = 0. \quad (119.1)$$

One may choose a representation of degree at most four after adding finitely many natural witnesses. For U , the represented set is Σ_1^0 -complete. On the hard input family, the never-hit property has no existential Diophantine representation over $\mathbb{N}$.

Proof. Exact rational evaluation and enumeration of times semidecide Hit_M^{ex} . Apply (112.1), using the repository's finite-polynomial interface. Completeness for U was proved in theorem 115.1.

For the quartic degree reduction, write $P_M = P_+ - P_-$, where P_+ and P_- have nonnegative integer coefficients. Compute the two polynomials by finite addition/multiplication circuits on natural inputs, introducing one natural variable for each intermediate result. Addition equations are linear, multiplication equations quadratic, and the final equality is linear. The sum of their squares has degree at most four. This familiar circuit reduction adds unique intermediate values for each assignment of the original MRDP witnesses; it does not make those original witnesses unique.

If never hitting on the hard family had an existential integer-polynomial representation with natural witnesses, enumeration of witness tuples and exact polynomial evaluation would semidecide $\mathbb{N} \setminus \mathcal{K}$. Together with the existing semidecider for $\mathcal{K}$, dovetailing would decide $\mathcal{K}$, a contradiction. $\square$

The fixed quartic in this theorem is an existence result. The supplied 2,105-term polynomial is instead the explicit bounded example from section 121; the two polynomials must not be confused.

119.1 Exact values, positive values, and approximation graphs

The positive-value graph of $\mathcal{P}$ on rational cone points is computably enumerable: enumerate until one positive iterate is found, bound the remaining tail, and compute the exact rational maximum. Canonical rational output coding then gives a semidecider for the positive graph, so MRDP represents it as well.

The full exact-value graph, including output zero, is not computably enumerable for U . If it were, restricting the output code to the rational number zero would semidecide the complete co-enumerable zero set. This contrasts with the graph of any fixed finite-precision approximation algorithm, which is decidable on its integer codes and therefore Diophantine. Effective real approximation, exact rational output, and exact positivity are three different specifications even though every rational input has a rational peak value.

Corollary 119.2 (No complete enumerable safety-certificate system). *For F_U , there is no sound, complete, computably checkable system of finite certificates establishing never hitting A_U for all rational initial points in the hard family.*

Proof. Enumerating finite certificates and checking them would enumerate every safe hard input, contradicting the non-enumerability of $\mathbb{N} \setminus \mathcal{K}$. $\square$

This is not a criticism of proof assistants or invariant methods. They can verify particular safety arguments and be complete on restricted classes. It is a limit on any claimed complete automatic method for this whole family, even though convergence and the unique attractor are already known.

120 Regularity, precision, and the limits of the construction

[source 14]

120.1 Smoothness at the apex is a real restriction

Proposition 120.1 (Homogeneous differentiability rigidity). *If $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is positively homogeneous of degree one and Frechet differentiable at zero, then G is linear. If a polynomial map on all of $\mathbb{R}^d$ is globally Lipschitz, it is affine; if it also fixes zero, it is linear.*

Proof. Positive homogeneity gives $G(0) = 0$ and $G(tx) = tG(x)$ for every $t > 0$. Differentiability at zero implies

$$DG(0)x = \lim_{t \downarrow 0} \frac{G(tx) - G(0)}{t} = G(x),$$

so G agrees with a linear map everywhere.

For a globally Lipschitz polynomial map, every partial derivative is a bounded polynomial: its absolute value is bounded by the global Lipschitz constant using difference quotients along coordinate directions. A nonconstant polynomial is unbounded on some line, as seen from a nonzero highest-degree homogeneous part. Therefore every partial derivative is constant, and the map is affine. Its constant term vanishes when it fixes zero. $\square$

This proposition explains why one cannot simply demand that our homogeneous construction become C^1 at the origin while retaining its nonlinear projective simulator. It does not classify all smooth contractions on a compact set. Nor does it settle all half-space questions for linear maps; those are distinct problems about signs of linear recurrence sequences.

120.2 Absolute precision and relative information

At a halting encoded state of radius $r = B^{-t}$, the signal is $\ell = r$. The distance in the ambient sup norm to the observation hyperplane is

$$\text{dist}_\infty(F^t(Y_e), \{\ell = 0\}) = \frac{B^{-t}}{4n - 1}. \quad (120.1)$$

Indeed, the dual norm of the coefficient vector of ℓ is its ℓ^1 norm $4n - 1$, giving this exact distance to a hyperplane. Thus an absolute observation error of fixed positive size cannot guarantee correct classification of arbitrarily late halts. The margin is positive at every individual encoded step, but it has no uniform positive lower bound over all possible halting times.

For comparison, if errors of norm at most η are added to state updates, the global $1/2$ contraction gives the standard estimate

$$\|\tilde{Y}_t - Y_t\|_\infty \leq 2^{-t} \|\tilde{Y}_0 - Y_0\|_\infty + 2\eta(1 - 2^{-t}).$$

This controls absolute trajectory error. It does not ensure a correct sign readout when the relevant margin shrinks as in (120.1). Consequently the theorem is about exact computation in a finitely described mathematical substrate, not a proposal for reliable unbounded computation in finite-precision hardware.

120.3 What remains unresolved

The construction neither gives a minimal dimension nor an optimized universal instruction table. It does not convert the exact reduction into a polynomial-time reduction on ordinary binary inputs. It does not establish robustness to persistent noise, a smooth global universal contraction, or a fixed-arity unique-witness representation of unbounded halting. None of those conclusions is needed for [theorem 111.1](#).

121 A worked example and executable verification

[source 14]

Consider the three-state program

State	Instruction
0	Decrement counter 1 if positive and go to 1; if zero go to 2.
1	Increment counter 2 and go to 0.
2	Halt.

It transfers the entire first counter into the second. Starting from $(j, c_1, c_2) = (0, 3, 1)$, it first halts at time 7 with counters $(0, 4)$. Here

$$m = 3, \quad n = 2, \quad d_M = 1, \quad L = 5, \quad B = 10, \quad K = 4, \quad D = 40.$$

The initial rational point and its canonical integer presentation are

$$Y_0 = (0, 1/8, 1/2, 1) = \frac{(0, 1, 4, 8)}{8}.$$

The trajectory is most legible when its common radial factor is displayed separately:

t	j_t	$c_{1,t}$	$c_{2,t}$	$\ell(F^t(Y_0))$
0	0	3	1	-3
1	1	2	1	-10^{-1}
2	0	2	2	$-3 \cdot 10^{-2}$
3	1	1	2	-10^{-3}
4	0	1	3	$-3 \cdot 10^{-4}$
5	1	0	3	-10^{-5}
6	0	0	4	$-3 \cdot 10^{-6}$
7	2	0	4	10^{-7}

At each row, $F^t(Y_0) = 10^{-t}(j_t/2, 2^{-c_{1,t}}, 2^{-c_{2,t}}, 1)$. Therefore $\mathcal{P}(Y_0) = 10^{-7}$, while the orbit still converges to zero.

The integer circuit uses $G = 9 \cdot 3 + 2 - 3 = 26$ ReLU gates. At $T = 7$, (118.4) gives

$$N_7 = (2 \cdot 26 + 5) \cdot 7 + 1 = 400.$$

The exporter retains four free input variables and these 400 natural witnesses. It generates 400 residual equations and expands their squared sum into a polynomial with exactly **2,105 nonzero monomials**, of total degree four. Both the factored equation system and the expanded polynomial are included, together with the exact satisfying assignment. This is an actual polynomial artifact, not merely pseudocode claiming to generate one.

121.1 What was run

The standard-library Python program was executed with

```
python verify_contraction.py --self-test --out verification
```

All arithmetic uses integers or `fractions.Fraction`; random test inputs are rational and the seed is fixed. The completed checks are:

Check	Cases
All increment/decrement/jump destinations for program sizes 2–4, legal tested counters 0–4; compare both real-formula and integer-circuit updates	5,000
Random rational global Lipschitz, homogeneity, and cone-scaling checks	1,000
Exhaustive ReLU-complementarity solutions for preactivations $-10, \dots, 10$	21
Exact example trajectory and target-point checks	15
Increase each example witness coordinate by one; reject the mutated certificate	400
Reject wrong first-hit horizons in $0, \dots, 9$	9

The expanded polynomial was checked at the canonical assignment and on a selection of mutated assignments. A second program, `check_export.py`, independently read the exported JSON, reconstructed the entire symbolic sum of squares, and confirmed exact equality with the exported quartic. It also checked the natural assignment and all 400 single-coordinate mutations. Zero-time and integer-input edge cases were checked separately. The machine-readable results are in `verification/test_report.json` and `verification/independent_checker_report.json`.

These tests exercise every instruction type and the sensitive zero/positive counter distinction. They do not exhaust all rational points, all programs, or all witnesses; [lemma 114.2](#) and [theorems 114.3](#) and [118.2](#) provide the corresponding universal proofs. In particular, the finite test set cannot certify universality of a source program. The worked program is deliberately small and is not universal.

121.2 Artifact format and reuse

`integer_circuit.json` records each gate's sparse integer linear preactivation, with input wires first and earlier positive ReLU wires after them. `certificate_system.json` records the quadratic residuals. `expanded_quartic.json` records the sum-of-squares polynomial as pairs `[coefficient, variable_index_list]`. A repeated variable index is a power; an empty index list denotes the constant monomial. `certificate_assignment.json` gives the natural values of all variables, with the four free inputs first. The example trace is stored separately as exact rational strings.

A user-supplied instruction table can be compiled by passing its JSON path, initial counters, and claimed first-hit time. Invalid instructions and incorrect first-hit claims are rejected. The code uses no computer-algebra package, SMT solver, floating-point solver, or network service.

[write] In this report the two programs are shipped as `code/14-strict-contractions-verify_contraction.py` and `code/14-strict-contractions-check_export.py`, the Makefile as `code/14-strict-contractions-Makefile`, and the files of `verification/` as `data/14-strict-contractions-*`. The commands above are the source's, for its delivered layout; the README gives the rerun recipe on a copy.

122 A concrete ProveIt formalization plan

[source 14]

The best integration boundary is a proved compiler theorem, not an axiom asserting that the new map is universal. A small sequence of independent modules would make the work auditable.

Counter semantics and encoding. Define the finite instruction type, deterministic configuration step, and absorbing halt convention. Prove the two numerical identities for encoded increment and decrement, including the $c = 0$ versus $c > 0$ split. The desired endpoint is the one-step theorem of [lemma 114.1](#).

Cone geometry and metric bound. Define R, H, F_M over $\mathbb{R}^4$. Prove cone preservation, nonexpansiveness of R , the explicit bound L , and homogeneity. Derive the iterate formula by induction. These proofs use ordered-field inequalities and elementary Lipschitz calculus, independently of computability theory.

Circuit semantics and reflection. Use an acyclic integer linear/ReLU circuit datatype. Prove that the compiler computes $S = KH$ on the integer cone. Prove the natural complementarity lemma once, then a generic unique-witness theorem for all circuits. A sparse-polynomial checker can validate the exported example, but the exporter itself need not become trusted code if its output is checked.

Computability and MRDP. Formalize exact rational circuit evaluation and the halting semidecider. This proves the appropriate `REPred` fact. Only then apply `Diophantine.mrdp`. The source-level theorem

already supplies ordinary integer polynomials and finite natural witnesses; no new arithmetic representation axiom is needed [60].

Peak values and negative results. Prove the uniform error bound for the finite maxima, followed by the Lipschitz and rationality statements. A separate computability argument then turns source-machine completeness into the exact-zero impossibility result. This separation prevents a numerical convergence proof from being mistaken for a decision procedure for zero.

The package contains no new Lean file advertised as compiling. The proposed module decomposition is a development plan, while the mathematical proofs and Python artifacts are the present deliverables.

123 The problem, the contribution, and its limits

[source 12]

123.1 Why another computational substrate is not enough

The assertion that an effectively presented machine’s halting set is Diophantine follows from MRDP once recursive enumerability has been proved. Replacing a Turing machine by an unusual machine does not, by itself, add much mathematics. A useful extension should control additional structure: regularity, finite-state implementation, quantitative convergence, witness complexity, or the difference between exact and finite-resolution observation.

This article studies all five in one construction. Its central phenomenon is

uniformly predictable convergence does not imply decidable exact extinction.

The phrase *exact extinction* always means reaching the point 0 in finitely many iterations, not merely approaching it. The target $\{0\}$ is closed but not open in $\mathbb{Z}_2$. That topological distinction is essential, not a technicality to be removed later.

The specific questions answered here are these. Can a scalar strict contraction with a finite-state digit implementation encode universal halting? Yes, for exact extinction. Which radial changes of clock retain finite-state realizability? Exactly those with eventually periodic added delay. Does arbitrarily fast convergence prevent this phenomenon? No. Does scalar analyticity near the attracting fixed point prevent it? Yes, by a uniform finite-horizon theorem. These are precise structural questions posed and resolved within this article; we do not assert that they were previously recognized open problems under these exact formulations.

123.2 Relation to ProveIt and to existing literature

The inspected ProveIt source exposes `Diophantine.mrdp`, which represents every recursively enumerable subset of $\mathbb{N}$ by one integer polynomial with a fixed finite number of natural-number witnesses. Its converse and its treatment of input zero are also explicit. The guide correctly distinguishes this existence interface from a practical polynomial-generation algorithm or a numerical witness bound [40, 41]. We use that interface only after proving recursive enumerability of our extinction predicate. We have inspected the relevant source and documentation, but have not rerun the repository’s Lean build or transitive axiom audit.

Dynamical simulation of computation is classical; Moore’s work is an early landmark [44]. The established correspondence between p -adic digit transducers and Lipschitz maps is developed by Anashin, who also gives a separate finiteness criterion [42]. In particular, being 1-Lipschitz must not be confused with being realizable by a *finite* transducer. We prove finiteness directly by constructing a bounded buffer.

There is also an important existing obstruction. Delvenne, Kůrka, and Blondel study effective symbolic systems with clopen observations; their Corollary 20 makes equicontinuous systems decidable in that framework [43]. Our map is equicontinuous and is *not* universal in that robust observation sense. We prove an elementary

finite-quotient version of the obstruction and locate all the hardness in exact singleton observation. For the formalization context, Carneiro’s paper formalizes the Diophantineness of exponentiation, the Matiyasevich component of MRDP, rather than being cited here as a formalization of every part of MRDP [45].

123.3 Status of the claims

The construction and theorems in Sections 124–131 are proved below. Finite checks of an implementation supplement those proofs but do not replace them. A concrete nonuniversal demonstration machine is exported in the archive. The universal existence theorem uses any fixed ordinary binary universal machine as the input to the proved compiler; the archive does not claim to contain a minimized or independently verified universal transition table.

The following distinctions will remain in force throughout:

Claim	Status and qualification
Strictly contracting universal exact extinction	Proved by a uniform compiler from a binary Turing machine.
Finite-state clock classification	Proved for the entire radial-clock family constructed here.
Scalar analytic finite horizon	Proved; an effective coefficient formula is given for rational polynomials.
Fixed-arity unbounded polynomial	Inherited from MRDP and ordinary quadratization; not numerically expanded here.
Bounded unique-witness quartic	Explicit construction, implementation, and exact finite checks. Its arity grows with the bounds.
Lean certification and literature priority	Not established in this article. Proposed formalization tasks are separated from completed work.

124 Binary tapes inside the odd 2-adic integers

[source 12]

124.1 Conventions

Write $\mathbb{N} = \{0, 1, 2, \dots\}$. For $x \in \mathbb{Z}_2$, let $\nu_2(x)$ be its 2-adic valuation, with $\nu_2(0) = +\infty$, and write $|x|_2 = 2^{-\nu_2(x)}$. Digits are always read *least significant first*. A natural number is viewed as a binary stream with an eventually zero tail.

Fix a deterministic binary Turing machine M with states $Q = \{0, \dots, m-1\}$, $m \geq 2$, an initial state q_0 , and one halting state q_h . For each $q \neq q_h$ and $b \in \{0, 1\}$, its total transition table gives

$$\delta(q, b) = (q', w, D), \quad w \in \{0, 1\}, \quad D \in \{\mathbf{L}, \mathbf{R}, \mathbf{S}\}.$$

Here $\mathbf{S}$ means that the head stays in place; the blank tape symbol is 0. Allowing stay moves is inessential. Put

$$r = \max\{1, \lceil \log_2 m \rceil\}, \quad K = r + 3. \quad (124.1)$$

We do not seek the smallest possible K .

A configuration is a triple (q, L, R) . The low bit of L is the cell immediately left of the head; subsequent bits extend leftward. The low bit of R is the scanned cell; subsequent bits extend rightward. Thus natural L, R describe a finite-support tape. Allowing arbitrary $L, R \in \mathbb{Z}_2$ extends the same rules to infinite binary tapes and makes the eventual map globally defined on a compact space.

124.2 The exact one-step formulas

Set

$$a = L \bmod 2, \quad b = R \bmod 2, \quad A = (L - a)/2, \quad B = (R - b)/2. \quad (124.2)$$

The quantities A, B belong to $\mathbb{Z}_2$. For the selected rule $(q', w, D) = \delta(q, b)$, the next tapes are

$$(L', R') = \begin{cases} (2L + w, B), & D = R, \\ (A, a + 2w + 4B), & D = L, \\ (L, w + 2B), & D = S. \end{cases} \quad (124.3)$$

For example, a left move makes the previous left-neighbor bit a the new scanned bit. The written bit w is then immediately to its right, followed by the old right tail B . This gives $R' = a + 2w + 4B$, including both factors that can easily be lost in an informal encoding.

When discussing a *padded machine trace*, we define the step at q_h to be the identity. The dynamical map below instead sends a halting configuration to zero. Keeping those two conventions distinct accounts for the one-step offset in all extinction statements.

124.3 Interleaving and the unit encoding

For $L = \sum_{i \geq 0} \ell_i 2^i$ and $R = \sum_{i \geq 0} \rho_i 2^i$, define

$$J(L, R) = \sum_{i \geq 0} \ell_i 2^{2i} + \sum_{i \geq 0} \rho_i 2^{2i+1}. \quad (124.4)$$

This is a bijection $\mathbb{Z}_2^2 \rightarrow \mathbb{Z}_2$ with an explicit digitwise inverse. Its restriction is a bijection $\mathbb{N}^2 \rightarrow \mathbb{N}$. Define

$$E(q, L, R) = 1 + 2q + 2^{r+1} J(L, R), \quad 0 \leq q < 2^r. \quad (124.5)$$

Every odd 2-adic integer has a unique such encoding: digit 0 is the unit marker, digits $1, \dots, r$ encode the state, and the remaining digits alternate the two tapes. State labels $m, \dots, 2^r - 1$, when present, are invalid labels, not additional computational states.

Definition 124.1 (Unit computation map). For an odd unit $u = E(q, L, R)$, put

$$G(u) = \begin{cases} 0, & q = q_h \text{ or } q \geq m, \\ E(q', L', R'), & q < m, q \neq q_h, \end{cases}$$

where the second line uses (124.3). Consequently, every nonzero value of G is odd.

Lemma 124.2 (Bounded loss of precision). For odd $u, v \in \mathbb{Z}_2$,

$$\nu_2(G(u) - G(v)) \geq \nu_2(u - v) - (r + 2). \quad (124.6)$$

Equivalently, G is 2^{r+2} -Lipschitz on the unit group.

Proof. The case $u = v$ is immediate. Put $n = \nu_2(u - v)$. If $n \leq r + 2$, the right-hand side is nonpositive, while the difference of two 2-adic integers has nonnegative valuation.

Suppose $n \geq r + 3$. The state headers and both low tape bits agree. In particular, the two inputs choose the same transition; if they choose the halting or invalid branch, both outputs are zero. Otherwise, their output state headers agree. Write c_j for the interleaved input tape digit of index j . Away from the first two tape digits, a right move sends even output tape index j to input index $j - 2$, and odd output index j to input index $j + 2$. A left move reverses these offsets; a stay move has offset zero. The few written boundary digits are fixed by the common rule and the common low bits. Thus every output tape digit at global position less than $n - 2$ is determined by common input digits at positions less than n . The outputs agree modulo 2^{n-2} . Since $r \geq 1$, this is at least the precision claimed in (124.6). $\square$

Remark 124.3. The deliberately generous precision loss makes the proof uniform across headers, invalid labels, and all three head directions. It is not a claim of an optimal Lipschitz constant for a particular machine.

125 The radial contraction compiler

[source 12]

125.1 A general lemma

Theorem 125.1 (Radial contraction). *Let $G: \mathbb{Z}_2^\times \rightarrow \mathbb{Z}_2$ satisfy $\nu_2(G(u) - G(v)) \geq \nu_2(u - v) - \lambda$ for some integer $\lambda \geq 0$. Let $h: \mathbb{N} \rightarrow \mathbb{N}$ satisfy $h(t) \geq t + \lambda + 1$. Define*

$$F_h(0) = 0, \quad F_h(2^t u) = 2^{h(t)} G(u) \quad (u \text{ odd}). \quad (125.1)$$

Then

$$|F_h(x) - F_h(y)|_2 \leq \frac{1}{2} |x - y|_2 \quad (x, y \in \mathbb{Z}_2). \quad (125.2)$$

Moreover, $\nu_2(F_h(x)) \geq \nu_2(x) + \lambda + 1$ for nonzero x , with the usual convention when $F_h(x) = 0$.

Proof. For inputs $x = 2^t u$, $y = 2^t v$ in the same shell,

$$\begin{aligned} \nu_2(F_h(x) - F_h(y)) &= h(t) + \nu_2(G(u) - G(v)) \\ &\geq h(t) + \nu_2(u - v) - \lambda \\ &\geq \nu_2(x - y) + 1. \end{aligned}$$

For distinct shells, say $\nu_2(x) = t < s = \nu_2(y)$, one has $\nu_2(x - y) = t$. Both output valuations are at least $t + \lambda + 1$, because $h(t) \geq t + \lambda + 1$ and $h(s) \geq s + \lambda + 1$. Their difference therefore has valuation at least $t + 1$. The comparison with zero follows from the same bound on the valuation of a single output. This proves all cases without requiring that h be monotone. $\square$

Apply the theorem with the map in Definition 124.1, $\lambda = r + 2$, and any *admissible clock*

$$h(t) \geq t + K. \quad (125.3)$$

All the maps below are therefore strict contractions. If $F_h(x) = x$, (125.2) applied to $x, 0$ forces $x = 0$. Also

$$\nu_2(F_h^j(x)) \geq \nu_2(x) + jK, \quad (125.4)$$

so every orbit converges to zero uniformly in $x \in \mathbb{Z}_2$.

125.2 Effectivity on an infinite digit stream

A formula involving $\nu_2(x)$ can conceal an uncomputable zero test on an arbitrary 2-adic name. This construction does not require such a test.

Proposition 125.2 (Finite-precision evaluation). *If h is total computable, then F_h is a computable map of $\mathbb{Z}_2$. Its residue modulo 2^k is computable from the input residue modulo 2^k . It also restricts to a total computable map $\mathbb{N} \rightarrow \mathbb{N}$.*

Proof. If the input residue is zero, (125.4) shows that the output residue is zero. Otherwise its valuation $t < k$ is known. Compute $h(t)$. If $h(t) \geq k$, output zero again. In the remaining case, the required precision of $G(u)$ is $k - h(t)$. Lemma 124.2 requires at most

$$k - h(t) + (r + 2) \leq k - t - 1$$

unit digits, already available from the input residue. Higher input digits can be filled with zero without changing the answer. On a natural-number input all tapes have finite support, the unit update produces a natural number, and the shift by $h(t)$ is an ordinary finite integer operation. $\square$

A rapidly growing clock can make full integer evaluation expensive because its output has many bits. This proposition is an effectivity statement, not a polynomial-time bound for arbitrary clocks.

125.3 Simulation and exact extinction

Theorem 125.3 (Clock-independent simulation). *Let $c_j = (q_j, L_j, R_j)$ be the run of M from a valid configuration c_0 , and let H be its first halting time, possibly infinite. Start from $x_0 = 2^{t_0} E(c_0)$ and define $t_{j+1} = h(t_j)$. For every $j \leq H$,*

$$F_h^j(x_0) = 2^{t_j} E(c_j). \quad (125.5)$$

If $H < \infty$, the first zero iterate occurs at time $H + 1$. If $H = \infty$, no iterate is zero. These extinction times are independent of the choice of admissible clock.

Proof. At time zero the formula is the definition of the input. If the current state is nonhalting, the unit update is precisely one machine step and is odd. Hence its prefactor $2^{h(t_j)}$ is its exact power of two, proving the inductive step. At the first halting configuration, the next unit update is zero. No preceding update can be zero. If there is no halting configuration, all unit updates remain odd forever. The case $H = 0$ gives extinction at time one, as stated. $\square$

For $n = 0$ extinction is already true at time zero. For a nonzero input with an invalid state label, extinction occurs at time one. Thus the entire set

$$\text{Ext}_M = \{n \in \mathbb{N} : \exists j \geq 0 \ F_h^j(n) = 0\} \quad (125.6)$$

is independent of h , including malformed natural-number inputs.

Corollary 125.4 (Complete exact extinction). *There is a fixed finite-state binary-machine table U such that, for every total computable admissible clock, Ext_U is Σ_1^0 -complete under computable many-one reductions. Its complement is Π_1^0 -complete. The completeness already holds on computably encoded odd initial configurations with a blank left tape.*

[write] As Section 123 states, the map is *not* universal in the robust clopen-observation sense of Delvenne, Kůrka and Blondel; Corollary 127.2 below locates all the hardness in exact singleton observation.

Proof. Choose a fixed binary universal Turing machine U . A program and its input can be translated effectively to a finite binary input tape R_e for U . Map that instance to $E(q_0, 0, R_e)$. By Theorem 125.3, this natural number becomes zero exactly when the simulated program halts. Conversely, iterating the total computable natural-number function F_h and stopping upon zero semidecides extinction. This gives completeness and, by complementation, the stated classification of nonextinction. $\square$

The word “finite-state” in the corollary describes U ’s finite control. The stronger assertion that the *entire digit map* F_h can be a finite transducer is proved next, for the clocks that permit it.

126 A finite transducer and an exact clock classification

[source 12]

126.1 The constant-drift realization

A synchronous binary Mealy transducer has finitely many internal states. At each step it reads one input bit, emits one output bit, and changes its state. Its infinite-stream interpretation is a map $\mathbb{Z}_2 \rightarrow \mathbb{Z}_2$.

Theorem 126.1 (Finite-state implementation). *For the clock $h(t) = t + K$, the map F_h is realized by a finite synchronous binary Mealy transducer. A direct construction uses at most*

$$2^{2r+8} + 6 \cdot 2^{r+5} + 2 \quad (126.1)$$

states before minimization. Its table can be generated effectively from the finite transition table of M .

Proof. An initial *leading-zero state* emits zero on every zero input and stays in that state. It therefore handles an arbitrarily large shell valuation without storing its value. On the first input one, the unit part begins. Index that digit by relative position zero. The required output, from this point onward, is $2^K G(u)$.

There is a finite bootstrap of length

$$I = K + r + 5 = 2r + 8.$$

It stores the input prefix. At relative position $i < K$ it emits zero. At position $i \geq K$ it emits digit $i - K$ of $G(u)$, computed by completing the stored prefix with zeros. This is correct by Lemma 124.2: the $i + 1$ known unit digits determine at least $i + 1 - (r + 2)$ output digits of G , while the requested digit requires only $i - K + 1 = i - r - 2$ digits. If the decoded state is halting or invalid, the transducer enters a permanent zero-output state.

For a nonhalting branch, the bootstrap determines the selected head movement. After processing relative position $I - 1$, it keeps only that movement, a parity bit, and the last $K + 2$ input bits. It discards the remaining prefix. For the next and all subsequent positions $i \geq I$, the output tape index is

$$j = i - K - r - 1 \geq 4.$$

There are no boundary writes at these indices. The source interleaved tape index is $j + \Delta$, where

Movement	j even	j odd
R	$\Delta = -2$	$\Delta = 2$
L	$\Delta = 2$	$\Delta = -2$
S	$\Delta = 0$	$\Delta = 0$

Consequently, the source unit bit is at position $i - K + \Delta$. Append the current bit to the $K + 2$ stored previous bits. If the oldest bit in this block is indexed zero, the desired source is at block position $2 + \Delta$, one of 0, 2, 4. Emit it, discard the oldest bit, and toggle the parity. This is a finite transition rule.

There are fewer than 2^I bootstrap states: the initial unit bit is one, so there are at most 2^{i-1} prefixes of length i at a bootstrap stage. There are at most $3 \cdot 2 \cdot 2^{K+2}$ steady states and two special states. Substituting $I = 2r + 8$ and $K = r + 3$ gives (126.1). Breadth-first enumeration of reachable states yields an explicit table. $\square$

Remark 126.2 (Where the unbounded memory resides). The machine does not acquire an unbounded internal state. Computation uses repeated application of a fixed finite transducer to words whose stored lengths may grow. After each iteration the transducer starts again in its designated initial state. This is not a claim that a one-pass finite automaton recognizes arbitrary recursively enumerable languages. The unbounded tape is in the external digit stream.

On natural inputs, one constant-drift iteration changes binary length by at most a constant depending on M . The head moves at most one cell, and interleaving shifts tape positions by at most two. Thus a direct streaming implementation takes linear time in the current word length; repeated iteration still allows unbounded computation.

126.2 Exactly which clocks are finite-state?

Definition 126.3. A sequence $d: \mathbb{N} \rightarrow \mathbb{N}$ is *eventually periodic* if some $A \geq 0$ and $P \geq 1$ satisfy $d(t+P) = d(t)$ for every $t \geq A$.

Theorem 126.4 (Clock classification). *Fix the unit computation map G of a machine with at least one nonhalting state. For any admissible clock h , the map F_h is realizable by a finite synchronous binary transducer if and only if*

$$d(t) = h(t) - t \tag{126.2}$$

is eventually periodic. No computability assumption on h is needed for this equivalence; eventual periodicity supplies computability on the finite-state side.

Proof. Necessity. Choose one fixed natural odd unit u_* encoding a valid nonhalting state. Its image $G(u_*)$ is odd. Suppose a finite transducer realizes F_h , and let s_t be its state after reading t zeros from its initial state. The sequence s_t is eventually periodic because it is obtained by repeatedly applying one function to a finite set.

On the input $2^t u_*$, the first t input bits are zero. The emitted prefix is also zero: the first nonzero output bit is at position $h(t)$, which is greater than t . After this prefix the remaining input is the same fixed stream of digits of u_* for every t . Therefore, if $s_t = s_{t+P}$, the two remaining output streams are identical. Their first one occurs, relative to the suffix start, at positions $h(t) - t$ and $h(t+P) - (t+P)$. Both are finite because $G(u_*)$ is odd. Hence $d(t) = d(t+P)$ for all sufficiently large t .

Sufficiency. An eventually periodic d has a finite range. While reading the initial zeros, a finite controller records t up to its preperiod and then records its residue modulo a period. It outputs zeros. At the first one, it knows which of the finitely many values $D = d(t)$ to use. Theorem 126.1 gives the unit-output stream $2^K G(u)$ with a finite implementation. Since $D \geq K$, prepend $D - K$ additional zero output digits using a fixed finite delay buffer. This realizes $2^D G(u)$ on the unit suffix. Combining the finitely many delay choices with the leading-zero controller is a finite transducer and produces $2^{t+D} G(u) = F_h(2^t u)$. On the all-zero input it produces zero forever. $\square$

Corollary 126.5. *Unbounded added delay excludes finite-state realizability. In particular, the quadratic clock $h(t) = (t+1)^2 + K$ is not finite-state, although it defines a computable strict contraction with exactly the same extinction set as the finite-state constant-drift map.*

Bounded added delay is not sufficient: a bounded computable but noneventually-periodic delay also fails the classification. Thus the boundary is periodic finite control, not merely a bound on buffer size. The theorem is specific to this radial family, not a classification of all contracting transducers.

127 Finite observation is easy; exact extinction is hard

[source 12]

127.1 Finite quotients

A 1-Lipschitz map $f: \mathbb{Z}_2 \rightarrow \mathbb{Z}_2$ induces a well-defined map f_k on $\mathbb{Z}/2^k\mathbb{Z}$: congruent inputs give congruent outputs. For a computable f , the finite table of f_k is computable. Therefore, reachability of any specified residue subset can be decided by exploring a finite directed graph. This elementary fact already prevents interpreting our hardness theorem as finite-resolution universality.

Our contractions give a much stronger bound.

Proposition 127.1 (Uniform nilpotence of every finite quotient). *For an admissible clock and any $k \geq 1$, put $N_k = \lceil k/K \rceil$. Then*

$$F_h^{N_k}(x) \equiv 0 \pmod{2^k} \quad \text{for every } x \in \mathbb{Z}_2. \quad (127.1)$$

The quotient map modulo 2^k is nilpotent with exponent at most N_k .

Proof. Every $x \in \mathbb{Z}_2$ has nonnegative valuation, and each nonzero iterate gains at least K by (125.4). After N_k steps its valuation is at least k . Zero iterates cause no exception. $\square$

Corollary 127.2 (A bounded clopen reachability test). *Let C be any clopen subset of $\mathbb{Z}_2$, given as a union of residue classes modulo 2^k . For a computably given input, deciding whether some iterate lies in C requires only the residues at times $0, 1, \dots, N_k$. The same conclusion holds for existential reachability from a given clopen initial set after choosing a common residue precision for both sets.*

Proof. At and after time N_k the residue is always zero. If the zero class is in C , time N_k already witnesses reachability; if it is not, later times cannot help. For a clopen initial set, enumerate its finitely many allowed initial residues and apply the same argument. $\square$

A finite-state observer of a fixed residue precision also sees a constant symbol from time N_k onward. Its own state sequence thereafter is ultimately periodic. Reachability, repeated visitation, and the usual finite-state acceptance conditions reduce to finite computations. This gives a direct explanation, within the present family, of its compatibility with the equicontinuity obstruction discussed in Section 123.

127.2 The quantifier distinction

For every $x \in \mathbb{Z}_2$, the following statement holds:

$$\forall k \exists j \quad F_h^j(x) \equiv 0 \pmod{2^k}. \quad (127.2)$$

Indeed, $j = N_k$ is a uniform witness independent of x . Exact extinction is the different statement

$$\exists j \forall k \quad F_h^j(x) \equiv 0 \pmod{2^k}. \quad (127.3)$$

For the universal compiler, (127.3) is undecidable on natural inputs. The same iterate must work at every precision. Interchanging those quantifiers would erase the entire computational content.

On a known natural-number orbit, testing equality to zero is an exact finite integer operation. On an arbitrary digit-stream name, equality to zero is not a finite-prefix observation. These are different input and observation conventions, and the theorems do not conflate them.

Corollary 127.3 (No computable bound on positive extinction times). *For the universal constant-drift map, there is no total computable $B: \mathbb{N} \rightarrow \mathbb{N}$ such that every input which eventually becomes zero already becomes zero by time $B(n)$.*

Proof. Such a function would decide extinction by evaluating the orbit up to $B(n)$, contrary to Corollary 125.4. The map fixes zero, so checking only the last iterate would also suffice. $\square$

Thus even knowing a uniform convergence modulus for *all* inputs does not furnish an input-dependent computable upper bound for those inputs that reach the limit exactly.

128 Arbitrary convergence rates and flatness

[source 12]

128.1 Prescribing a convergence modulus

Theorem 128.1 (Fast clocks with unchanged exact computation). *Let $b: \mathbb{N} \rightarrow \mathbb{N}$ be any total computable nondecreasing function. There is a total computable admissible clock such that, for all $x \in \mathbb{Z}_2$ and $j \geq 1$,*

$$\nu_2(F_h^j(x)) \geq b(j). \quad (128.1)$$

The exact extinction set and the first extinction time of every natural input are unchanged from the constant-drift construction.

Proof. Choose

$$h(t) = \max\{t + K, b(t + 1)\}.$$

Along a nonzero orbit the valuations satisfy $t_{j+1} = h(t_j)$ and $t_j \geq t_0 + jK \geq j$. Hence, for $j \geq 1$, $t_j \geq b(t_{j-1} + 1) \geq b(j)$. If an orbit has already become zero, the inequality holds with infinite valuation. Clock independence is Theorem 125.3 together with the immediate treatment of invalid labels and zero. $\square$

This result controls the number of iterations to achieve a precision, not the bit complexity of one iteration. The latter can increase substantially under the clock change. It also does not say that an arbitrary prescribed clock remains finite-state; Theorem 126.4 states exactly when that is possible.

128.2 Infinite-order asymptotic flatness

Proposition 128.2 (A flat but computational contraction). *For the clock $h(t) = (t + 1)^2 + K$, and every integer $d \geq 1$,*

$$\lim_{x \rightarrow 0, x \neq 0} \frac{F_h(x)}{x^d} = 0 \quad \text{in } \mathbb{Q}_2. \quad (128.2)$$

In particular, F_h is differentiable at zero with derivative zero. For a universal machine its natural-input extinction set is still Σ_1^0 -complete.

Proof. For $x = 2^t u$ with u odd and $G(u) \neq 0$, the latter is odd, so

$$\nu_2\left(\frac{F_h(x)}{x^d}\right) = h(t) - dt = (t + 1)^2 + K - dt \longrightarrow +\infty.$$

If $G(u) = 0$, the quotient is zero. The estimate is uniform over units and proves the limit. The derivative statement is the case $d = 1$. Completeness follows from clock independence. $\square$

Remark 128.3 (What flatness does not say). Equation (128.2) is asymptotic, or Peano, flatness at one point. It is not a proof that the map is globally C^∞ , nor that iterated classical derivatives exist at all nearby points. Digit interleaving on nonzero shells introduces its own regularity questions. No such stronger smoothness claim is used below.

128.3 An exact differentiability classification

The clock also gives a simple necessary and sufficient condition for the ordinary 2-adic derivative at the attracting fixed point.

Theorem 128.4 (Differentiability versus finite control). *For the machine compiler, let $d(t) = h(t) - t$. Then F_h is differentiable at zero if and only if*

$$d(t) \longrightarrow +\infty \quad (t \longrightarrow \infty). \quad (128.3)$$

When this holds, its derivative is zero. Consequently, none of the finite-state maps in Theorem 126.4 is differentiable at zero. More generally, for each integer $a \geq 1$,

$$F_h(x) = o(x^a) \text{ as } x \rightarrow 0 \iff h(t) - at \longrightarrow +\infty. \quad (128.4)$$

In particular, flatness of every positive integral order is equivalent to $h(t)/t \rightarrow +\infty$ as $t \rightarrow \infty$.

Proof. Along the ray $x = 2^t u_h$ for a fixed halting unit u_h , the difference quotient $F_h(x)/x$ is identically zero. Thus, if a derivative exists, it must be zero. Along a fixed nonhalting unit u_n , one has

$$\nu_2\left(\frac{F_h(2^t u_n)}{2^t u_n}\right) = h(t) - t = d(t).$$

Convergence of this quotient to zero forces (128.3). Conversely, for every unit u , the quotient is either zero or has exactly that valuation, independently of u . Condition (128.3) therefore gives convergence uniformly over all units and proves the existence of the derivative. Eventually periodic d is bounded, proving the assertion about finite-state maps.

The identical argument with x^a in the denominator proves (128.4). If $h(t)/t \rightarrow \infty$, then $h(t) - at \rightarrow \infty$ for each fixed a . Conversely, applying (128.4) at every integer a shows that $h(t)/t$ eventually exceeds every fixed real bound: choose an integer a larger than that bound and divide $h(t) - at$ by t . $\square$

For example, constant or eventually periodic added delay gives finite control but no derivative at zero. The clock $h(t) = 2t + K$ gives derivative zero but fails $o(x^2)$, since $h(t) - 2t = K$. The quadratic clock is flat at every order. All three cases preserve exactly the same machine computation and natural-input extinction predicate. These classifications concern the present radial family; they do not rule out different finite-state or smooth encodings with other properties.

129 Scalar analytic rigidity and explicit finite horizons

[source 12]

The preceding examples are continuous contractions and may be arbitrarily flat at zero, yet they are not analytic there. This is a genuine boundary for scalar maps.

Theorem 129.1 (Analytic extinction horizon). *Let p be prime and let $f: \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ satisfy $f(0) = 0$ and*

$$|f(x) - f(y)|_p \leq p^{-\kappa} |x - y|_p \quad (x, y \in \mathbb{Z}_p) \quad (129.1)$$

for an integer $\kappa \geq 1$. Suppose f is analytic on a neighborhood of zero. Then there is a finite integer N such that, for all $x \in \mathbb{Z}_p$,

$$(\exists j \geq 0 \ f^j(x) = 0) \iff f^N(x) = 0. \quad (129.2)$$

More precisely, either f vanishes identically on a neighborhood of zero and every point becomes zero in uniformly bounded time, or a sufficiently small neighborhood contains no nonzero root of f and no orbit can first reach zero after its uniform entry time into that neighborhood.

Proof. Choose a ball on which f is represented by a convergent power series. If that series is zero, there is $B \geq 0$ such that f vanishes on $p^B \mathbb{Z}_p$. From (129.1), $\nu_p(f^j(x)) \geq j\kappa$ for every $x \in \mathbb{Z}_p$. Thus all points enter $p^B \mathbb{Z}_p$ by time $N_0 = \lceil B/\kappa \rceil$ and become zero by time $N_0 + 1$.

Otherwise, because $f(0) = 0$, its series has a first nonzero term of degree $m \geq 1$. On a sufficiently small ball, factor it as

$$f(z) = z^m (a_m + zH(z)), \quad a_m \neq 0,$$

where H is analytic and bounded on a smaller closed ball. Shrink that ball to $p^B \mathbb{Z}_p$ so that $|zH(z)|_p < |a_m|_p$ throughout it. The strong triangle inequality gives $|a_m + zH(z)|_p = |a_m|_p \neq 0$. Consequently, $f(z) = 0$ on this ball only when $z = 0$.

All orbits enter the ball by $N = \lceil B/\kappa \rceil$, and it is forward invariant by contraction. If $f^N(x) \neq 0$, then the no-nonzero-root property shows inductively that every later iterate is also nonzero. Conversely, if an earlier iterate was zero, all subsequent iterates, including the N th, are zero. This proves (129.2). $\square$

129.1 An effective polynomial version

Analytic existence of a small ball is not automatically an algorithm for finding it from an arbitrary coefficient oracle. A finite polynomial presentation removes that issue.

Corollary 129.2 (A coefficient-computable cutoff). *Suppose a nonzero polynomial*

$$f(z) = \sum_{j=m}^d a_j z^j \in \mathbb{Q}[z], \quad a_m \neq 0, \quad m \geq 1,$$

maps $\mathbb{Z}_p$ to itself and satisfies (129.1). Set

$$B = 1 + \max \left(\{0\} \cup \left\{ \left\lfloor \frac{\nu_p(a_m) - \nu_p(a_j)}{j - m} \right\rfloor : j > m, a_j \neq 0 \right\} \right), \quad N = \lceil B/\kappa \rceil. \quad (129.3)$$

Then (129.2) holds. Exact extinction for rational inputs in $\mathbb{Z}_p$, and hence for natural-number inputs, is decidable by N exact iterations. The extinction basin in $\mathbb{Z}_p$ has at most d^N elements.

Proof. For $z \neq 0$ with $\nu_p(z) \geq B$ and every $j > m$ with $a_j \neq 0$,

$$\nu_p(a_j z^j) - \nu_p(a_m z^m) = \nu_p(a_j) - \nu_p(a_m) + (j - m)\nu_p(z) > 0.$$

The lowest-degree nonzero term therefore has strictly smaller valuation than every other term and cannot cancel. There is no nonzero root in $p^B \mathbb{Z}_p$, so the second case of Theorem 129.1 applies. Rational arithmetic and a zero test are effective. Finally, the basin is the zero set of the nonzero polynomial f^N (composition, not pointwise power), whose degree is d^N . A nonzero polynomial over $\mathbb{Q}_p$ has at most its degree many roots. $\square$

The zero polynomial has horizon one. Formula (129.3) is a safe bound, not generally the least one. It assumes that the self-map and contraction hypotheses have been established; it is not itself a general algorithm for verifying those hypotheses.

For example, for $p = 2$ the polynomial

$$f(z) = 2z(z - 1)$$

is a $1/2$ -contraction on $\mathbb{Z}_2$, since $f(x) - f(y) = 2(x - y)(x + y - 1)$. The formula gives $B = N = 1$. Indeed, the only points whose orbits ever become zero are 0 and 1. After one step every orbit is even; the even ball contains no nonzero root of f . This example exhibits a nontrivial extinction basin and an exact finite horizon without any computability pathology.

129.2 Why the constructed maps cannot be analytic

Proposition 129.3. *For every admissible clock, the map F_h associated to a machine with a halting state and a nonhalting state is not analytic near zero.*

Proof. Choose odd natural units u_h and u_n encoding a halting and a nonhalting state, respectively. For all $t \geq 0$,

$$F_h(2^t u_h) = 0, \quad F_h(2^t u_n) \neq 0.$$

Both input sequences approach zero. A nonzero analytic germ has an isolated zero there by the factorization in Theorem 129.1; an identically zero germ vanishes on a whole neighborhood. The two displayed sequences rule out both possibilities. $\square$

Remark 129.4 (Dimension matters). The proof uses the scalar factorization $f(z) = z^m$ times a local unit. In several variables an analytic zero fiber need not have an isolated nonzero-free neighborhood. The theorem is not a multivariate analytic undecidability obstruction. That extension is a separate research question, not an omitted case of the proof.

130 The Diophantine representation and what MRDP contributes

[source 12]

Theorem 130.1 (Clock-invariant Diophantine extinction). *For a fixed machine M , there are an integer $v \geq 0$ and an integer polynomial $P_M(n, w_1, \dots, w_v)$ such that, for every natural input n and every total computable admissible clock,*

$$\exists j \ F_h^j(n) = 0 \iff \exists w \in \mathbb{N}^v \ P_M(n, w) = 0. \quad (130.1)$$

The same polynomial can be used for every such clock. It can be chosen of degree at most four after adding finitely many witnesses. For universal M , the complement of the represented set is not Diophantine over $\mathbb{N}$.

Proof. Use the constant-drift map. Its extinction set is recursively enumerable by exact natural-number iteration. Apply the MRDP theorem in the form provided by ProveIt [40, 41]. The entire set is clock-independent, so the resulting representation serves every clock.

For the degree reduction, write the resulting integer polynomial as $P^+ - P^-$ with P^+, P^- having nonnegative integer coefficients. Evaluate both with finite circuits using natural constants, addition, and multiplication. Introduce one natural variable for every gate. Each gate is constrained by an equation of degree at most two, and the two final gate values are constrained to agree. The sum of the squares of all these residuals is a single polynomial of degree at most four. Its natural zeros are exactly the original zeros with their circuit evaluations. This adds only a fixed finite number of variables.

If the universal extinction complement were Diophantine, enumerating its finite witness tuples would semidecide it. Since extinction itself is semidecidable, dovetailing the two procedures would decide extinction, contradicting Corollary 125.4. $\square$

The quartic conclusion is standard circuit quadratization, not a newly improved universal degree bound. The original MRDP witnesses can be nonunique. Making circuit values unique *conditional on those witnesses* does not make the entire representation single-fold.

There is also a practical distinction. The proof above identifies the correct semidecidable predicate and the repository theorem to apply. It does not print all coefficients of a universal unbounded polynomial or supply a numerical value for v . The explicit compiler in the next section addresses a different, quantitatively controlled task: a bounded trace with a unique witness.

131 A clock-independent unique-witness quartic compiler

[source 12]

131.1 The bounded input convention

Fix M , a time bound $T \geq 0$, and an input-bit bound $S \geq 0$. These are *compiler parameters*. The public polynomial input is one natural number n . A valid initial input has the form

$$n = E(q_0, 0, R), \quad 0 \leq R < 2^S. \quad (131.1)$$

Writing $R = \sum_{i=0}^{S-1} d_i 2^i$ with $d_i \in \{0, 1\}$ gives

$$n = 1 + 2q_0 + 2^{r+2} \sum_{i=0}^{S-1} d_i 4^i. \quad (131.2)$$

The factor 2^{r+2} , rather than 2^{r+1} , reflects the odd positions of the right-tape bits in the interleaving.

Theorem 131.1 (Explicit bounded certificate). *There is an effectively constructed integer polynomial $C_{M,T,S}(n, w)$ of degree at most four, with*

$$V = (3m + 6)T + 3S + m + 4 \quad (131.3)$$

$$R_{\text{eq}} = (4m + 7)T + 3S + 2m + 7 \quad (131.4)$$

natural witness variables and quadratic-or-linear squared residuals, respectively, such that the following are equivalent:

- (i) n is a valid input (131.1) and M halts on that input within T machine steps;
- (ii) there exists $w \in \mathbb{N}^V$ with $C_{M,T,S}(n, w) = 0$.

For each fixed n , there is at most one such witness. On valid initial inputs, these conditions are also equivalent to $F_h^{T+1}(n) = 0$ for every admissible clock. The residual coefficients are bounded in magnitude by a constant depending only on M , not on T, S . The sparse description size is $O_M(T + S + 1)$, and the accepting witness coordinates have bit length $O_M(T + S + 1)$.

We give the full construction and proof rather than treating bounded trace encoding as a black box.

131.2 Input bits and bounded-coefficient packing

Introduce the S bits d_i , and two Horner sequences $x_0, \dots, x_S$ and $y_0, \dots, y_S$. Use the residuals

$$d_i(d_i - 1) = 0 \quad (0 \leq i < S), \quad (131.5)$$

$$x_0 = y_0 = 0,$$

$$x_{i+1} - 2x_i - d_{S-1-i} = 0 \quad (0 \leq i < S), \quad (131.6)$$

$$y_{i+1} - 4y_i - d_{S-1-i} = 0 \quad (0 \leq i < S), \quad (131.7)$$

$$n - 1 - 2q_0 - 2^{r+2}y_S = 0. \quad (131.8)$$

Here $x_S = R$ and $y_S = \sum_i d_i 4^i$. Horner evaluation avoids inserting coefficients such as 4^{S-1} directly into the polynomial system. The only power of two appearing in (131.8) depends on the fixed machine.

131.3 Configurations and one-hot control

Introduce L_t, R_t for $0 \leq t \leq T$, and state bits $s_{t,q}$ for $0 \leq t \leq T$, $q \in Q$. The constraints are

$$L_0 = 0, \quad R_0 - x_S = 0, \quad (131.9)$$

$$s_{t,q}(s_{t,q} - 1) = 0, \quad \sum_{q \in Q} s_{t,q} - 1 = 0, \quad (131.10)$$

$$s_{0,q} - \mathbf{1}_{\{q=q_0\}} = 0. \quad (131.11)$$

All variables range over $\mathbb{N}$. Thus (131.10) says that precisely one state is selected at each time.

For every $t < T$, introduce a_t, b_t, A_t, B_t and impose

$$L_t - 2A_t - a_t = 0, \quad R_t - 2B_t - b_t = 0, \quad (131.12)$$

$$a_t(a_t - 1) = 0, \quad b_t(b_t - 1) = 0. \quad (131.13)$$

These enforce the unique quotient and remainder upon division by two. There is no hidden natural-number truncated subtraction in the equations: all residuals are ordinary integer polynomials evaluated on natural variables.

131.4 Selectors and simultaneous transition equations

For each $t < T$, $q \in Q$, and $c \in \{0, 1\}$, introduce a selector $e_{t,q,c}$ and impose

$$e_{t,q,0} - s_{t,q}(1 - b_t) = 0, \quad (131.14)$$

$$e_{t,q,1} - s_{t,q}b_t = 0. \quad (131.15)$$

They force exactly one selector to be one and the others to be zero. Additional Boolean equations for the selectors are unnecessary.

For a nonhalting table entry (q, c) , let $\tau_{q,c}$ be its target state, and let $(U_{q,c}, W_{q,c})$ be the affine pair of tape expressions in (124.3), evaluated on the time- t variables. For $q = q_h$ set

$$\tau_{q_h,c} = q_h, \quad U_{q_h,c} = L_t, \quad W_{q_h,c} = R_t.$$

This is the identity padding of a halting trace. Impose

$$s_{t+1,p} - \sum_{\tau_{q,c}=p} e_{t,q,c} = 0 \quad (p \in Q), \quad (131.16)$$

$$L_{t+1} - \sum_{q,c} e_{t,q,c} U_{q,c} = 0, \quad (131.17)$$

$$R_{t+1} - \sum_{q,c} e_{t,q,c} W_{q,c} = 0. \quad (131.18)$$

Every expression is of degree at most two. The potentially signed term $1 - b_t$ is used inside an integer polynomial and is zero or one on every solution. Finally impose

$$s_{T,q_h} - 1 = 0. \quad (131.19)$$

Let $\mathcal{R}$ be the ordered list of all the residuals just specified. The promised polynomial is the literal sum of squares

$$C_{M,T,S}(n, w) = \sum_{R \in \mathcal{R}} R(n, w)^2. \quad (131.20)$$

The use of R as a residual in this display is local; it is not a tape variable.

131.5 Soundness, completeness, and uniqueness

Proof of Theorem 131.1: semantic assertions. A zero of (131.20) over natural witnesses makes every integer residual zero, because a sum of nonnegative integer squares vanishes only when each square does. The bit and Horner equations force the input packing (131.2), with $0 \leq x_S < 2^S$. The initial tape and state equations therefore specify precisely the claimed initial configuration.

At each time, the one-hot equations specify one state. Division equations and Boolean remainders specify the scanned bit and both tape tails. Equations (131.14)–(131.15) select precisely the applicable transition. Equations (131.16)–(131.18) are then exactly that transition, including identity padding at halt. By induction, the trace is the deterministic padded run of M . Equation (131.19) implies that this run has reached its halt state by T . This is soundness.

Conversely, if the machine halts within T , fill the variables with the input digits, the two Horner computations, and the actual padded run. Use the ordinary quotients and remainders for A_t, B_t, a_t, b_t , and the selected transition indicators for $e_{t,q,c}$. Every residual vanishes. This proves completeness.

For uniqueness, the public value n fixes y_S by (131.8), and a base-four expansion of fixed length S with digits in $\{0, 1\}$ is unique, including leading zero digits. Thus it fixes all d_i , then all x_i, y_i , and then the initial configuration. The quotient-remainder equations have unique natural solutions. The selectors are fixed by the current state and scanned bit. The next configuration is fixed by the update equations. Induction fixes the entire witness tuple. Identity padding introduces no choice even if the machine halts early.

The equivalence with $F_h^{T+1}(n) = 0$ on valid initial inputs follows from the precise time offset in Theorem 125.3. The formulation includes $T = 0$ and $S = 0$: an empty input-bit list encodes $R = 0$, and a zero-step trace accepts exactly when the initial state is halting. $\square$

131.6 Exact counts and complexity

Witness family	Number
Input bits d_i	S
Horner accumulators x_i, y_i	$2(S + 1)$
Tape values L_t, R_t	$2(T + 1)$
One-hot states $s_{t,q}$	$m(T + 1)$
Tape splits a_t, b_t, A_t, B_t	$4T$
Transition selectors $e_{t,q,c}$	$2mT$
Total	$(3m + 6)T + 3S + m + 4$

Residual family	Number
Input Boolean, Horner, packing, initial tapes	$3S + 5$
All state Boolean and one-hot equations	$(m + 1)(T + 1)$
Initial state equations	m
Tape divisions and remainder Boolean equations	$4T$
Selectors	$2mT$
State and tape updates	$(m + 2)T$
Final halt	1
Total	$(4m + 7)T + 3S + 2m + 7$

Each residual has a number of terms bounded in terms of m alone. Its coefficients are bounded in terms of r, m and the fixed machine table. Squaring and summing gives degree at most four and $O_M(T + S + 1)$ sparse terms before collection, hence no more after collection. The theorem claims bounded coefficients for the residual representation; it does not need an additional bound on collected coefficients of an arbitrarily chosen expanded presentation.

During T steps, each tape's binary length increases by at most T from its initial maximum, up to a fixed additive constant. Tape and quotient values therefore have $O(S + T + 1)$ bits. The base-four Horner values have at most $2S$ bits, and control variables are Boolean. This proves the stated coordinate bit bound. Since there are $O_M(T + S + 1)$ coordinates, a full witness has $O_M((T + S + 1)^2)$ bits under a straightforward explicit encoding. Linear coordinate count must not be misreported as linear total bit size.

131.7 Why this is not a single-fold MRDP solution

For each *fixed pair* T, S , the polynomial has at most one accepting witness. But increasing either bound changes the number of variables and the polynomial itself. The theorem is a family of bounded representations, not one fixed-arity representation of unbounded halting. Replacing the bounds by free variables would require a new compression argument, whose auxiliary witnesses might cease to be unique. No such argument is assumed or proved here.

The clock independence is nevertheless useful: even an extremely fast radial clock can create huge dynamical integers while the certificate stores only the normalized machine trace. The certificate does not need to store the clock valuations or the full radial factors at every time.

132 A worked example and reproducible exact checks

[source 12]

132.1 The demonstration machine

The archive contains a three-state *nonuniversal* machine. Its initial state is 0, its halt state is 2, and its nonhalting rules are

State	Read	Next state	Write	Move
0	0	1	1	L
0	1	0	0	R
1	0	2	1	S
1	1	2	0	L

Here $r = 2$, $K = 5$. Start with $L = 0$, $R = 7$, whose three low bits are one. Then

$$J(0, 7) = 2 + 8 + 32 = 42, \quad E(0, 0, 7) = 1 + 8 \cdot 42 = 337.$$

The machine erases the run of three ones, moves left after encountering a blank, and halts in five machine steps. Under the constant-drift map the exact orbit is

Iteration j	$F^j(337)$	$\nu_2(F^j(337))$
0	337	0
1	2592	5
2	17408	10
3	32768	15
4	70254592	20
5	2852126720	25
6	0	$+\infty$

The rapidly increasing ordinary integer sizes are compatible with rapidly decreasing 2-adic norms. The last nonzero iterate encodes the halting state; the next is zero.

Breadth-first generation gives 1,795 reachable states for the demonstration transducer, below the generic bound 4,866 from (126.1). For $T = 5$, $S = 3$, the compiled certificate has

$$V = 91, \quad R_{\text{eq}} = 117.$$

Its expanded quartic has 818 nonzero monomials. The archive contains both the residual presentation and the expanded sparse polynomial, together with the exact accepting witness for $n = 337$.

132.2 The implementation

The implementation uses only Python's standard library and exact integers. `code/padic_machine.py` defines the tape machine, encoding, full-value and residue evaluation, and the finite Mealy compiler. `code/quartic_certificate.py` constructs sparse residual polynomials and the unique deterministic candidate witness. `code/verify.py` runs the tests and regenerates the example files. The JSON outputs describe exact polynomial expressions and transition tables; they are not fitted numerical approximations.

From the archive directory, run

```
python3 code/verify.py
```

The deterministic seed is 20260930. The suite tests thirteen machine tables: the demonstration table and three generated tables at each of $m = 2, 3, 4, 5$. It exercises all head directions, unused header labels, zero input, residues at several precisions, and time/input bounds beginning at zero. A full integer evaluator has a configurable output-bit safety limit; residue evaluation avoids materializing an unnecessarily large radial shift.

Check family	Passed instances
Interleaving round trips	4,096
Configuration round trips	16,384
Finite transducer versus full-value evaluation	13,312
Strict-contraction congruences	43,264
Valuation drift	6,643
Residue iteration consistency	17,472
Finite-quotient nilpotence	9,984
Certificate shape and count formulas	48
Certificate truth on bounded inputs	504
Single-coordinate witness perturbations rejected	9,917
Public-input perturbations rejected	101
Nonlinear-clock simulation comparisons	96
Expanded quartic equality check	1
Total	121,822

A perturbation test is not an exhaustive proof of uniqueness, and a finite congruence test is not a proof on all of $\mathbb{Z}_2$. Their roles are to detect implementation errors and check the correspondence with the mathematical construction. Uniqueness is proved by the deterministic reconstruction in Section 131; contraction and finiteness are proved independently in Sections 125 and 126.

[write] In this report the programs are shipped as `code/12-twoadic-contraction-padic_machine.py`, `code/12-twoadic-contraction-quartic_certificate.py` and `code/12-twoadic-contraction-verify.py`, and the files of `examples/` and the verification report as `data/12-twoadic-contraction-*`. The command above is the source's, for its delivered layout; the README gives the rerun recipe on a copy.

133 A concrete formalization route for ProveIt

[source 12]

The useful integration target is not a fresh proof of MRDP. It is a verified chain from a finite machine table to the predicate consumed by the existing theorem. The following modular route separates finite arithmetic from 2-adic analysis.

Binary tape arithmetic. Define the finite transition table and the padded natural configuration step. Prove the three equations (124.3) against an elementary list-of-bits tape semantics. Include explicit theorems for halt padding, finite support, and the one-step extinction offset.

Encoding and residues. Define interleaving first on naturals, prove its inverse and the odd-unit header decomposition, then prove compatibility of the residue maps. Construct the 2-adic map from compatible finite quotients, or use the existing 2-adic digit API ([write] that is, Mathlib's `PadicInt`; ProveIt has no 2-adic development of its own) after checking its exact conventions. This avoids introducing a noncomputable global zero test into an executable component.

Radial clock and simulation. Formalize Theorem 125.1 separately from the machine encoding. Its proof has only the equal-valuation and unequal-valuation cases. Next prove the simulation invariant with an explicit clock sequence and the equivalence of extinction predicates for all admissible clocks.

Finite transducer correctness. Use an inductive type for leading, bootstrap, steady, and zero-output states. Prove the streaming invariant for the bounded buffer. The state bound is a finite counting theorem; it

is logically separate from the streaming correctness theorem. The necessity half of Theorem 126.4 needs only eventual periodicity of an orbit of a function on a finite set.

The bounded quartic. Represent residuals as multivariate integer polynomials. A finite sum-of-squares lemma reduces zeros to the residual conjunction. Soundness and completeness can follow the time-index induction already given. Uniqueness should be a separate theorem on witness tuples, not an informal consequence of a Boolean simulator test.

The unbounded MRDP endpoint. Establish that natural-number extinction is an **REPred**; the existing `Diophantine.mrdp` then yields the fixed finite polynomial. A clean theorem should quantify the polynomial before the natural input and before any admissible-clock choice. The clock-independence lemma is what makes this order possible.

Analytic rigidity. The proof needs local power-series factorization, a bound for the tail, and uniform entry into a small ball. The polynomial corollary can be formalized earlier using finite valuation comparisons, without waiting for the general analytic infrastructure. It is a good independently useful target.

These are proposed modules and proof obligations. No accompanying Lean file is represented as having been built, and the Python tests do not certify a Lean theorem. The inspected public MRDP interface is the only repository theorem invoked as an inherited endpoint.

134 [write] Two routes to one phenomenon: sources 14 and 12

Sources 14 and 12 were written independently, from different pins, and neither cites the other. Each builds a globally strict contraction with an effectively known attractor and shows that an exact transient observation is Σ_1^0 -complete while every observation separated from the attractor is decidable. Neither is a special case of the other, and they do not contradict each other. The table compares them term by term.

Item	Source 14 (real piecewise-linear)	Source 12 (2-adic)
Space and metric	$\mathbb{R}^4$ with the sup norm; invariant compact polytope $\mathcal{C}$	$\mathbb{Z}_2$ with the 2-adic metric
Contraction	globally 1/2-Lipschitz; on the cone the radius is divided by $B \geq 8$	$ F(x) - F(y) _2 \leq \frac{1}{2} x - y _2$; valuation drift at least $K = r + 3$ per step
Attractor	unique fixed point 0; zero entropy; unique invariant measure δ_0 (Proposition 115.4)	unique fixed point 0 (no entropy or measure theorem; its Question 80 asks about the basin)
Source machine	two-counter (Minsky) program, counters as 2^{-c}	binary Turing machine, tapes interleaved in the odd unit part
Where computation lives	projective ratio z/r ; the radius r is the clock (Theorem 113.1)	odd unit part u of $x = 2^t u$; the valuation t is the clock (Theorem 125.1)
Hard predicate	reaching an open half-space $\ell > 0$ that touches the apex (Theorem 115.1)	exact extinction, reaching the fixed point itself (Corollary 125.4)
Point reachability	decidable (Theorem 115.2); 0 is never reached from a nonzero cone point	reaching 0 is the hard predicate
Decidable observations	targets at positive distance from the apex (Proposition 115.3); every positive peak threshold (Proposition 116.3)	every clopen, finite-quotient observation (Proposition 127.1, Corollary 127.2)
No computable bound	no computable positive lower bound on peaks (Corollary 116.4)	no computable bound on extinction times (Corollary 127.3)
Halting time becomes	the amplitude $B^{-\tau}$ of the first positive signal	the valuation of the last nonzero iterate
Clock freedom	one fixed clock (division by B)	any admissible $h(t) \geq t + K$; finite-state exactly when $h(t) - t$ is eventually periodic (Theorem 126.4)
Finite realization	integer zero-bias ReLU circuit with $9m + 2d_M - 3$ gates (Proposition 117.1)	finite synchronous Mealy transducer (Theorem 126.1)
Regularity at the attractor	a positively homogeneous map differentiable at 0 is linear, so the construction cannot be C^1 at the apex (Proposition 120.1)	differentiable at 0 exactly when $h(t) - t \rightarrow \infty$, and can be flat to all orders (Theorem 128.4, Proposition 128.2)
Analytic obstruction	a globally Lipschitz polynomial map is affine; linear half-space problems are not settled	a scalar p -adic contraction analytic near 0 has a uniform extinction horizon (Theorem 129.1)
Bounded certificate	traces the integer-scaled dynamics: $(2G + 5)T + 1$ witnesses and equations (Theorem 118.2)	traces the source machine: $(3m + 6)T + 3S + m + 4$ witnesses (Theorem 131.1)
Worked example	three-state counter program, $T = 7$: 400 witnesses, 2,105 monomials	three-state machine, $T = 5$, $S = 3$: 91 witnesses, 117 residuals, 818 monomials
Unbounded	MRDP (Theorem 119.1)	MRDP, clock-invariant (Theorem 130.1)

The apparent opposition. In source 12 reaching the fixed point is hard; in source 14 point reachability is decidable. The two statements do not conflict. Source 12's map collapses whole halting shells onto 0, so it is not injective, and extinction is an exact singleton observation. Source 14's map is positively homogeneous and never reaches 0 from a nonzero cone point, so its hardness must sit in an open region; that region, the half-space $\ell > 0$, reaches all the way to the apex. The common principle is that a hard target must accumulate at the attractor: $\{0\}$ is closed but not open in $\mathbb{Z}_2$, and the halting shells' union is open with 0 in its closure; source 14's hard half-space touches the apex. Observations separated from the attractor are decidable in both.

The regularity mirror. The regularity results are mirror images. Source 12 attains differentiability, and even flatness to all orders, at the fixed point, while scalar analyticity forces a uniform extinction horizon. Source 14 proves that its homogeneous route cannot be differentiable at the apex at all. Source 12’s analytic horizon theorem is thus the 2-adic counterpart of source 14’s Question 62 and of Part III’s Question 25, which it partly answers in the scalar 2-adic case.

Exact versus robust, and Part III. Part III’s map is Lipschitz but expanding (its emergency multiplier is three), and its robust rejection is a positive per-step noise radius. The two contractions here have no positive margin to exploit: source 14’s margin $B^{-t}/(4n - 1)$ tends to zero (equation (120.1)), and source 12’s finite quotients are nilpotent in $\lceil k/K \rceil$ steps, so every positive-precision observation is trivial. Part III certifies robust rejection by finite cuts; these sources show that, for exact observation, even a global contraction with an explicit rate does not help. Source 12 is a data point for the negative direction of Part III’s Question 27 (a substrate on which exact simulation destroys every positive margin), although it does not formalize Part III’s noise-radius semantics on $\mathbb{Z}_2$. Source 15 is the exact-precision counterpart on Part III’s own kind of substrate, rational piecewise-affine networks: without a noise margin, precision is unbounded memory (Theorems 141.4 and 141.6).

135 What is being resolved, and what is not

[source 15]

A generic appeal to the Matiyasevich–Robinson–Davis–Putnam theorem shows that any effectively simulable computational substrate has a Diophantine acceptance relation. That observation is useful, but by itself says little about the shape, size, or arithmetic meaning of the certificate. For neural computation the more informative questions are these: can one certify a trajectory without ambiguous activation bits, avoid degree inflation, preserve exact rational semantics, and identify what changes when the weights themselves become unknown?

This article answers those questions for the saturated-linear activation

$$\sigma(u) = \max(0, \min(1, u)). \quad (135.1)$$

The main constructions are summarized in table 8. In each statement, “quadratic” means degree at most two; the domain of the witnesses is $\mathbb{N}$, including zero. A polynomial is not claimed to be nonnegative on all of $\mathbb{R}^m$ when its proof uses nonnegative coordinates.

Problem	Representation or bound	Essential qualification
Fixed rational input, T steps	One quadratic; $4nT$ uniquely determined witnesses	Arity grows with T
First acceptance at T	Same construction, plus T slacks	No absorbing halt state needed
Polyhedral real input, fixed rational weights	Projective quadratic; rational witness exists whenever a real one does	Finite time and linear specifications
Bounded rational tied-weight synthesis	Four-layer problem equivalent to $\text{H10}(\mathbb{Q})$; explicit quartic lift	One output must be positive
Strict-target rational synthesis	Real feasibility equals rational feasibility; decidable	All target conditions strict
First acceptance with denominator height H	$T + 1 = O_S(H^n(1 + \log H)^{ S -1})$	S contains the denominator primes
Universal fixed rational network	No computable accepting-run precision ceiling	Exact arithmetic, not fixed precision

Table 8: The main results and the hypotheses that prevent misleading interpretations.

135.1 Relationship to ProveIt

The repository was inspected at revision

526c2557f2c6173054794f873a9624ebef616510.

Its Hilbert-tenth-problem documentation gives the public theorem `Diophantine.mrdp`: for a recursively enumerable set $S \subseteq \mathbb{N}$ there are a fixed finite number of natural witnesses and an integer polynomial representing membership in S . It also documents the converse, bounded-trace interfaces, and separate article-specific representation bounds [63, 64]. The guide explicitly distinguishes this existence interface from a practical polynomial generator with numerical size bounds.

Our contribution is a substrate-specific front end. The local certificate exposes every scaled neuron value and every saturation deficit. The unbounded acceptance predicate can subsequently be passed to the repository’s MRDP interface. Neither the repository build nor a Lean formalization of the new front end was executed for this article. The integration plan in section 142 keeps these different obligations separate.

135.2 Earlier work and the scope of novelty

Rational saturated-linear recurrent networks are a classical universal model: Siegelmann and Sontag provide an autonomous simulation with a rational initial encoding and a coordinate indicating termination [65]. Neural complementarity formulations also predate this article; for example, Aydinoglu, Fazlyab, Morari, and Posa represent ReLU-controlled dynamics using complementarity systems [66]. We do not claim either universality or the complementarity idea as new.

Bounded-time piecewise-affine reachability has an established complexity theory [67], and real-weight neural training has $\exists\mathbb{R}$ -completeness results [68]. The proposed refinements here are the kink-safe unique natural witnesses, the two carefully distinguished scalings, the restricted four-layer rational synthesis reduction with a direct quartic lift, and the denominator-prime refinement of the precision bound. These are explicit statements proved below; a comprehensive priority search and independent proof review remain necessary before making publication-level novelty claims.

Most importantly, $H10(\mathbb{Q})$ is not known to be decidable or undecidable. A 2026 survey still identifies this rational-field case as open, despite major progress for finitely generated rings [69, Question 6.2]. Our synthesis result is an equivalence and a barrier, not a resolution of that open problem.

136 A unique quadratic certificate for a saturated scalar

[source 15]

Definition 136.1. For $D > 0$ and $U \in \mathbb{R}$, put

$$[U]_D = \max(0, \min(D, U)).$$

For nonnegative variables y, z, a, b , define

$$K_D(U; y, z, a, b) = (y + z - D)^2 + (y - U - a + b)^2 + ya + zb. \quad (136.1)$$

Lemma 136.2 (Canonical clamp certificate). *For each $D > 0$ and $U \in \mathbb{R}$, the zero set of $K_D(U; \cdot)$ in $\mathbb{R}_{\geq 0}^4$ consists of the single point*

$$y = [U]_D, \quad z = D - y, \quad a = \max(-U, 0), \quad b = \max(U - D, 0). \quad (136.2)$$

If D and U are integers, all four coordinates are natural numbers.

Proof. Every summand of (136.1) is nonnegative on the specified domain. A zero therefore satisfies

$$y + z = D, \quad y - U - a + b = 0, \quad ya = 0, \quad zb = 0. \quad (136.3)$$

In particular $0 \leq y \leq D$.

If $0 < y < D$, both y and z are positive, so $a = b = 0$ and $y = U$. If $y = 0$, then $z = D > 0$, whence $b = 0$ and $U = -a \leq 0$. If $y = D$, then $a = 0$ and $U = D + b \geq D$. These three cases force (136.2), and substitution verifies that point directly. At $U = 0$ and $U = D$, both excess variables are zero. Thus the thresholds create no extra witnesses. Integrality follows from the displayed formulas. $\square$

The terms ya and zb are complementarity conditions. Squaring them would raise the degree to four for no reason: the witness domain already makes them nonnegative. This simple observation is the degree-saving step used throughout the fixed-weight constructions.

Remark 136.3 (No branch selector is needed). A lower/middle/upper activation label is ambiguous at a threshold if the corresponding branch intervals are closed. The four numbers in (136.2) are not ambiguous. We use branch itineraries later only as an existence proof for rational points, never as part of the canonical fixed-input witness.

Remark 136.4 (Domain dependence). Over arbitrary signed witnesses, $ya + zb$ can be negative and cancel the squares. All subsequent single-polynomial equivalences retain the natural-number domain. Replacing each natural variable by a sum of four integer squares is possible, but changes the degree and witness count. Those changed bounds are not the bounds claimed here.

137 Prescribed rational trajectories: one quadratic, unique witnesses

[source 15]

137.1 The network model and its canonical scale

Fix $n \geq 1$, an integer matrix $A \in \mathbb{Z}^{n \times n}$, an integer bias vector $b \in \mathbb{Z}^n$, and a positive integer B . The synchronous autonomous network is

$$x_i^{t+1} = \sigma \left(\frac{\sum_j A_{ij} x_j^t + b_i}{B} \right), \quad x^t \in [0, 1]^n. \quad (137.1)$$

Every rational-weight, rational-bias network of this form admits such a common B , by clearing denominators. The chosen B need not be minimal.

The initial state is fixed rational data

$$x_i^0 = p_i / D_0, \quad D_0 \geq 1, \quad 0 \leq p_i \leq D_0, \quad p_i \in \mathbb{Z}.$$

For a prescribed horizon $T \in \mathbb{N}$, let

$$D_t = D_0 B^t \quad (0 \leq t \leq T). \quad (137.2)$$

All D_t are constants computed by the compiler. In particular, (137.2) does not introduce exponentiation into the Diophantine language.

137.2 The explicit polynomial

Set $X_{0,i} = p_i$ as constants. For every $0 \leq t < T$ and $1 \leq i \leq n$, introduce four natural variables

$$X_{t+1,i}, \quad C_{t+1,i}, \quad L_{t,i}, \quad H_{t,i},$$

and put

$$U_{t,i} = \sum_{j=1}^n A_{ij} X_{t,j} + b_i D_t. \quad (137.3)$$

The trace polynomial is

$$P_{\mathcal{N},p,D_0,T} = \sum_{t=0}^{T-1} \sum_{i=1}^n K_{D_{t+1}}(U_{t,i}; X_{t+1,i}, C_{t+1,i}, L_{t,i}, H_{t,i}). \quad (137.4)$$

Theorem 137.1 (Canonical finite-trace representation). *The integer polynomial (137.4) has degree at most two and exactly $4nT$ designated witness coordinates. It is nonnegative on their natural-number domain, has exactly one natural zero, and that zero satisfies*

$$X_{t,i}/D_t = x_i^t \quad (0 \leq t \leq T), \quad (137.5)$$

where x^t is the unique run of (137.1) from p/D_0 .

Proof. Each $U_{t,i}$ is affine in the witness variables, so every summand has degree at most two. Nonnegativity follows from lemma 136.2.

Construct the witness chronologically. The $X_{0,i}$ are fixed. Given all $X_{t,i}$, equation (137.3) gives an integer $U_{t,i}$, and lemma 136.2 gives a unique natural quadruple at each neuron. Its state coordinate obeys

$$\frac{X_{t+1,i}}{D_{t+1}} = \sigma \left(\frac{U_{t,i}}{D_{t+1}} \right) = \sigma \left(\frac{\sum_j A_{ij}(X_{t,j}/D_t) + b_i}{B} \right).$$

Induction proves (137.5) and constructs a zero of the whole sum. Conversely, any zero makes every nonnegative block vanish. Starting at $t = 0$, the scalar uniqueness lemma forces each quadruple in turn, so no second zero exists. When $T = 0$, the polynomial is zero on the unique empty tuple; the statement remains valid. $\square$

137.3 Acceptance and first acceptance

Choose a halt coordinate h , with acceptance defined by $x_h = 1$. Adding

$$(X_{T,h} - D_T)^2 \quad (137.6)$$

to (137.4) gives a polynomial with one natural zero precisely when the network accepts at time T , and no natural zeros otherwise. For first acceptance, also introduce $s_0, \dots, s_{T-1} \in \mathbb{N}$ and add

$$\sum_{t=0}^{T-1} (D_t - X_{t,h} - 1 - s_t)^2. \quad (137.7)$$

Corollary 137.2 (First-hit certificate). *The polynomial (137.4) plus (137.6) and (137.7) has degree at most two and $4nT + T$ witness coordinates. It has exactly one natural zero if first acceptance occurs at T , and none otherwise.*

Proof. The trace part forces the unique trajectory. Because $X_{t,h}$ is an integer between zero and D_t , a natural solution of $D_t - X_{t,h} - 1 - s_t = 0$ exists exactly when $X_{t,h} < D_t$, and its value is uniquely $D_t - X_{t,h} - 1$. The final square enforces acceptance. No assertion that acceptance persists at later times is needed. $\square$

137.4 Witness and coefficient sizes

Proposition 137.3 (Finite-horizon precision bound). *Let*

$$R_0 = \max \left(1, B, \max_i (|b_i| + \sum_j |A_{ij}|) \right).$$

Every coordinate of the canonical zero, including the first-hit slacks when present, is at most $R_0 D_0 B^T$. Its binary length is consequently at most $1 + \lceil \log_2(R_0 D_0 B^T + 1) \rceil$.

Proof. State and complement coordinates are at most $D_{t+1} \leq D_0 B^T$. The bound $0 \leq X_{t,j} \leq D_t$ gives $|U_{t,i}| \leq R_0 D_t$. The two excesses are $\max(-U_{t,i}, 0)$ and $\max(U_{t,i} - D_{t+1}, 0)$, each at most $R_0 D_t$. Finally $0 \leq s_t < D_t$ whenever a first-hit witness exists. $\square$

The structured expression contains $2nT$ squared affine residuals and $2nT$ unsquared complementarity products, before acceptance conditions are added. In a dense implementation, writing the affine residuals takes $O(n^2 T)$ coefficient occurrences. Expanding their squares can take $O(n^3 T)$ monomial occurrences; storing the residuals is usually preferable. Coefficient bit length grows polynomially in T and the binary description of the rational data. This is a finite-horizon statement: it gives no computable bound on T for an eventually accepting universal computation.

Example 137.4 (A complete first-hit trace). Consider

$$x^{t+1} = \sigma((6x^t - 1)/4), \quad x^0 = 3/5.$$

The first value equal to one occurs at $T = 4$:

$$\frac{3}{5}, \quad \frac{13}{20}, \quad \frac{29}{40}, \quad \frac{67}{80}, \quad 1.$$

The integer data are shown in [table 9](#). There are sixteen trace witnesses and four first-hit slacks, namely 1, 6, 21, 51. The complete expanded quadratic and its twenty-coordinate zero are included in `examples/fixed_trace.json`.

t	D_t	X_t	U_{t-1}	C_t	H_{t-1}
0	5	3	—	—	—
1	20	13	13	7	0
2	80	58	58	22	0
3	320	268	268	52	0
4	1280	1280	1288	0	8

Table 9: The canonical scale is not reduced at every step. All lower excesses L_{t-1} are zero.

[write] Relation to Part III’s neural corollary. Theorem [137.1](#) certifies the finite runs of *any* concrete rational clipped network, and $\sigma(u) = \text{ReLU}(u) - \text{ReLU}(u - 1)$, so the same unique-witness quadratic also certifies a concrete rational ReLU network once it is rewritten with clipped units. It is therefore the certification layer that an extraction of the network of Corollary [31.2](#) would need. It does not extract that network: Question [26](#) remains open.

138 Projective quadratic certificates for real-input reachability

[source 15]

The preceding canonical scale starts from a specified rational input. A finite reachability question may instead ask whether *some real input* in a rational polyhedron reaches a target. A preselected grid can miss such a trajectory. The correct replacement is an existential common denominator, not arbitrary rounding.

138.1 Specifications and rationalization

Fix the rational network, the finite horizon T , and a finite conjunction of rational affine equalities, weak inequalities, and strict inequalities in the coordinates of $x^0, \dots, x^T$. The variables remain in $[0, 1]^n$ at every time. This includes rational polyhedral input regions and target regions; it also allows linear conditions at intermediate times.

Lemma 138.1 (Rational finite-trajectory witnesses). *Such a specification has a real trajectory if and only if it has a rational trajectory, meaning all coordinates of all x^t are rational.*

Proof. The reverse implication is immediate. Given a real trajectory, choose a lower, middle, or upper activation branch for each of its nT evaluations. The branch equations and inequalities are respectively

$$z \leq 0, y = 0; \quad 0 \leq z \leq 1, y = z; \quad z \geq 1, y = 1.$$

Here z is the rational affine preactivation. Intersect these constraints with the specification and the state cube. The result is a nonempty bounded set defined by rational affine equalities and weak or strict inequalities.

For completeness, such a set contains a rational point. Replace each strict condition $\ell_j(x) > 0$ by $\ell_j(x) \geq \eta$, add $0 \leq \eta \leq 1$, and maximize η . The given point makes some $\eta > 0$ feasible. A maximum is attained on the resulting compact rational polytope, and an optimal vertex has rational coordinates. Its x -coordinates satisfy the original strict conditions. If there are no strict conditions, any vertex of the nonempty compact rational polytope suffices. Rationality of a vertex follows by solving a full-rank system of active rational linear equations. All chosen activation relations hold at this rational point, so it is a genuine trajectory. $\square$

138.2 The projective lift

Introduce $Q \geq 1$ and write $x_i^t = X_{t,i}/Q$. We eliminate a positivity inequality by setting $Q = 1 + r$ with $r \in \mathbb{N}$. For every state coordinate, introduce a complement $C_{t,i}$ and impose

$$X_{t,i} + C_{t,i} = Q. \quad (138.1)$$

For each neuron-time pair put

$$V_{t,i} = \sum_j A_{ij} X_{t,j} + b_i Q$$

and impose

$$BX_{t+1,i} - V_{t,i} - L_{t,i} + H_{t,i} = 0, \quad (138.2)$$

$$X_{t+1,i} L_{t,i} = 0, \quad C_{t+1,i} H_{t,i} = 0. \quad (138.3)$$

These are the scalar clamp relations with total output scale BQ , after multiplying the state and complement by B ; since $B > 0$, the unsquared products in (138.3) are equivalent to those rescaled complementarity conditions.

Clear denominators in each affine specification. If $\ell(x) = c_0 + \sum c_{t,i} x_i^t$ has integer coefficients, its homogenization is

$$\ell^h(X, Q) = c_0 Q + \sum c_{t,i} X_{t,i}.$$

An equality becomes $\ell^h = 0$. A weak inequality $\ell \geq 0$ becomes $\ell^h - s = 0$, and a strict inequality $\ell > 0$ becomes $\ell^h - 1 - s = 0$, with one new natural slack in each case.

Theorem 138.2 (Projective quadratic reachability). *For fixed rational weights and a finite affine specification, one can construct an integer polynomial of degree at most two whose natural zero set is nonempty exactly when a real satisfying trajectory exists. With k specified inequalities, the construction uses*

$$2n + 4nT + k + 1 \quad (138.4)$$

witness coordinates. Its terms are the squares of the linear residuals just described, plus the unsquared products in (138.3).

Proof. A natural zero forces every nonnegative summand to vanish. Dividing the state coordinates by $Q = 1 + r > 0$ gives the network equations by the scalar lemma. The bound equations place all states in the cube. The specification is recovered by dividing each homogenized relation by Q ; in particular, $1 + s > 0$ gives a strict inequality.

Conversely, [lemma 138.1](#) supplies a rational satisfying trajectory. Choose a common denominator Q for its finitely many state coordinates. Then $X = Qx$ and $C = Q - X$ are natural. With V as above, choose

$$L_{t,i} = \max(-V_{t,i}, 0), \quad H_{t,i} = \max(V_{t,i} - BQ, 0).$$

These are integers and satisfy all transition and complementarity relations. A strictly positive homogenized integer is at least one, so all the specification slacks also exist. The count is $2n(T+1)$ states and complements, $2nT$ excesses, k inequality slacks, and r . $\square$

There is intentionally no uniqueness assertion: multiplying a successful common denominator produces further representations of the same rational trajectory. Moreover, different inputs may satisfy the specification.

138.3 An explicit small-denominator bound

Theorem 138.3 (A determinant bound). *Let $m = n(T+1)$. After choosing an activation itinerary and clearing all linear constraints, suppose every coefficient and right-hand side has absolute value at most $M \geq 1$. If the resulting specification is feasible and contains no strict inequalities, it has a trajectory with a common denominator*

$$Q \leq m!M^m.$$

With strict inequalities, it has one with

$$Q \leq (m+1)!M^{m+1}.$$

The bound M includes the coefficient 1 of the auxiliary margin variable.

Proof. In the closed case, take a vertex of the bounded feasible polytope. There are m linearly independent active constraints, including equations represented as pairs of inequalities when necessary. Their integer coefficient matrix E has nonzero determinant. Cramer's rule expresses every coordinate with common denominator $|\det E|$. The Leibniz formula bounds this determinant by $m!M^m$. Lower-dimensional polytopes cause no exception: at a vertex the active normals still span the full ambient coordinate space.

For strict constraints use the auxiliary η construction in the proof of [lemma 138.1](#), and choose a vertex of the positive optimal face. This is a vertex of the entire polytope in $m+1$ variables. The same argument bounds a common denominator for both the trajectory and η . $\square$

If T is presented in unary, the bit length of this bound is polynomial in the expanded linear-system description. This recovers the usual short rational-witness mechanism behind finite piecewise-affine reachability; we do not claim a new complexity classification [67]. The determinant bound concerns a successful rational trajectory, not a uniform denominator for every point in an input region.

Example 138.4 (A preselected grid can miss the answer). For $x' = \sigma(2x - 1/2)$ and the target $x' = 1/3$, the input must be $x = 5/12$. A grid with denominator two contains no such input. With $Q = 1 + r$, a suitable seven-variable projective polynomial is

$$\begin{aligned} & (X_0 + C_0 - Q)^2 + (X_1 + C_1 - Q)^2 \\ & + (2X_1 - 4X_0 + Q - L + H)^2 + X_1L + C_1H + (3X_1 - Q)^2. \end{aligned}$$

At $Q = 12$ its zero is

$$(X_0, C_0, X_1, C_1, L, H, r) = (5, 7, 4, 8, 0, 0, 11).$$

Every positive multiple of twelve also supplies a scale, with the associated coordinates scaled accordingly. The file `examples/projective_trace.json` contains the expanded polynomial.

139 The rational parameter-synthesis frontier

[source 15]

The preceding rationalization theorem depends critically on fixed weights. When an unknown weight multiplies an unknown activation, activation cells are no longer polyhedra in all unknowns. The obstruction is not merely numerical conditioning: rational feasibility can now express a major Diophantine decision problem.

139.1 The precise synthesis model

Definition 139.1 (Bounded tied-weight synthesis). An instance consists of a finite feedforward network with four non-input layers. Every node uses σ . There is a single fixed input equal to one. Each edge weight is either a fixed rational number or one of the named unknown parameters $\theta_1, \dots, \theta_k$. The parameters must belong to $\mathbb{Q} \cap [0, 1]$ and may be reused on multiple edges; such reuse is called a *tie*. Biases are fixed rationals. Two output nodes, denoted o and p , are distinguished. The question is whether there are parameter values with

$$o = 0, \quad p > 0. \quad (139.1)$$

The number of nodes is part of the instance. Four is a depth bound, not a claim of fixed width or a fixed universal architecture. Parameters do not change during an evaluation, and all uses of the same named parameter must agree exactly.

Theorem 139.2 (Four-layer rational synthesis equivalence). *The decision problem in definition 139.1 and $\text{H10}(\mathbb{Q})$ are computably many-one equivalent. In the reduction from $\text{H10}(\mathbb{Q})$, unknown weights occur only in the first two layers. All parameters lie in $[0, 1]$, all biases and remaining weights can be integers, and there is only one strictly positive output condition.*

Using arithmetic circuits, both reductions have polynomial description size. In particular, the lower reduction avoids expansion of high-degree polynomials after homogenization.

We prove the theorem in three steps, giving every transformation explicitly.

139.2 Step one: a bounded quadratic presentation of a rational zero

Let $P \in \mathbb{Z}[z_1, \dots, z_m]$ be the input to $\text{H10}(\mathbb{Q})$. Introduce variables for the wires of an arithmetic circuit computing P . Its constraints can be chosen from

$$z_k = z_i + z_j, \quad z_k = z_i - z_j, \quad z_k = z_i z_j, \quad z_i = c, \quad z_{\text{out}} = 0, \quad (139.2)$$

where $c \in \mathbb{Z}$. A solution of $P = 0$ extends to a solution of these constraints by evaluating the circuit. Conversely, a solution of the constraints gives $P = 0$ by induction through the circuit.

Even when a sparse polynomial uses exponents in binary, repeated squaring produces a circuit of polynomial size. Thus no unbounded expansion is hidden in this step.

Lemma 139.3 (Compactification with one positive scale). *A finite circuit system (139.2) has a rational solution if and only if a corresponding finite system of quadratic integer equations has a solution in parameters*

$$a_i, v \in \mathbb{Q} \cap [0, 1], \quad v > 0.$$

The transformation is explicit and increases description size only polynomially.

Proof. For every circuit wire put

$$u_i = 2a_i - 1, \quad z_i = u_i/v. \quad (139.3)$$

Translate the relations in (139.2) into

$$u_k - u_i - u_j = 0 \quad \text{for addition,} \quad (139.4)$$

$$u_k - u_i + u_j = 0 \quad \text{for subtraction,} \quad (139.5)$$

$$vu_k - u_i u_j = 0 \quad \text{for multiplication,} \quad (139.6)$$

$$u_i - cv = 0 \quad \text{for a constant,} \quad (139.7)$$

$$u_{\text{out}} = 0. \quad (139.8)$$

After substituting $u_i = 2a_i - 1$, these are integer polynomials of degree at most two. For example, the multiplication equation is

$$2va_k - v - 4a_i a_j + 2a_i + 2a_j - 1 = 0. \quad (139.9)$$

Given rational values for the original wires, choose a positive rational $v \leq 1$ so small that $|vz_i| \leq 1$ for every wire. Such a choice exists because there are only finitely many wire values. Set $a_i = (1 + vz_i)/2$. All parameters lie in $[0, 1]$ and satisfy the new equations. Conversely, $v > 0$ makes (139.3) legitimate, and division by v or v^2 recovers each original circuit relation. Every transformed equation has bounded degree and constant-size support relative to its circuit gate. $\square$

The condition $v > 0$ is essential. At $v = 0$, the affine chart disappears, and the homogeneous multiplication constraints no longer reconstruct the original rational wire values. We retain that condition explicitly rather than silently admitting points at infinity.

139.3 Step two: realize the quadratic system in four layers

Let $R_1(\theta), \dots, R_E(\theta)$ be the quadratic polynomials supplied by lemma 139.3; the parameter list θ includes all a_i and v . Let k be the number of parameters and let q be the number of distinct degree-two monomials occurring in the R_ℓ .

Lemma 139.4 (Four-layer realization). *There is a network of the form in definition 139.1 with*

$$2k + q + 2E + 3 \quad (139.10)$$

non-input neurons whose outputs satisfy

$$o = 0 \iff R_1(\theta) = \dots = R_E(\theta) = 0, \quad p = v.$$

Proof. The four layers are as follows.

Layer one. For each parameter, set $h_i = \sigma(\theta_i \cdot 1) = \theta_i$. This uses k neurons.

Layer two. Copy each h_i , using weight one, and for each needed monomial $\theta_i \theta_j$ create the neuron

$$h_{ij} = \sigma(\theta_i h_j) = \theta_i \theta_j.$$

Both factors lie in $[0, 1]$, so clipping does not change the product. The use of θ_i here is tied to its occurrence in layer one. This layer uses $k + q$ neurons, including square monomials when $i = j$.

Layer three. Every R_ℓ is now an integer affine expression in the second-layer copies and products. Create

$$r_\ell^+ = \sigma(R_\ell), \quad r_\ell^- = \sigma(-R_\ell)$$

for each equation, and one additional copy of v . Since $\sigma(s) = 0$ if and only if $s \leq 0$, the two detectors are simultaneously zero exactly when $R_\ell = 0$. Saturation at large values introduces no false zeros. There are $2E + 1$ neurons in this layer.

Layer four. Define

$$o = \sigma \left(\sum_{\ell=1}^E (r_\ell^+ + r_\ell^-) \right), \quad p = \sigma(v) = v.$$

Every summand entering o is nonnegative, so $o = 0$ precisely when all detectors vanish. Adding the four layer sizes gives (139.10). $\square$

Together, [lemmas 139.3](#) and [139.4](#) prove the reduction $\text{H10}(\mathbb{Q}) \leq_m$ bounded tied-weight synthesis. The construction uses products only where parameters serve as weights; no neuron is given a nonstandard multiplication activation.

139.4 Step three: reduce rational synthesis back to a polynomial

Lemma 139.5 (Nonnegativity over the rationals). *For $r \in \mathbb{Q}$, one has $r \geq 0$ if and only if $r = t_1^2 + t_2^2 + t_3^2 + t_4^2$ for some $t_i \in \mathbb{Q}$.*

Proof. The reverse implication follows from the ordering of $\mathbb{Q}$. For the forward implication, write $r = a/b$ with $a \in \mathbb{N}$ and $b \geq 1$ integral. Lagrange’s four-square theorem gives integers k_i with $ab = \sum k_i^2$. Taking $t_i = k_i/b$ proves the claim, including $a = 0$. $\square$

Completion of theorem 139.2. Introduce rational variables for all parameters, node values, and the scalar clamp auxiliaries. For each node use the relations [\(136.3\)](#) with $D = 1$ and the node’s preactivation U . That preactivation is polynomial of degree at most two in the unknowns: each edge is a fixed rational weight times a state or a named parameter times a state. Impose nonnegativity of the clamp variables and both θ_i and $1 - \theta_i$.

The resulting equations and inequalities describe exactly the rational network graph by [lemma 136.2](#). Enforce $o = 0$. The graph already makes $p \geq 0$, so the strict condition $p > 0$ is equivalent over $\mathbb{Q}$ to the existence of a rational s with $ps = 1$.

Replace every nonnegativity condition by the four-square equation in [lemma 139.5](#). Clear rational coefficients in each equation, and combine all equations $F_j = 0$ as $\sum_j F_j^2 = 0$. Over the ordered field $\mathbb{Q}$, this single integer polynomial vanishes exactly when all equations hold. The transformations add only polynomially many variables and coefficient bits. They are effective and yield an instance of $\text{H10}(\mathbb{Q})$. This proves the other reduction and completes the theorem. $\square$

139.5 Why real feasibility cannot replace rational feasibility

Example 139.6 (An irrational-only parameter fit). Use a tied parameter $\theta \in [0, 1]$, form

$$h = \sigma(\theta) = \theta, \quad z = \sigma(\theta h) = \theta^2,$$

and demand that both $\sigma(2z - 1)$ and $\sigma(1 - 2z)$ be zero. These requirements are equivalent to $2\theta^2 = 1$. A real solution exists, but no rational solution exists: a reduced fraction a/b would give $2a^2 = b^2$, forcing both a and b even. Padding by copying neurons gives four layers if desired.

Consequently, a real-arithmetic feasibility engine cannot by itself settle rational synthesis. The established $\exists\mathbb{R}$ -completeness results for real-weight training concern a different coefficient domain and different architectural restrictions [\[68\]](#). The present reduction must not be read as a theorem about ordinary untied training, nor as an undecidability proof for it.

Corollary 139.7 (Trading strict positivity for an unbounded gain). *The lower reduction can replace $p > 0$ by an exact output condition equal to one if a single additional unknown nonnegative, unbounded rational weight λ is allowed. The other parameters remain in $[0, 1]$, and four layers still suffice.*

Proof. After layer one supplies v , add a second-layer neuron $g = \sigma(\lambda v)$ and copy it through the remaining layers. For $v \in [0, 1]$, a rational $\lambda \geq 0$ with $g = 1$ exists exactly when $v > 0$: choose $\lambda = 1/v$ in the forward direction. At $v = 0$, the output is zero for every finite λ . The rational upper reduction works with the extra nonnegativity constraint on λ . $\square$

This corollary does not remove a hypothesis for free. It exchanges an open output condition for an unbounded parameter. The all-bounded, closed-target variant is a separate question.

139.6 An exact-versus-open-target dichotomy

There is also a useful positive result on the other side of the exact-zero boundary. It does not require a four-layer depth restriction.

Theorem 139.8 (Strict-target rational synthesis is decidable). *Consider any finite feedforward clipped network with fixed rational constants and finitely many tied parameters in $\mathbb{Q} \cap [0, 1]$. Suppose that every target condition is a strict polynomial inequality, with rational coefficients, in the parameters and the output values. Then rational feasibility is equivalent to real feasibility and is decidable.*

In particular, for a fixed rational $\varepsilon > 0$, replacing $o = 0$, $p > 0$ by $o < \varepsilon$, $p > 0$ produces a decidable problem in this model.

Proof. As a function of its real parameters in the cube, each node value is continuous: this follows inductively from continuity of addition, multiplication, and σ . Every strict target condition therefore pulls back to a relatively open subset of the parameter cube. Their finite intersection is relatively open. The rational points of the cube are dense, so any nonempty such intersection contains a rational parameter point. Evaluation at rational parameters is rational, proving the equivalence.

The graph of σ is a finite union of sets given by polynomial equations and inequalities. Thus real feasibility for the entire finite network is an existential sentence over the real field with rational constants. Real algebraic decision procedures decide its truth [70]. By the first part, their answer is also the answer to the rational problem. $\square$

The theorem requires *all* target conditions to be strict; one cannot retain an unrelated exact equality and invoke density. The only weak parameter restrictions used in the proof are the full cube bounds. Arbitrary extra algebraic restrictions on the parameters would destroy the density argument. Parameter ties are harmless because each distinct name is already a single independent coordinate of that cube.

This gives a concrete boundary rather than only a warning: in the specified finite architecture model, open output tolerance permits a decision procedure, whereas exact zero together with one positive output can encode all of H10($\mathbb{Q}$). Neither statement asserts an efficient practical algorithm.

140 A direct quartic natural-number certificate for synthesis

[source 15]

The reverse reduction in [theorem 139.2](#) gives a rational polynomial. For direct use with a natural-number Diophantine library, one can instead lift the entire parameter network to one common integer scale. The construction exposes exactly where degree four enters.

140.1 Homogenizing parameters and activations together

For clarity, allow an edge weight to be an affine rational function of the parameters,

$$w_e(\theta) = \frac{a_{e,0} + \sum_a a_{e,a} \theta_a}{B},$$

with all $a_{e,a} \in \mathbb{Z}$ and a common $B \geq 1$. Biases have the form b_v/B . This slightly larger class includes every network constructed above. The fixed input is one. Choose a common scale $Q = 1 + r$ and write

$$\theta_a = \Theta_a/Q, \quad x_v = X_v/Q.$$

Introduce complements $\hat{\Theta}_a$ and C_v , with

$$\Theta_a + \hat{\Theta}_a = Q, \quad X_v + C_v = Q. \tag{140.1}$$

The integer polynomial representing a preactivation numerator is

$$U_v = \sum_{e:u \rightarrow v} \left(a_{e,0}Q + \sum_a a_{e,a}\Theta_a \right) X_u + b_v Q^2. \quad (140.2)$$

For the input node, use $X_{\text{input}} = Q$ as an expression rather than an extra variable. The actual preactivation is $U_v/(BQ^2)$.

For each non-input node introduce $L_v, H_v \in \mathbb{N}$ and impose

$$BQX_v - U_v - L_v + H_v = 0, \quad (140.3)$$

$$X_v L_v = 0, \quad C_v H_v = 0. \quad (140.4)$$

The output conditions become

$$X_o = 0, \quad X_p - 1 - s = 0, \quad s \in \mathbb{N}. \quad (140.5)$$

Theorem 140.1 (Explicit quartic lift). *Square and sum all residuals in (140.1), (140.3), and (140.5), and add the unsquared products in (140.4). After substituting $Q = 1 + r$, the resulting integer polynomial has degree at most four, is nonnegative on the natural-number orthant, and has a natural zero if and only if the rational synthesis instance has a solution.*

For a network with k bounded parameters and N non-input nodes, the construction uses $2k + 4N + 2$ natural witness coordinates.

Proof. The preactivation numerator (140.2) has degree at most two. The residual (140.3) therefore has degree at most two, so its square has degree at most four. Every remaining squared residual is linear before squaring, and the two complementarity products per node have degree two. All summands are nonnegative on natural coordinates.

Suppose the polynomial is zero. Then every residual and every complementarity product vanishes. Since $Q \geq 1$ and $B \geq 1$, the bounds and gate conditions give the clamp relations at scale BQ^2 , with output BQX_v and complement BQC_v . Multiplication by the positive factor BQ does not alter a zero complementarity condition. The scalar lemma implies

$$X_v/Q = \sigma(U_v/(BQ^2)).$$

Induction in the feedforward order recovers the evaluation of the rational parameter network. The output conditions give $o = 0$ and $p > 0$.

Conversely, take a rational satisfying parameter assignment and evaluate all nodes exactly. All activations are rational. Choose a common denominator Q for all parameters and activations, and set $\Theta_a = Q\theta_a$, $X_v = Qx_v$. The complements are natural. Each U_v is an integer; choose

$$L_v = \max(-U_v, 0), \quad H_v = \max(U_v - BQ^2, 0).$$

The clamp identities make all node conditions hold. Because $x_p > 0$, X_p is a positive integer, so $s = X_p - 1$ is natural. This constructs a zero. The count consists of two coordinates for each parameter, four for each non-input node, r , and the positive-output slack. $\square$

140.2 Where the quadratic/quartic distinction comes from

In the fixed-weight projective compiler, the transition residual is

$$BX' - AX - bQ - L + H,$$

which is affine in the unknown integers. In parameter synthesis, it is

$$BQX' - U(\Theta, X, Q) - L + H,$$

which is quadratic. Squaring the latter gives degree four. The increase is not caused by approximation, by activation branch bits, or by an appeal to the full MRDP theorem. It records the multiplication of an unknown weight and an unknown signal.

If Q is replaced by kQ in a valid rational lift, parameter and activation numerators scale by k , while L_v, H_v scale by k^2 . The positive-output slack changes from $X_p - 1$ to $kX_p - 1$. These rescalings provide infinitely many natural zeros for a successful rational assignment; no uniqueness is claimed for this compiler.

Combining [theorems 139.2](#) and [140.1](#) identifies an explicitly generated family of quartic natural-number feasibility instances equivalent to $H10(\mathbb{Q})$. Here the instance includes its generating network or the displayed structured residual presentation. No claim is made that recognizing such presentations from an arbitrary expanded polynomial is automatic. Nor does the equivalence make the family known to be undecidable.

Example 140.2 (A reproducible synthesis instance). Take the circuit with input wires x, y , a constant wire $c = 2$, a product wire $m = xy$, and output wire $z = m - c$. It computes $xy - 2$. The rational solution $x = 2, y = 1$ gives wire values $(2, 1, 2, 2, 0)$. Choose $v = 1/4$ and

$$(a_x, a_y, a_c, a_m, a_z, v) = \left(\frac{3}{4}, \frac{5}{8}, \frac{3}{4}, \frac{3}{4}, \frac{1}{2}, \frac{1}{4} \right).$$

There are four compactified constraints, two distinct quadratic monomials, and six parameters. Formula [\(139.10\)](#) gives $12 + 2 + 8 + 3 = 25$ non-input neurons. Their outputs are $(o, p) = (0, 1/4)$. A common scale for these parameters and all evaluated nodes is $Q = 32$. The quartic compiler uses $2 \cdot 6 + 4 \cdot 25 + 2 = 114$ natural coordinates. The complete structured and expanded polynomial, together with a zero, is in `examples/rational_synthesis.json`.

141 Denominator height as computational memory

[source 15]

The state cube is bounded, but the set of rational points in it is infinite. The resource that prevents this model from being a finite-state machine is precision, not the magnitude of the neuron values.

141.1 Reduced common denominators and prime support

Definition 141.1. For $x \in \mathbb{Q}^n$, define the reduced common denominator

$$q(x) = \min\{q \geq 1 : qx \in \mathbb{Z}^n\}.$$

This is the least common multiple of the reduced coordinate denominators, not the largest individual denominator. For a finite run, define

$$H_T = \max_{0 \leq t \leq T} q(x^t).$$

An integer is S -smooth if all of its prime divisors belong to a finite prime set S ; the integer one is S -smooth for every S .

Lemma 141.2 (Denominator support invariant). *For the rational network [\(137.1\)](#) with initial denominator D_0 ,*

$$q(x^t) \mid D_0 B^t.$$

Consequently all denominator primes belong to the finite set of prime divisors of $D_0 B$.

Proof. If the coordinates of x^t have a common denominator dividing $D_0 B^t$, then each affine preactivation divided by B has denominator dividing $D_0 B^{t+1}$. Clipping either leaves that rational value unchanged or replaces it by zero or one. Induction starts at the given initial state. $\square$

Lemma 141.3 (Distinct states before first acceptance). *If a deterministic autonomous network first reaches an accepting state at time T , then $x^0, \dots, x^T$ are pairwise distinct.*

Proof. Suppose $x^i = x^j$ with $0 \leq i < j \leq T$. Determinism gives $x^{i+k} = x^{j+k}$ for every $k \geq 0$. Taking $k = T - j$ gives an accepting state at $T - (j - i) < T$, contradicting first acceptance. The argument uses only that acceptance is a predicate of the state. $\square$

141.2 A prime-sensitive time–height inequality

Theorem 141.4 (First-hit height bound). *Suppose first acceptance occurs at time T in an n -neuron network. Put $H = H_T$, and let S contain all denominator primes along the run. Then*

$$T + 1 \leq \sum_{\substack{1 \leq q \leq H \\ q \text{ } S\text{-smooth}}} (q + 1)^n. \quad (141.1)$$

If $S = \{p_1, \dots, p_r\}$ is nonempty, then

$$T + 1 \leq \frac{2^n}{1 - p_1^{-n}} H^n \prod_{j=2}^r \left(1 + \lfloor \log_{p_j} H \rfloor\right). \quad (141.2)$$

For fixed n, S , this is $O_S(H^n(1 + \log H)^{r-1})$. If S is empty, $T + 1 \leq 2^n$.

Proof. For a fixed common denominator q , the grid $\{0, 1/q, \dots, 1\}^n$ has $(q + 1)^n$ points. Every state in the run belongs to the grid of its own reduced common denominator, which is at most H and has prime support in S . Counting the union by the sum of its grid sizes may overcount, but is an upper bound. By lemma 141.3, at least $T + 1$ different states must be counted, proving (141.1).

For the sharper estimate, use $(q + 1)^n \leq 2^n q^n$ for $q \geq 1$. Write an S -smooth q uniquely as $q = sp_1^k$, where s has prime factors only in $\{p_2, \dots, p_r\}$. For a fixed $s \leq H$, let $K = \lfloor \log_{p_1}(H/s) \rfloor$. The finite geometric series gives

$$\sum_{k=0}^K (sp_1^k)^n \leq \frac{(sp_1^K)^n}{1 - p_1^{-n}} \leq \frac{H^n}{1 - p_1^{-n}}.$$

There are at most $\prod_{j=2}^r (1 + \lfloor \log_{p_j} H \rfloor)$ choices for the exponents forming s . This proves (141.2), with the empty product equal to one when $r = 1$. If S is empty, only $q = 1$ occurs. $\square$

For a single denominator prime, the bound gives the especially direct necessary condition

$$H \geq \left(\frac{1 - p_1^{-n}}{2^n} (T + 1) \right)^{1/n}.$$

Thus, up to an additive constant depending on n, p_1 , at least $\log_2(T + 1)/n$ bits of common-denominator precision must occur somewhere before first acceptance. For several fixed primes the logarithmic product in (141.2) is the additional factor, rather than the full extra power of H in the cruder count over all denominators.

Corollary 141.5 (Integral weights do not create new precision). *If $B = 1$, then every state lies in the fixed grid of denominator D_0 . Any first acceptance time satisfies $T + 1 \leq (D_0 + 1)^n$.*

Proof. By lemma 141.2, $q(x^t) \mid D_0$ for every t , so all states belong to that one grid. Apply lemma 141.3. $\square$

141.3 An elementary stack illustration

The classical base-four binary-stack encoding used in rational neural universality is [65]

$$\text{enc}(s_1 \cdots s_k) = \sum_{i=1}^k \frac{2s_i + 1}{4^i}, \quad \text{enc}(\emptyset) = 0. \quad (141.3)$$

For these encodings, a nonempty stack with top bit zero lies in $[1/4, 1/2)$, and one with top bit one lies in $[3/4, 1)$. Consequently

$$\begin{aligned} \text{nonempty}(x) &= \sigma(4x), & \text{top}(x) &= \sigma(4x - 2), \\ \text{push}_b(x) &= (x + 1 + 2b)/4, & \text{pop}(x) &= \sigma(4x - 1 - 2\sigma(4x - 2)). \end{aligned}$$

The push identity follows by shifting the base-four expansion. For a nonempty stack, the inner saturation extracts its first bit, after which $4x - (1 + 2s_1)$ is exactly the tail encoding; the outer saturation also maps the empty-stack case to zero. These formulas explain how a bounded rational state can carry an arbitrarily long finite word. They illustrate the memory representation, not a replacement for the full finite-control universality proof.

141.4 No computable precision ceiling on accepting computations

Theorem 141.6 (Accepting precision has no computable uniform ceiling). *There exist a fixed rational saturated-linear autonomous network and a computable rational initial-state encoding $e \mapsto x^0(e)$ such that no total computable function $f(e)$ bounds H_T for every encoded input that first accepts at some time T .*

Proof. Choose the fixed network and effective encoding supplied by the autonomous universality theorem of Siegelmann and Sontag [65, Theorem 2], applied to a universal partial computable function. Its halt coordinate first becomes one precisely for the accepting machine inputs. Rational network steps and the equality test for that coordinate are effectively computable by exact arithmetic.

Suppose such a total computable f exists, and replace it by $\max(1, f)$ if necessary. On input e , compute $H = f(e)$ and

$$M(H) = \sum_{q=1}^H (q + 1)^n.$$

Simulate the network for $M(H)$ steps, including the initial acceptance test. If it accepts, answer yes; otherwise answer no.

To justify the no answer, an eventually accepting run would, by the assumed bound, have reduced common denominators at most H through first acceptance. There are at most $M(H)$ possible states of that height in the cube. By lemma 141.3, first acceptance would occur at $T < M(H)$, and would already have been seen. The algorithm would decide universal halting, a contradiction. $\square$

The hypothetical precision bound was required only on accepting runs; no bound on nonaccepting runs was assumed. The same argument excludes a total computable upper bound on the number of precision bits, since exponentiating such a bit bound would give a computable denominator bound. It also excludes a computable bound depending only on input length that works for every accepting input of that length.

There is no conflict with proposition 137.3. That proposition bounds precision as a function of a *given horizon*. The theorem above rules out a bound computable from the encoded input alone that covers every successful, unbounded-duration computation.

142 What can and cannot be compressed into the MRDP back end

[source 15]

142.1 The honest unbounded statement

Fix a rational network, an effective rational encoding of natural inputs, and a decidable acceptance predicate on rational states. Exact simulation semidecides eventual acceptance: inspect the initial state, then enumerate time steps until an accepting state appears. The accepted input set is therefore recursively enumerable. The repository's MRDP interface gives one fixed integer polynomial $F(e, w_1, \dots, w_m)$ such that

$$\text{input } e \text{ eventually accepts} \iff \exists w \in \mathbb{N}^m F(e, w) = 0$$

[64]. For the universal network this is a uniform Diophantine representation of a complete computably enumerable set.

This conclusion does not preserve the specific degree-two normal form, the $4nT$ count, or uniqueness of the finite trace witnesses. The MRDP polynomial has fixed arity, whereas the explicit trajectory compiler has one block for every simulated neuron-time pair. Describing the latter as a single-fold fixed-arity representation of unbounded halting would be a mistake. No assertion about the general finite-fold or single-fold Diophantine representation questions is proved here.

142.2 An obstruction specific to the complementarity normal form

Proposition 142.1 (Decidability of the finite affine-complementarity form). *Natural feasibility is decidable for any given finite system of rational affine equations and inequalities, together with finitely many constraints of the form $u_i v_i = 0$, where u_i, v_i are natural-coordinate variables. Equivalently, it is decidable for a polynomial supplied as a sum of squares of affine residuals and nonnegative coordinate products, with natural witnesses and explicitly encoded affine slacks.*

Proof. For every product $u_i v_i = 0$, choose one of the two possibilities $u_i = 0$ or $v_i = 0$. Enumerate all finitely many such choices. After making a choice, the remaining constraints are linear integer equations and inequalities in natural variables, a decidable fragment of Presburger arithmetic. The original system is feasible exactly when one of these finitely many linear systems is feasible. A product with a positive rational coefficient has the same zero condition; a zero coefficient can be discarded. $\square$

Hence no total computable translation of arbitrary machine inputs into this finite normal form can characterize unbounded halting: following the translation by the proposition's decision procedure would decide halting. This explains structurally why the finite quadratic front end cannot simply retain all its local properties after uniform unbounded compression. The claim concerns the stated normal form, not an unproved general classification of all quadratic Diophantine equations.

142.3 A concrete formalization plan for ProveIt

A sound integration should establish the following modules in dependency order. These are proposed proof obligations, not declarations claimed to have been compiled.

Scalar semantics. Define the saturated-linear map and prove lemma 136.2 first over an ordered field, then derive the integral specialization. In Lean, separate integer residual evaluation from the coercions of natural witnesses. The nonnegative-sum-zero argument should explicitly include the unsquared products.

Canonical finite execution. Represent a rational network by the integer data (A, b, B) and prove the scale invariant. Define the T -step witness constructor recursively and show that decoding is inverse to zero-set recognition. Prove the first-hit adapter separately, including $T = 0$ and an initially accepting state.

Polynomial extraction. Build the finite residual family as multivariate integer polynomials and verify the degree bound. The $4nT$ count should be a theorem about the chosen finite indexing type, rather than an informal count of displayed symbols. Test whether array sharing can reduce the stored expression size without changing the polynomial semantics.

Unbounded acceptance. Prove computability of exact rational iteration and recursive enumerability of the acceptance set. Only then invoke `Diophantine.mrdp` or the corresponding library-level Diophantine predicate. Keep this theorem distinct from a numerical degree/witness theorem for a specific universal polynomial.

Synthesis and rational algebra. Formalize arithmetic circuits, the bounded chart $z_i = (2a_i - 1)/v$, and the four-layer realization. The reverse reduction needs a rational four-square lemma and effective polynomial assembly, not a decision oracle for $H10(\mathbb{Q})$. The direct quartic lift is a separate natural-witness theorem.

The existing repository documentation distinguishes its audited endpoints, trace interfaces, and article-specific results [63, 64]. Maintaining that distinction will prevent an implementation test or a semidecidability proof from being misreported as an audited explicit polynomial construction.

143 Exact implementation and validation

[source 15]

The accompanying code uses Python’s standard library only. Rational values are represented by `fractions.Fraction`; sparse polynomials have integer coefficients and sorted tuples of variable names as monomials. There are no floating-point comparisons, numerical optimizers, or external solver calls.

The main module implements the scalar witness, the canonical trajectory compiler, the projective example, arithmetic-circuit compactification, the four-layer network construction, and the quartic synthesis compiler. Each certificate is stored both as its structured residual presentation and, for the supplied examples, as an expanded sparse polynomial. A proposed natural witness is evaluated with a domain check before any polynomial zero test is accepted.

The deterministic test suite completed **53,472 assertions**. Its principal checks are summarized in table 10. The full log is `verification_report.txt`; the random seed is 20260930.

Check	Count
Canonical scalar witnesses	234
Scalar uniqueness checks in a declared finite box	32,997
Random rational network trace instances	240
Independent rational transition comparisons	972
Single-coordinate trace mutations rejected	13,550
Binary words checked through length eight	511
Projective scales checked	72
Quartic synthesis-coordinate mutations rejected	114
Prime-support counting inequalities checked	840

Table 10: Selected checks. Counts are not claims of exhaustive verification of the infinite theorems. Some objects support several assertions.

The scalar enumeration checks both thresholds and the transition from unsaturated to saturated behavior. The trace tests include $B = 1$, zero steps, negative coefficients, and exact rational saturation. Every single-coordinate increment, and every valid decrement, of the tested canonical trace witnesses produces a positive certificate value. The synthesis example checks the expanded polynomial as well as the structured residuals, so a residual-list success is not being confused with a correct polynomial expansion.

These tests verify the implementation on the specified finite samples. They do not replace the inductive and algebraic proofs, establish global uniqueness by testing alone, or prove the absence of a rational solution in the irrational-only example. That absence was proved arithmetically in [example 139.6](#). No Lean execution is included in the package.

[write] In this report the programs are shipped as `code/15-neural-precision-neural_certificates.py` and `code/15-neural-precision-test_certificates.py`, and the report, manifest and examples as `data/15-neural-precision-*`; the source's text and its package listing use the delivered paths. The README gives the rerun recipe on a copy.

144 Source 14 appendix: dependency and claim ledger

Claim	Mathematical dependency	Present verification
Two-counter halting completeness	Minsky's classical representation theorem, applied to a complete partial semidecider	External theorem, exact source instruction convention checked
General homogeneous lift	Finite rational polyhedral formulas; elementary Lipschitz estimates	Proof in theorem 113.1
Four-coordinate contraction	Explicit hats, clipped local maps, and $L = \max(2n + 1, 4)$	Proof; 5,000 instruction cases and 1,000 rational tests
Point/region separation	Radial clock and source halting completeness	Proof in section 115
Peak rationality and zero hardness	Tail bound and exact simulation	Proof in section 116 ; exact example checked
Bounded unique quartics	Integer circuit and natural complementarity	Proof; expanded example and 400 rejected mutations
Unbounded fixed polynomial	Computable enumeration and MRDP	Classical theorem; actual ProveIt interface inspected, not rebuilt here
Unbounded degree four	Natural addition/multiplication circuit flattening	Proof; no explicit universal coefficient list claimed
Historical firstness	Would require a broader priority investigation	Not claimed
Lean verification of new results	Requires the modules proposed in section 122	Not performed

145 Source 14 appendix: minimal decision and approximation algorithms

The point algorithm for a nonzero cone input, after handling the exceptional cases described in [theorem 115.2](#), is:

```
PointHit(F, B, y, z):
  r := radius(y); s := radius(z)
  if s <= 0 or s > r: return false
```

```

t := 0
while r > s:
    r := r / B
    t := t + 1
if r != s: return false
return iterate_exactly(F, y, t) == z

```

The peak approximation algorithm is:

```

PeakInterval(F, B, ell, y, T):
    r := radius(y)
    a := 0
    for t = 0, ..., T-1:
        a := max(a, ell(y))
        y := F(y)
    return [a, max(a, r / B^T)]

```

The width is at most rB^{-T} . When its lower endpoint is positive, a sufficiently longer finite run will produce a singleton interval and hence the exact rational peak. When the true peak is zero, this singleton event need not occur. Adding an unsupported “eventually it must” termination claim would reintroduce the halting problem.

146 Source 12 appendix: compact algorithms and file formats

146.1 Natural-number evaluation

The essential evaluator is the following; the archive adds validation and an output-size guard.

```

def F(n, clock):
    if n == 0:
        return 0
    t = valuation_2(n)
    u = n >> t
    q, left, right = decode_unit(u)
    if q == halt_state or q >= state_count:
        return 0
    q1, left1, right1 = machine_step(q, left, right)
    return encode_unit(q1, left1, right1) << clock(t)

```

For residue evaluation modulo 2^k , reduce the input first and avoid the shift if $h(t) \geq k$. A zero residue is answered by zero. This is valid because the output depends only on that residue, not because an unknown higher tail is assumed absent.

146.2 Polynomial representation

A sparse monomial is stored as an integer coefficient and a list of variable names, with repeated names representing powers. The empty factor list represents a constant. A residual is a list of such terms. The certificate JSON includes the public parameter name `n`, the full ordered witness-variable list, the time and input bounds, the machine table, and the residual list. Its polynomial is exactly the sum of squares of those residuals. The expanded example uses the same monomial format without the residual grouping.

The witness file records a natural value for every declared witness and the public input separately. Evaluation checks that the set of supplied witness names agrees exactly with the declaration. It then evaluates all residuals with exact integer arithmetic.

147 Source 12 appendix: proof audit and scope boundaries

The following dependency summary is intended to make the most vulnerable steps easy to inspect.

Result	Essential dependency or boundary
Unit Lipschitz estimate	Common state and scanned bit at high precision; interior tape-digit displacement at most two.
Global contraction	Separate same-shell and distinct-shell estimates; no monotonicity assumption on the clock.
Computable 2-adic map	Residue procedure never needs a global zero test.
Simulation	Nonzero unit updates are odd; halting-to-zero uses one extra iteration.
Finite realization	Bounded bootstrap and FIFO; not inferred from Lipschitz continuity alone.
Clock necessity	Eventual periodicity of internal states along the all-zero input and a fixed nonhalting suffix.
Clock sufficiency	Finite leading-zero controller and a finite family of fixed output delays.
Analytic horizon	Scalar local factorization and uniform entry into a small invariant ball; effectivity stated separately.
Bounded quartic uniqueness	Unique input digits, unique natural division, and deterministic identity-padded trace.
Unbounded quartic	MRDP followed by standard gate equations; no uniqueness of original MRDP witnesses.

Three claims are deliberately absent. We do not claim robust clopen universality. We do not claim global smoothness from asymptotic flatness. We do not claim a fixed-arity single-fold representation from a family whose arity grows with T and S . These absences are mathematical scope boundaries, not missing steps in the theorems that were actually stated.

148 Source 12 appendix: source provenance and reproducibility

Repository inspection was pinned to `b6bf6406a4c49017fedc6fba068b36c634987150`. The relevant inspected paths are

```

Computability/HilbertTenthProblem/README.md
Computability/HilbertTenthProblem/Lean/MRDP.md
Computability/HilbertTenthProblem/Lean/Diophantine/MRDP.lean

```

The source guide and actual public theorem agree that the polynomial and finite witness dimension are chosen before the natural input, including input zero. Their claims about repository build and axiom audits are not substituted here for a new local build.

The literature check consulted the primary sources listed below, including the automata/valuation interface and the clopen-observation obstruction. This is enough to situate the construction and avoid the most immediate misstatements, but not enough to certify priority for every theorem or combination of properties. A submission-ready novelty claim would require a broader review of iterated transducers, computational p -adic dynamics, and termination for regularity-restricted dynamical systems.

The archive's `verification_report.json` records the actual completed finite checks and the exact demonstration orbit. The Python code is independent of a network connection, computer algebra package, or Lean installation. The article can be rebuilt by two runs of `pdflatex article.tex`, or by `latexmk -pdf article.tex`.

149 Source 15 appendix: package layout and reproduction

The package contains the following principal files:

```
article.tex
article.pdf
README.md
verification_report.txt
source_manifest.json
code/neural_certificates.py
code/test_certificates.py
examples/fixed_trace.json
examples/projective_trace.json
examples/rational_synthesis.json
```

To rerun the exact checks, use Python 3.10 or newer:

```
python code/test_certificates.py
```

This recreates the example JSON files and the verification report. To build the article, use a standard TeX distribution with the packages named in its preamble:

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

No bibliography database, shell escape, external image, or downloaded font is required. The bibliography is contained in the source. The package does not contain copies of the cited research papers or of the ProveIt source tree.

150 Source 15 appendix: dependency and claim audit

Statement	Proof dependency and status
Scalar certificate	Elementary ordered arithmetic; complete proof in lemma 136.2 .
Canonical trace and first hit	Scalar lemma and induction; explicit compiler and exact examples supplied.
Projective real-input certificate	Rational polytope vertices, Cramer's rule, and the scalar lemma; fixed rational weights essential.
Four-layer synthesis equivalence	Arithmetic circuits, bounded positive chart, detector construction, and the four-square theorem; ties and positive output explicit.
Strict-target decidability	Continuity, rational density in the parameter cube, and classical real algebraic decidability; no extra equality targets.
Quartic natural lift	Common rational denominator and scaled clamp relations; multiple witnesses are expected.
Prime-support precision bound	Denominator invariant, distinct-state argument, and geometric-series estimate; no heuristic counting.
No computable precision ceiling	Classical rational neural universality and halting undecidability; not a new universality theorem.
Uniform unbounded polynomial	Application of the documented MRDP interface after proving recursive enumerability; does not inherit finite-horizon degree or uniqueness.

Statement	Proof dependency and status
Implementation validation	53,472 finite exact-arithmetic assertions; not an independent formal verification of the paper.
Novelty	Proposed refinements and synthesis reduction; priority and independent review not established.

151 Conclusions of Part VII

151.1 Source 14

The construction separates three tasks that are easy to conflate: locating a limit, predicting an exact transient observation, and representing a finite computation arithmetically. The limit is the known point zero and is approached at an explicit geometric rate. A single half-space observation can still encode the halting problem. Every bounded observation has a canonical quartic certificate, while the unbounded existential description uses MRDP and does not inherit uniqueness for free.

The strongest operational lesson is that global stability is not an exact verification theorem. Even rationality of every peak value, a computable Lipschitz bound, and a uniform effective approximation scheme do not supply a rational-input zero test. The proofs identify where the missing information lives: in arbitrarily small positive amplitudes and in the singular projective decoding at the attracting apex.

151.2 Source 12

The main construction separates three ideas that are often bundled together: local numerical stability, finite-state digit realization, and exact termination of an unbounded computation. A fixed finite transducer can induce a global strict contraction on $\mathbb{Z}_2$ while its iterated natural-number dynamics has a complete recursively enumerable extinction set. Every finite quotient is nevertheless nilpotent with an explicit precision-dependent bound.

The radial clock exposes a second separation. It changes convergence speed without changing the exact computation, and it is finite-state exactly when its added delay is eventually periodic. Differentiability at zero instead requires the added delay to tend to infinity. Arbitrarily strong asymptotic flatness at the attracting fixed point does not remove the hardness, whereas scalar analyticity does: an analytic contraction has a uniform finite extinction horizon, explicitly computable for rational polynomials under the stated hypotheses.

The Diophantine conclusion is not only an appeal to universality. The article provides an executable bounded compiler with exact counts, clock-independent normalized traces, and unique witnesses. Its boundary with fixed-arity unbounded representation is stated explicitly. The most concrete next milestone is a Lean-verified universal instantiation of the finite transducer compiler together with the bounded quartic correctness and uniqueness theorem.

151.3 Source 15

Exact rational neural computation exposes two different arithmetic resources. For a fixed network, the nonlinear-looking activation is controlled by a canonical complementarity relation, and every finite trajectory has a single quadratic natural-number certificate with unique witnesses. Even real-input finite reachability stays within rational polyhedral geometry, so a projective quadratic lift remains exact.

When the parameters are unknown rationals, weight–signal multiplication changes the geometry. Four clipped layers, bounded tied parameters, and a single positive output suffice to express all of $H10(\mathbb{Q})$; the direct integer lift is quartic. Replacing exact output fitting by wholly strict target conditions restores rational density and a decision procedure.

Finally, bounded state magnitude is not bounded computational memory. First acceptance forces distinct rational states, and their denominator support gives a quantitative precision requirement. In a fixed universal network, no computable precision ceiling can cover every accepting input. These statements supply a rigorous, explicit neural front end for the repository’s Diophantine program, while preserving the distinction between proved constructions, existing foundational theorems, and genuine open arithmetic questions.

152 Further research questions of Part VII

[write] Source 14 posed ten questions, source 12 twelve and source 15 twelve; all are printed. A [write] note after a question names the question of Parts I–IV, or of another source, that it duplicates or bears on.

152.1 Source 14: strict contractions

The following questions are left open by this report. Unless explicitly linked to an external source, they are proposed project questions rather than claims about the complete published open-problem landscape.

Research question 59 (Dimension and observation minimality). Can the point/half-space separation be realized by a rational continuous strict contraction in three coordinates, or in two, while keeping a single fixed linear observation and computable rational encodings? The present construction spends three coordinates on the cube simulator and one on the radial clock. A solution must preserve continuity, not silently use a piecewise discontinuous state selector. [write] Compare Question 24.

Research question 60 (An explicit small universal instance). Choose a published universal source program, normalize its exact input and halt conventions, and compile the complete instruction table. What are the

best resulting values of m , B , circuit depth, number of gates, and coefficient heights? Small-register-machine literature provides useful starting points, but register count and instruction set must be checked rather than inferred from the word “universal” [59]. [write] Compare source 12’s Question 69.

Research question 61 (Encoding cost). Can a different continuous cube simulator give a polynomial-time map from ordinary binary source inputs to rational initial vectors of polynomial bit length while retaining the four-coordinate global contraction theorem? The family 2^{-2^e} proves computable hardness but is quantitatively very expensive. An efficient solution would strengthen the complexity meaning of the substrate considerably.

Research question 62 (Smooth compact contractions). Can a C^∞ contraction on a compact rational region realize a comparable exact reachability separation with a natural finite effective description? The homogeneous route cannot simply be smoothed at zero, by [proposition 120.1](#). A successful construction would need to abandon exact degree-one homogeneity or change the encoding mechanism. [write] Compare Question 25 and source 12’s Question 73.

Research question 63 (Structural subclasses). Which additional restrictions force half-space reachability to be decidable: monotonicity in the coordinatewise order, nonnegative neural weights, a bounded number of activation patterns along every orbit, or a genuine Bellman-operator structure? The present hats and zero tests use signed coefficients. The Bellman results [57] concern a structurally narrower class and a different target specification. [write] Compare, in spirit, Question 21.

Research question 64 (Relative-precision semantics). Suppose admissible observation errors shrink proportionally to the radius. Which machine encodings remain reliable, and what additional dependence on the counter magnitudes is needed? The control-state margin is radial, but large counters are represented by very small ratios. A fault-tolerance claim must analyze those ratios as well as the final halt readout. [write] Compare Question 31.

Research question 65 (Geometry of the zero set). The zero set of $\mathcal{P}$ is effectively closed, while the positive locus is locally rational piecewise affine. What geometric restrictions on a computably compact subset of $\mathcal{C}$ make rational zero membership decidable? Can one give useful sufficient conditions in terms of a finite polyhedral decomposition or an effective lower bound on positive peaks? [write] Compare Question 32 and source 12’s Question 80.

Research question 66 (Certificate optimization). Can the exact gate count $G = 9m + 2d_M - 3$ and first-hit witness count $(2G + 5)T + 1$ be reduced without losing canonical witnesses? In particular, can state selection share more ReLU gates, and can the two natural variables per gate be replaced by a smaller globally canonical encoding? Any claimed improvement should provide both a count and a unique-witness proof at zero-valued gates. [write] Compare Question 2 and source 12’s Question 76.

Research question 67 (A verified end-to-end compiler). Can ProveIt prove, from one counter-program syntax tree, the conjunction of exact simulation, the global metric bound, the half-space observation lemma, and the semantics of the exported quartic? This would turn a collection of existence arguments into a reusable certified substrate compiler. A useful first milestone is the three-state transfer example, followed by a fixed universal source with a separately verified interpreter theorem. [write] Compare Question 18.

Research question 68 (Fixed-arity witness behavior). Which restricted source-program classes admit fixed-arity unbounded Diophantine encodings with controlled witness multiplicity? Bounded trace uniqueness alone is insufficient. A meaningful advance would identify a new restricted class where the additional MRDP-style witnesses can be controlled, rather than claiming a general single-fold result from deterministic semantics. [write] Compare Question 19 and source 12’s Question 77.

152.2 Source 12: contracting 2-adic dynamics

The following questions are proposed research directions, not assertions that each is a documented open problem in the literature. The first group has relatively short, checkable targets; the later group asks for genuinely stronger theory.

Research question 69 (A completely audited universal finite transducer). Instantiate the compiler with a small, independently verified binary universal machine. Export and minimize the reachable transducer, and prove its equivalence to the machine in Lean. What is the smallest state count obtainable while retaining the explicit $1/2$ -contraction property? [write] Compare Questions 18 and 23.

The present archive supplies the generic compiler and a nonuniversal example. A universal instantiation with a traceable source machine and an end-to-end correctness proof would turn the existence result into a particularly useful benchmark.

Research question 70 (The state-delay tradeoff). How far can $K = r + 3$ and the exponential bootstrap bound be reduced? Is there a lower bound relating the maximum negative tape-digit displacement, header complexity, contraction factor, and minimal transducer state count?

An immediate experiment is to minimize generated tables for families of machines, then identify which stored prefix bits are redundant. A theorem should distinguish a smaller implementation from an improved worst-case bound.

Research question 71 (Automatic coefficient descriptions). Can one compute a compact van der Put coefficient description for the constant-drift compiler directly from the source table, and derive the clock classification through the corresponding automatic-sequence kernel?

Anashin’s finiteness criterion provides a natural alternative language [42]. The target is an explicit coefficient algorithm or a quantitative kernel bound, not a second citation for finiteness alone.

Research question 72 (Beyond eventual periodic clocks). For computable noneventually-periodic delay, what is the least additional streaming memory needed to realize F_h ? Can clock growth or descriptive complexity yield sharp lower bounds for pushdown, counter, or log-space transducers?

The finite-state case is completely settled by Theorem 126.4. The question is the next level of a memory hierarchy, not whether bounded but aperiodic delay is finite-state.

Research question 73 (Regularity between flatness and analyticity). Is there a globally C^1 , or a suitably defined globally C^∞ , strict contraction of $\mathbb{Z}_p$ whose natural-input exact extinction set is recursively enumerable complete? Which notions of non-Archimedean smoothness force a finite extinction horizon? [write] Compare Question 25, which Theorem 129.1 partly answers in the scalar 2-adic case.

Our asymptotically flat map does not answer this. A productive first step is to analyze differentiability on nonzero shells and identify the precise obstruction caused by digit interleaving.

Research question 74 (Several analytic variables). For a polynomial or analytic map $f: \mathbb{Z}_p^d \rightarrow \mathbb{Z}_p^d$ that is a global strict contraction fixing zero, must exact extinction of rational inputs be decidable? Does a uniform horizon exist under a finite-fiber or isolated-zero-fiber hypothesis?

The scalar argument extends immediately when a sufficiently small neighborhood has no nonzero point in $f^{-1}(0)$. The hard issue is what happens without that hypothesis. Positive-dimensional zero fibers prevent simply repeating the scalar proof.

Research question 75 (Effective analytic presentations). For which representations of an analytic contraction can the radius B in Theorem 129.1 be computed uniformly? Can a verified tail bound be converted into a practical extinction algorithm with a useful complexity estimate?

The rational-polynomial case is settled here. Infinite series require explicit effective control of tails and of a first nonzero coefficient; an arbitrary coefficient stream need not supply that information.

Research question 76 (Smaller bounded unique-witness certificates). Can one reduce the leading coefficient $3m + 6$ in the witness count while preserving uniqueness, degree four, bounded residual coefficients, and linear sparse size? Can any substantial machine class admit degree-three single-polynomial certificates with comparably controlled witnesses? [write] Compare Questions 2 and 17, and the sharp-degree question of the sibling report `canonical-diophantine-certificates`.

A promising concrete direction is binary rather than one-hot control, but transition selection and preservation of degree two in the residuals must be handled explicitly. Eliminating variables by substitution can increase degree or coefficient size.

Research question 77 (Unbounded compression without spurious multiplicity). What additional hypotheses on a computational substrate permit a fixed-arity finite-fold representation whose witnesses retain a canonical trace interpretation? Can the bounded certificates be compressed in any nontrivial restricted class without introducing uncontrolled witness multiplicity? [write] Compare Question 19 and the restricted-class compression questions of the sibling report `canonical-diophantine-certificates`.

This is deliberately separated from the proven bounded single-fold result. No general fixed-arity single-fold MRDP claim follows from the construction.

Research question 78 (Growing-precision observers). At time j , let an observer inspect the residue modulo $2^{k(j)}$ rather than at a fixed precision. Which growth rates and input-length allowances recover the simulated configuration, and what exact resource bounds result?

For constant drift, the unit marker itself moves to position $jK + t_0$. An observer whose precision remains below that position sees only zero. Above it, recovering the full finite tape additionally requires its current length. This gives a concrete starting point for a hierarchy of observation costs.

Research question 79 (Other primes and other digit spaces). Develop an equally explicit compiler over $\mathbb{Z}_p$ for arbitrary primes, or over formal power-series rings over finite fields. Which parts of the clock classification depend only on a finite digit alphabet and a valuation, and which parts require the binary tape encoding?

The radial contraction argument is valuation-theoretic, while the specific unit map and quartic packing use binary digits. Separating those layers should make a generalization substantially cleaner than redoing the entire proof for each prime.

Research question 80 (Measure and density of exact extinction). For a fixed universal compiler, what can be said about the Haar measure of its extinction basin in $\mathbb{Z}_2$, and about natural or logarithmic densities of its natural-number extinction set? Can prescribed computability properties be obtained while keeping the constant-drift map finite-state? [write] Compare Question 32.

The set A of halting or invalid units is clopen. Its shell union $H = \bigcup_{t \geq 0} 2^t A$ is open, and the full extinction basin is $\{0\} \cup \bigcup_{j \geq 0} F_h^{-j}(H)$. Thus its part away from zero is open, while zero need not be an interior point. Undecidable membership alone does not determine the measure or its computability. A proof must analyze this construction rather than identify it informally with a halting probability.

152.3 Source 15: precision is memory

The questions below are specific to the constructions of this article. They are not assertions that every neighboring literature problem remains open. In particular, established real-training complexity results should not be relabeled as unanswered rational questions.

Research question 81 (Formal, source-preserving certificate extraction). Can ProveIt extract the actual sparse polynomial and its witness constructor from a checked finite-network specification, with machine-checked proofs of degree, variable count, and inverse decoding? A useful first milestone is the scalar lemma plus the twenty-variable example, followed by the $T = 0$ -safe induction. [write] Compare Question 1.

Research question 82 (Fewer canonical coordinates). Can the $4nT$ trace witness count be reduced while keeping an explicitly quadratic, orthant-nonnegative polynomial and unique witnesses at both thresholds? Eliminating $C = D - X$ syntactically makes $(D - X)H$ potentially negative away from valid states, so that naive elimination invalidates the single-sum proof. A reduction needs a different global nonnegativity argument or a different certificate architecture. [write] Compare Question 2.

Research question 83 (Rational synthesis without ties). Which tie patterns are actually necessary in [theorem 139.2](#)? Can a comparably explicit $H10(\mathbb{Q})$ reduction be made with untied edge parameters, or can selected sparse tie classes be proved decidable? Copies enforced by extra exact equations must be counted as part of the architecture, not assumed automatically.

Research question 84 (Closed targets with all parameters bounded). Can the positive-chart condition $v > 0$ be eliminated while retaining a compact rational parameter cube, only closed output conditions, and a fixed small depth? The unbounded-gain corollary does not answer this: it replaces positivity with noncompactness. Understanding this trade-off would sharpen the arithmetic boundary of exact rational synthesis.

Research question 85 (Depth, width, and bilinear structure). Can the four-layer reduction be lowered to three non-input layers without increasing the parameter domain or strengthening the target language? What happens when the number of parameter-activation product nodes is bounded, or their incidence graph is acyclic in a stronger sense than the feedforward order? A useful classification should track these products, not just count all neurons.

Research question 86 (Effective open-target bounds). [Theorem 139.8](#) gives a decision procedure but no attractive complexity bound or denominator estimate. Can one give useful rational witness-height bounds in terms of depth, coefficient height, and a specified positive margin? Bounds should distinguish an explicit margin from mere strict feasibility, which may have arbitrarily small margins across instances.

Research question 87 (Canonical projective scales). Can a rational-trajectory certificate enforce the least common denominator without introducing an uncontrolled multiplicity of greatest-common-divisor or Bezout witnesses? The current projective construction is intentionally nonunique. A canonical scale would be valuable for counting and extraction, but does not follow from the canonical fixed-input theorem.

Research question 88 (Sharpness of the prime-support height estimate). For a fixed rational network and denominator-prime set S , how close can first-hit runs come to the order $H^n(\log H)^{|S|-1}$? The proof counts all admissible grid points, not only states dynamically reachable by one affine-clipping map. Matching constructions or stronger dynamical restrictions would reveal how much of the arithmetic state space can actually be used as memory.

Research question 89 (Precision growth for a chosen universal network). For a completely specified universal network and binary input encoding, compare the maximum accepting-run denominator height on inputs of length at most ℓ with the associated halting-time maximum. The no-ceiling result proves noncomputability of any total upper bound, but does not give a precise growth equivalence or eventual domination theorem.

Research question 90 (Proof-producing safety rather than counterexamples). The projective quadratic encodes the *existence* of a finite counterexample. Can a complementary certificate system give concise independently checkable proofs that all inputs in a rational polytope are safe through a fixed horizon? Rational polyhedral separation is available on individual activation cells; a useful construction must also account for coverage of all cells without silently replacing a universal statement by an existential one. [write] Compare [Question 23](#).

Research question 91 (Other continuous-looking computational substrates). Which exact rational systems with piecewise-affine transitions admit the same canonical scalar-style witnesses? ReLU and fixed-threshold clipping are natural starting points. Contact dynamics, reset systems, and hybrid models require care when a physical transition relation is multivalued: uniqueness of a numerical output need not imply uniqueness of contact or event witnesses. [write] Compare [Question 27](#).

Research question 92 (Cost of fixed-arity compression). Starting from this transparent neural front end, what exact coefficient, operation-count, and witness bounds result from a specified ProveIt universal-polynomial route? These are concrete engineering-and-proof questions. They should be answered for the chosen route without claiming that the finite-horizon uniqueness theorem settles general finite-fold Diophantine representation. [write] Compare [Question 9](#) and the fixed-arity compression question of the sibling report [canonical-diophantine-certificates](#).

A Reproduction notes and algorithmic contracts

A.1 Manuscript 08: using the delivered files

[write] Manuscript 08’s appendix listed the files of its archive. In this report they are:

Shipped path (delivered path)	Purpose
<code>article.tex</code> (08’s <code>article.tex</code> , now merged)	Source of this report; 08’s own PDF is not shipped.
<code>code/08-probabilistic-laws-normalization.py</code> (<code>code/normalization.py</code>)	Exact budget evaluator and residual compiler.
<code>code/08-probabilistic-laws-quantum_exact.py</code> (<code>code/quantum_exact.py</code>)	Exact ring arithmetic and small state vectors.
<code>code/08-probabilistic-laws-prefix_allocator.py</code> (<code>code/prefix_allocator.py</code>)	Canonical labeled prefix allocation.
<code>code/08-probabilistic-laws-verify.py</code> (<code>code/verify.py</code>)	Reproducible test and artifact generator.
<code>data/08-probabilistic-laws-verification.json</code> (<code>artifacts/verification.json</code>)	Test counts, versions, and limitations.
<code>README.md</code>	The report README (08’s delivered README is replaced).

The two files beginning `data/08-probabilistic-laws-normalization_quartic_N2_m2` contain the residual system with example assignment and the full expanded polynomial. The file `data/08-probabilistic-laws-HTH_prefix_allocation.json` records the finite prefix-code example. The source rebuilt its artifacts with `python code/verify.py` from its archive root; the README gives the equivalent command on a copy with the delivered layout. The Python scripts require Python 3.10 or newer and SymPy. Tests were run with Python 3.13.5 and SymPy 1.14.0. A newly generated report records the versions actually used. The source code uses zero-based coordinate indices; the mathematical exposition uses indices $1, \dots, m$. The earliest-maximum rule is the same ordering in both conventions.

A.2 Manuscript 07: algorithmic contracts and reproduction notes

A.2.1 Rational Gram search

The search routine uses a common-denominator presentation. At height h , it considers denominators $1 \leq q \leq h$ and integer column vectors in $[-h, h]^s$. Candidate columns are first filtered by the equations

$$\|v_j\|^2 = q^2 Q_{jj},$$

and then by the mutual inner products $v_i^\top v_j = q^2 Q_{ij}$. A successful tuple gives $B = q^{-1}(v_1, \dots, v_d)$. Every rational matrix has such a representation for some finite h , so Proposition 15.4 proves termination when $s = d + 3$ and no height cap is imposed. Repeating earlier candidates is inefficient but harmless.

The implementation checks positive definiteness first. A finite height cap raises a timeout-style exception and is explicitly not interpreted as nonexistence. The same routine may be called with a smaller s , but then the general termination guarantee no longer applies.

A.2.2 Polynomial serialization

The expanded sample polynomial is stored as a list of integer coefficients and monomials. A monomial is a sorted list of variable names, with a name repeated according to its exponent. Thus

```
{"coefficient": -2, "monomial": ["x", "x", "y"]}
```

means $-2x^2y$. The file also lists the free input names and existential witness names. The symbolic checker includes the free H^+, H^- coordinates in the degree computation, which verifies the uniform degree claim rather than only a specialization to constant matrix entries.

A.2.3 Independent checks worth repeating

The most failure-sensitive identities are the completeness equation, the order of multiplication in the word evaluator, the fact that a zero source word may need one extra target letter, and the uniqueness of the natural sign split. These are all tested in the package and proved in the text. A useful independent review should also inspect that the inactive right block columns are zero and that no restriction on the allowable target outcome language is silently being imposed.

[write] The primary artifacts of manuscript 07 were the manuscript source and PDF, the two Python files, and three example JSON files. Here the manuscript is merged into `article.tex`, its PDF is not shipped, and the Python and JSON files are shipped under the prefixed names of the README. The source compiled with a standard pdfL^AT_EX installation, and its test command is `python code/run_checks.py` in the delivered layout. The build does not fetch the repository, run Lean, or require a network connection.

B The exact PWA evaluation recipe

The reference compiler evaluates the global extension without expanding a large list of affine matrices. Locate a point in a rectangular cell with lower corner a , upper corner b , and normalized coordinates $t_i = (x_i - a_i)/(b_i - a_i)$. Choose a deterministic ordering $t_{\sigma(1)} \geq t_{\sigma(2)} \geq t_{\sigma(3)}$. Let $v_0 = a$ and form v_j by changing coordinates $\sigma(1), \dots, \sigma(j)$ to their upper endpoints. Then

$$F(x) = (1 - t_{\sigma(1)})F(v_0) + (t_{\sigma(1)} - t_{\sigma(2)})F(v_1) + (t_{\sigma(2)} - t_{\sigma(3)})F(v_2) + t_{\sigma(3)}F(v_3). \quad (\text{B.1})$$

All four coefficients are nonnegative and sum to one. At ties, the formula agrees on the common simplex face. At rectangular boundaries, it agrees with the adjacent cell formula. This makes endpoint behavior explicit, including points exactly on $1/3$, $2/3$, or a terminal slab boundary.

The Python compiler first checks terminal slab membership at a vertex, then active emergency membership, then active low-coordinate digit membership. Because the relevant prescribed sets are disjoint, this order is a convenient implementation convention rather than a mathematical priority rule. Vertices outside all prescribed sets get the rejecting sink value. The cache stores rational vertex values only.

C A finite-certificate pseudocode contract

A proposed safety certificate contains a mesh index n , an ordering of the vertices consistent with the public grid convention, and a bit for each vertex. An untrusted producer may generate it by any method. A trusted checker can use the following contract.

```

N := D * 2^n
V := all d-dimensional N-grid vertices, plus the exact source if absent
T := grid vertices belonging to a target box
lambda := (L + 1) / N
reject unless every certificate entry is 0 or 1
reject unless the source entry is 1 and every target entry is 0
for each ordered pair (u, v) in V x V:
    if ||v - f(u)||_infinity < lambda:
        reject if bit[u] = 1 and bit[v] = 0
accept, certifying rho >= (L + 1) / (2*N)

```

The inputs f, H, L, D are not arbitrary unchecked text. The checker assumes a validated map description, a valid global Lipschitz bound, a target inside the cube, and a denominator-clearing integer. These form its specification. The companion implementation checks several syntactic conditions but cannot prove an externally supplied Lipschitz bound for an arbitrary Python callable. For maps produced by the TM compiler, lemma 27.2 supplies the bound.

The accepted certificate guarantees safety for every per-step budget strictly below the reported radius bound. This final strictness is not an implementation detail: critical-value attainment can fail, and the radius lower bound alone does not settle the equality case.

D Source and verification notes

The repository was inspected at the pinned commit printed in Section 1. Bibliography entries for repository sources use that commit rather than a moving branch. The references on weakly computable reals, probabilistic-program hardness, exact Clifford+ T arithmetic, and max-relative entropy provide the established context; the proofs of the specific budget and compiler statements appear in full in this manuscript.

No claim is made that the literature search establishes first publication of every consequence proved here. In particular, general c.e.-to-Diophantine existence, fair-coin representation of lower semicomputable output laws, and the entropy formulas are not advertised as new discoveries. The primary new deliverable of this work is the explicit synthesis of these ideas with the canonical finite normalization certificate, including its verified implementation and its carefully separated infinite semantics.

E Proof audit, provenance, and non-claims

E.1 Dependencies

The core construction and quantitative results use elementary finite graph theory, real infima, rational affine geometry, and the standard existence of a binary universal Turing machine. MRDP is used only to pass from an enumerable finite-certificate predicate to one fixed polynomial. The recurrent ReLU corollary additionally uses the cited max-min representation theorem. Earlier dynamics literature motivates and contextualizes the questions; its qualitative robustness results are not substituted for the proofs of the explicit estimates here.

E.2 Points checked in the mathematical argument

The real infimum need not be attained. The source is never inadvertently rounded. Target rounding works for closed boxes with aligned endpoints, including degenerate ones. The graph is complete, so a bottleneck minimum and finite cut minima exist. Strict versus non-strict edge thresholds are kept separate. Integer polynomial subtraction is signed. The quartic family has varying arity, while the MRDP polynomial has fixed arity.

The Turing update formulas map entire guard boxes into the cube. Terminal slabs have genuine neighborhoods, and the extension is continuous across all faces. The rejected computation has a finite-prefix error estimate followed by an invariant rejecting slab. Emergency timing is indexed so that a first rejection at T can be bypassed at time T , not one step too late. The universal map is fixed before the enumerable pair is selected. The original machine's halting time is never silently identified with the universal interpreter's rejection time.

E.3 Limits of the claim

No named classical open conjecture is declared solved. The proposed three-dimensional joint theorem package is presented with proofs, but without an exhaustive novelty certification or independent peer review. No

minimal dimension, optimal Lipschitz constant, optimal polynomial degree, optimized network size, or finite-fold MRDP result is asserted. No ordinary polynomial map, smooth flow, chemical network, or physical device is claimed to implement the construction without additional work.

The proofs and code were prepared with ChatGPT for this request. All reported finite tests were actually run on the included code. No Lean verification, repository-wide audit, expanded universal transition table, extracted ReLU weight list, or expanded fixed MRDP polynomial is included. The archive contains the precise ingredients that were produced, rather than placeholders masquerading as those further artifacts.

F [write] Provenance of this report

This report was built from three manuscripts of batch 60, delivered in the arrival commit `ccfb084ad` and placed in the placement commit `7498484af`. All three are dated 30 September 2026 and pin ProveIt at `f608f1cb3c5be8a736df1328c01b293aabfccf4e`.

No.	Archive	Contribution
07	<code>quantum_diophantine_research</code> (22-page A4 PDF, not shipped)	Rational Gram packing, undecidability for seven rational 4×4 outcomes, bijective bounded-history quartics, the fixed MRDP polynomial for <code>QMort_{4,7}</code> , probability gaps and perturbation bounds, nonnegative decidability; 32,418 exact checks; ten questions.
08 (base)	<code>ProveIt_Diophantine_Quantum_Normalization</code> (29-page Letter PDF, not shipped)	The front matter, Part I, Sections 11, 12 and 33; the explicit single-fold normalization quartic; twelve questions.
09	<code>ProveIt_Diophantine_Barriers</code> (26-page Letter PDF, not shipped)	Part III: horizon-free radius approximation, cut duality and complete quartic safety certificates, the three-dimensional universal trichotomy and its arithmetic classification, transfer and ReLU corollaries; 5,660 exact checks; ten questions.

Where the merge had to choose:

- Manuscript 08 is the base: its framework (output laws of classical and quantum programs) contains 07’s quantum setting as one kind of program, and 09 is a separate spine. 08’s section order is kept, except that its quantum front end and operator budget open Part II beside 07 rather than closing Part I.
- 07’s and 09’s introductions are split: their repository paragraphs are printed in Section 1, the rest opens their Part. 07’s implementation, formalization, question and conclusion sections, and 09’s test, formalization, question and conclusion sections, are printed with 08’s in Part IV and the appendices; 09’s worked example stays in Part III. The three abstracts and status statements are printed in Section 1.
- Every result, proof, example, remark, question and limitation of the three sources is printed. No result was duplicated across the sources, so no second routes arise.
- Labels were prefixed `pqc:` (08), `pqc:qm:` (07) and `pqc:cb:` (09); labels added here are `pqc:conv:`, `pqc:part:`, `pqc:app:provenance`, `pqc:qm:rem:chainformal` and the `pqc:q:`, `pqc:qm:q:`, `pqc:cb:q:` labels on research questions.
- Symbols and macros: Section 2.4. File names: shipped names throughout; 08’s archive table and 07’s and 09’s reproduction paragraphs were rewritten for them.
- Bibliography: the three lists are merged. 07’s and 08’s entries named `matiyasevich` are different works (the 1993 book, the 1971 paper) and both are kept; 07’s entry for ProveIt’s MRDP guide is kept apart from 08’s entry for `MRDP.lean`. 09 cited the same two repository files at the moving `main` branch; those citations now point to the pinned entries, with 09’s remark that the guide was last modified on 24 September 2026.

F.1 [write] Batch 62: Parts V–VII

Eight manuscripts of batch 62, all dated 30 September 2026, were added as Parts V–VII. They arrived in three commits and were placed, with their code and data, in the placement commit `adeddcecc`; their manuscripts, READMEs, PDFs and checksum manifests are not shipped and survive in the arrival commits. They are called here by their shipped file prefixes, sources 10–17, which continue this report’s own sequence after 07–09.

Source	Batch 62	Archive	Arrival	Pin
10	06	Diophantine_Certificates_for_Coherent_Computation	b57c0b5ff	ccfb084ad
11	07	proveit_rational_rotations_research	b57c0b5ff	b6bf6406a
12	08	Diophantine_Contracting_2Adic	b57c0b5ff	b6bf6406a
13	09	Quartic_Probability_Landscapes	b57c0b5ff	b998f70c6
14	11	strict_contractions_research	6d04e1e38	526c2557f
15	12	Precision_Is_Memory_Diophantine_Neural_Computation	cb1646dc4	526c2557f
16	13	dense_rotations_research	cb1646dc4	e18718e83
17	14	ProveIt_Reversible_Equilibria	cb1646dc4	e18718e83

Their contributions:

- Source 10: Part VI: unique-witness contraction certificates over $\mathbb{Z}[e^{i\pi/4}]$, the $\mathbb{Q}(\sqrt{2})$ comparator, exact circuit probabilities with counts and heights, graph-state branches, the $\text{NP} = \text{coNP}$ limit, the Cauchy-name obstruction, positive and almost-sure halting of quantum controllers; twelve projects.
- Source 11: Part VI: the shared core’s main proofs (denominator rigidity, polynomial-time decoder, matrix membership), the Miller instantiation (polynomial-time membership, c.e.-complete state transfer, order-five recurrence, no recursive precision bound), the state-transfer quartic; seven questions.
- Source 12: Part VII: the 2-adic 1/2-contraction with c.e.-complete exact extinction, Mealy realization, clock classification, flatness, the analytic horizon theorem, the bounded quartic; twelve questions.
- Source 13: Part V: well-conditioned sparse quartics, rounding and topology, the degree-two/degree-three counting frontier, left-c.e. realization with uniform constants, quantifier normal forms, finite mean, substrate transfer; ten questions.
- Source 14: Part VII: the rational CPWL 1/2-contraction of $\mathbb{R}^4$, c.e.-complete half-space reachability, decidable point reachability, the peak function, the ReLU compiler and first-hit quartic; ten questions.
- Source 15: Part VII: trace quadratics for clipped networks, projective certificates, synthesis equivalent to $\text{H10}(\mathbb{Q})$, strict-target decidability, denominator height as memory; twelve questions.
- Source 16: Part VI: the shared core’s second routes (finite-field tables, residue decoder, rank- d embedding, Tarski net), the complete instantiation (local completeness, precision gap, depth of degree $0'$, dimension obstruction), the matrix-synthesis quartic; ten questions.
- Source 17: Part V: reversible chains with a sharp common gap, rationality versus eventual periodicity, one-defect halting, the rational/Liouville dichotomy, Thue–Morse transcendence, prefix quartics; twelve questions.

The title pages, not printed, read: source 10 “Research manuscript prepared for Vladimir Reshetnikov”; source 11 “Research note prepared for the ProveIt project”; source 12 an AI-assisted research article; source 13 “Prepared for Vladimir Reshetnikov’s ProveIt research program”; source 14 “Research report for the ProveIt project”; source 15 “Research article prepared for the ProveIt project”; sources 16 and 17 research drafts “prepared for Vladimir Reshetnikov”. Wording that names an AI assistant or a company is not reproduced; all eight are AI-assisted and unrefereed.

Where the merge had to choose:

- *Placement.* The eight are exact-probability, exact-gate or continuous-state work, this report’s remit. They form three new Parts after Part IV, so that no existing section, theorem or equation renumbers, and so that the 07/13 core and the 08/11 comparison are each stated once. Each Part opens with its conventions and ends with its conclusions and questions; each source’s appendices are printed inside its Part.
- *Sources 11 and 16* share one arithmetic core, printed once in Section 84. Source 11’s statement and proof are used where both prove the same result, because its decoder is polynomial-time and its approximation bound is explicit; source 16’s finite-field tables, left-peeling decoder, block-expansion argument and Tarski net are kept as marked second routes. The two free-group embeddings index generators differently and both are kept. Merged statements carry both sources’ labels. Three pairs of questions are merged into one environment each, with both texts.
- *Sources 14 and 12* reach one phenomenon by independent routes; both are printed in full, with a written term-by-term comparison (Section 134).
- *Re-derivations* of Parts I–III (source 13’s interior threshold table, source 17’s almost-sure cuts, source 14’s no-gap corollary, source 10’s positive and almost-sure halting) are cited to the earlier statement and kept as second routes; only source 13’s table was not reprinted, apart from its three rows that Part I lacks. Source 10’s two examples are also Part II’s example and are printed with a pointer to it.
- *Research questions.* Sources 11 and 16 posed theirs as subsections and source 10 as a numbered list; source 11’s and 16’s are numbered here as research questions. Source 13’s numbered paragraphs and source 10’s list are kept in their forms. Every question of the eight sources is printed.
- *Labels.* Each source’s labels are prefixed `pqc:cc:` (10), `pqc:rr:` (11), `pqc:ad:` (12), `pqc:ql:` (13), `pqc:sc:` (14), `pqc:pm:` (15), `pqc:dr:` (16) and `pqc:rv:` (17); written material uses `pqc:gates:`, `pqc:contr:` and `pqc:conv:b62-`. Every label of the eight manuscripts is kept. Labels were added to thirteen existing research questions and to one existing example; no existing label was renamed.
- *Symbols* renamed in the new material are listed in the conventions of each Part: source 10’s profile $(L_{\text{leaf}}, T_{\text{mul}}, O)$; source 11’s quaternions a, b , spin map Φ and evaluation η ; source 13’s level η_* ; source 14’s ReLU (the source printed ρ) and exact predicates Hit^{ex} , Safe^{ex} . Source 17’s environment b and source 13’s thresholds a/b keep their letters and are disambiguated by the Part V table.
- *Corrections.* Source 12’s equation for R_{eq} lost a backslash in the delivered source and printed R_{meq} ; it is printed as intended. Source 12 gave the Delvenne–Kürka–Blondel article’s pages as those of its author version; the journal pages are added. Source 11’s phrase “the companion guide” now names the file. Source 17’s citations of this report are internal cross-references. Escaped underscores inside file names are unescaped.
- *Bibliography.* The eight lists are appended to this report’s. An entry for the same work in two batch-62 sources is printed once (Higman, Bogopolski–Ventura, Bell–Potapov, Moore’s 1990 letter, Matiyasevich’s 1970 note), and the Kaminski–Katoen and Giles–Selinger entries of sources 10 and 13 are this report’s existing entries. Different editions of Tarski’s decision method (sources 15 and 16) and the sources’ pinned ProveIt entries are kept separately. Source 17’s entry for this report’s README is omitted.

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- [12] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics **7**, Springer, 1973. Chapter IV, *Quadratic Forms over $\mathbb{Q}_p$ and over $\mathbb{Q}$* , pp. 27–47. doi:10.1007/978-1-4684-9884-4_4. Classical background for the rational quadratic-form inputs; these are not originality claims of this manuscript.
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